ORGANIC REACTION MECHANISM-II (ORM-II)

(I) Electrophilic Substitution reaction (S_E)

(II) Free radical substitution " " (S_{FR})

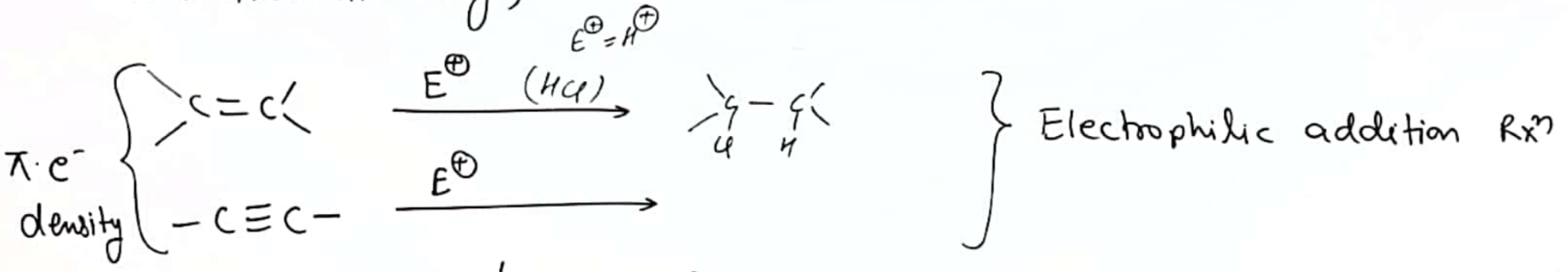
(III) Electrophilic addition " " (A_E)

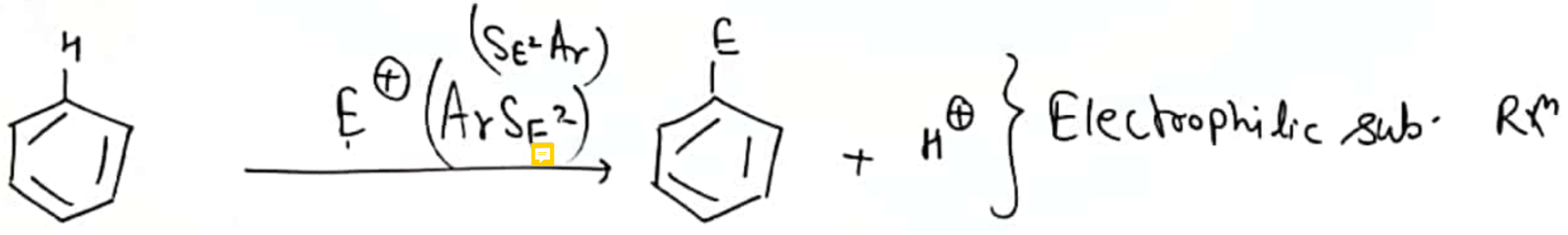
(IV) Free radical addition reactions (A_{FR})

ORGANIC REACTION MECHANISM-II (ORM-II)

(I) Electrophilic Substitution reaction (S_E)

It is characteristic reaction of aromatic compounds (due to π e⁻ density on aromatic ring.)





ORGANIC REACTION MECHANISM-II (ORM-II)

(I) Electrophilic Substitution Reactions (S_E)



if $E^{\oplus}$ = $\overset{\oplus}{N}O_2$ (nitronium ion) ⇒ Nitration

if $E^{\oplus}$ = SO_3 ⇒ Sulphonation.

$E^{\oplus}$ = $X^{\oplus}$ (Halonium ion) ⇒ Halogenation

$E^{\oplus}$ = $R^{\oplus}$ ⇒ Friedel craft alkylation.

$E^{\oplus}$ = $R-\overset{O}{\overset{\|}{C}}\oplus$ = " " acylation.

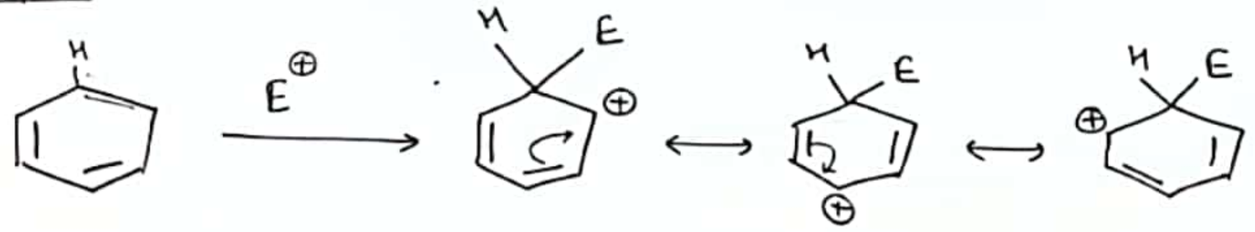
$E^{\oplus}$ = $\overset{\oplus}{N}O$ (nitrosonion ion) ⇒ Nitrosogation. etc

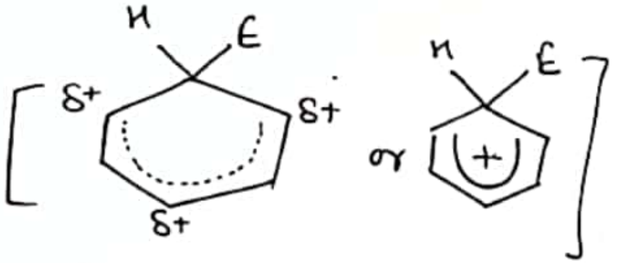
ORGANIC REACTION MECHANISM-II (ORM-II)

(I) Electrophilic Substitution Reactions (S_E)

Mechanism for $Ar S_N 2$

Step 1





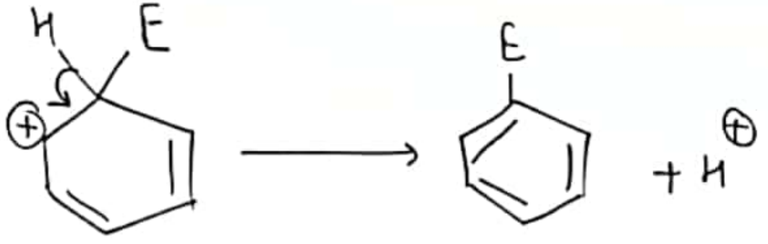
⇒ Resonance hybrid

⇒ Arenium ion

⇒ σ complex

⇒ Wehland intermediate

Step 2



ORGANIC REACTION MECHANISM-II (ORM-II)

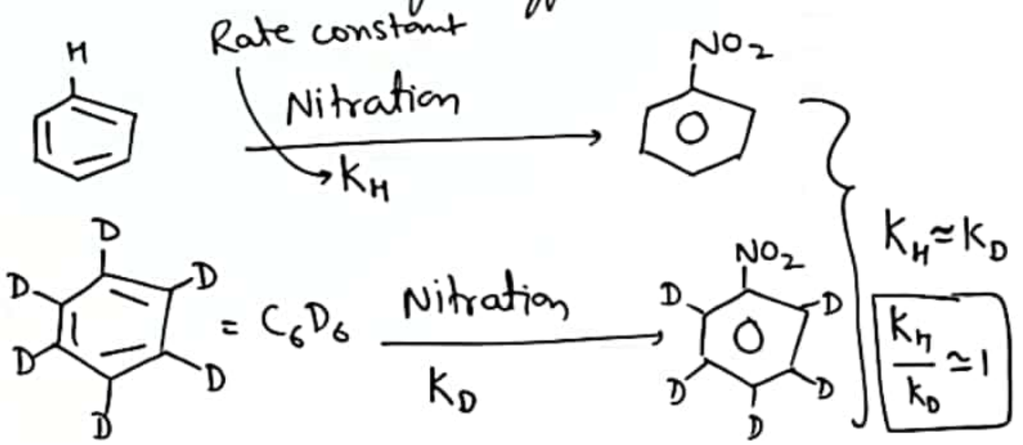
(I) Electrophilic Substitution Reactions (S_E)

⇒ Generally I^{st} Step = RDS (Slowest step) but for sulphonation and iodination → Step-2 = RDS

⇒ Generally Kinetic Isotopic Effect (KIE) is not applicable for S_E Rxn but for sulphonation and iodination, this effect is applicable

Kinetic Isotopic Effect:-

Effect on reaction rate/reactivity due to presence of diff. isotopes.



ORGANIC REACTION MECHANISM-II (ORM-II)

(I) Electrophilic Substitution Reactions (S_E)

C–H < C–D

$C_6H_6 \xrightarrow[K_H]{\text{sulphonation}}$

$C_6D_6 \xrightarrow[K_D]{\text{Sulphonation}}$

$K_H > K_D$

$\frac{K_H}{K_D} > 1$

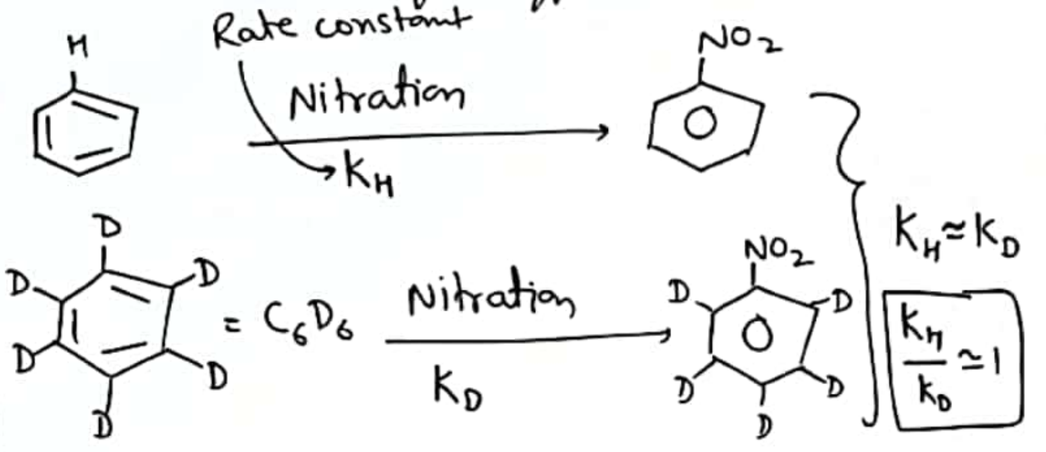
KIE applicable

⇒ When C–H is dissociates in RDS then KIE is applicable

(due to C–D > C–H Bond energy)

Kinetic Isotopic Effect:–

Effect on reaction rate/reactivity due to presence of diff. isotopes.



No KIE

ORGANIC REACTION MECHANISM-II (ORM-II)

(I) Electrophilic Substitution Reactions (S_E)

Factors affecting reactivity of S_E Rx^n

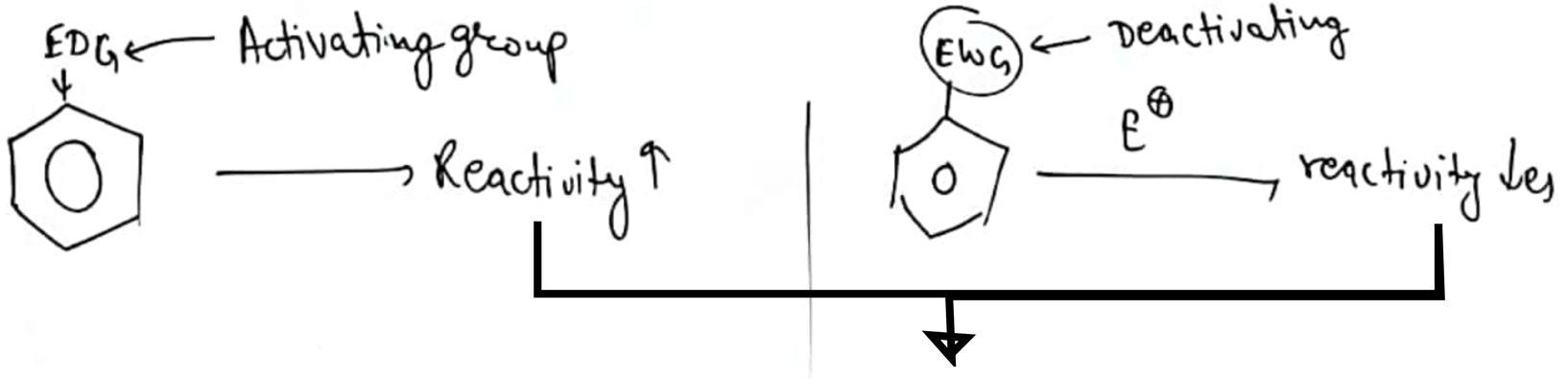


⇒ Reactivity^{SE} $\propto$ Stability of intermediate (arenium ion)

⇒ " , $\propto$ e^- density on aro. ring of reactant

⇒ " " ∴ ↑es in presence of electron donating group (+M, +H, +I)

⇒ " " " ↓es " " ∴ electron withdrowing group (−M −I)

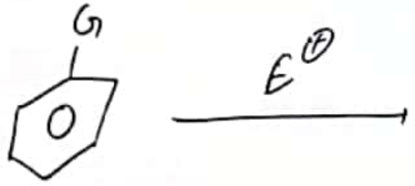


ORGANIC REACTION MECHANISM-II (ORM-II)

(I) Electrophilic Substitution Reactions (S_E)

Note :-

For substituted aromatic compounds reactivity for S_E reaction and suitable position for incoming electrophile, depent upon substituents which, is/are present on aromatic ring of reactant.



ORGANIC REACTION MECHANISM-II (ORM-II)

(I) Electrophilic Substitution Reactions (S_E)

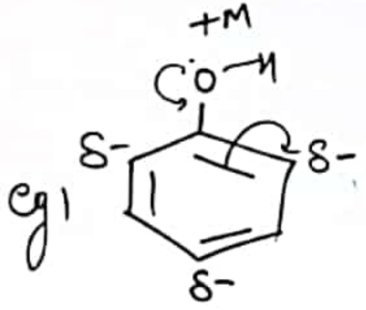
Ortho and para directing & Activating groups	meta directing & Deactivating group.	O, P directing & Deactivating
$-\ddot{O}^{\ominus}$, $-\ddot{N}H_2$, $-NHR$, $-NR_2$, $-OH$, $-\ddot{O}R$, $-\ddot{N}H-\overset{\overset{O}{\|}}{C}-R$, $-\ddot{O}-\overset{\overset{O}{\|}}{C}-R$	$-\overset{\oplus}{N}H_3$, $-\overset{\oplus}{N}R_3$, $-NO_2$, $-CN$, $-CHO$, $-COR$, $-SO_3H$, $-COOH$, $-COCl$, $-COOR$, $-CONH_2$, $-CX_3$	$-\ddot{F}:$, $-\ddot{Cl}:$, $-\ddot{Br}:$, $-\ddot{I}:$, $-\ddot{N}=O$, $-\overset{\overset{O}{\|}}{\underset{\underset{:}{O}}{S}}-OH$ (deactivating due to -I; o, p directing due to +M effect); $-CH_2-X$, $-CH=CH-COOH$ etc., $-CHX_2$
$-C_6H_5$ (-Ar), $-CH=CH_2$, $-C\equiv CH$, -Alkyl ($3^\circ > 2^\circ > 1^\circ$)		

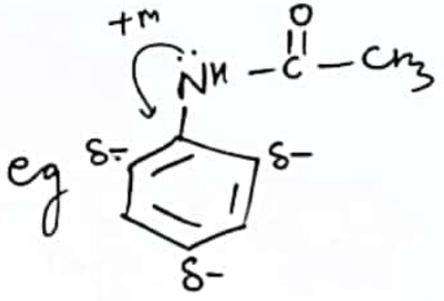
F, Cl, Br, I, $-N=O$, $-\overset{\overset{O}{\|}}{S}-OH$ — $-I > +M$

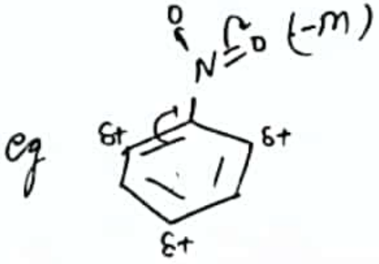
ORGANIC REACTION MECHANISM-II (ORM-II)

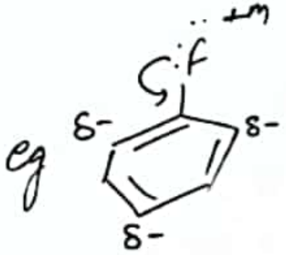
(I) Electrophilic Substitution Reactions (S_E)

Explanation for directing influence.

eg 

eg 

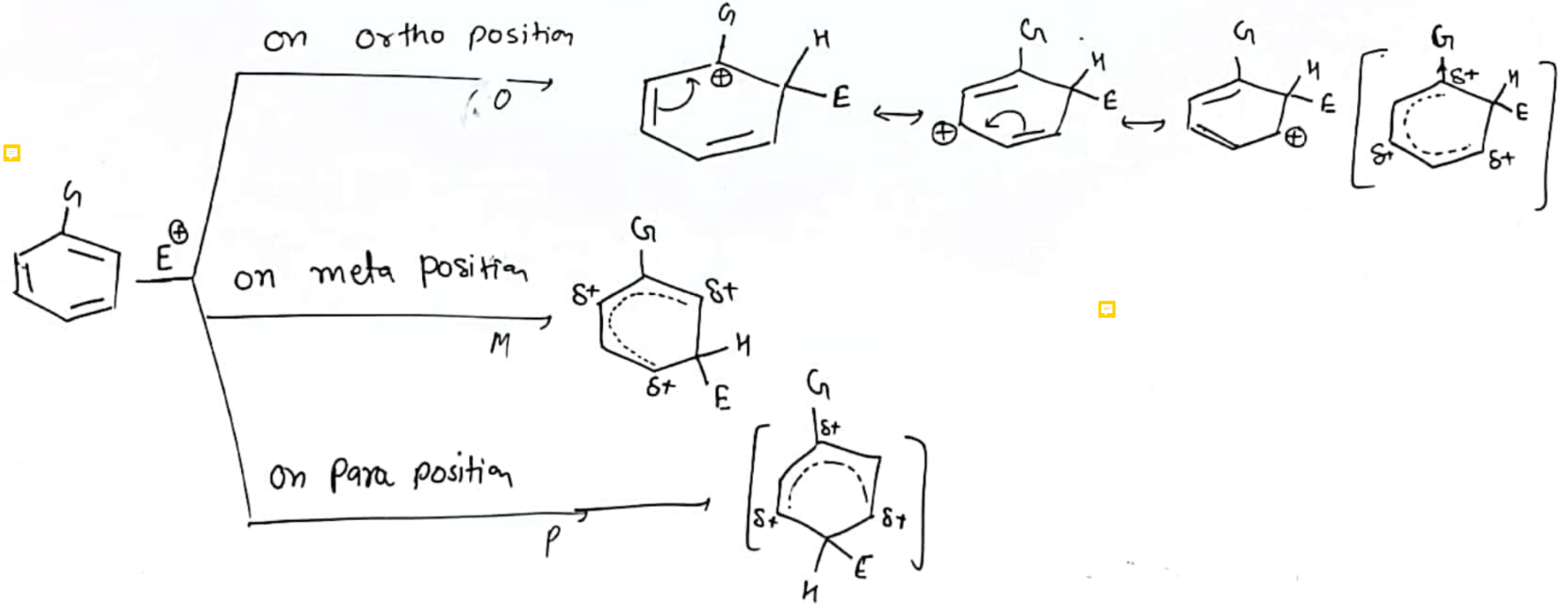
eg 

eg 

ORGANIC REACTION MECHANISM-II (ORM-II)

(I) Electrophilic Substitution Reactions (S_E)

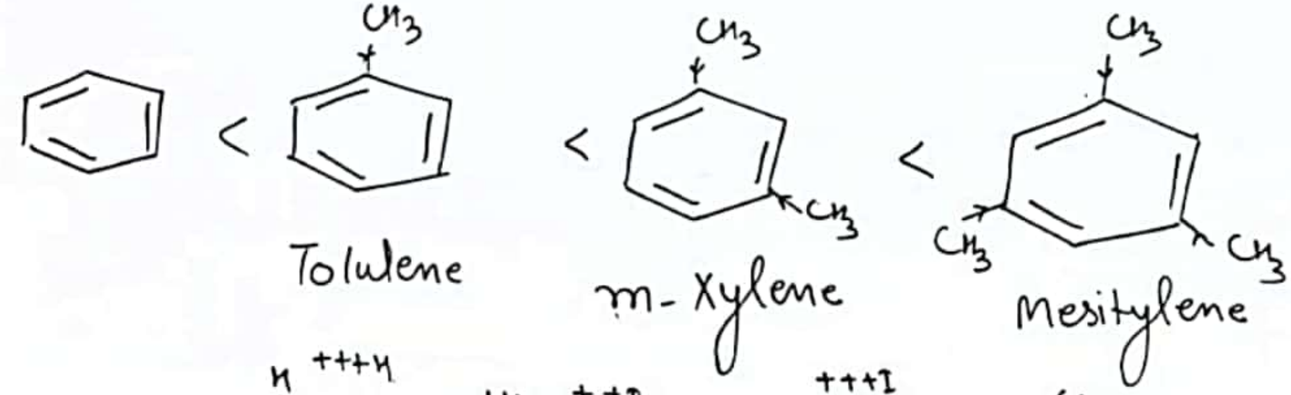
Explanation for directing influence.

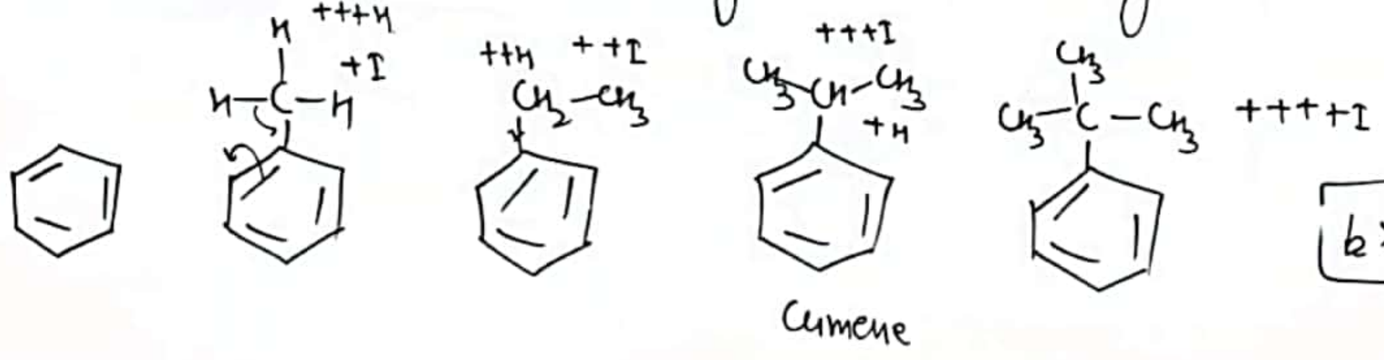


ORGANIC REACTION MECHANISM-II (ORM-II)

(I) Electrophilic Substitution Reactions (S_E)

Q. Give order of reactivity for S_E Rx^n

① 

② 

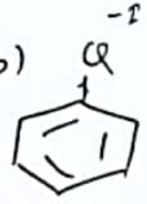
b > c > d > e > a

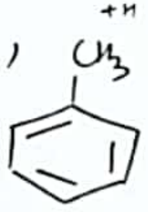
ORGANIC REACTION MECHANISM-II (ORM-II)

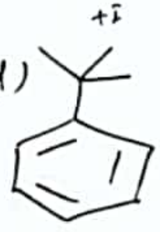
(I) Electrophilic Substitution Reactions (S_E)

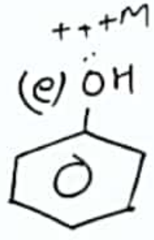
Q. Give order of reactivity for S_E R_X^n

③ (a) 

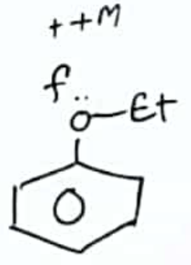
(b) 

(c) 

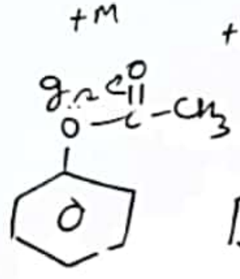
(d) 

(e) 

Phenol (Carbolic acid)



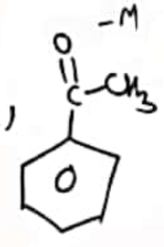
(Phenetol)



Phenyl acetate



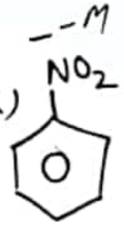
Aniline

(i) 

Acetophenone

(j) 

Styrene

(k) 

(l) 

Biphenyl

$+M > +H > +I > \dot{C} > -I > -M$

h > e > f > g > d > j > c > d > a > b > i > k

Activated ring — Deactivated

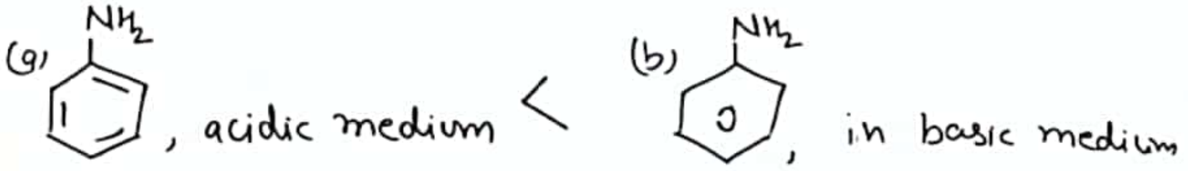
+M: $-O^- > -NH_2 > -NHR > -NR_2 > -OH > -OR > -NH-\overset{O}{\overset{\|}{C}}-R > -O-\overset{O}{\overset{\|}{C}}-R$ > $-Ph > -CH=CH > -C\equiv C-H > -F > -Cl > -Br > -I$

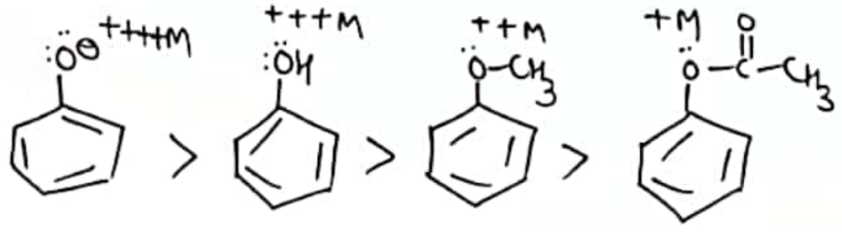
ORGANIC REACTION MECHANISM - II (ORM - II)

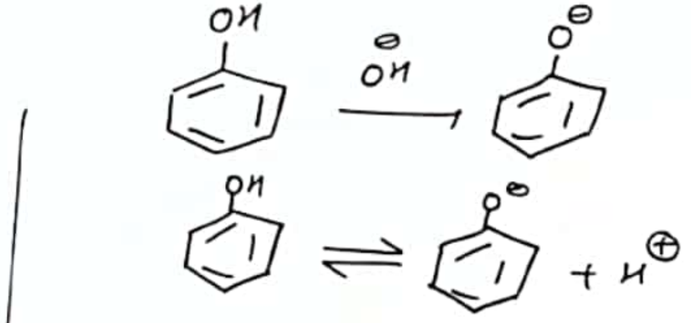
(I) Electrophilic Substitution Reactions (S_E)

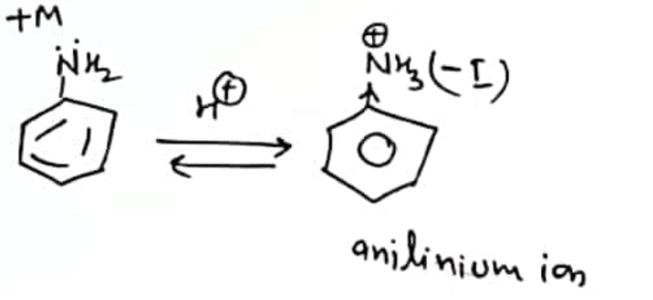
Q. Give order of reactivity for electrophilic substitution Rxn.

(4) 

(5) 

(6) 





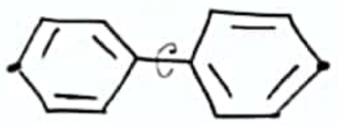
anilinium ion

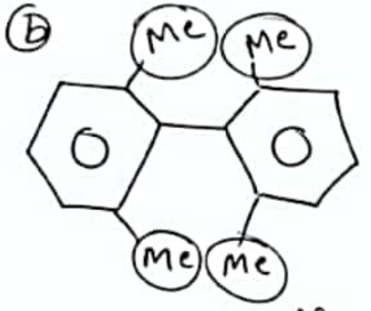
ORGANIC REACTION MECHANISM - II (ORM - II)

(I) Electrophilic Substitution Reactions (S_E)

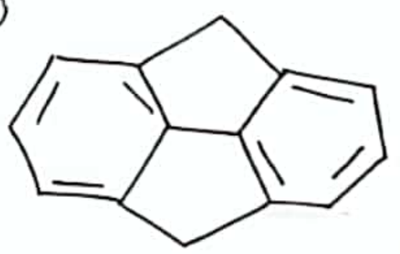
Q. Give order of reactivity for electrophilic substitution Rxn.

(7)

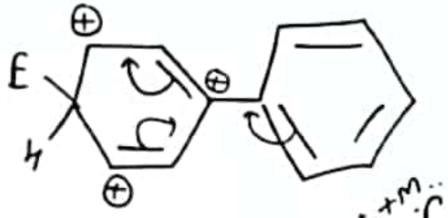
ⓐ 

ⓑ 

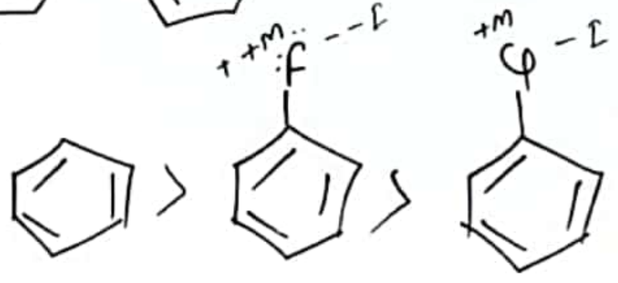
(SIR) effect

ⓒ 

c > a > b



(8)

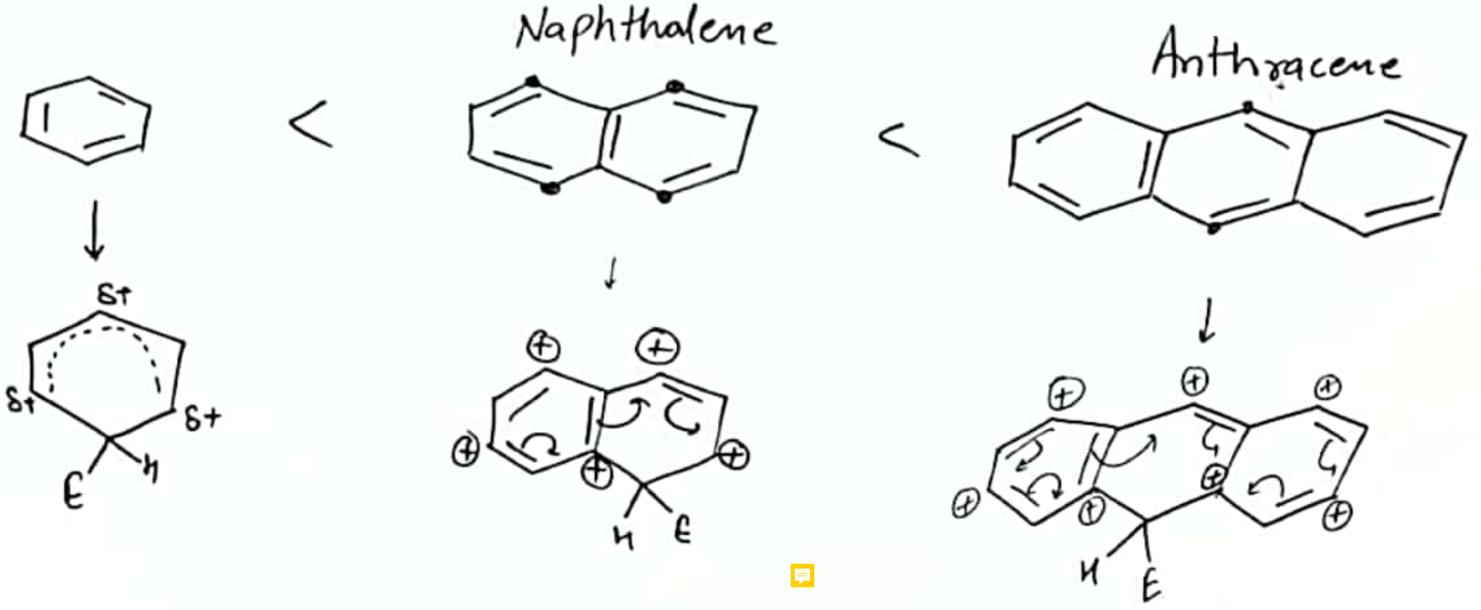


ORGANIC REACTION MECHANISM - II (ORM - II)

(I) Electrophilic Substitution Reactions (S_E)

Q. Give order of reactivity for electrophilic substitution Rxn.

eg 9

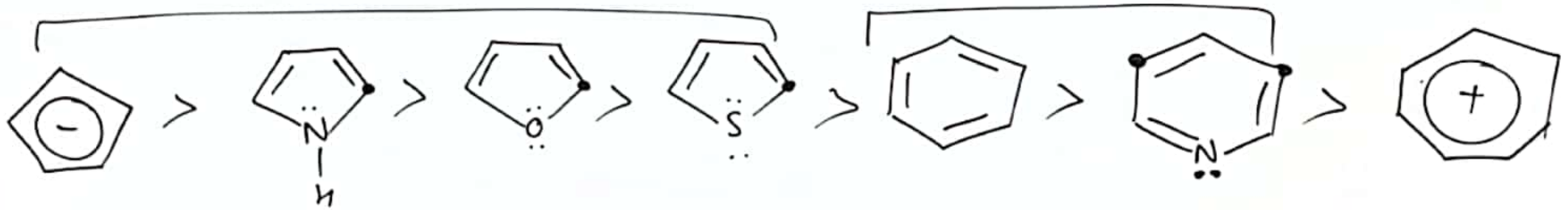


ORGANIC REACTION MECHANISM - II (ORM - II)

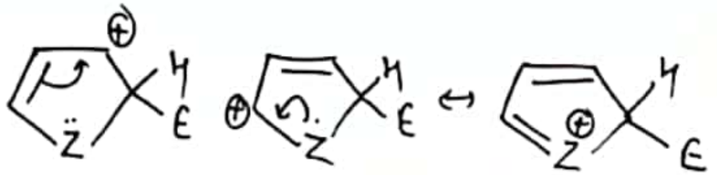
(I) Electrophilic Substitution Reactions (S_E)

Q. Give order of reactivity for electrophilic substitution Rxn.

(10) $6\pi e^-$ Systems:-

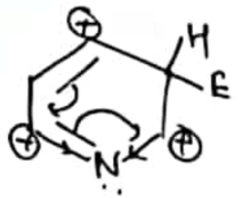


Pyrrol



+M

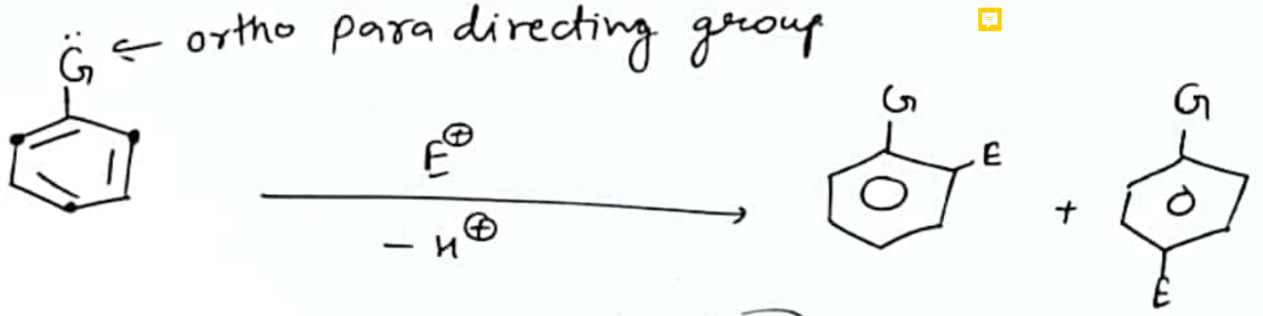
$-\ddot{N}H_2 > -\ddot{O}H > -\ddot{S}H$ +M



ORGANIC REACTION MECHANISM - II (ORM - II)

(I) Electrophilic Substitution Reactions (S_E)

Factors affecting of ortho, para ratio.

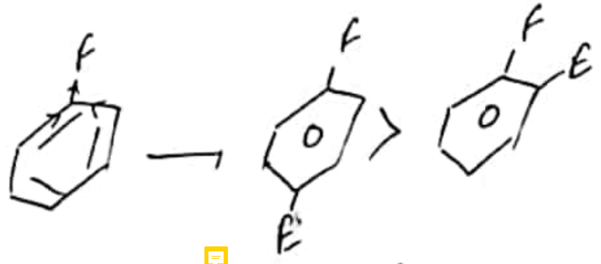


① Probability factor ⇒ $\boxed{\frac{O}{P} = \frac{2}{1}}$

② Steric hindrance ⇒ P > O

③ H-Bonding.

④ Electronic effect



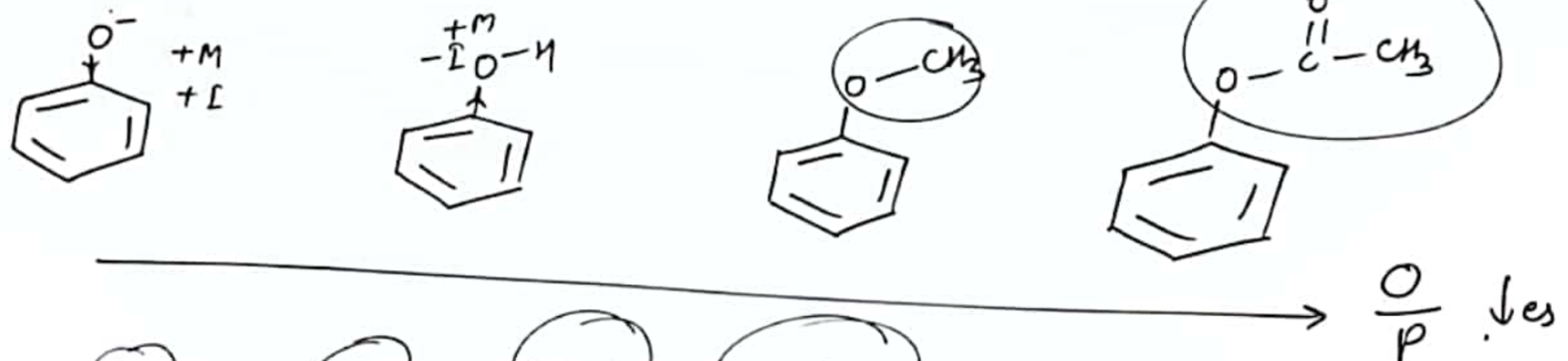
⑤ Temp.

ORGANIC REACTION MECHANISM - II (ORM - II)

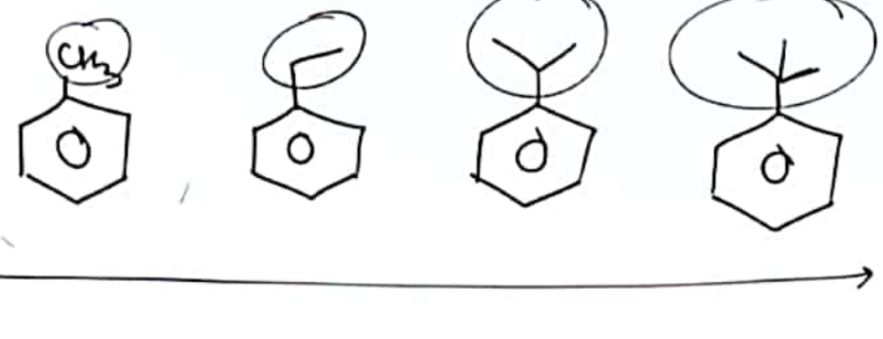
(I) Electrophilic Substitution Reactions (S_E)

Ortho - Para ratio :-

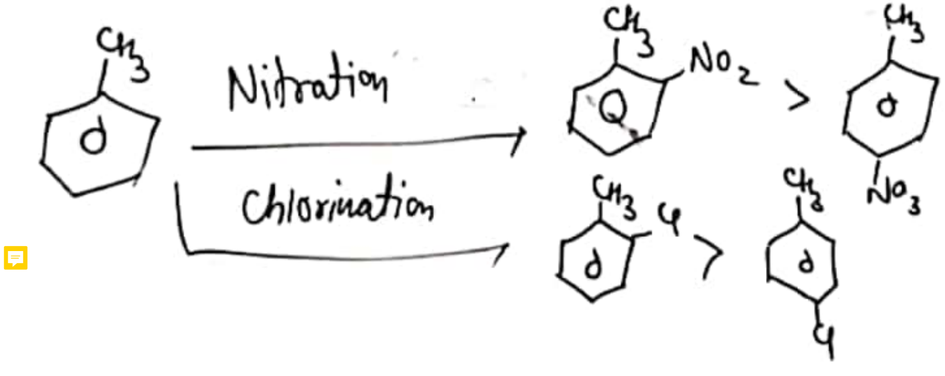
eg 1



eg 2



Note

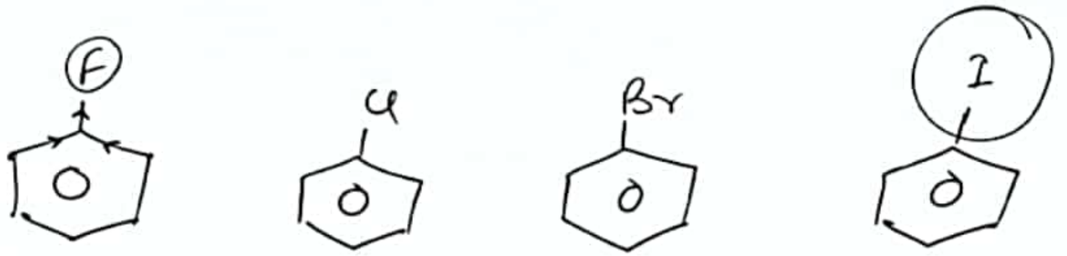


ORGANIC REACTION MECHANISM - II (ORM - II)

(I) Electrophilic Substitution Reactions (S_E)

Ortho - Para ratio :-

*eg 3



(due to -I effect)



ORGANIC REACTION MECHANISM - II (ORM - II)

(I) Electrophilic Substitution Reactions (S_E)

Q. Give most suitable position for electrophilic attack:-

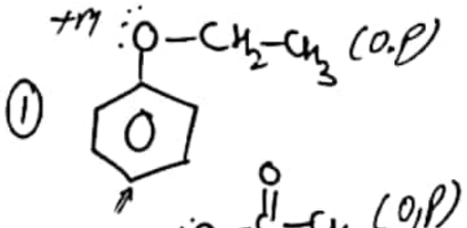
Hint ⇒

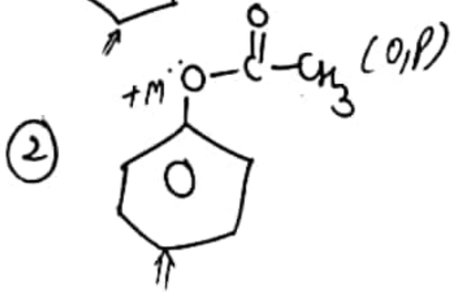
① Activating group > Deactivating group.

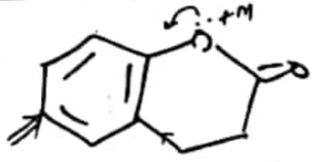
② O,P. directing grp. > meta directing group.

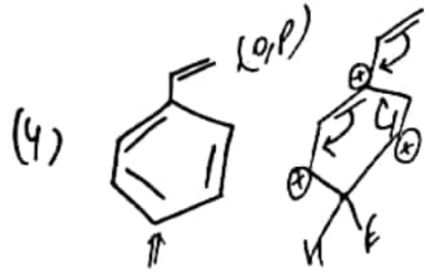
③ more activating group > lesser activating grp

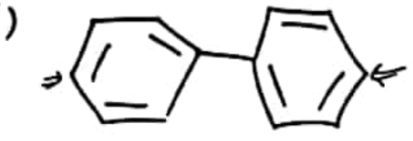
④ more deactivating group. > lesser deactivating group. (When both groups are meta directing)

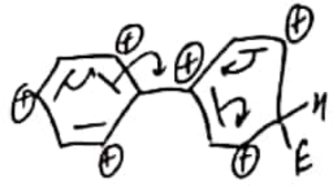
(1) 

(2) 

(3) 

(4) 

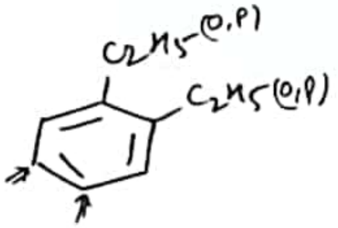
(5) 



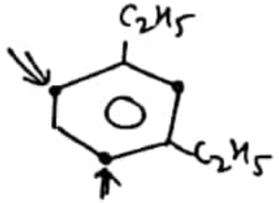
ORGANIC REACTION MECHANISM - II (ORM - II)

(I) Electrophilic Substitution Reactions (S_E)

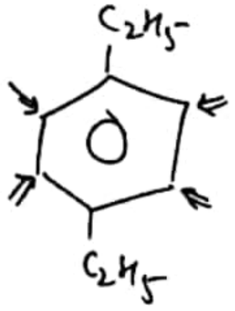
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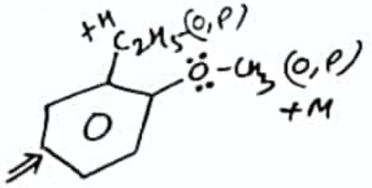
eg 7



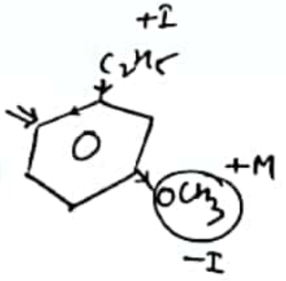
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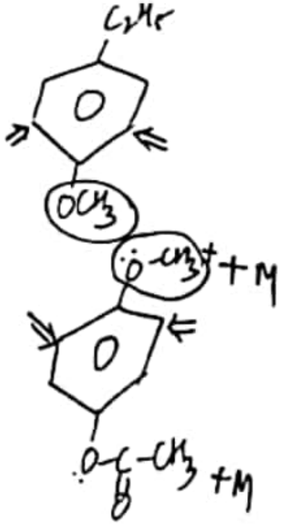
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(10)

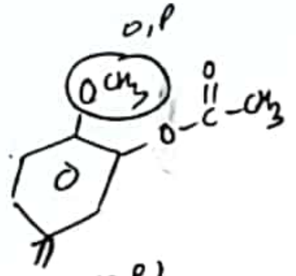


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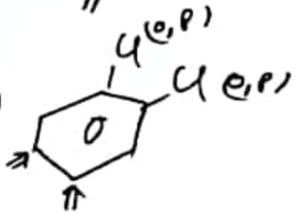


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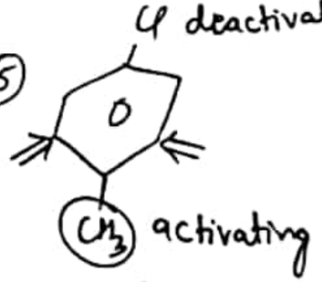
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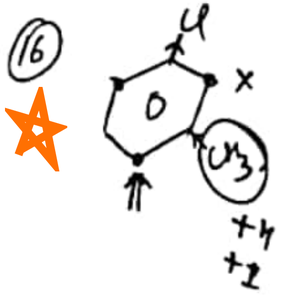
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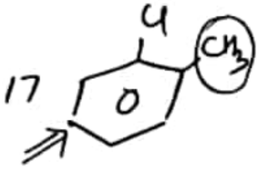
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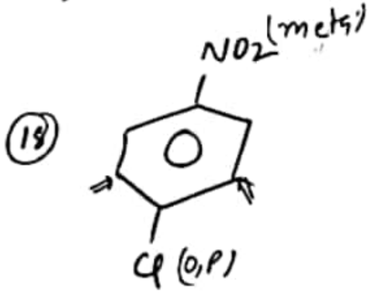
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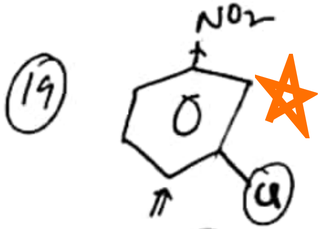
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(19)



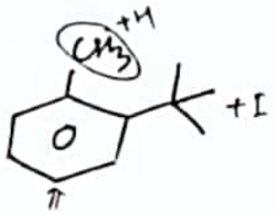
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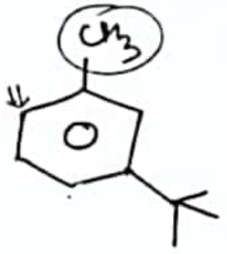
ORGANIC REACTION MECHANISM - II (ORM - II)

(I) Electrophilic Substitution Reactions (S_E)

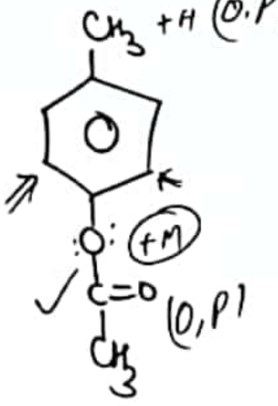
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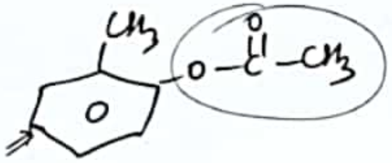
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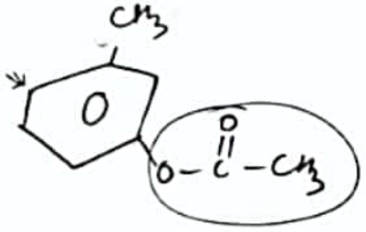
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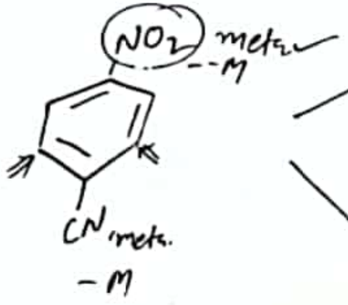
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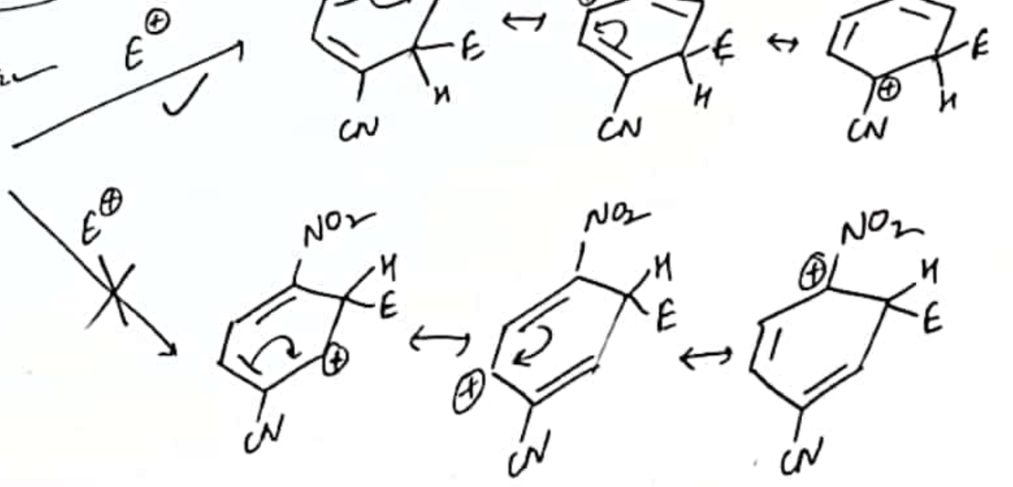


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26



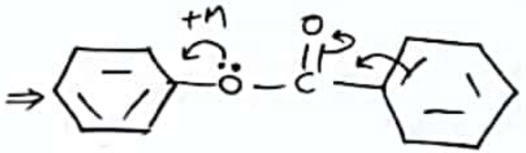


ORGANIC REACTION MECHANISM - II (ORM - II)

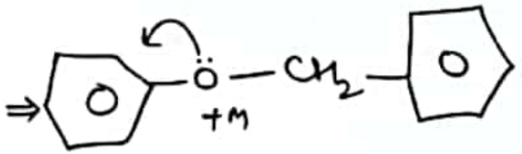
(I) Electrophilic Substitution Reactions (S_E)

Q Give most suitable position for electrophilic attack

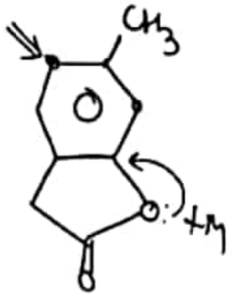
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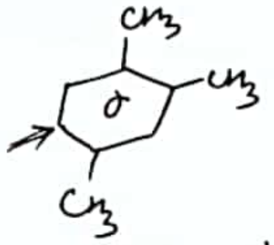
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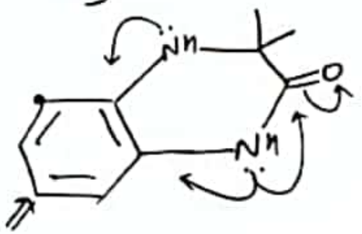
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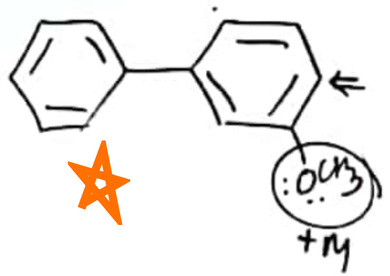
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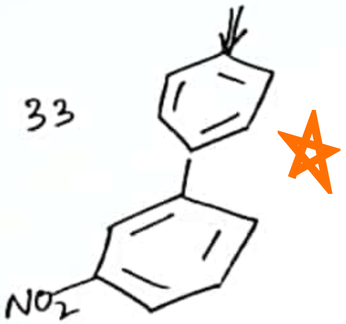
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32



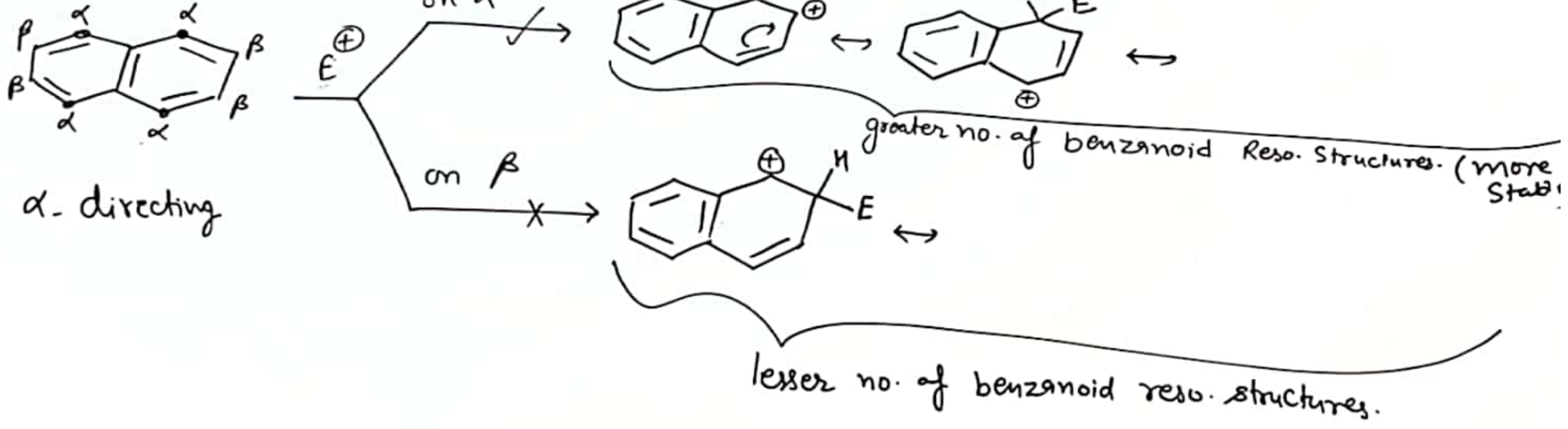
33



ORGANIC REACTION MECHANISM - II (ORM - II)

(I) Electrophilic Substitution Reactions (S_E)

Q Give most suitable position for electrophilic attack



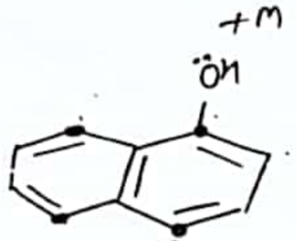
α - directing

ORGANIC REACTION MECHANISM - II (ORM - II)

(I) Electrophilic Substitution Reactions (S_E)

Q Give most suitable position for electrophilic attack

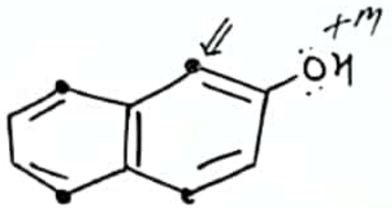
35



α-Naphthol

36

Imp



β-Naphthol

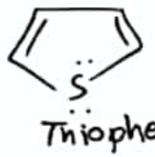
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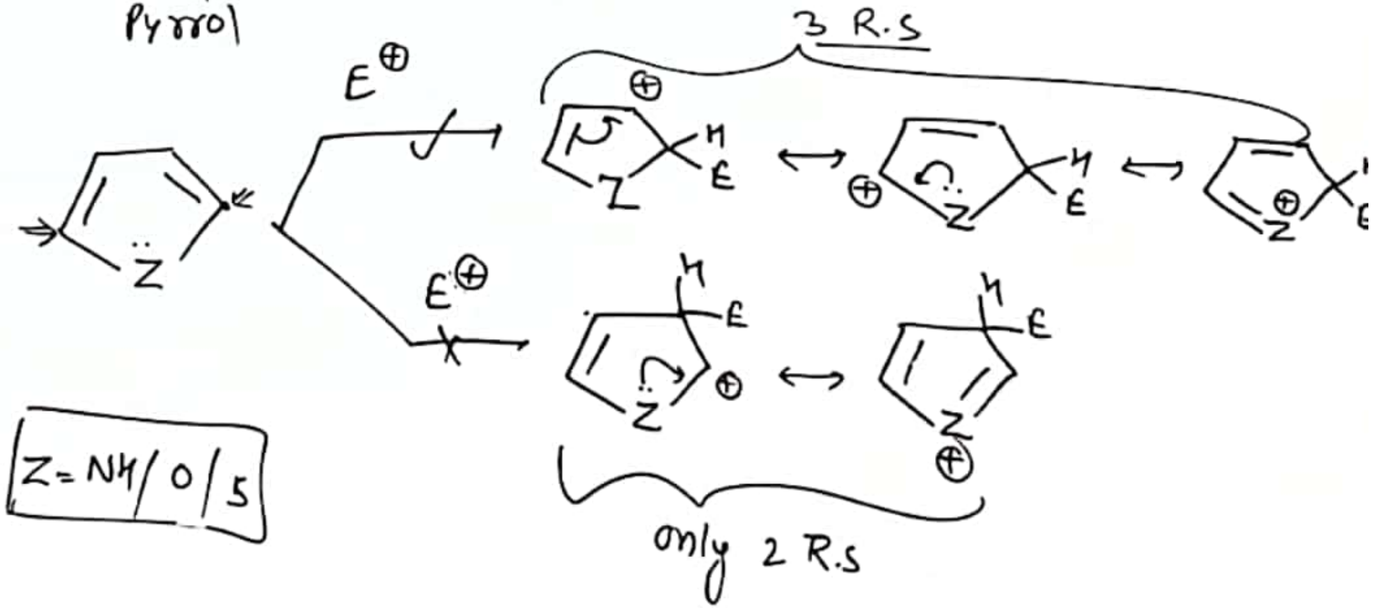
Pyrrol



Furan



Thiophene



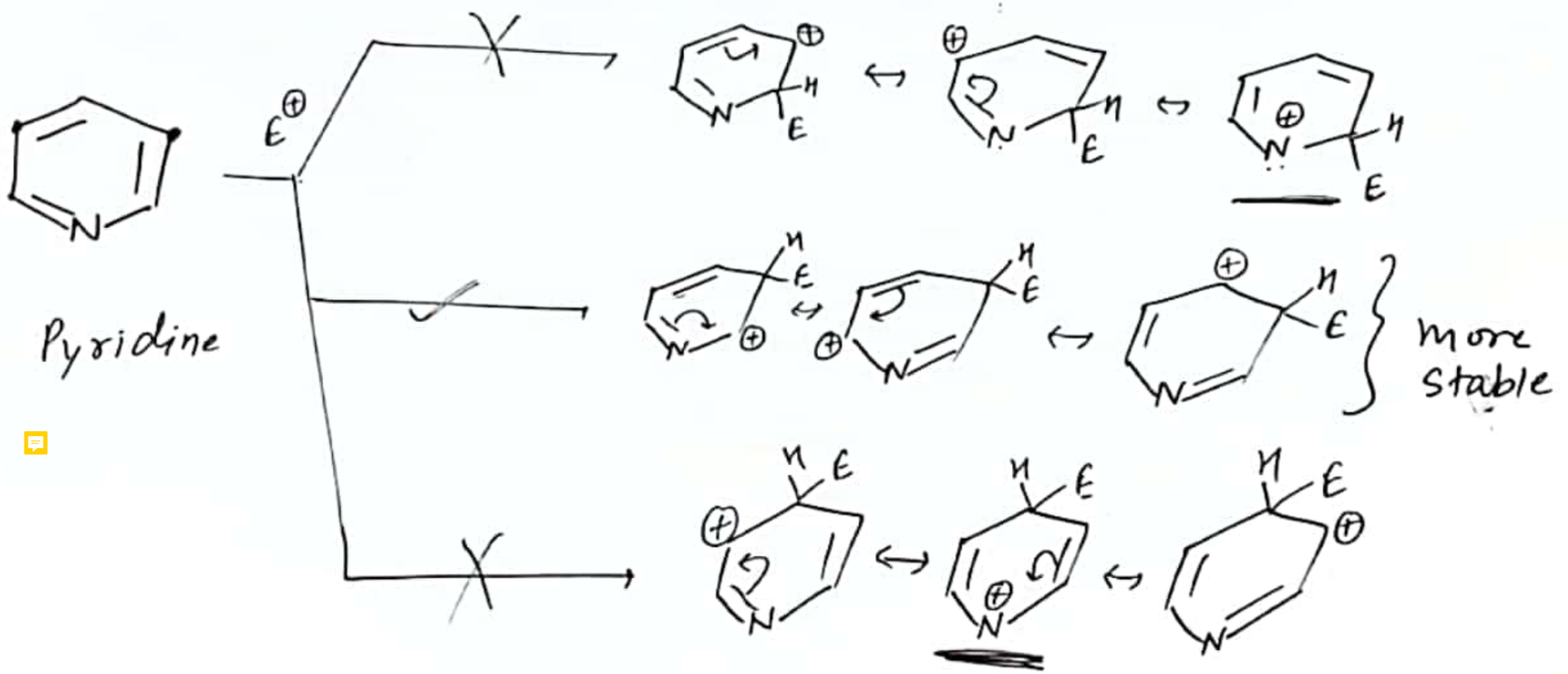
Z = NH / O / S

ORGANIC REACTION MECHANISM - II (ORM - II)

(I) Electrophilic Substitution Reactions (S_E)

Q Give most suitable position for electrophilic attack

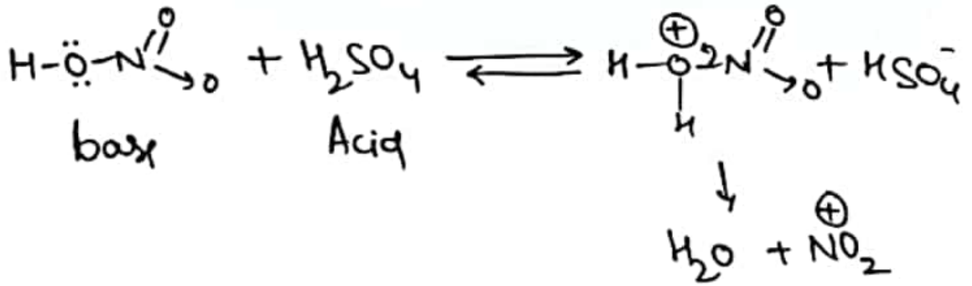
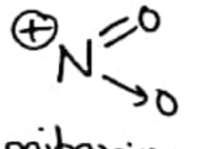
38



ORGANIC REACTION MECHANISM - II (ORM - II)

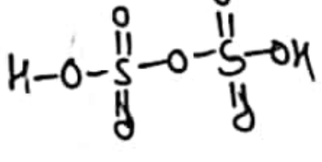
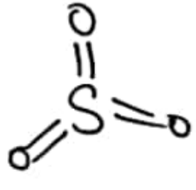
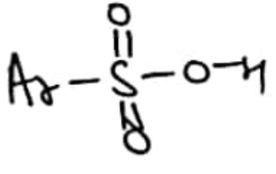
(I) Electrophilic Substitution Reactions ($Ar S_E$) :-

Examples of $Ar S_E$

Reaction	Reagents	Electrophile	Product (Ar-E)
① Nitration	(a) Conc. HNO_3 + Conc. H_2SO_4 :- 	 nitronium	$Ar-NO_2$
	(b) fuming HNO_3 :- $2HNO_3 \rightleftharpoons NO_2^{\oplus} + NO_3^{\ominus} + H_2O$		
	(c) $N_2O_5 \rightleftharpoons NO_2^{\oplus} + NO_3^{\ominus}$		
	(d) $NO_2^{\oplus}(CF_3COO^{\ominus})$ or $NO_2^{\oplus}(BF_4^{\ominus})$ etc.		

Electrophilic Substitution Reactions ($Ar S_E$) :-

Examples of $Ar S_E$

Reaction	Reagents	Electrophile	Product (Ar.
② Sulphonation	(a) conc. H_2SO_4 :- $2H_2SO_4 \rightleftharpoons H_3O^{\oplus} + SO_3 + HSO_4^{\ominus}$ (b) Oleum = $H_2S_2O_7 \rightleftharpoons H_2SO_4 + SO_3$ 		

ORGANIC REACTION MECHANISM - II (ORM - II)

(I) Electrophilic Substitution Reactions ($Ar S_E$) :-

Examples of $Ar S_E$

Reaction	Reagents	Electrophile	Product (Ar-E)
(3) Halogenation	(a) $\ddot{X}-\ddot{X} + AlX_3 \longrightarrow [X\overset{\oplus}{-}X-\overset{\ominus}{A}lX_3] \longrightarrow X^{\oplus} + (X-\overset{\ominus}{A}lX_3)$ Lewis acid $[AlX_3, FeX_3, BF_3, ZnCl_2, TiCl_4$ etc. Reactivity $\rightarrow F_2 > Cl_2 > Br_2 > I_2$ (d) $Cl-Cl + AlCl_3 \longrightarrow Cl^{\oplus} + (AlCl_4^{\ominus})$ (c) $Br-Br$ (excess) $+ Fe \longrightarrow FeBr_3$ $FeBr_3 + Br_2 \longrightarrow Br^{\oplus}(FeBr_4^{\ominus})$	$X^{\oplus}$ Halonium ion.	Ar-X

ORGANIC REACTION MECHANISM - II (ORM - II)

(I) Electrophilic Substitution Reactions ($Ar S_E$) :-

Examples of $Ar S_E$

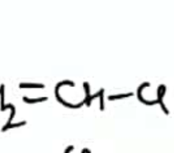
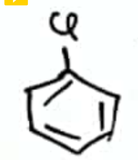
Reaction	Reagents	Electrophile	Product (Ar-E)
(3) Halogenation	(a) $I_2 \xrightarrow[\text{oxidising agent}]{HNO_3} 2I^{\oplus}$ $I^{\delta+}-Cl^{\delta-} + AlCl_3 \longrightarrow I^{\oplus} + AlCl_4^{\ominus}$ (b) $Cl-Cl + AlCl_3 \longrightarrow Cl^{\oplus} + (AlCl_4^{\ominus})$ (c) $Br-Br + Fe \longrightarrow FeBr_3$ (excess) $FeBr_3 + Br_2 \longrightarrow Br^{\oplus}(FeBr_4^{\ominus})$	$X^{\oplus}$ Halonium ion.	Ar-X

ORGANIC REACTION MECHANISM - II (ORM - II)

(I) Electrophilic Substitution Reactions ($Ar S_E$) :—

$R-\ddot{O}-H + BF_3 \rightarrow R-\overset{H}{\overset{|}{\overset{\oplus}{O}}}-BF_3^{\ominus} \rightarrow R^{\oplus} + HO-BF_3^{\ominus}$

Examples of $Ar S_E$

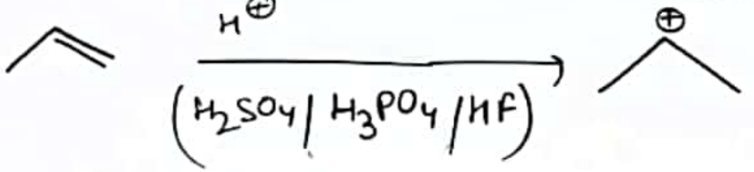
Reaction	Reagents	Electrophile	Product (Ar-E
(D) Friedel craft alkylation	(a) $R-\ddot{Cl} + AlCl_3 \rightarrow [R-\overset{\oplus}{Cl}-\overset{\ominus}{AlCl_3}] \rightarrow R^{\oplus} + AlCl_4^{\ominus}$ (lewis acid)	$R^{\oplus}$	Ar-R
	Note: $R-Cl \neq$ { $CH_2=CH-Cl$, $R-C\equiv C-Cl$,    }		
	(b) $R-\ddot{O}-H \xrightarrow[(H_2SO_4 / H_3PO_4 / HF / BF_3)]{H^{\oplus}\ (acid)} R-\overset{\oplus}{\overset{H}{\overset{|}{O}}}-H \rightarrow R^{\oplus} + H_2O$		
	$X \rightarrow$ (HCl, HBr, HI)		

ORGANIC REACTION MECHANISM - II (ORM - II)

(I) Electrophilic Substitution Reactions ($Ar S_E$) :—

$$R-\ddot{O}-H + BF_3 \rightarrow R-\overset{H}{\underset{\oplus}{O}}-BF_3^{\ominus} \rightarrow R^{\oplus} + HO-BF_3^{\ominus}$$

Examples of $Ar S_E$

Reaction	Reagents	Electrophile	Product (Ar-E
(D) Friedel craft alkylation	(C)  X (HCl, HBr, HI)	$R^{\oplus}$	Ar-R

Note Rearrangement is possible in $R^{\oplus}$ of this reaction.

Note $FeCl_3$ } = No Rearrangement
Lewis acid

$$R-Cl \xrightarrow{FeCl_3} R-\overset{\oplus}{Cl}-\overset{-}{FeCl_3}$$

ORGANIC REACTION MECHANISM - II (ORM - II)

(I) Electrophilic Substitution Reactions ($Ar S_E$) :—

$R-\ddot{O}-H + BF_3 \rightarrow R-\overset{H}{\underset{\oplus}{O}}-BF_3^{\ominus}$

$\hookrightarrow R^{\oplus} + HO-BF_3^{\ominus}$

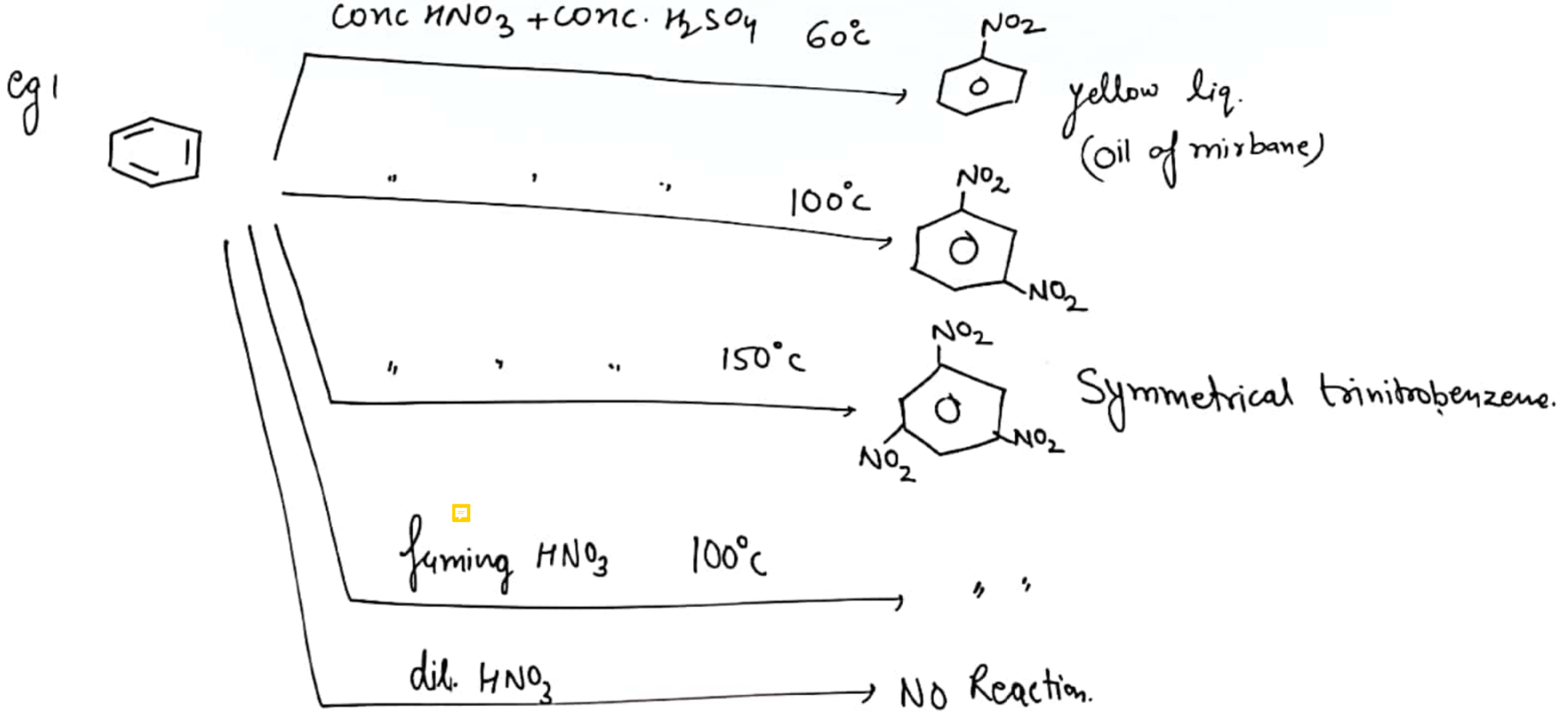
Examples of $Ar S_E$

Reaction	Reagents	Electrophile	Product (Ar-E
(E) Friedel craft acylation	ⓐ $R-\overset{O}{\overset{\|\|}{C}}-Cl + AlCl_3 \rightarrow R-\overset{O}{\overset{\|\|}{C}}^{\oplus} + AlCl_4^{\ominus}$	Acyliniom ion $R-C\equiv O^{\oplus} \leftrightarrow R-\overset{\oplus}{C}=O$	$Ar-\overset{O}{\overset{\|\|}{C}}-R$ ketone
	ⓑ $R-\overset{O}{\overset{\|\|}{C}}-O-\overset{O}{\overset{\|\|}{C}}-R + AlCl_3 \rightarrow R-\overset{O}{\overset{\|\|}{C}}^{\oplus} \quad R-\overset{O}{\overset{\|\|}{C}}-O-AlCl_3^{\ominus}$		
	ⓒ $R-\overset{O}{\overset{\|\|}{C}}-O-R' + AlCl_3 \rightarrow R-\overset{O}{\overset{\|\|}{C}}^{\oplus} \quad R'-O-AlCl_3^{\ominus}$ minor		
	ⓓ $R-\overset{O}{\overset{\|\|}{C}}-\ddot{O}-H \xrightarrow[H_2SO_4 / H_3PO_4 / PPA]{H^{\oplus} \text{ strong}} R-\overset{O}{\overset{\|\|}{C}}^{\oplus} + H_2O$		

ORGANIC REACTION MECHANISM - II (ORM - II)

(I) Electrophilic Substitution Reaction (A_2S_E)

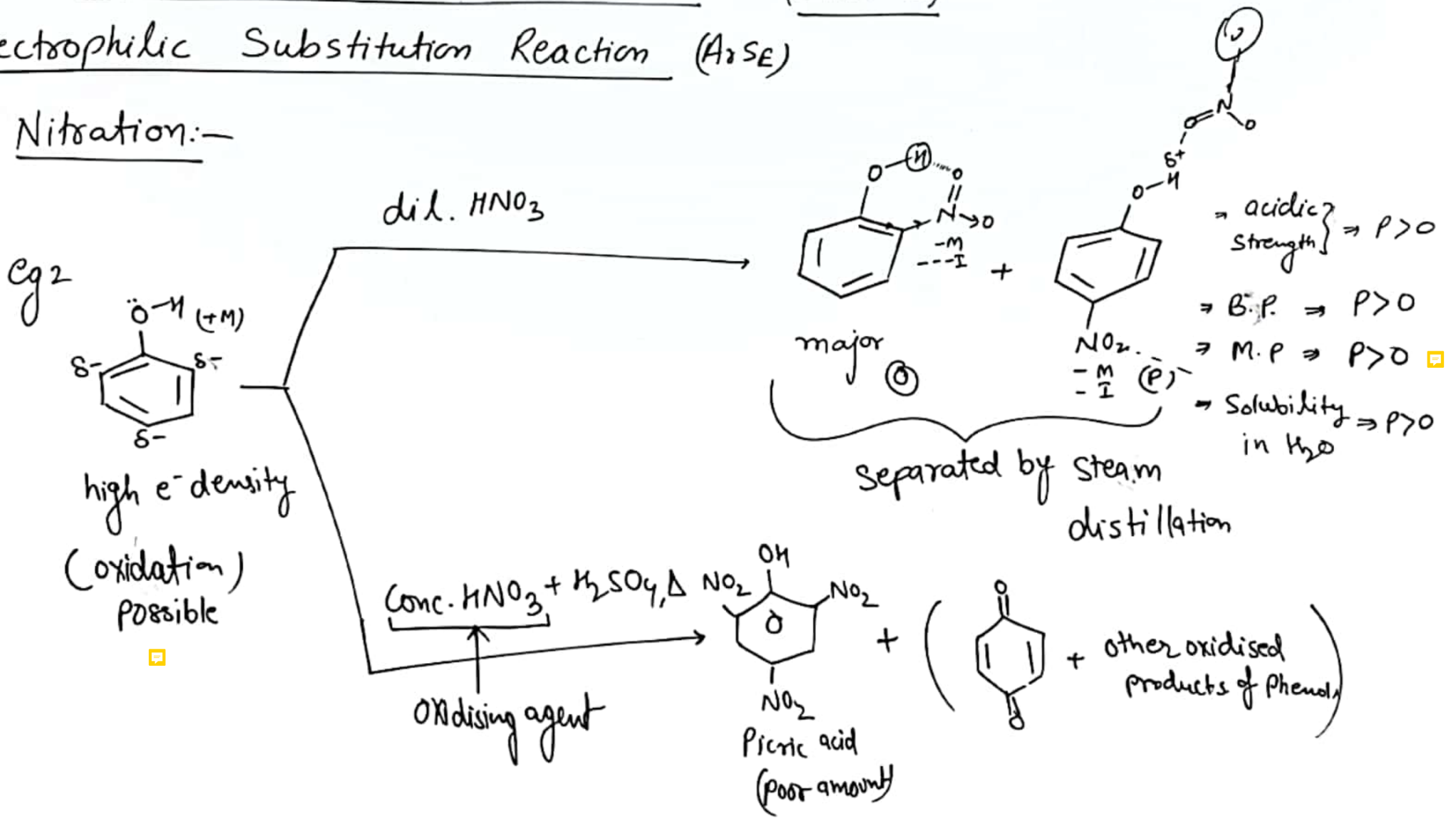
[A] Nitration:—



ORGANIC REACTION MECHANISM - II (ORM - II)

(I) Electrophilic Substitution Reaction (A_rS_E)

[A] Nitration :—

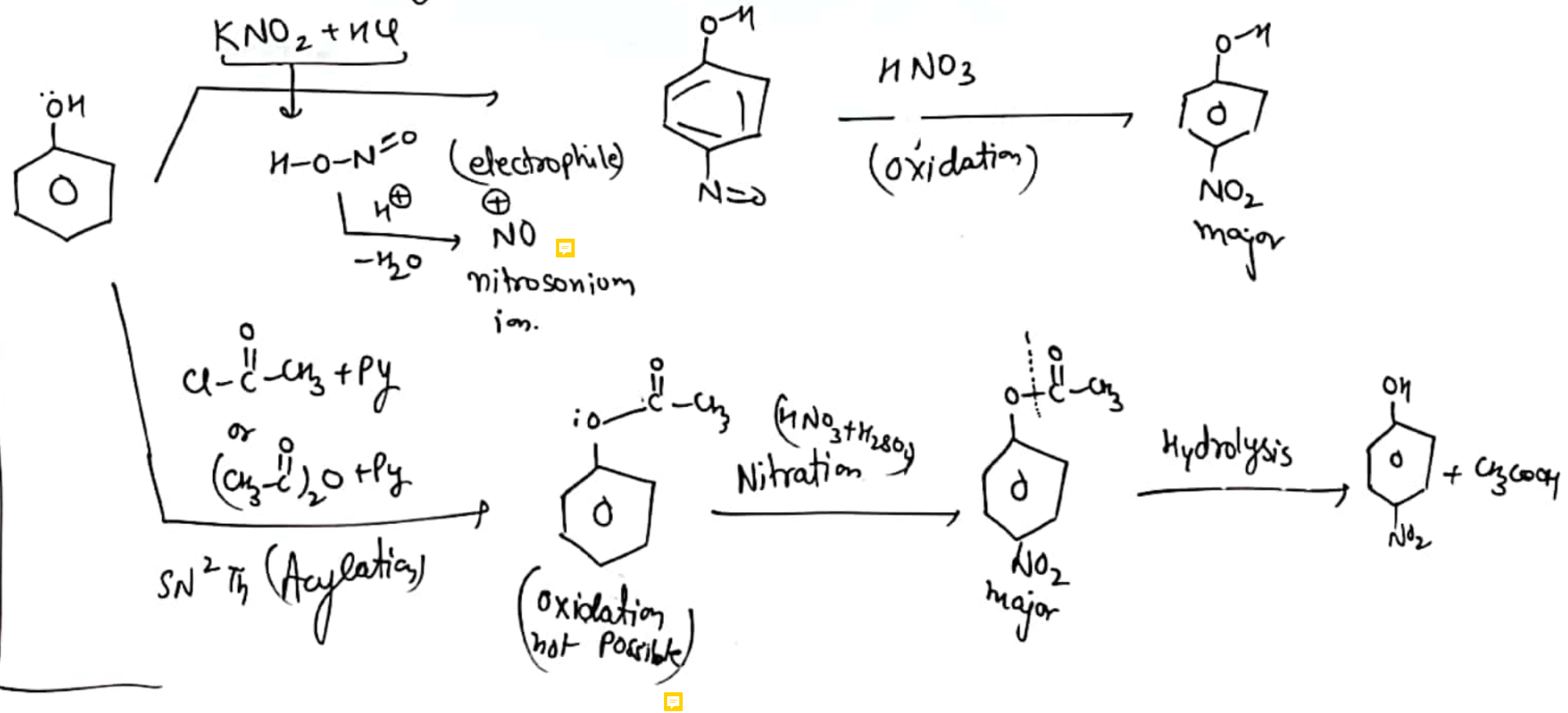


ORGANIC REACTION MECHANISM - II (ORM - II)

(I) Electrophilic Substitution Reaction (A_rS_E)

[A] Nitration:—

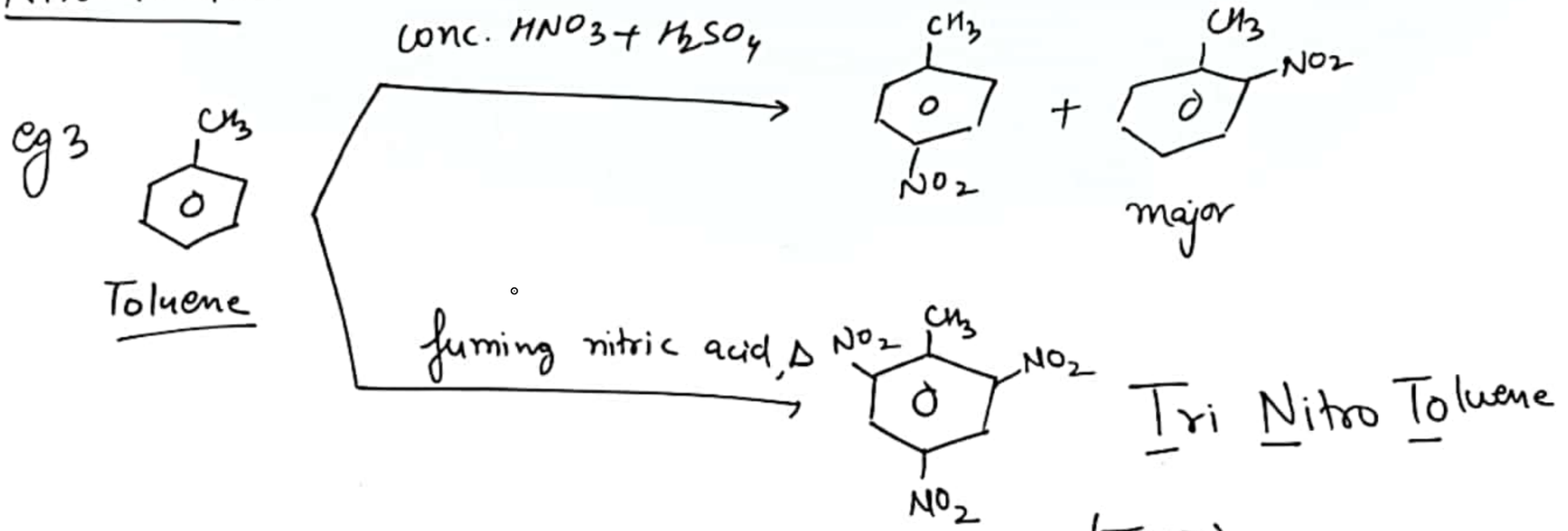
indirect method for formation of p-nitrophenol as major product

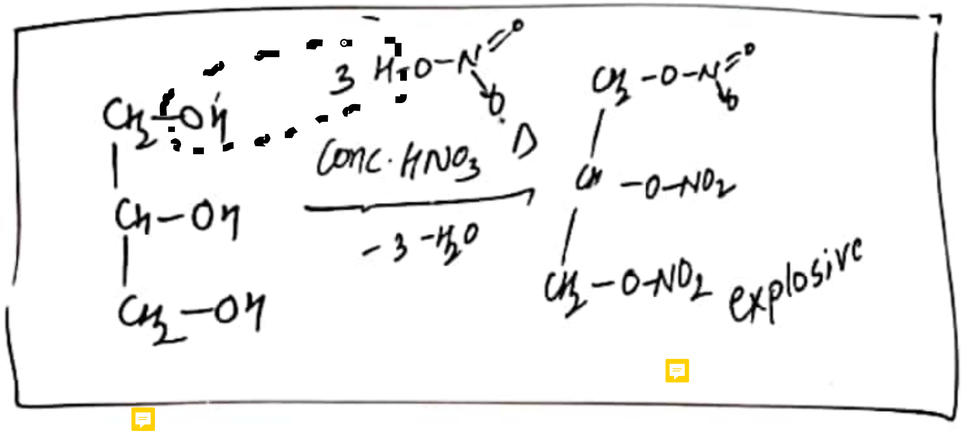


ORGANIC REACTION MECHANISM - II (ORM - II)

(I) Electrophilic Substitution Reaction (A_2S_E)

[A] Nitration:—

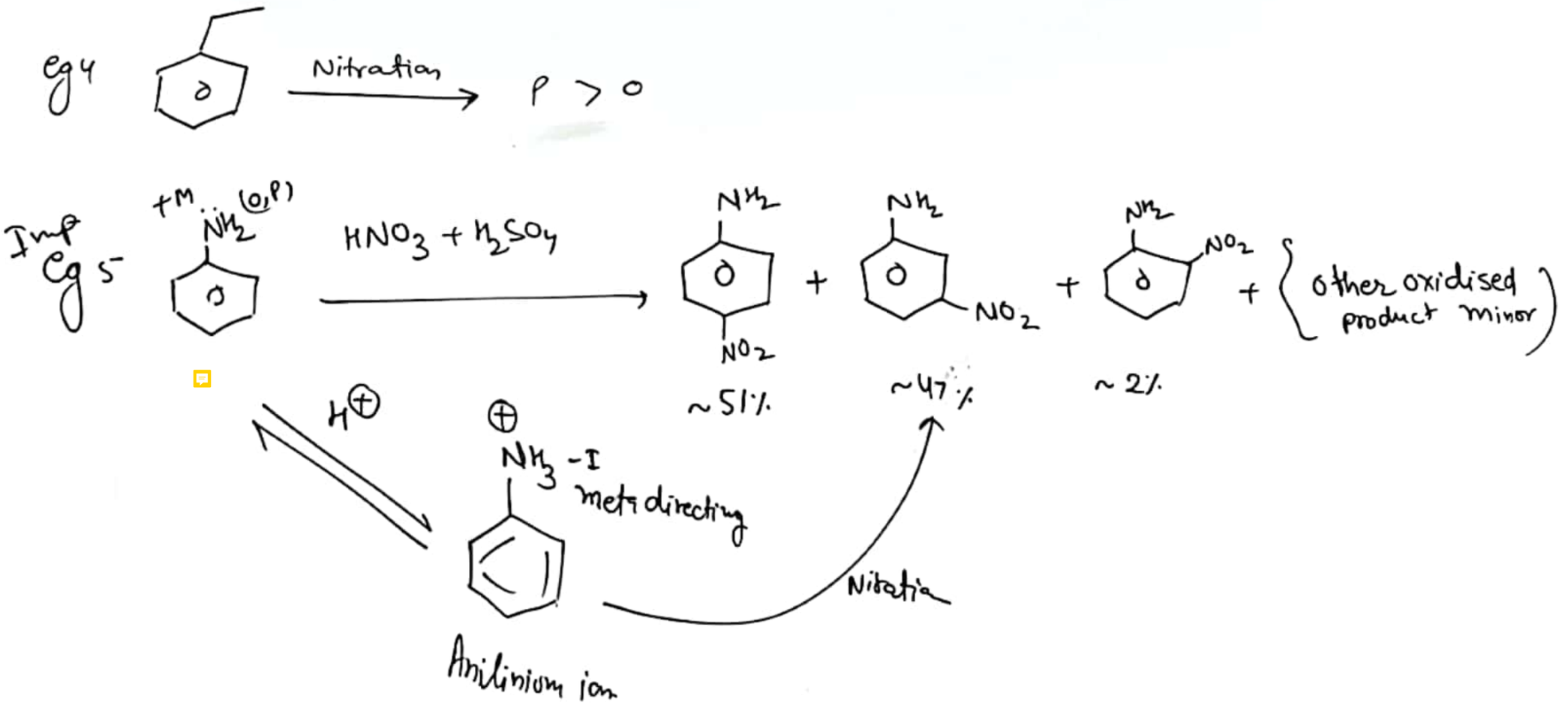




ORGANIC REACTION MECHANISM - II (ORM - II)

(I) Electrophilic Substitution Reaction (A_2S_E)

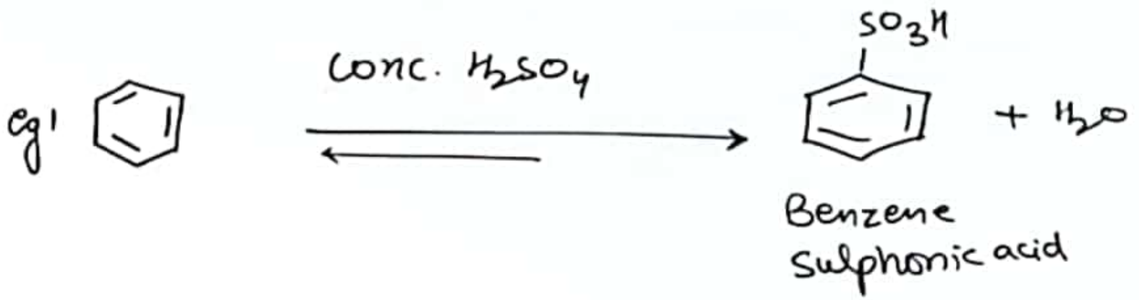
[A] Nitration:—



ORGANIC REACTION MECHANISM - II (ORM-II)

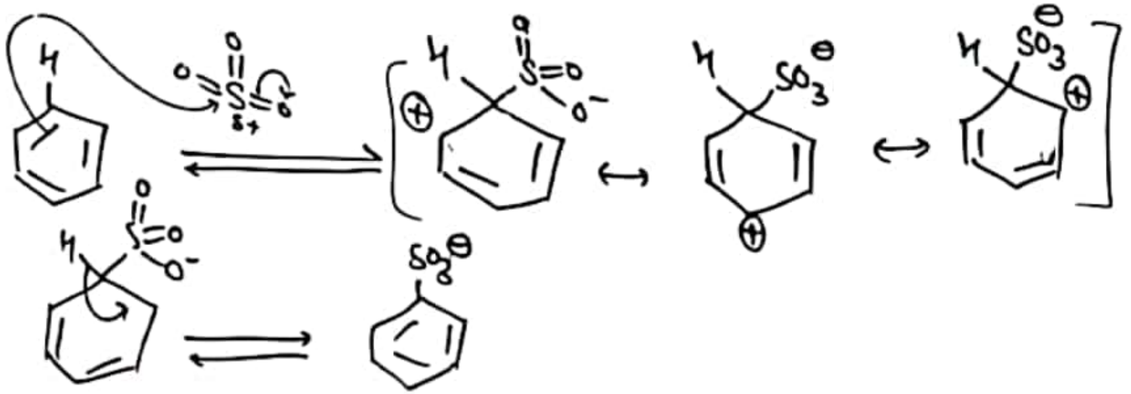
(I) Electrophilic substitution reactions (Ar_SE)

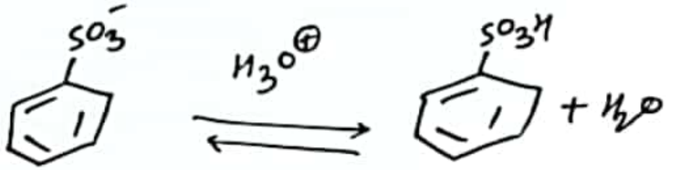
(B) Sulphonation:-

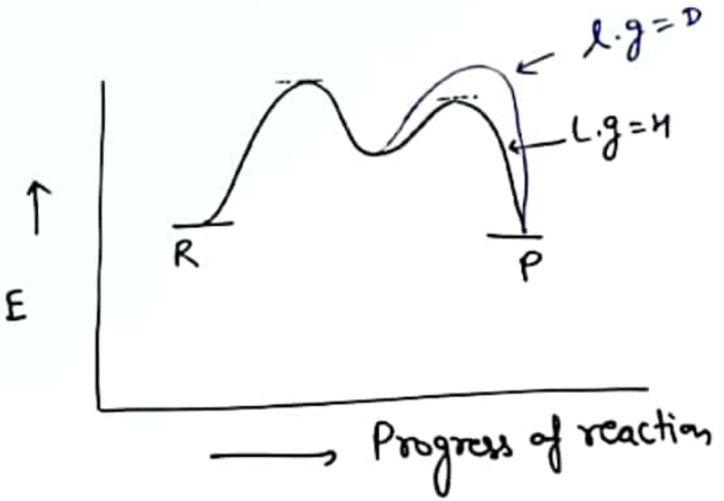


mechanism

$$2H_2SO_4 \rightleftharpoons H_3O^{\oplus} + HSO_4^{\ominus} + SO_3$$







⇒ Sulphonation is reversible reaction

(due to $(E_a)_{forward} \simeq (E_a)_{Backword}$)

⇒ It's backword reaction is called desulphonation

competishun

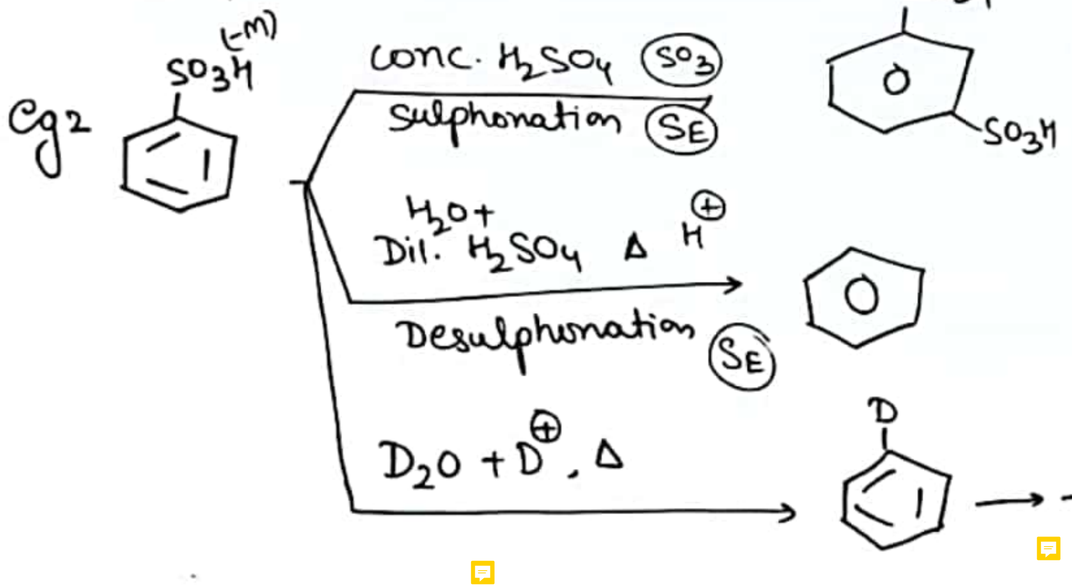
ORGANIC REACTION MECHANISM – II (ORM-II)

(I) Electrophilic substitution reactions ($Ar S_E$)

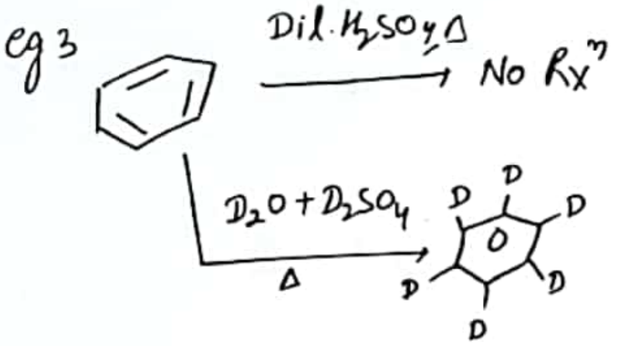
⇒ Sulphonation ⇒ conc. H_2SO_4

⇒ Desulphonation ⇒ dil. H_2SO_4, Δ

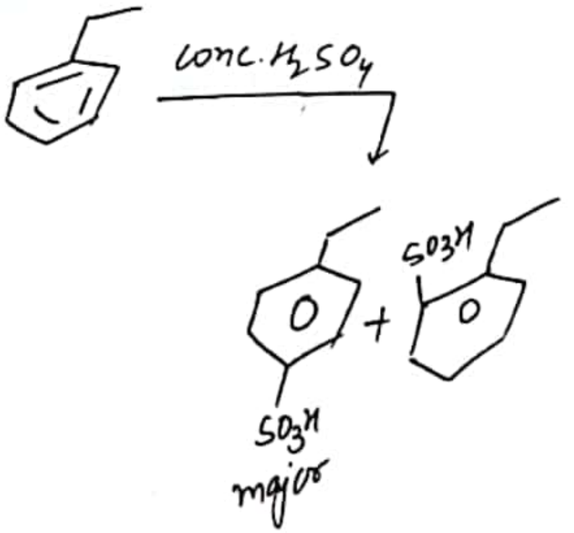
Eg 2



Eg 3



Eg 4

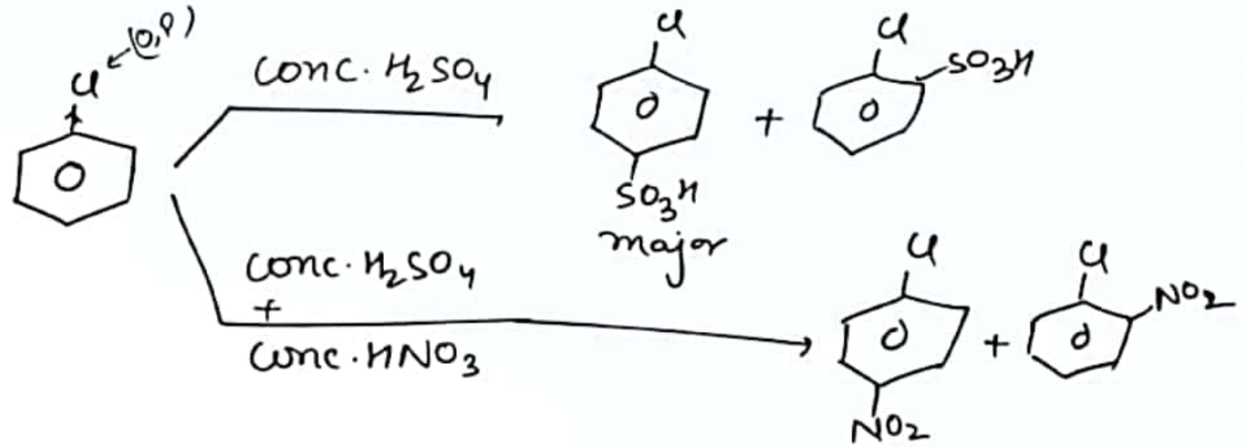


competishun
Power of real guns

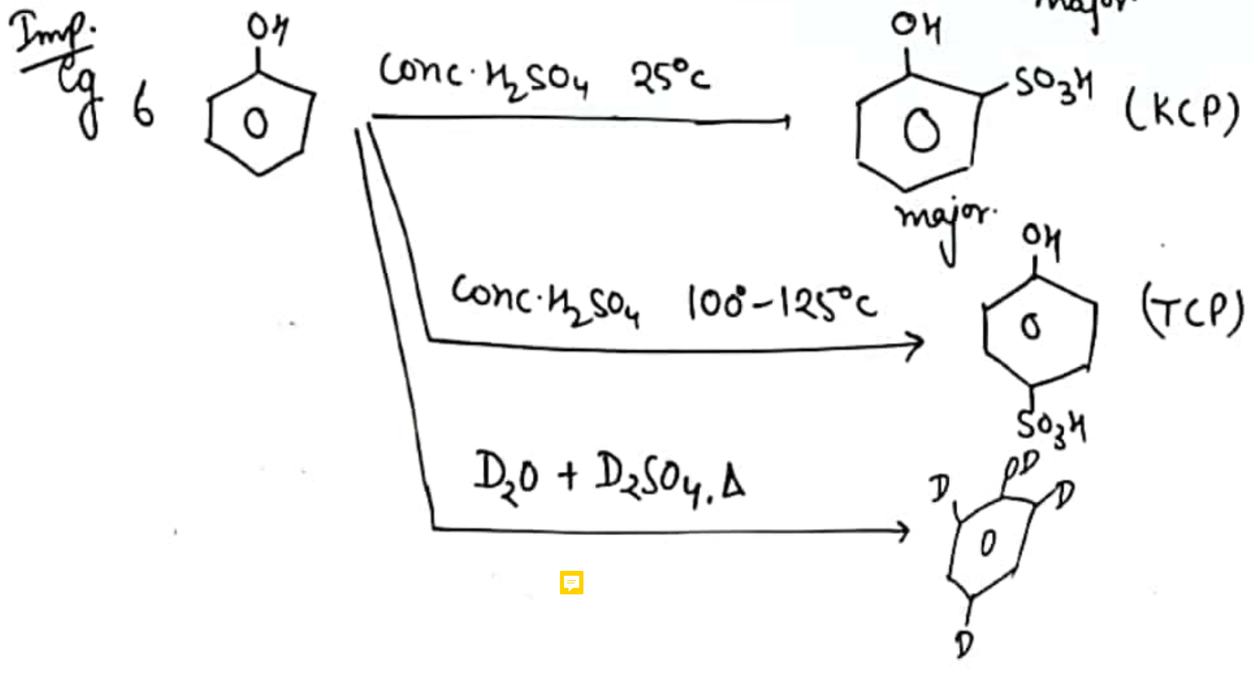
ORGANIC REACTION MECHANISM - II (ORM-II)

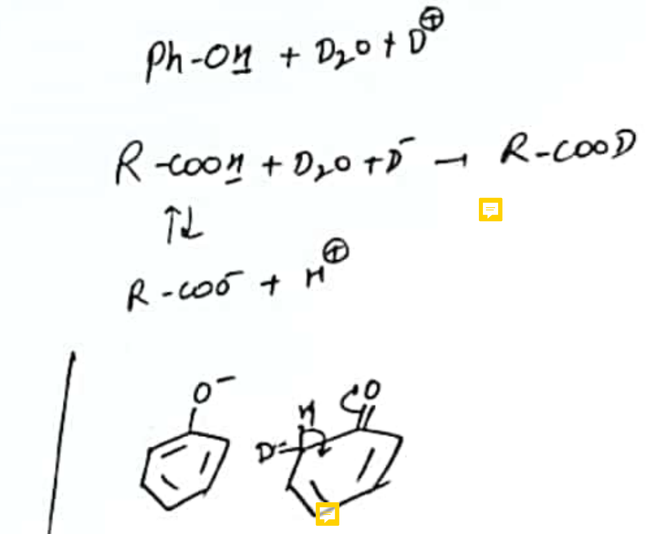
(I) Electrophilic substitution reactions ($Ar S_E$)

Eg 5



Imp. Eg 6





competishun

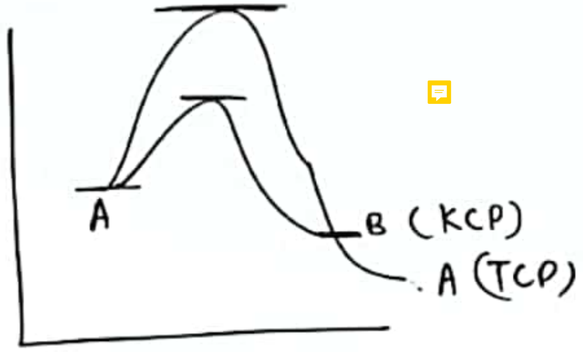
ORGANIC REACTION MECHANISM - II (ORM-II)

(I) Electrophilic substitution reactions (A_rS_E)

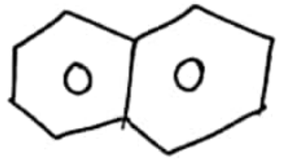
KCP = Kinetic control product ⇒ major at lower temp.

TCP = Thermodyanamic control product ⇒ major at higher temp.

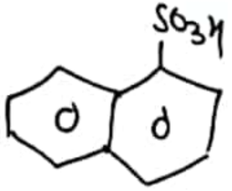




Eg:

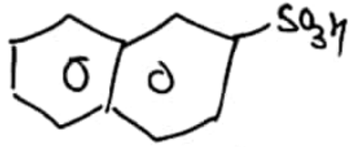


Sulphonation →



major at 80°C (KCP)

+



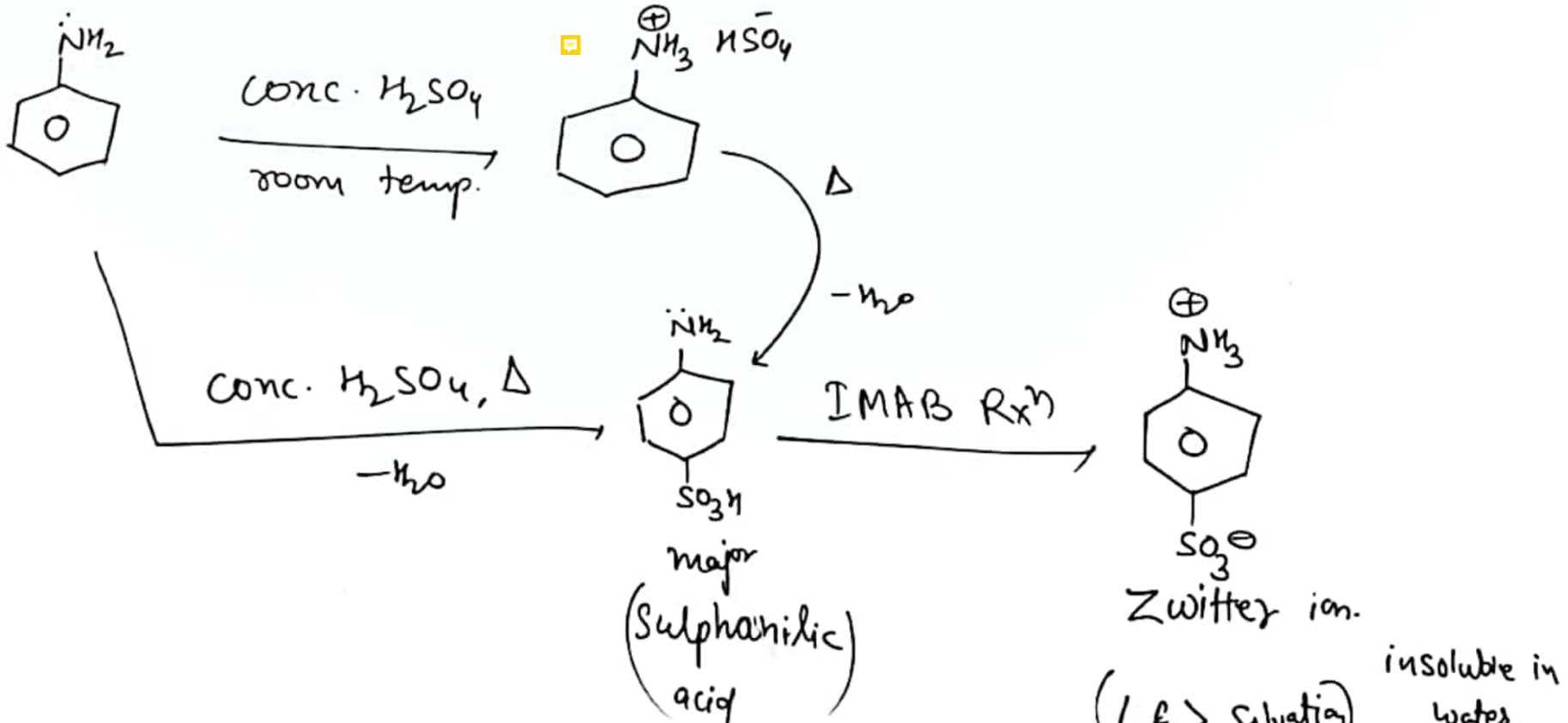
major at 160°C (TCP)

competishun
Power of real guruji

ORGANIC REACTION MECHANISM - II (ORM-II)

(I) Electrophilic substitution reactions ($Ar S_E$)

eg :



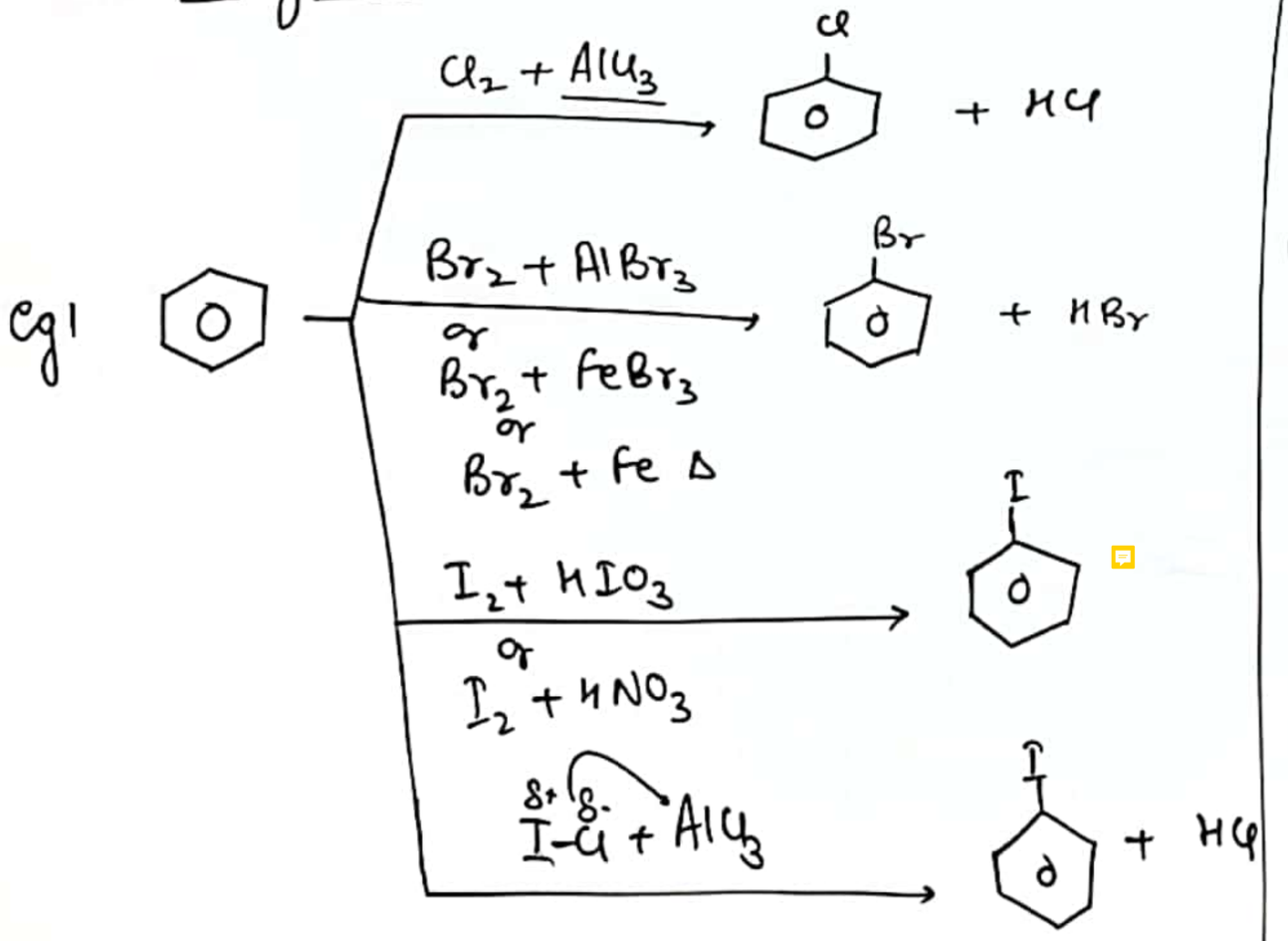
competishun
Power of real guide

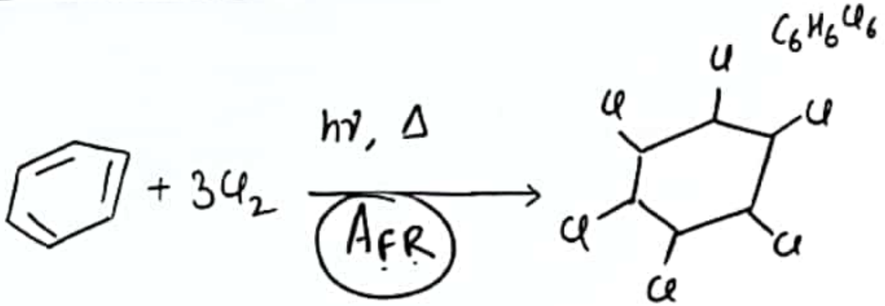
ORGANIC REACTION MECHANISM-II

ORM-II

(I) Electrophilic substitution Reactions (S_E)

(c) Halogenation:-



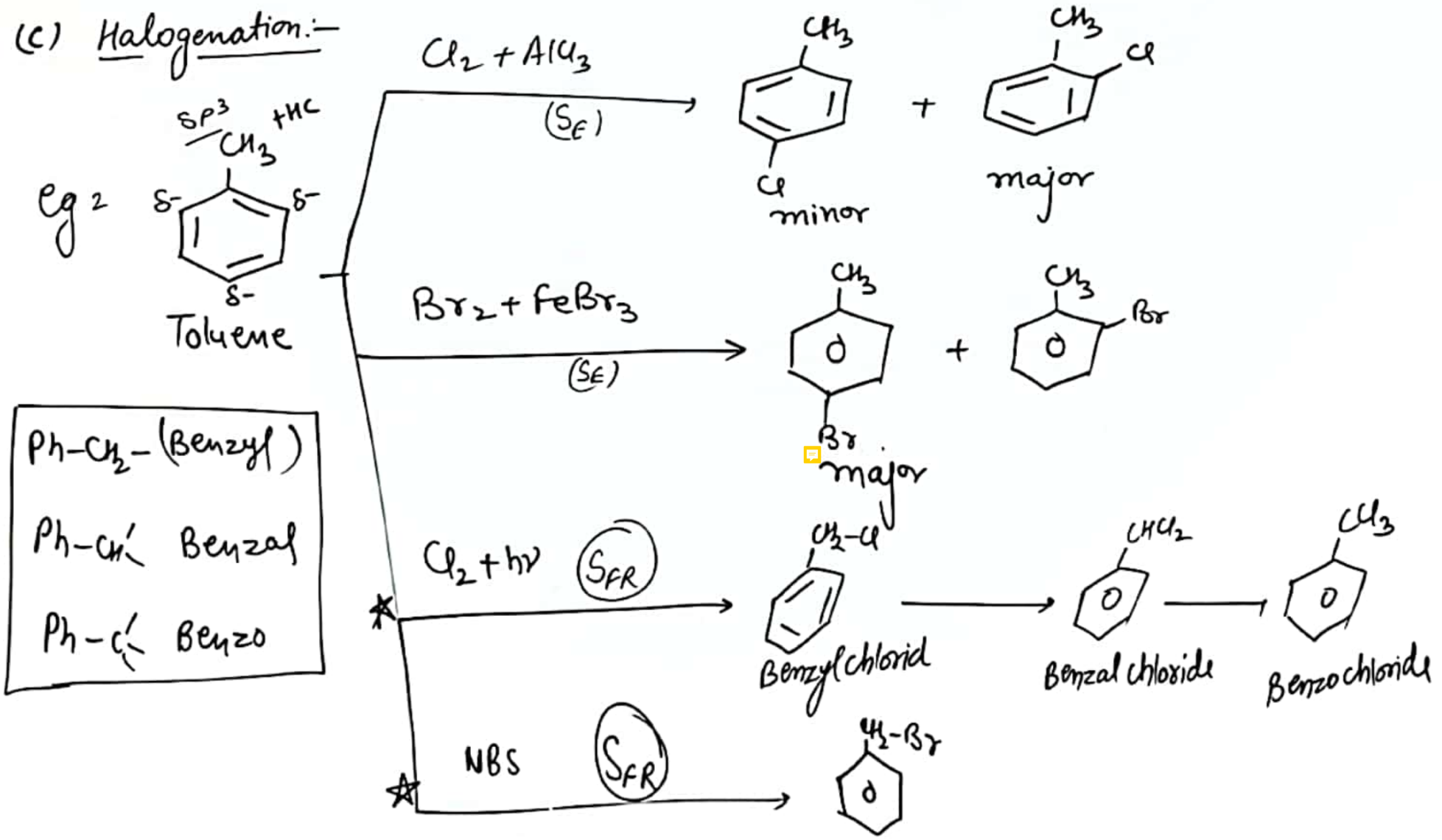


⇒ Benzene Hexa Chloride
⇒ BHC
⇒ 666
⇒ Gamaxene
⇒ Lindane

ORGANIC REACTION MECHANISM-II ORM-II

(I) Electrophilic substitution Reactions (S_E)

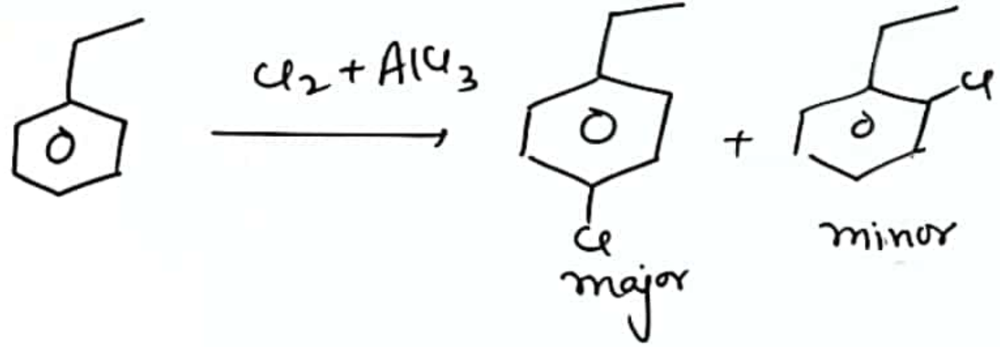
(c) Halogenation:-

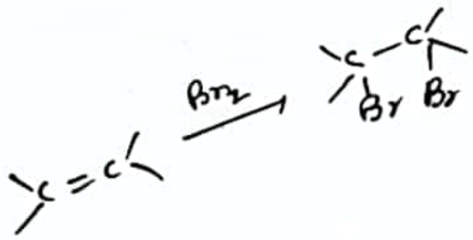


ORGANIC REACTION MECHANISM-II ORM-II

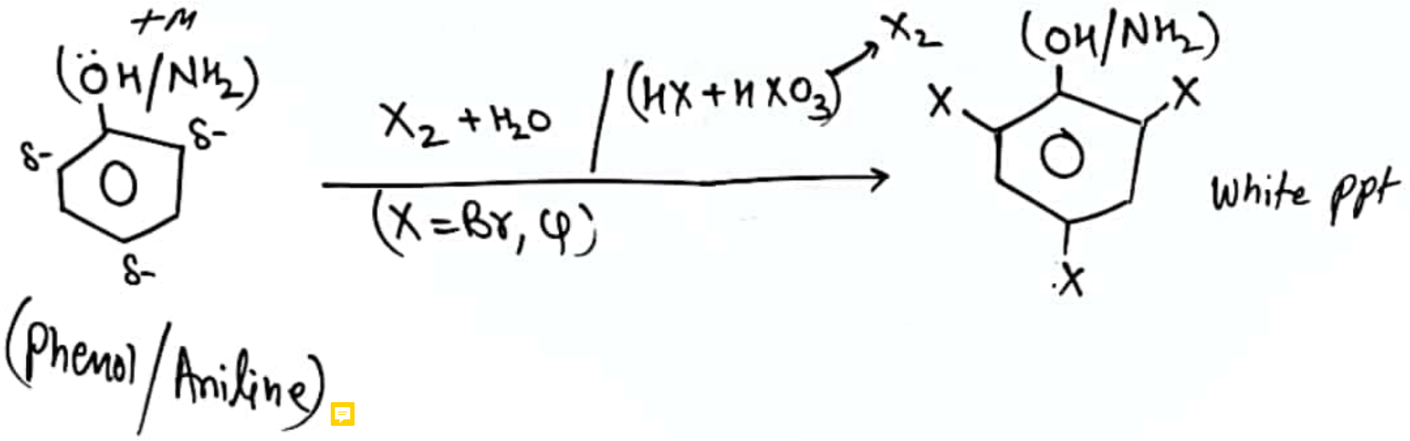
(I) Electrophilic substitution Reactions (S_E)

eg 3





eg 4



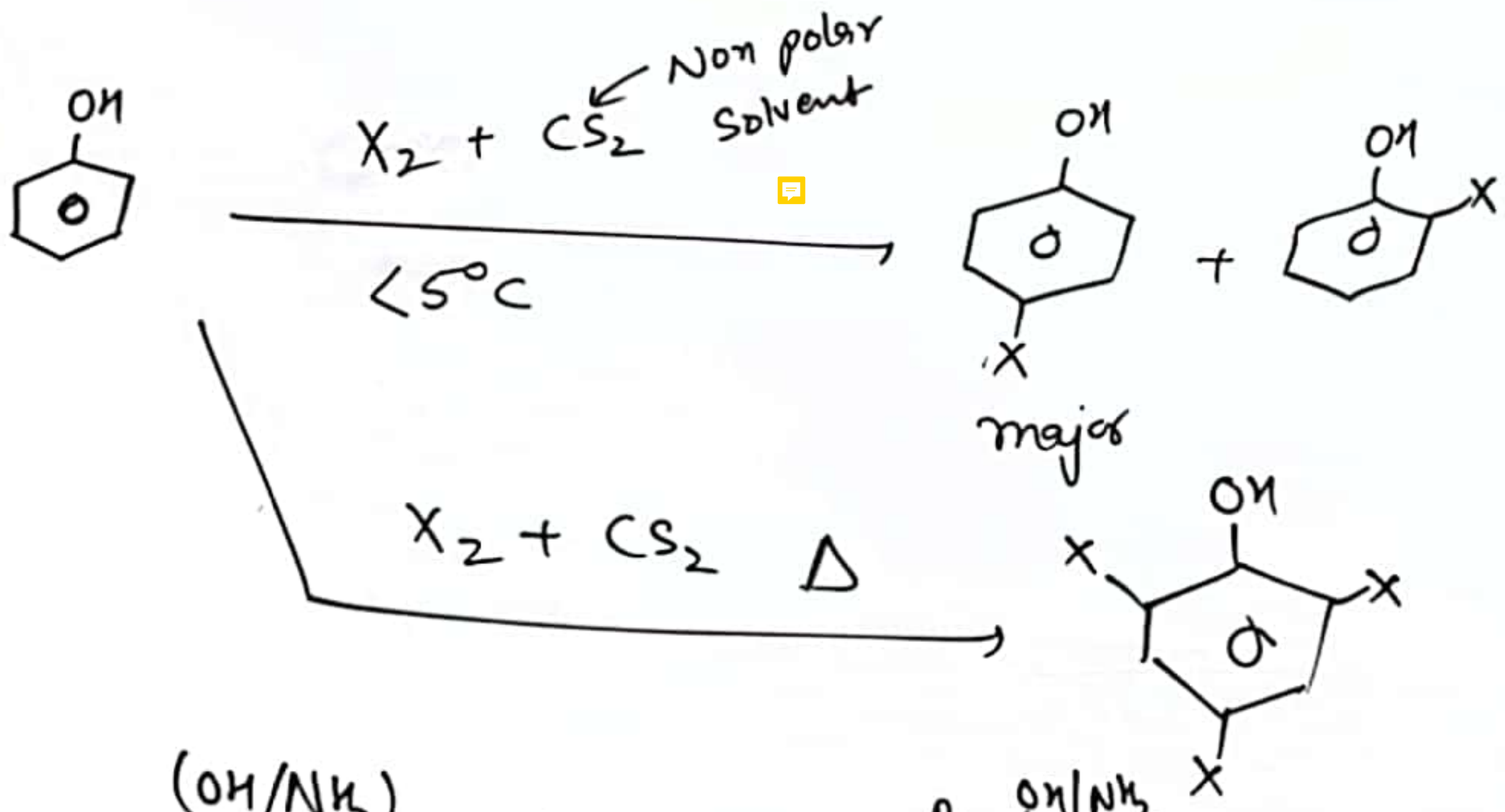
(Phenol / Aniline)

ORGANIC REACTION MECHANISM-II ORM-II

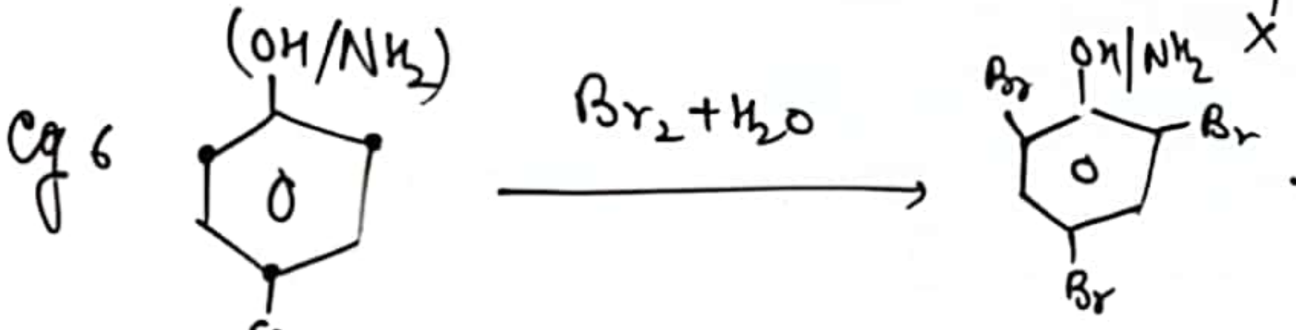
Electrophilic Substitution Reactions (S_E)

$Ph\text{-}OH \rightleftharpoons \underline{Ph\text{-}O^-} + H^{\oplus}$

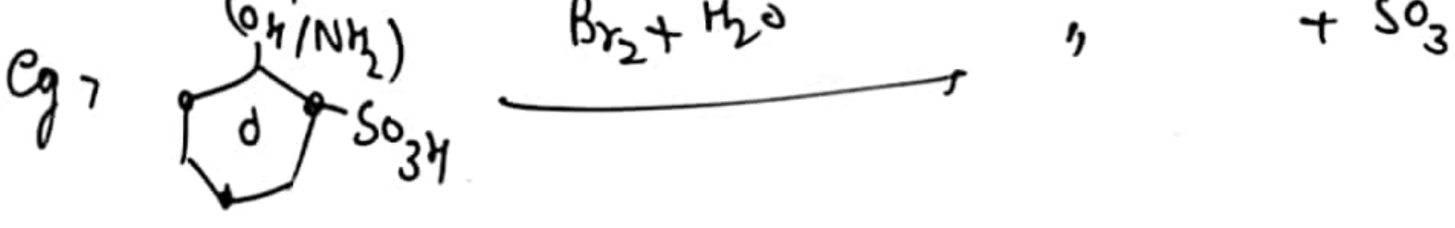
eg 5



eg 6



eg 7

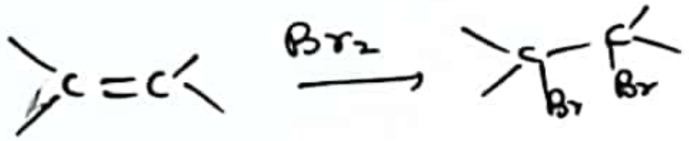


ORGANIC REACTION MECHANISM-II

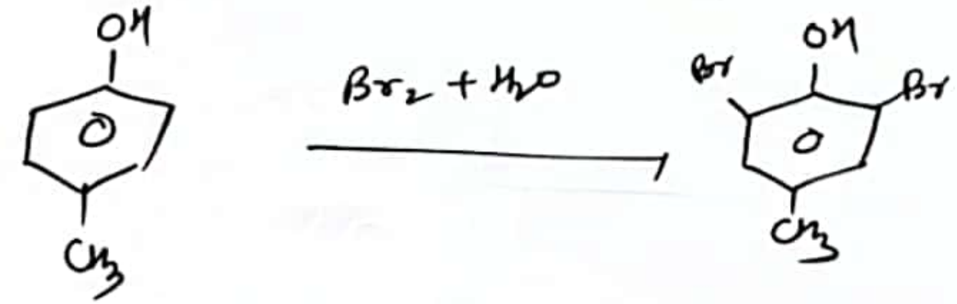
ORM-II

(I) Electrophilic substitution Reactions (S_E)

$$Ph\text{-}OH \rightleftharpoons \underline{Ph\text{-}O^-} + H^{\oplus}$$

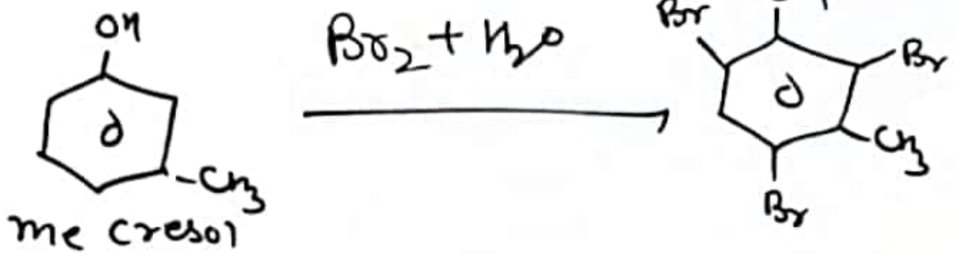


Eg 8



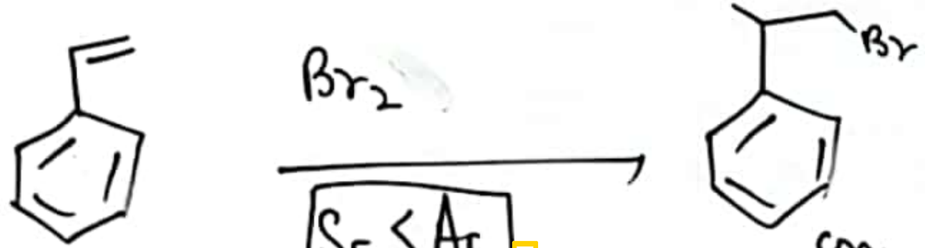
P-cresol

Eg 9



me cresol

Eg 10



$S_E < A_E$

Eg 11 Cinnamic acid



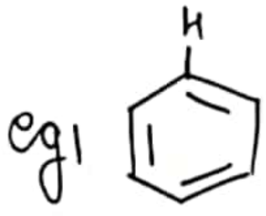
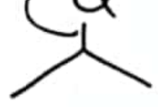
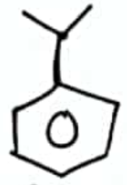
ORGANIC REACTION MECHANISM - II (ORM - II)

(I) Electrophilic substitution reactions (S_E)

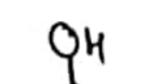
(D) Friedel craft alkylation reaction:-

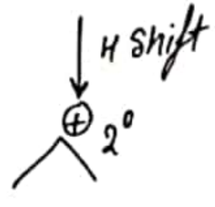
$R^{\oplus}$ = Electrophile

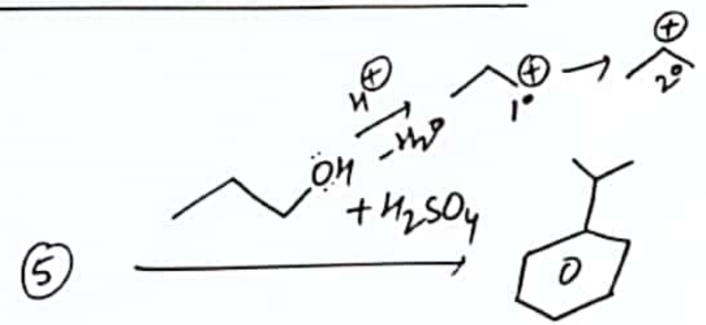
① $CH_3-Cl + AlCl_3 \longrightarrow$  $+ HCl$

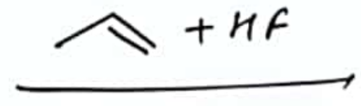
②  (egl) $+$  $+ AlCl_3 \longrightarrow$  Cumene $+ HCl$

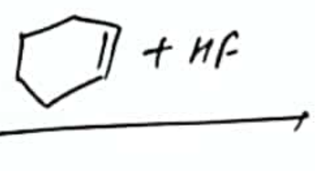
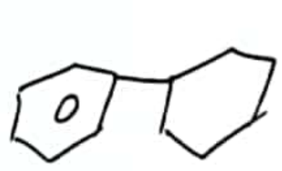
③  $+ AlCl_3 \longrightarrow$ " $+ HCl$

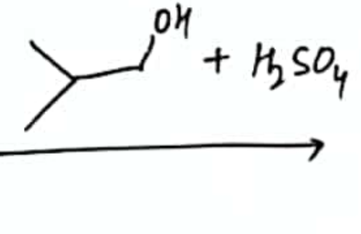
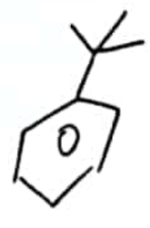
④  $+ HF \longrightarrow$ 

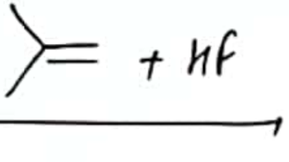


⑤ 

⑥  $\longrightarrow$ "

⑦  $\longrightarrow$ 

⑧  $\longrightarrow$ 

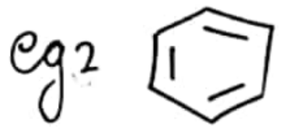
⑨  $\longrightarrow$ "

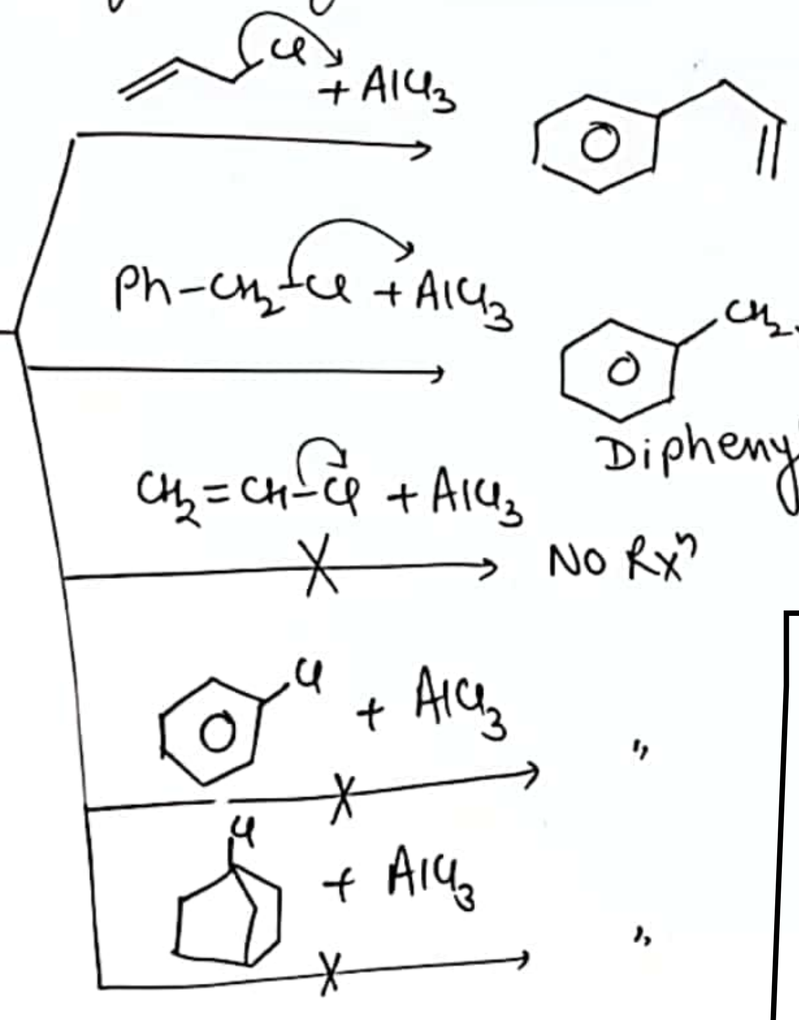
ORGANIC REACTION MECHANISM - II (ORM - II)

(I) Electrophilic substitution reactions (S_E)

(D) Friedel craft alkylation reaction:-

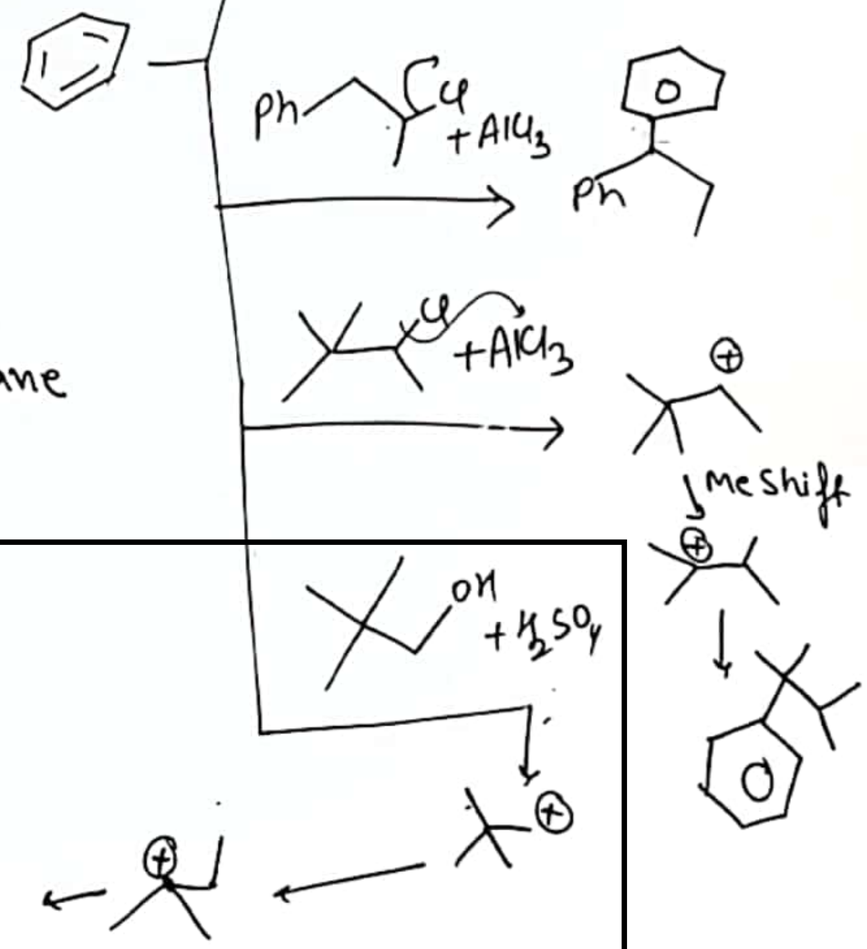
eg2





Diphenylmethane

eg3

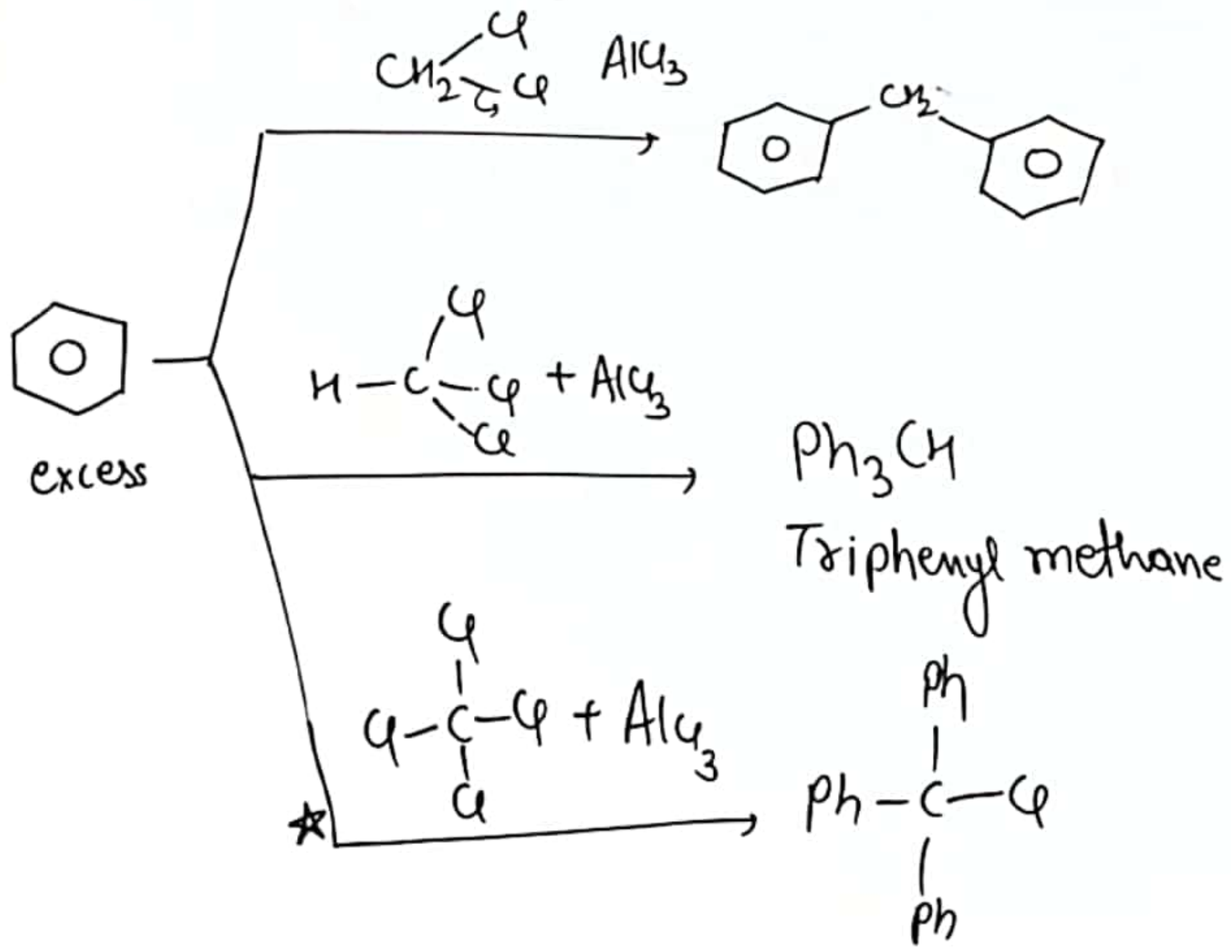


ORGANIC REACTION MECHANISM - II (ORM - II)

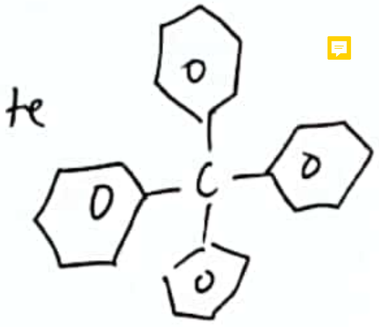
(I) Electrophilic substitution reactions (S_E)

(D) Friedel craft alkylation reaction:-

eg 4



Note



unstable

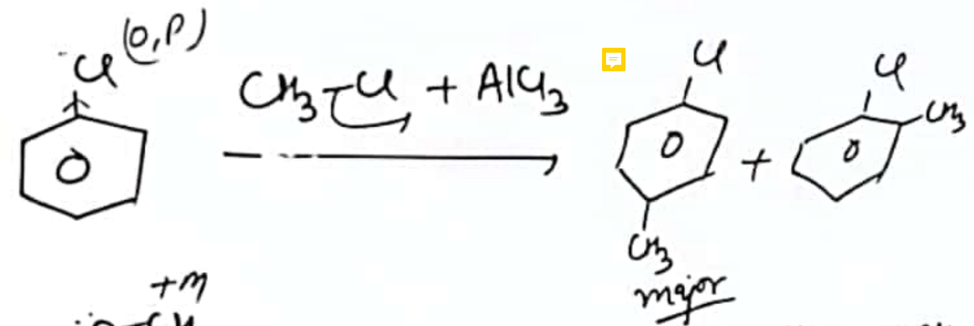
due to steric hindrance

ORGANIC REACTION MECHANISM - II (ORM - II)

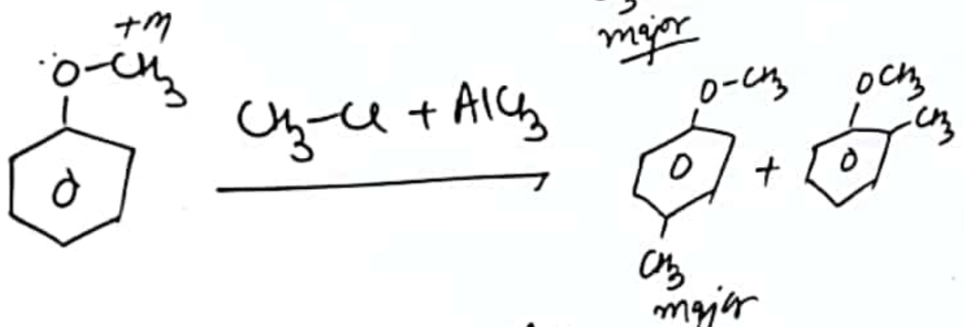
(I) Electrophilic substitution reactions (S_E)

(D) Friedel craft alkylation reaction:-

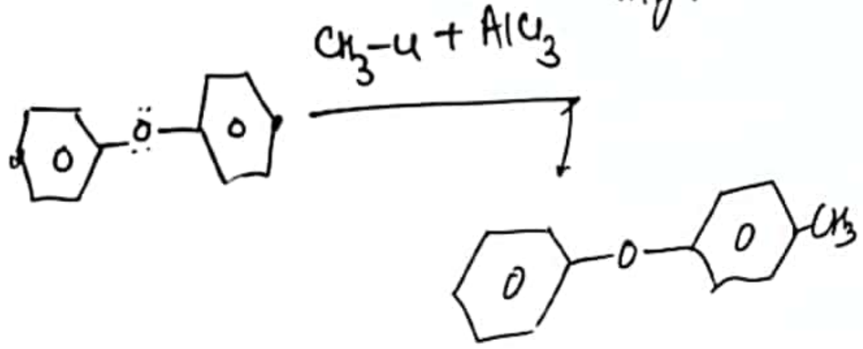
eg 5



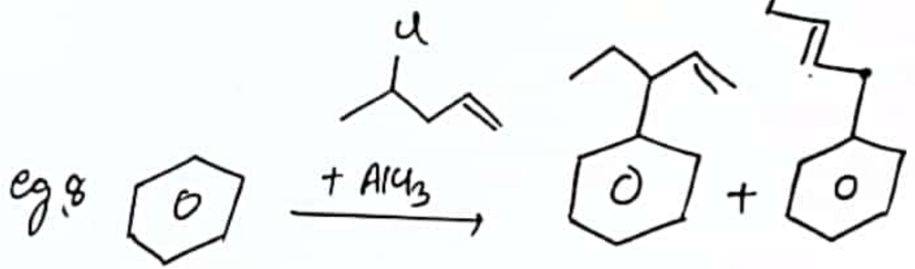
eg 6

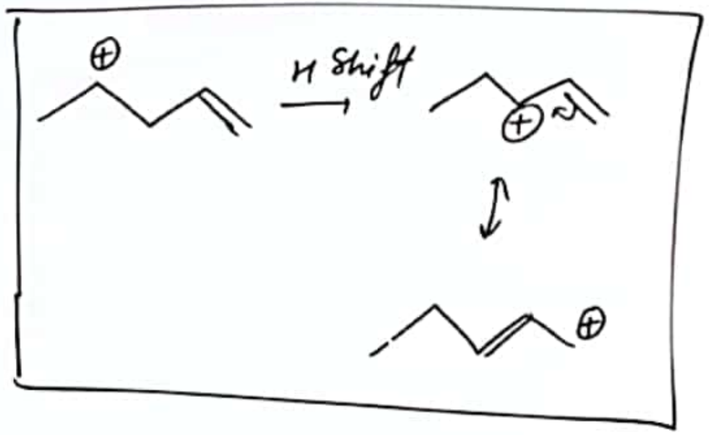


eg 7



eg 8



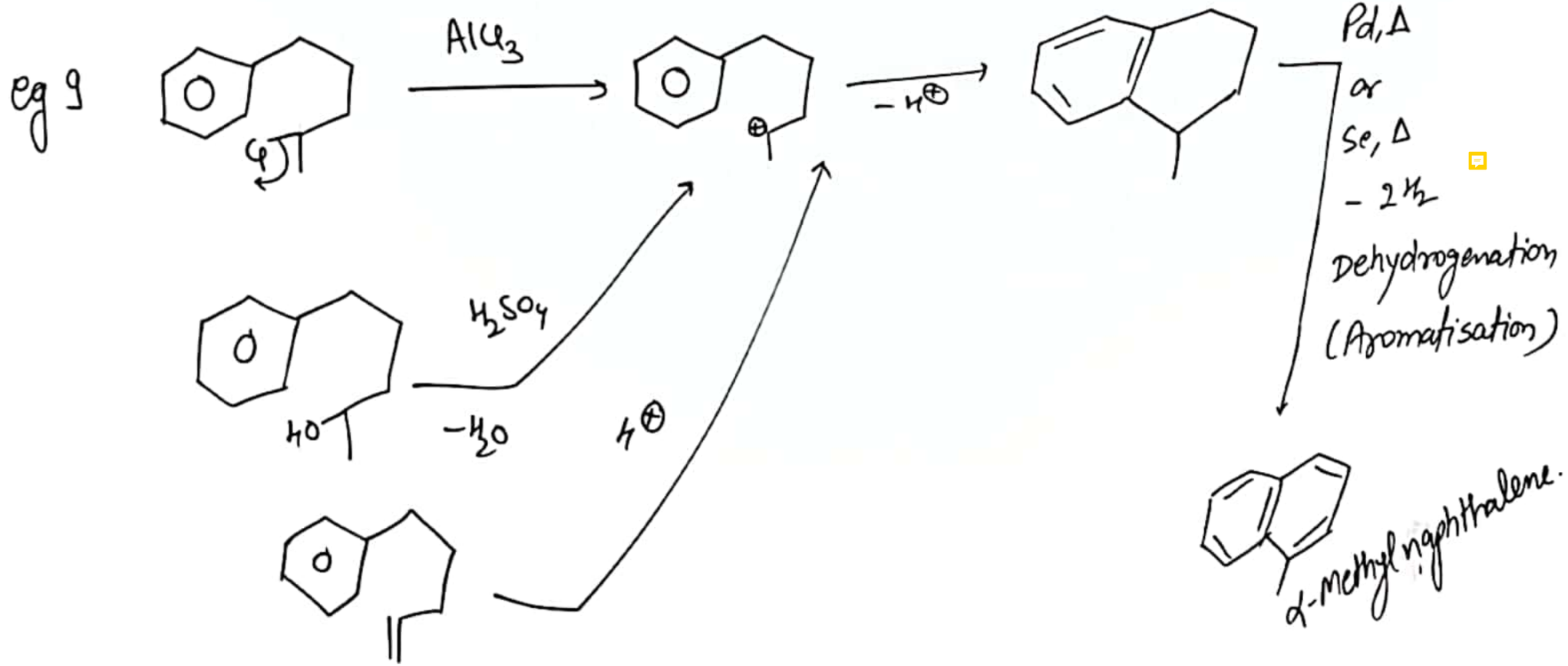


ORGANIC REACTION MECHANISM-II (ORM-II)

(I) Electrophilic substitution reactions (S_E)

(D) Friedel craft alkylation reaction:-

Intramolecular F.C. alkylation:-



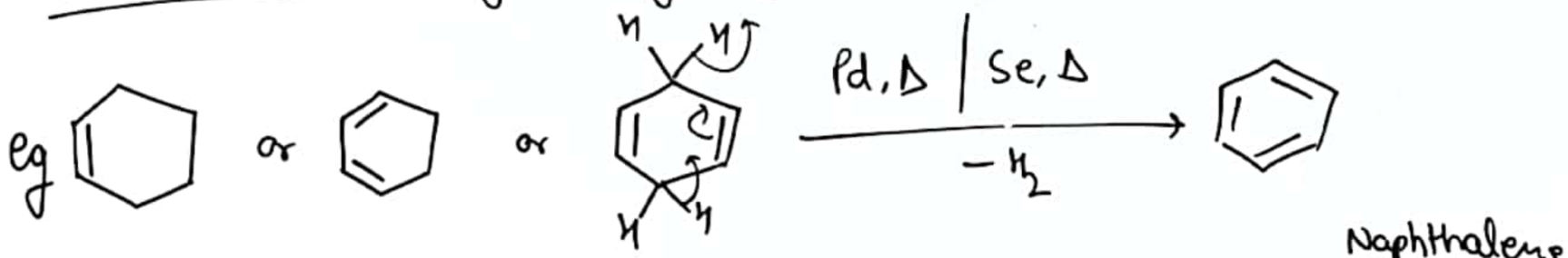
ORGANIC REACTION MECHANISM-II (ORM-II)

(I) Electrophilic substitution reactions (S_E)

(D) Friedel craft alkylation reaction:-

#Intramolecular F.C. alkylation:-

Aromatisation by dehydrogenation:-



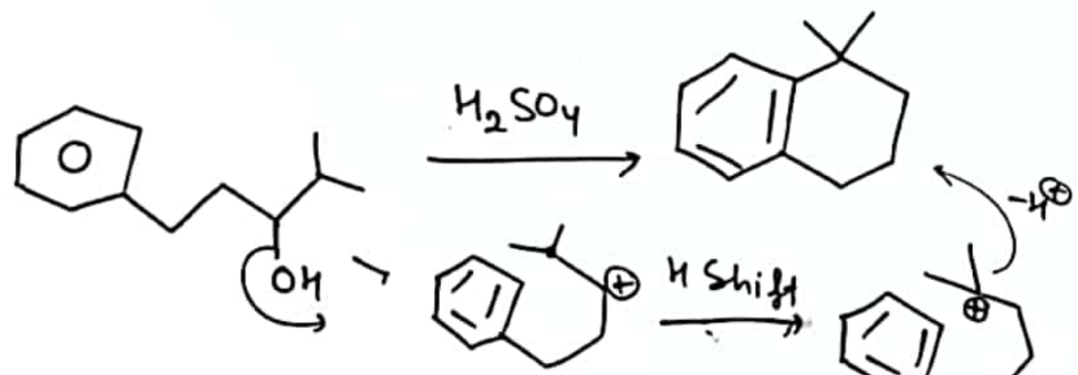
ORGANIC REACTION MECHANISM - II (ORM - II)

(I) Electrophilic substitution reactions (S_E)

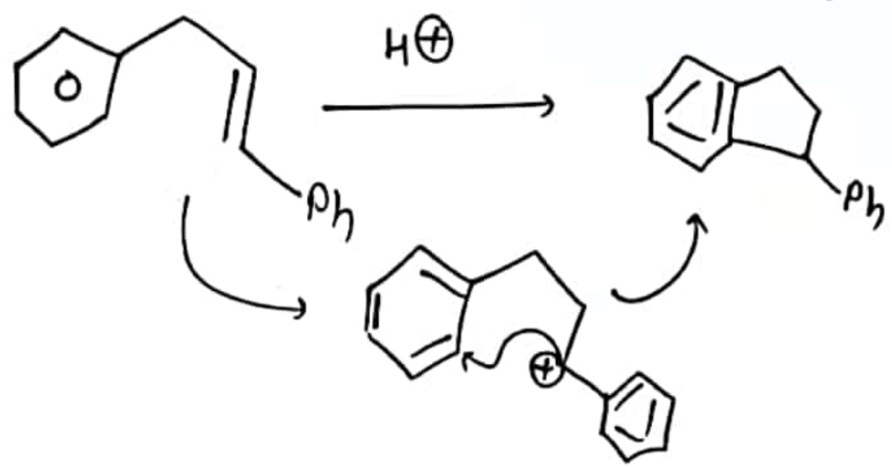
(D) Friedel craft alkylation reaction:-

Intramolecular F.C. alkylation:-

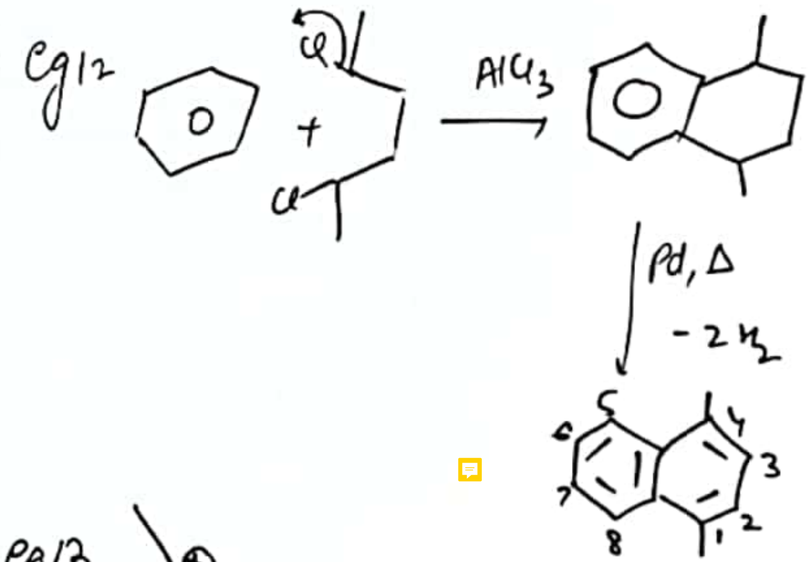
eg 10



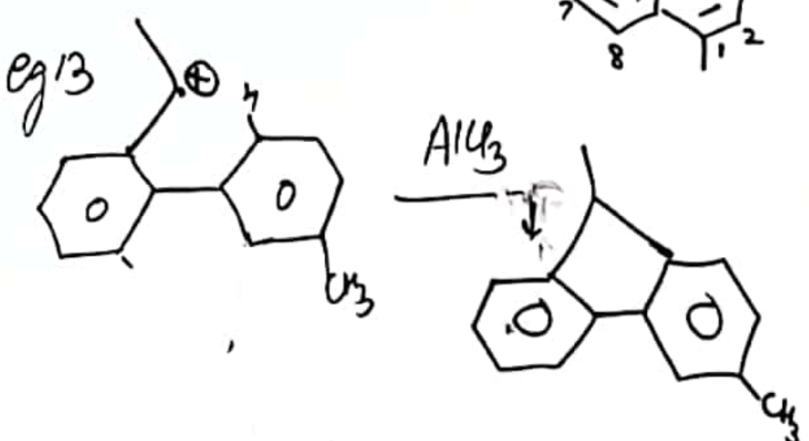
eg 11



eg 12



eg 13



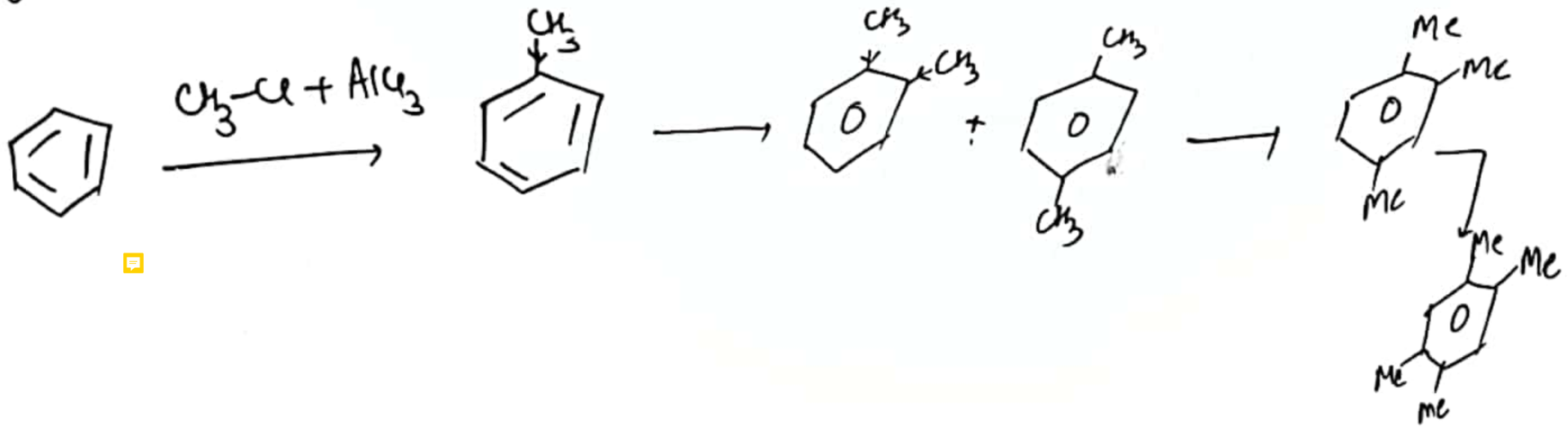
ORGANIC REACTION MECHANISM-II (ORM-II)

(I) Electrophilic substitution reactions (S_E)

(D) Friedel craft alkylation reaction:-

Note :-

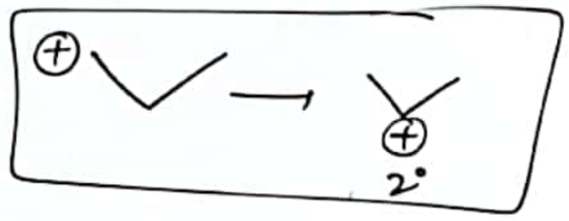
In F.C. alkylation polyalkylation take place because alkylated products are more stable than reactant ring.



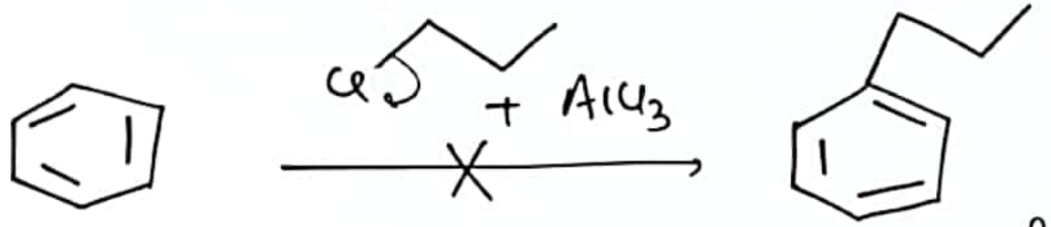
ORGANIC REACTION MECHANISM - II (ORM - II)

(I) Electrophilic substitution reactions (S_E)

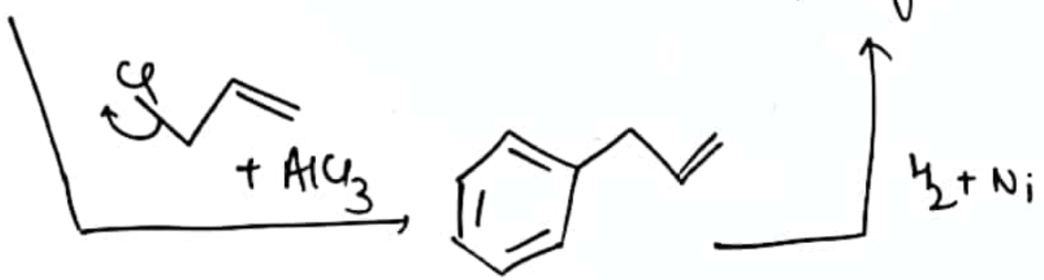
(D) Friedel craft alkylation reaction:-



Q



n-propyl benzene

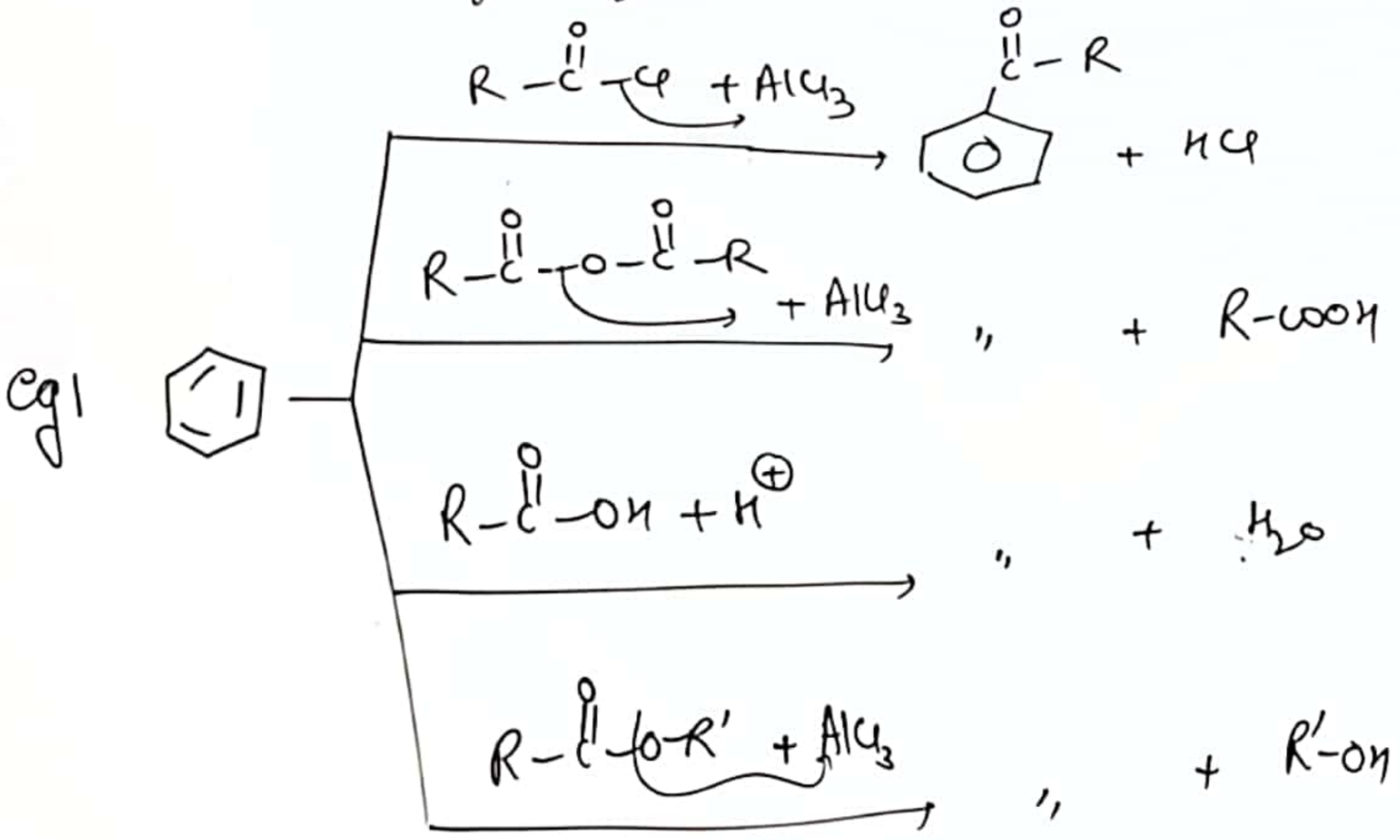


ORGANIC REACTION MECHANISM - II (ORM - II)

(I) Electrophilic substitution reactions (S_E)

(E) Friedel craft acylation:-

$E^{\oplus} = R-\overset{\overset{O}{\|}}{C}{}^{\oplus}$

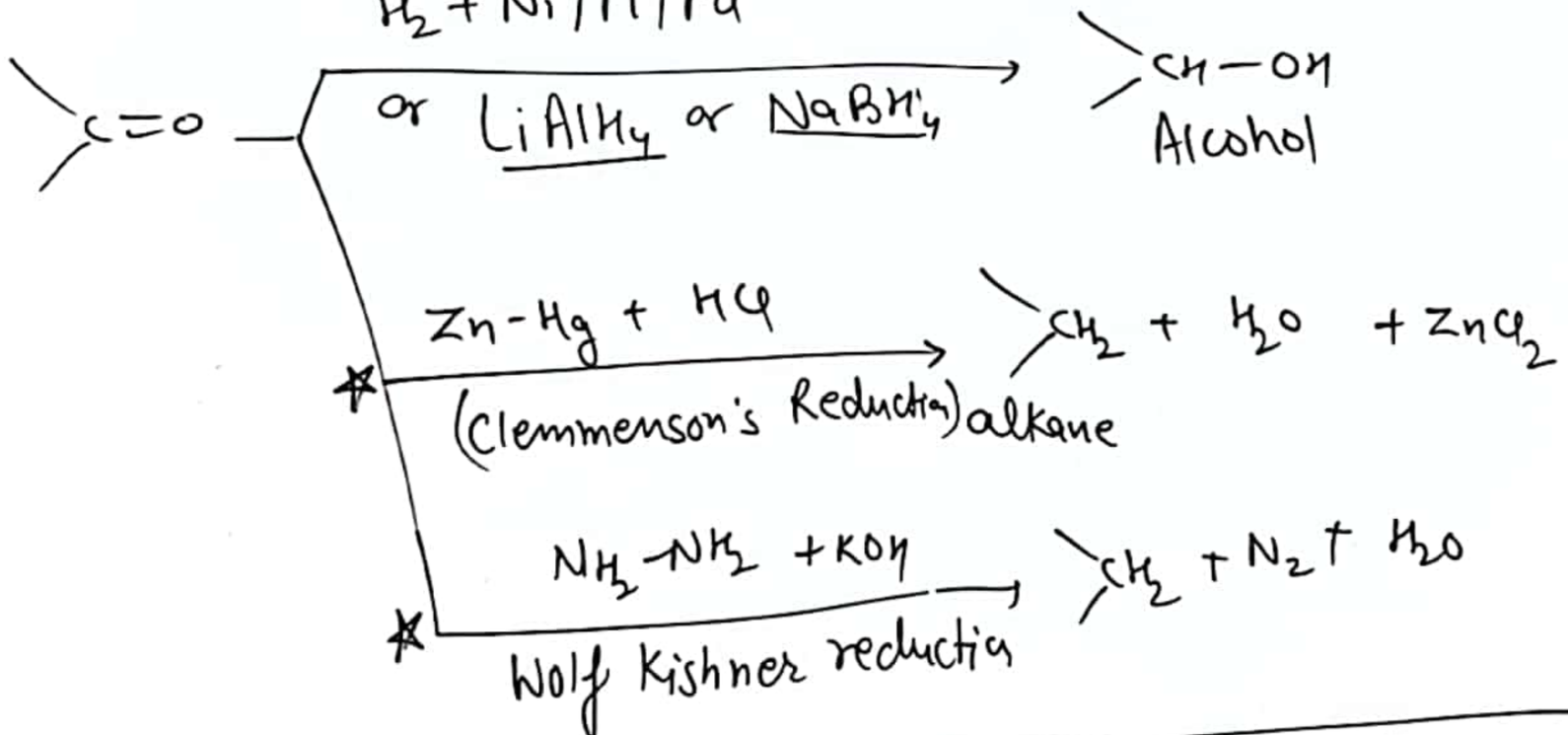


ORGANIC REACTION MECHANISM-II (ORM-II)

(I) Electrophilic substitution reactions (S_E)

(E) Friedel craft acylation:-

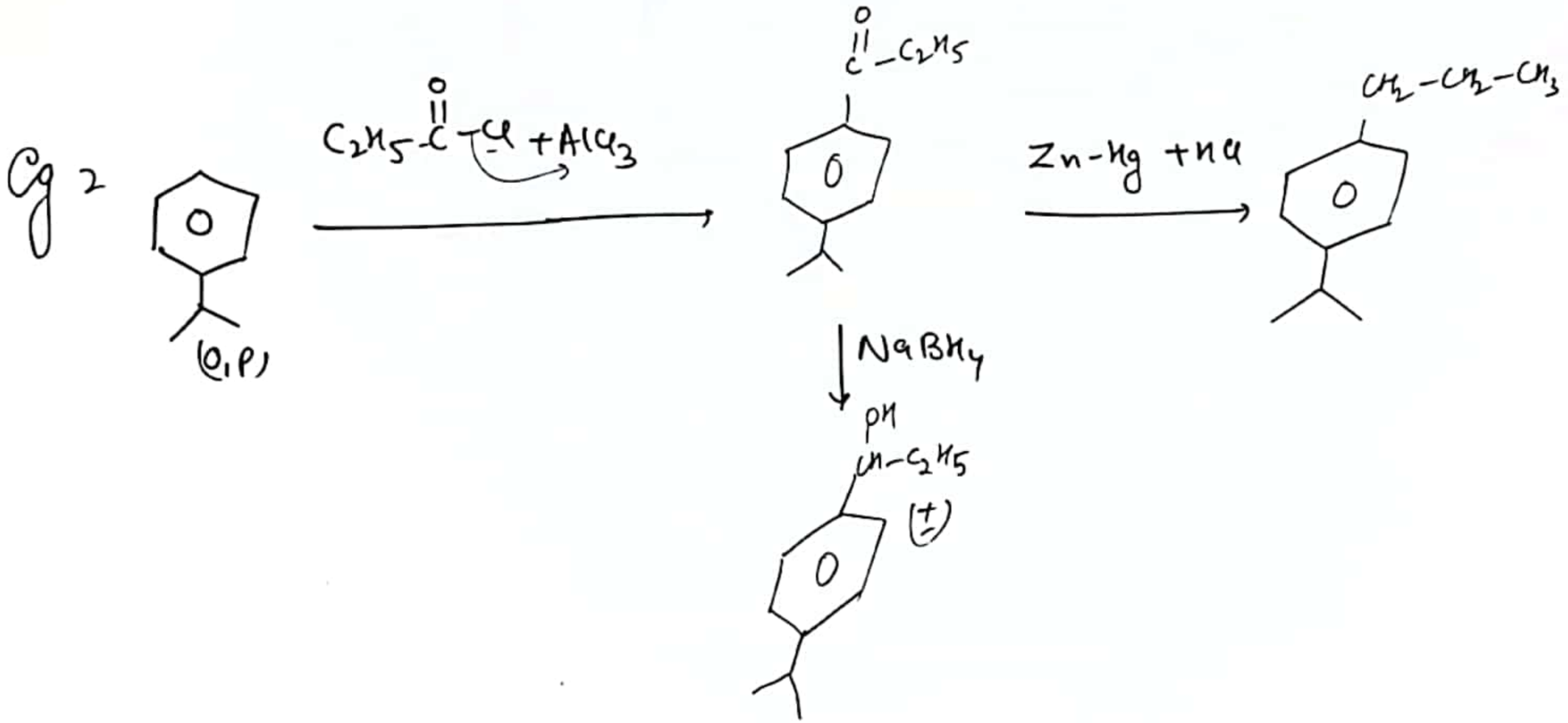
Reduction of aldehydes and Ketones:-



ORGANIC REACTION MECHANISM - II (ORM - II)

(I) Electrophilic substitution reactions (S_E)

(E) Friedel craft acylation:-

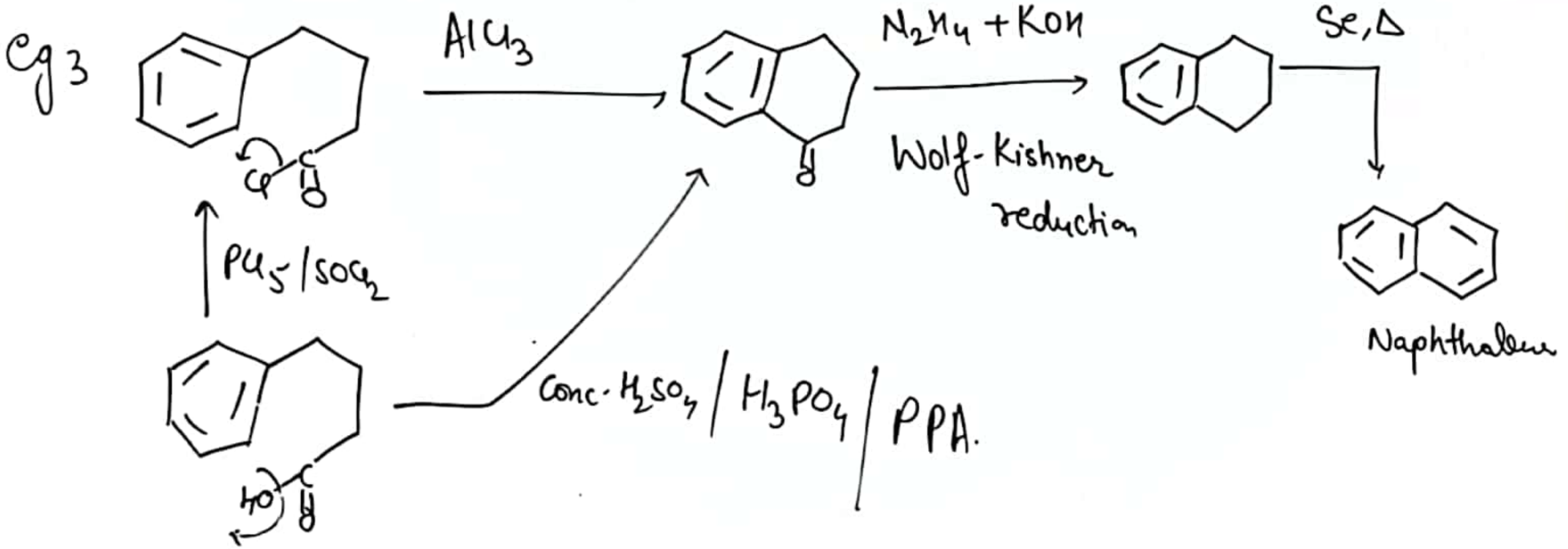


ORGANIC REACTION MECHANISM - II (ORM - II)

(I) Electrophilic substitution reactions (S_E)

(E) Friedel craft acylation:-

Intramolecular F.C. acylation.



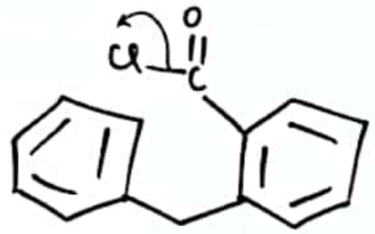
ORGANIC REACTION MECHANISM - II (ORM - II)

(I) Electrophilic substitution reactions (S_E)

(E) Friedel craft acylation:-

Intramolecular F.C. acylation.

eg 4



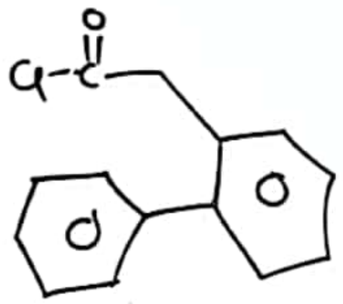
(i) $AlCl_3$

(ii) Zn-Hg + HCl

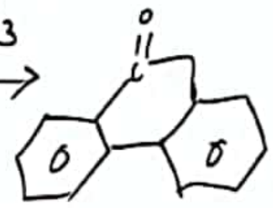
(ii) Se, Δ



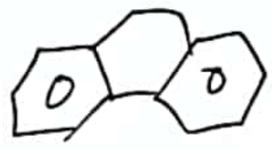
eg 5



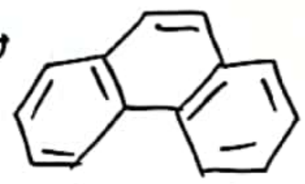
(i) $AlCl_3$



N_2H_4 + KOH



Se Δ



Phenanthrene.

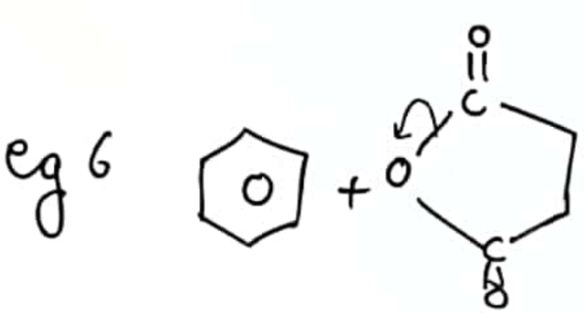
ORGANIC REACTION MECHANISM - II (ORM - II)

(I) Electrophilic substitution reactions (S_E)

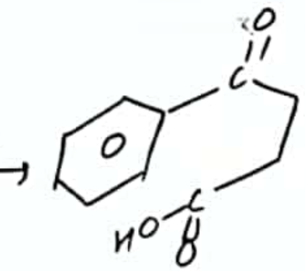
(E) Friedel craft acylation:-

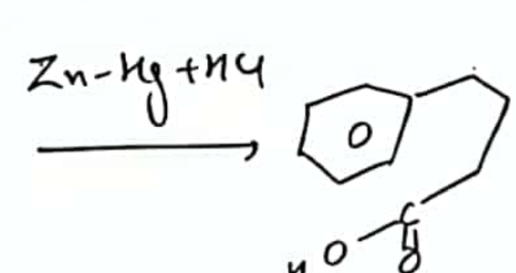
Intramolecular F.C. acylation.

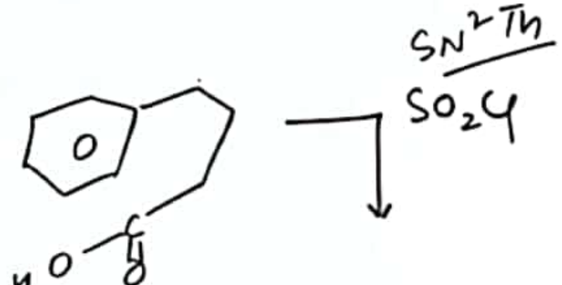
eg 6

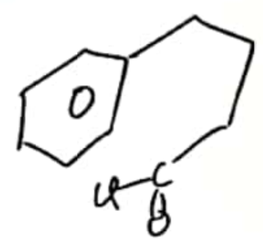


Succinic anhydride

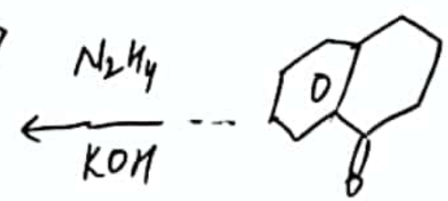


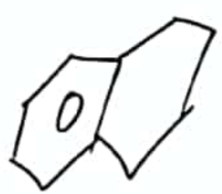














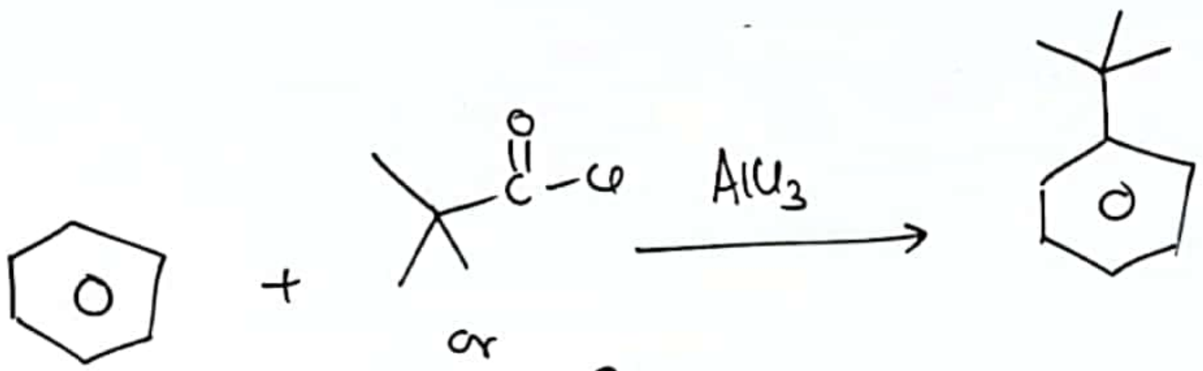
ORGANIC REACTION MECHANISM - II (ORM - II)

(I) Electrophilic substitution reactions (S_E)

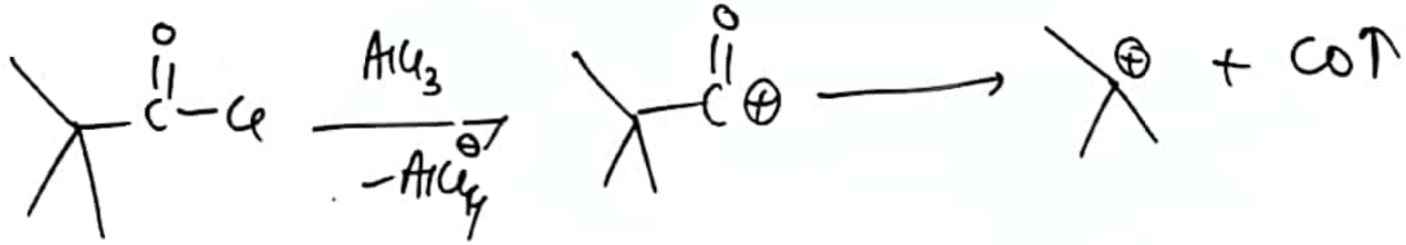
(E) Friedel craft acylation:-

Intramolecular F.C. acylation.

Imp.



Hint



ORGANIC REACTION MECHANISM-II (ORM-II)

(I) Electrophilic substitution reactions (S_E)

Limitation of Friedel craft reactions (alkylation - acylation)

ⓐ Both F.C reactions are not possible in presence of strong deactivating group (all meta directing group)

$-\overset{\oplus}{N}H_3$, $-\overset{\oplus}{N}R_3$, $-NO_2$ $-CN$ $-CHO$ $-\overset{O}{\overset{\|}{C}}-R$, $-COOH$, $-COCl$, $-CX_3$ etc.

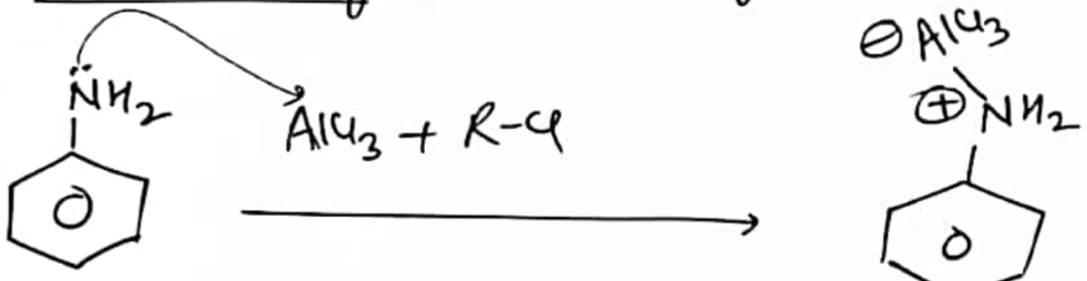
$\Rightarrow R\overset{\oplus}{} = \overset{\oplus}{C}$ = weak electrophile

ⓑ Both F.C. reactions are not possible in presence of strong activating group ($-\ddot{O}^{\ominus}$, $-OH$, $-\ddot{N}H_2$, $-\ddot{N}HR$, $-\ddot{N}R_2$), $AlCl_3$ can form complex with above group

ORGANIC REACTION MECHANISM-II (ORM-II)

(I) Electrophilic substitution reactions (S_E)

Limitation of Friedel craft reactions (alkylation - acylation)



Now deactivated ring

Note :- In presence of following group F.C. reactions are possible

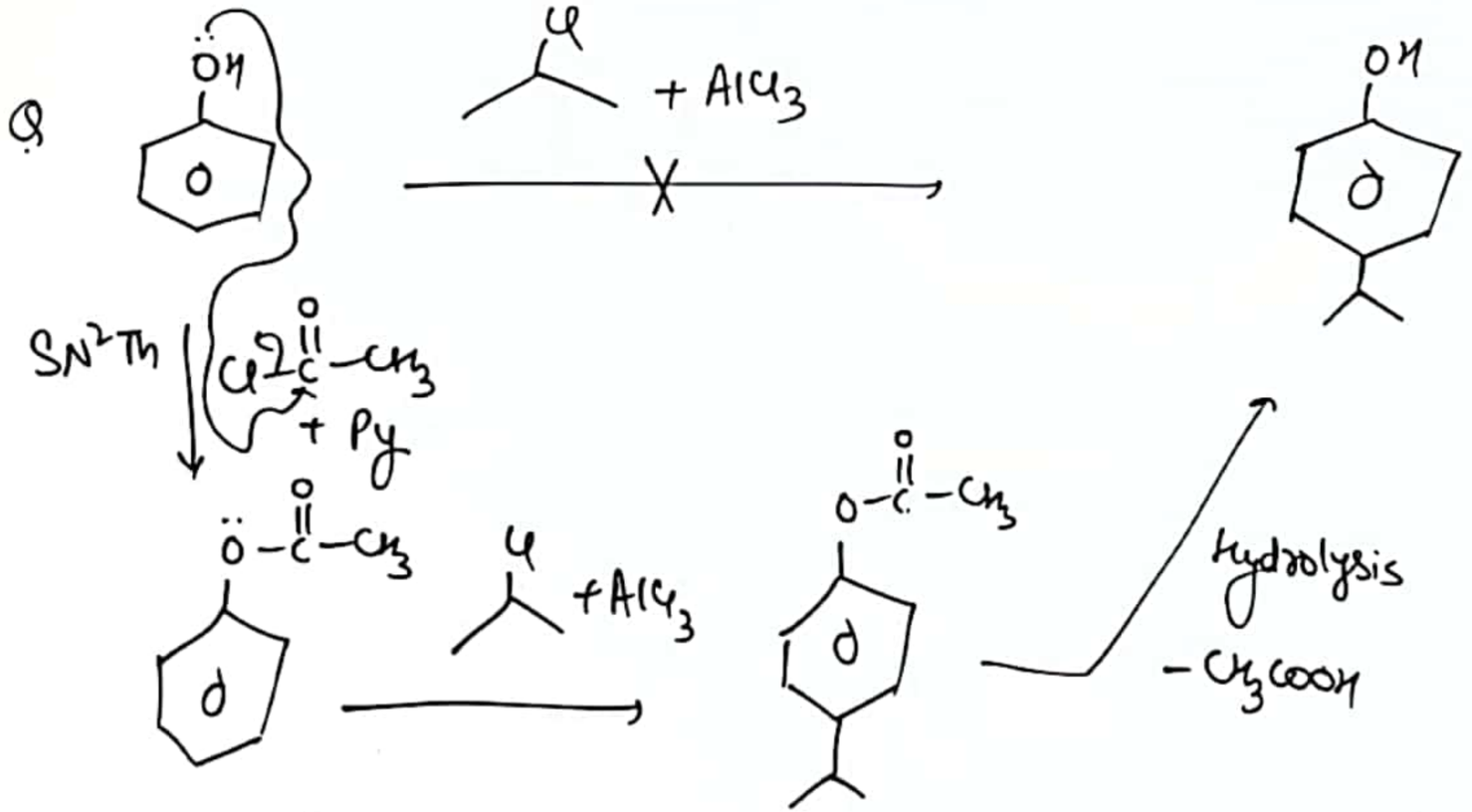
Alkyl (1°, 2°, 3°), $-CH=CH_2$, $-C\equiv C-R$, -Ar, $-X$ (F, Cl, Br, I)

$-\ddot{O}-\overset{O}{\overset{\|}{C}}-R$ $-NH-\overset{O}{\overset{\|}{C}}-R$ etc

ORGHNIC REHCTION MECHHNISM-II (ORM-II)

(I) Electrophilic substitution reactions (S_E)

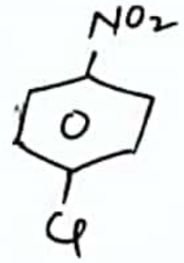
Limitation of Friedel craft reactions (alkylation - acylation)



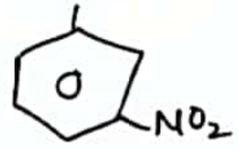
ORGANIC REACTION MECHANISM-II (ORM-II)

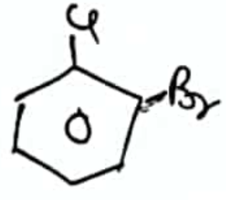
(I) Electrophilic substitution reactions (S_E)

Q

①  → → → → → 



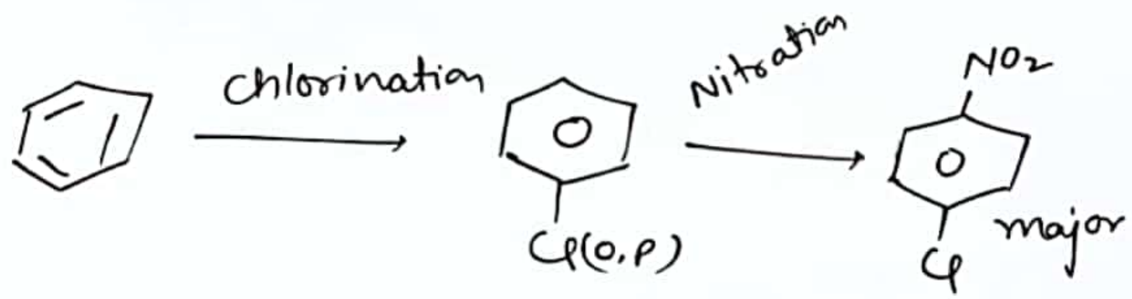
②  → → → 

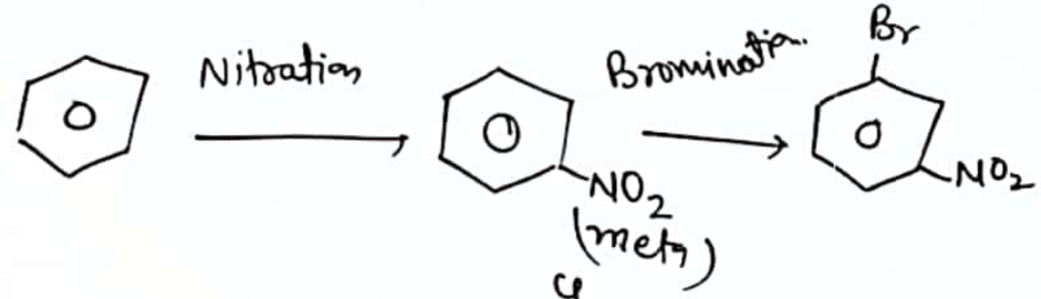
③  → → → 

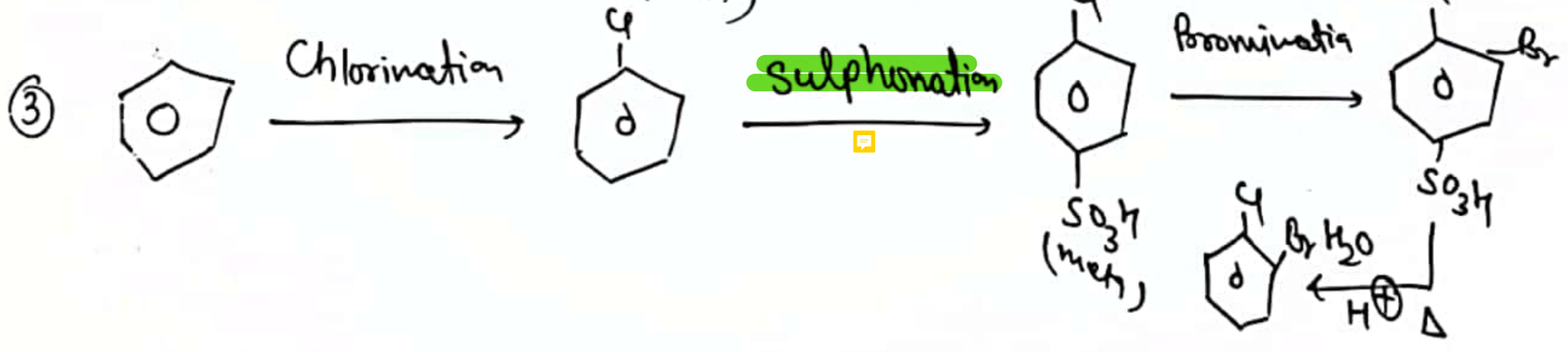
ORGANIC REACTION MECHANISM - II (ORM - II)

Q

Sol:

① 

② 

③ 

ORGANIC REACTION MECHANISM-II (ORM-II)

(II) FREE RADICAL SUBSTITUTION REACTIONS (S_{FR})

[A] Halogenation on saturated carbon.

(B) Halogenation on allylic and benzylic position with NBS / NCS

[C] Kolbe's electrolysis of carboxylic acid salt

[D] Hunsdeker's reaction.

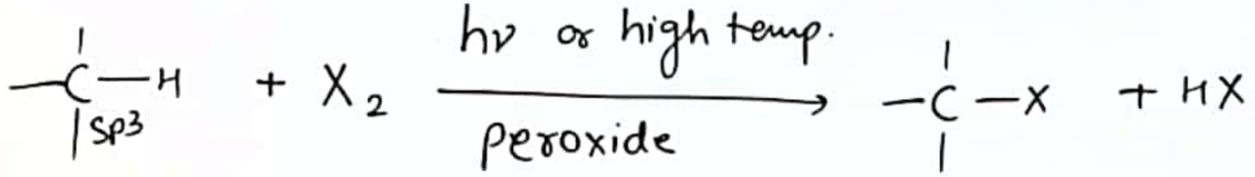
(E) Nitration of alkane

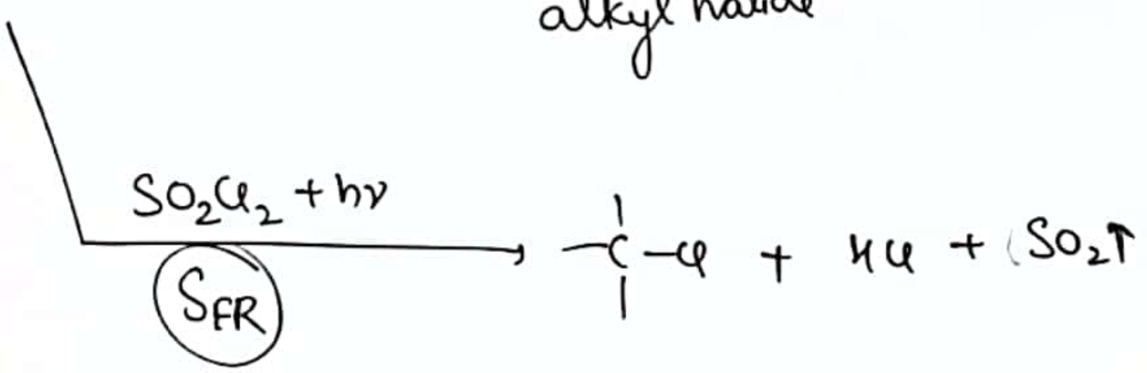
(F) Sulphonation of alkane etc

ORGANIC REACTION MECHANISM-II (ORM-II)

(II) FREE RADICAL SUBSTITUTION REACTIONS (S_{FR})

[A] Halogenation on saturated carbon.





⇒ Reactivity of X_2 ⇒ $F_2 > Cl_2 > Br_2 > I_2$

highly exothermic

Reversible reaction (in presence of oxidising agent HIO_3, HNO_3, HgO)

ORGANIC REACTION MECHANISM - II (ORM - II)

(II) FREE RADICAL SUBSTITUTION REACTIONS (S_{FR})

[A] Halogenation on saturated carbon.

$$R-H + I_2 \rightleftharpoons R-I + \underline{HI}$$

↑ strong Reducing agent

$$HI + \underline{HIO_3} \longrightarrow I_2 + H_2O$$

⇒ Reactivity w.r.t. C-H bond ⇒ 3° C-H > 2° C-H > 1° C-H

ORGANIC REACTION MECHANISM - II (ORM - II)

(II) FREE RADICAL SUBSTITUTION REACTIONS (S_{FR})

[A] Halogenation on saturated carbon.

Bond energy $F_2 > Cl_2 > Br_2 > I_2$ ✗

$Cl_2 > Br_2 > F_2 > I_2$ ✓

Mechanism (Chain reaction)

(i) **Chain initiation step:-**

$$X\!-\!X \xrightarrow[\text{Homolysis}]{h\nu \text{ or } \Delta \ [\text{Endo}]} 2X^{\bullet}$$

(ii) **Chain propagation step**

- $R\!-\!H + X^{\bullet} \xrightarrow{[RDS]} R^{\bullet} + H\!-\!X$ [X = F = Exothermic ; X = Cl, Br, I = Endothermic]

$R\!-\!H + X^{\bullet} \xrightarrow{\text{high } E_a} R\!-\!X + H^{\bullet}$

$R^{\bullet} + X\!-\!X \longrightarrow R\!-\!X + X^{\bullet}$ (Exothermic)

(iii) **Chain termination**

$X^{\bullet} + X^{\bullet} \longrightarrow X_2$

$R^{\bullet} + X^{\bullet} \longrightarrow R\!-\!X$

$R^{\bullet} + R^{\bullet} \longrightarrow R\!-\!R$

} Exothermic

$H\!-\!F > \underset{|}{C}\!-\!H$ Bond energy

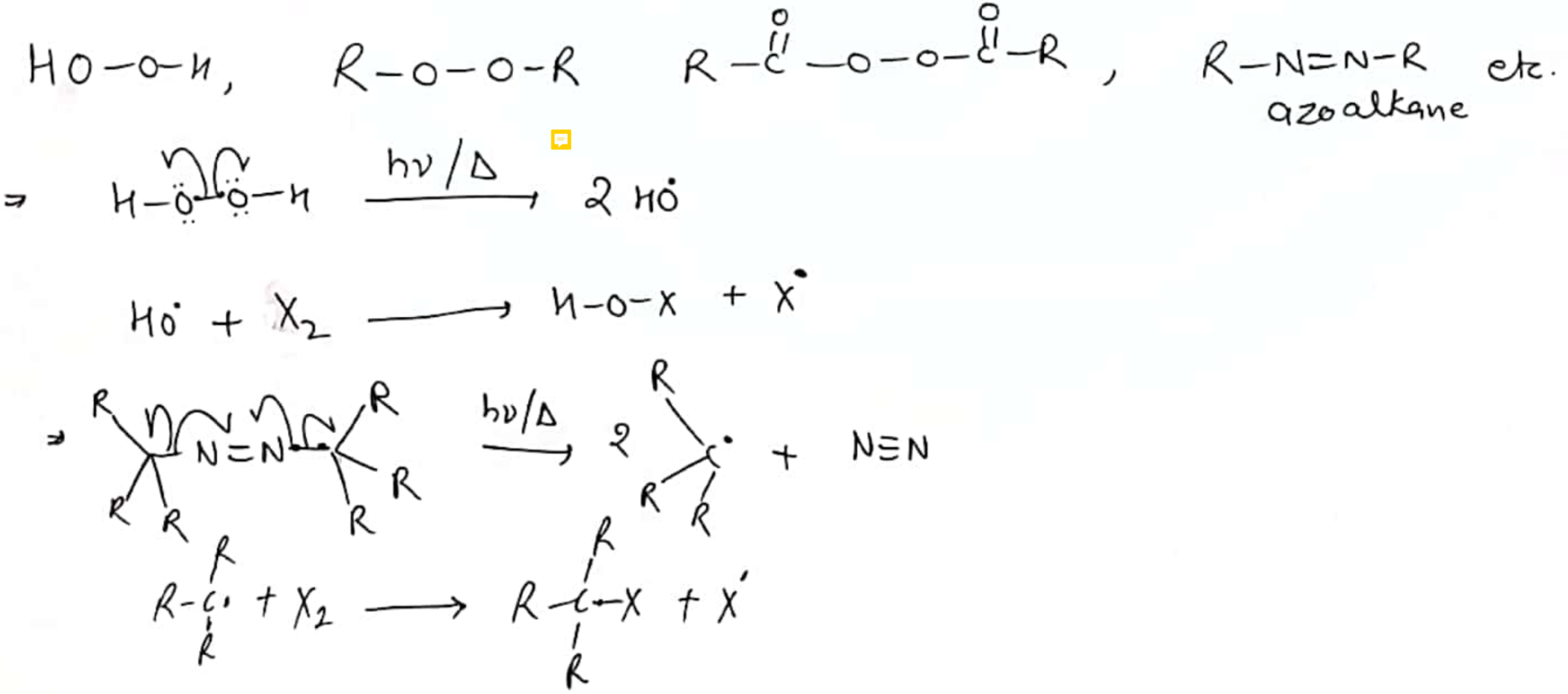
$C\!-\!H > C\!-\!Cl > C\!-\!Br > C\!-\!I$

ORGANIC REACTION MECHANISM – II (ORM-II)

(II) Free Radical Substitution Reactions :-

(A) Halogenation on Saturated carbon :-

Inhitiators for free radical reaction

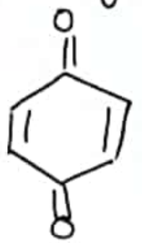


ORGANIC REACTION MECHANISM – II (ORM-II)

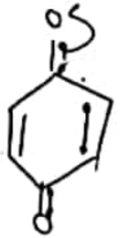
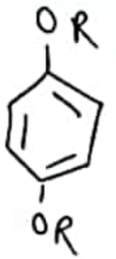
(II) Free Radical Substitution Reactions :-

(A) Halogenation on Saturated carbon :-

Inhibitors for free radical reaction :-

O_2 ,  Benzoquinone

⇒ $R^{\cdot} + O=O \longrightarrow R-O-O^{\cdot} \xrightarrow{R^{\cdot}} R-O-O-R$

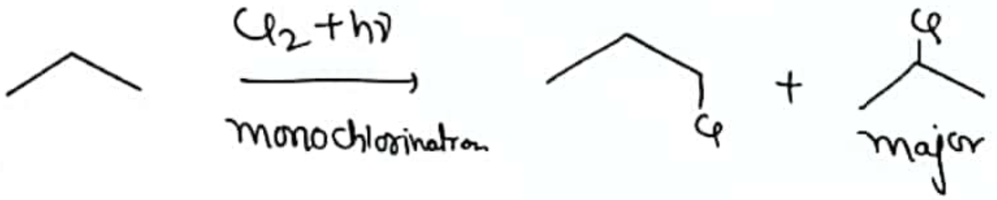
⇒  $+ 2R^{\cdot} \longrightarrow$ 

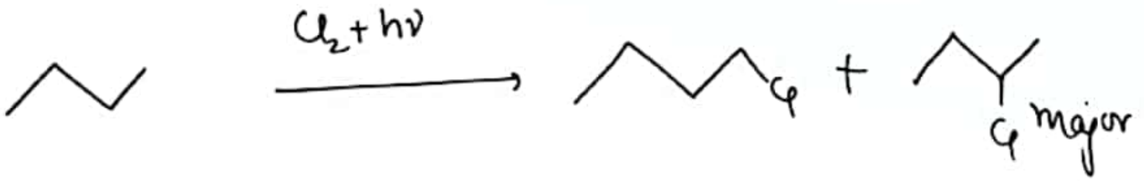
ORGANIC REACTION MECHANISM – II (ORM-II)

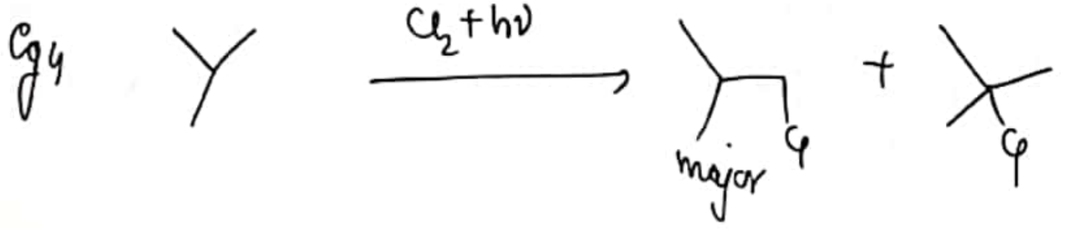
(II) Free Radical Substitution Reactions :-

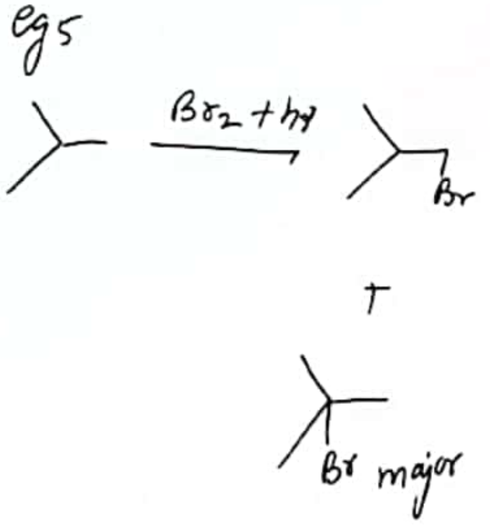
(A) Halogenation on Saturated carbon :-

eg 1 $CH_4 \xrightarrow[R_2O_2]{Cl_2 + h\nu} CH_3-Cl \xrightarrow[-HCl]{Cl_2} CH_2Cl_2 \xrightarrow[-HCl]{Cl_2} CHCl_3 \xrightarrow[-HCl]{Cl_2} CCl_4$

eg 2 

eg 3 

eg 4 

eg 5 

ORGANIC REACTION MECHANISM – II (ORM-II)

(II) Free Radical Substitution Reactions :-

(A) Halogenation on Saturated carbon :-

Note :- **Regioselective reaction :-** Reactions in which more than Structural isomeric products form in diff. amount.

⇒ Factors affecting regioselectivity of halogenation.

(a) Stability of carbon free radical ⇒ $3^\circ > 2^\circ > 1^\circ$

(b) Reactivity of C–H bond :- 3° C–H > 2° C–H > 1° C–H

(c) Availibility of same hydrogens : Probability factor.

(d) Reactivity and selectivity of $X^\cdot$ → Reactivity ⇒ $F^\cdot > Cl^\cdot > Br^\cdot > I^\cdot$
→ Selectivity ⇒ $F^\cdot < Cl^\cdot < Br^\cdot < I^\cdot$

ORGANIC REACTION MECHANISM – II (ORM-II)

(II) Free Radical Substitution Reactions :-

(A) Halogenation on Saturated carbon :-

Relative ratio of replacement of one hydrogen (for alkane)

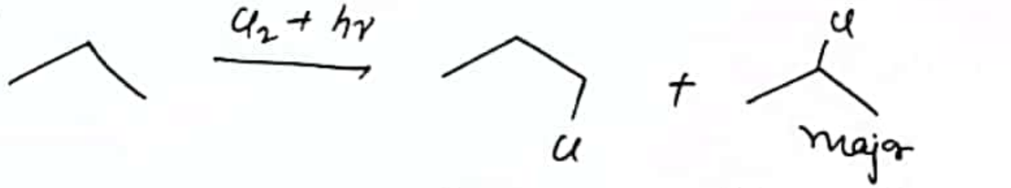
$1^\circ H : 2^\circ H : 3^\circ H$

$F^\cdot \Rightarrow 1 : 1 : 1$

$(25^\circ C)\ Cl^\cdot \Rightarrow 1 : 3.8 : 5$

$(125^\circ C)\ Br^\cdot \Rightarrow 1 : 82 : 1600$

eg1

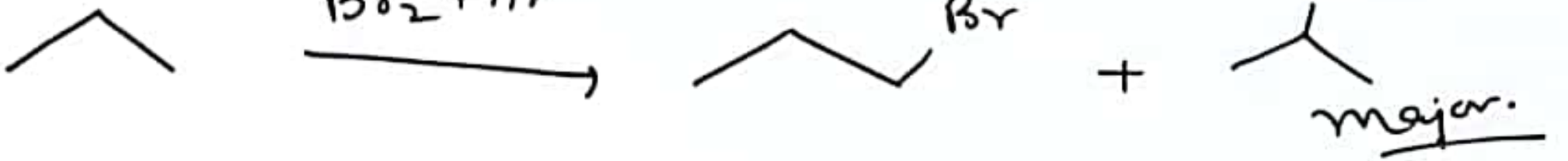


$1 \times 6 = 6$ $\qquad$ $3.8 \times 2 = 7.6$

$6 : 7.6$

$\Rightarrow \% \frac{6}{13.6} \times 100$ $\qquad$ $\% = \frac{7.6}{13.6} \times 100$

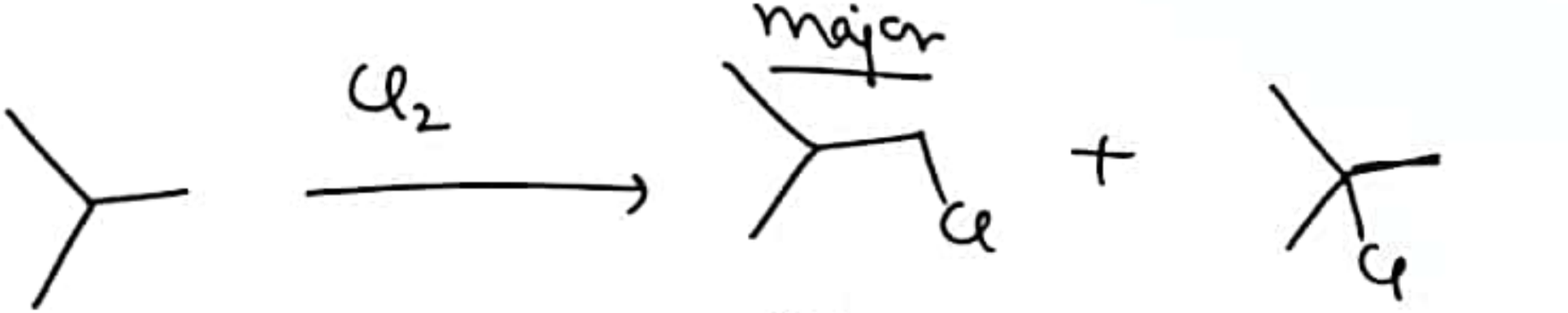
2



$1 \times 6 = 6$ $\qquad$ $82 \times 2 = 164$

$6 \;:\; 164$

g3



$1 \times 9 = 9$ $\qquad$ $5 \times 1 = 5$

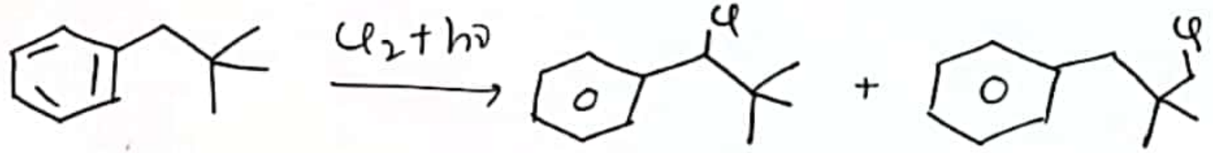
$9 \;:\; 5$

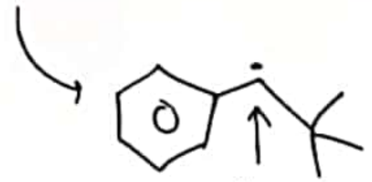
ORGANIC REACTION MECHANISM – II (ORM-II)

(II) Free Radical Substitution Reactions :-

(A) Halogenation on Saturated carbon :-

eg 5

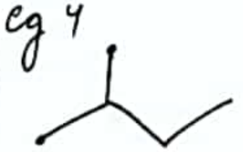


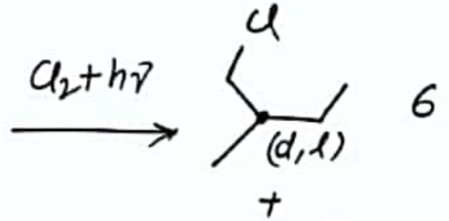


3.8 × 2 = 7.6 9

7.6 : 9

eg 4





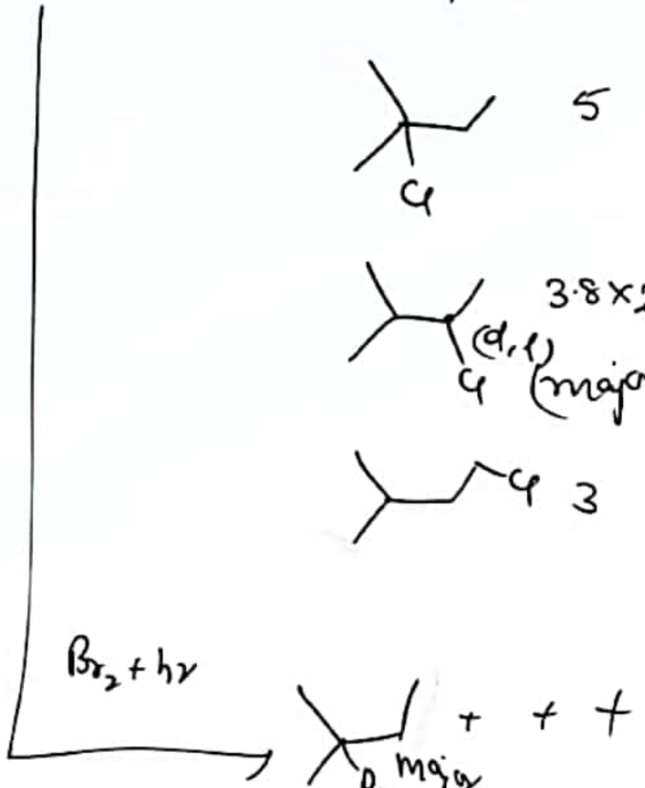
6

+

5

3.8 × 2 = 7.6

3



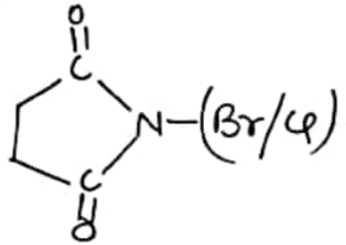
ORGANIC REACTION MECHANISM - II (ORM - II)

(II) FREE RADICAL SUBSTITUTION REACTION (S_{FR})

(B) Allylic and Benzylic halogenation with NBS / NCS :-

NBS = N-Bromo succinamide

NCS = N-Chloro succinamide

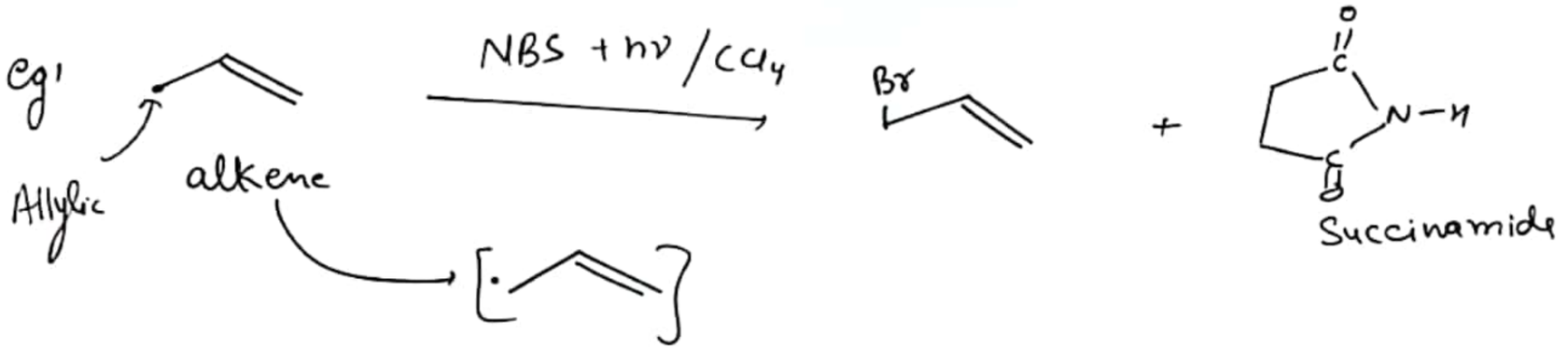


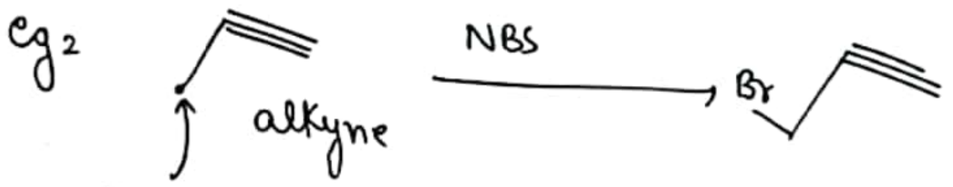
with NBS and NCS, monohalogenation take place on allylic, benzylic, propargylic and α-position of carbonyl compound (aldehyde & ketone)

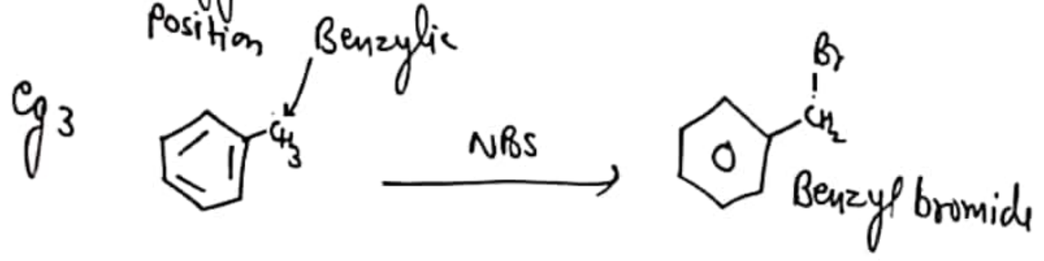
ORGANIC REACTION MECHANISM - II (ORM - II)

(II) FREE RADICAL SUBSTITUTION REACTION (S_{FR})

(B) Allylic and Benzylic halogenation with NBS / NCS :-





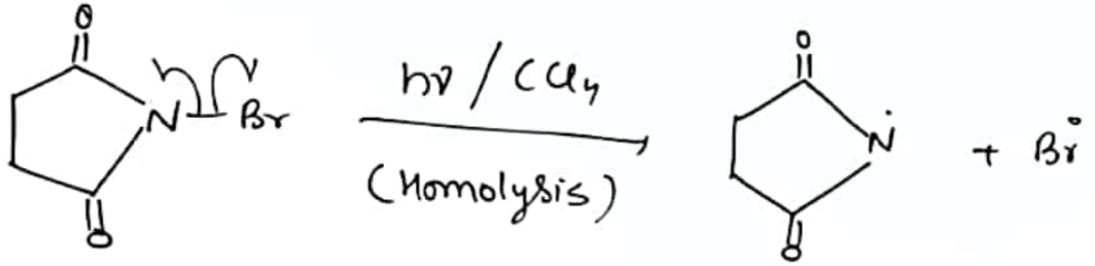


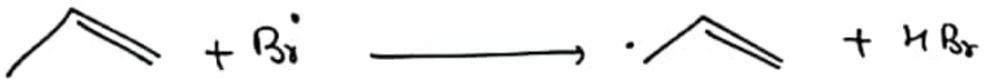
ORGANIC REACTION MECHANISM - II (ORM - II)

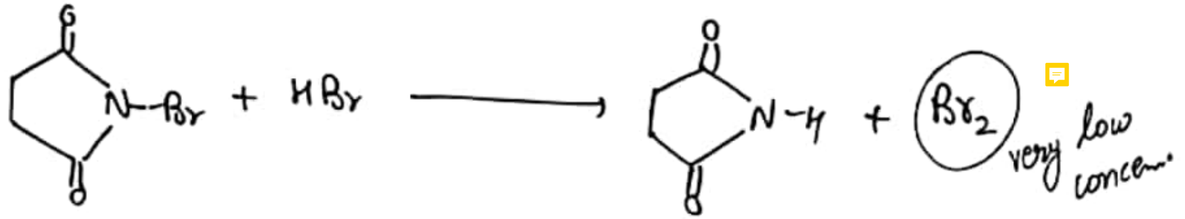
(II) FREE RADICAL SUBSTITUTION REACTION (SFR)

(B) Allylic and Benzylic halogenation with NBS / NCS :-

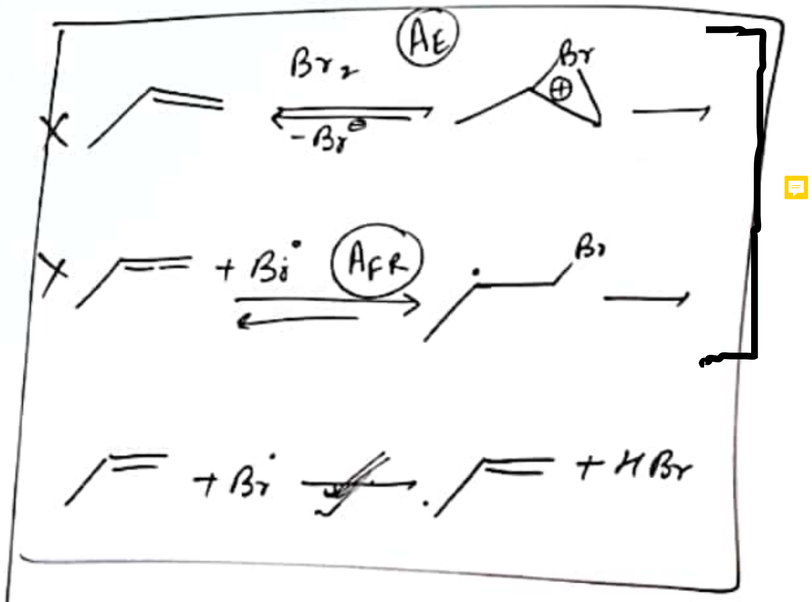
Mechanism

(i) 

(ii) 

(iii) 

(iv) 

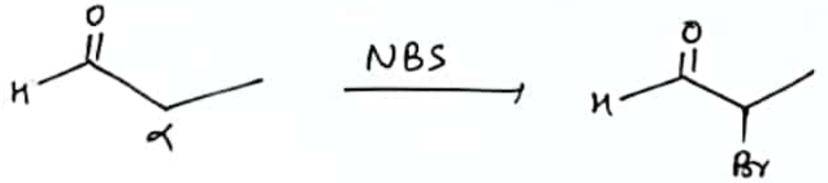


ORGANIC REACTION MECHANISM - II (ORM - II)

(II) FREE RADICAL SUBSTITUTION REACTION (S_{FR})

(B) Allylic and Benzylic halogenation with NBS / NCS :-

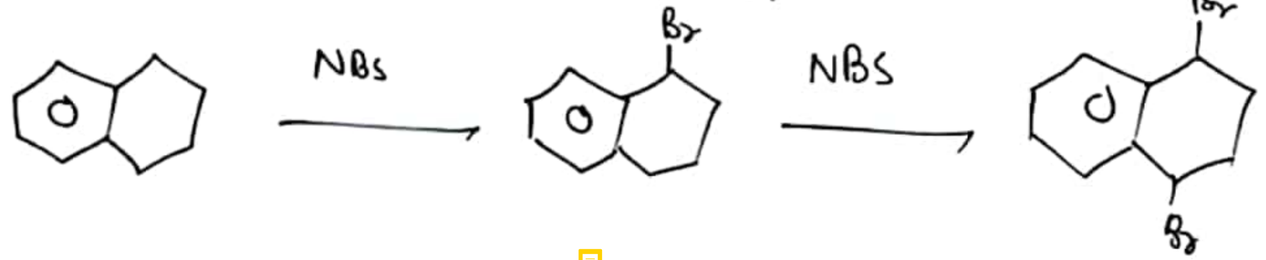
eg 4



eg 5



eg 6

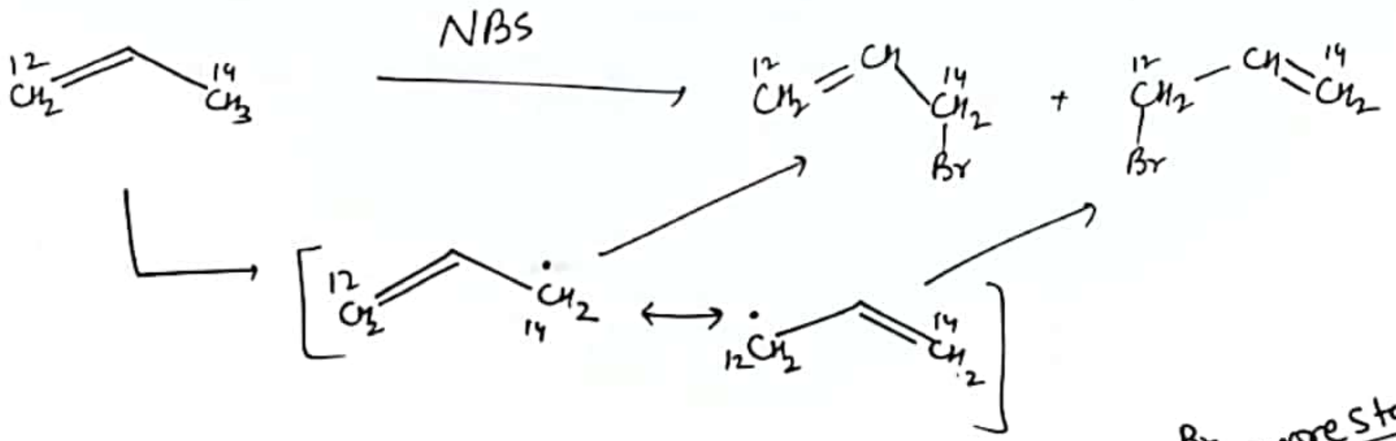


ORGANIC REACTION MECHANISM - II (ORM - II)

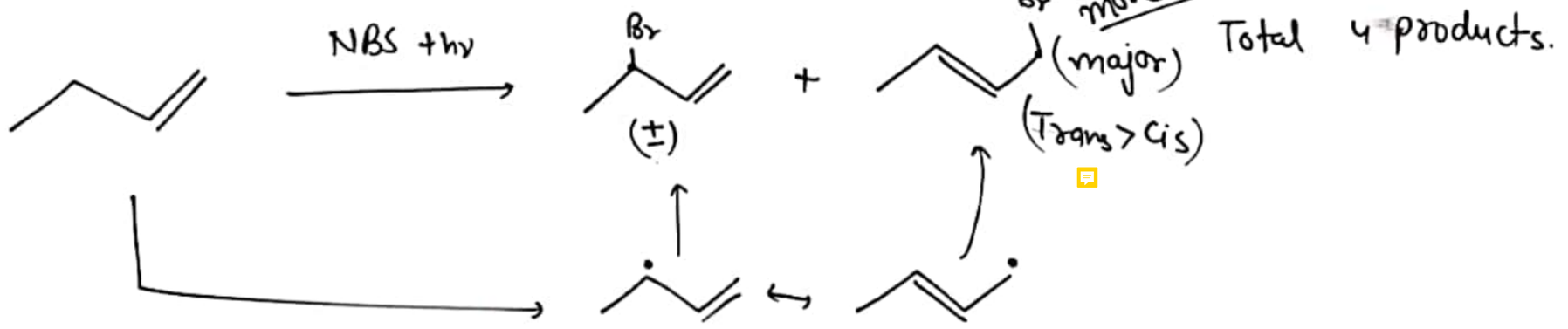
(II) FREE RADICAL SUBSTITUTION REACTION (S_{FR})

(B) Allylic and Benzylic halogenation with NBS / NCS :-

Eg 7



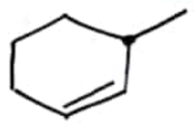
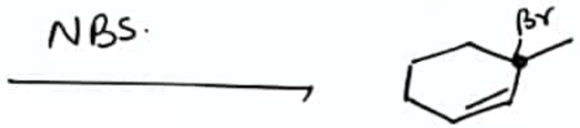
Eg 8



ORGANIC REACTION MECHANISM - II (ORM - II)

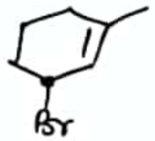
(II) FREE RADICAL SUBSTITUTION REACTION (S_{FR})

(B) Allylic and Benzylic halogenation with NBS / NCS :-

eg  NBS. 

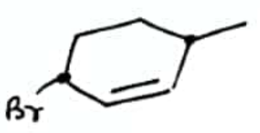
(±)

major

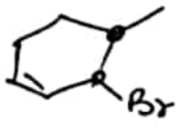


(±)

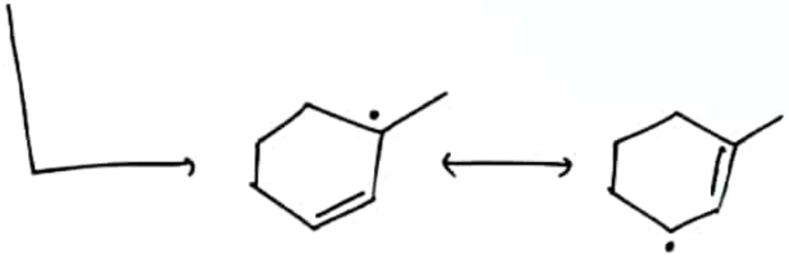
$2^n = 2^2 = 4$

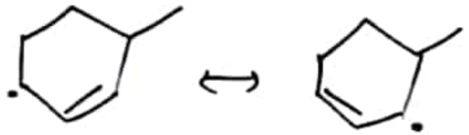


4 product



4 products.

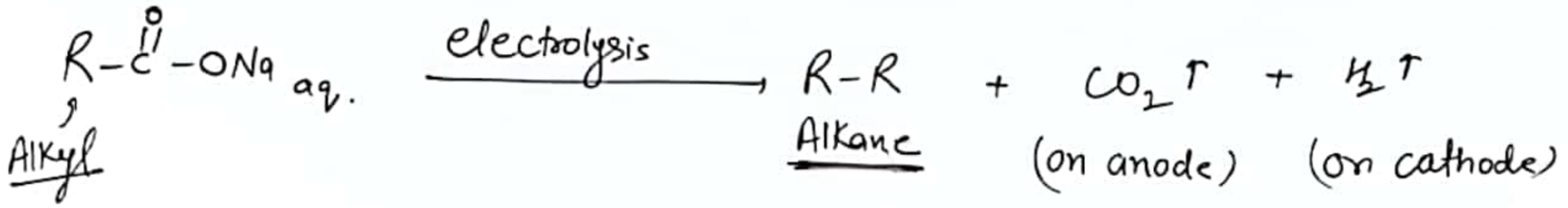




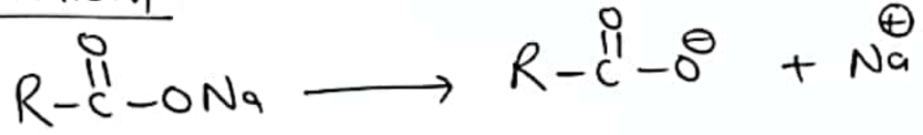
ORGANIC REACTION MECHANISM - II (ORM - II)

(II) FREE RADICAL SUBSTITUTION REACTION (S_{FR})

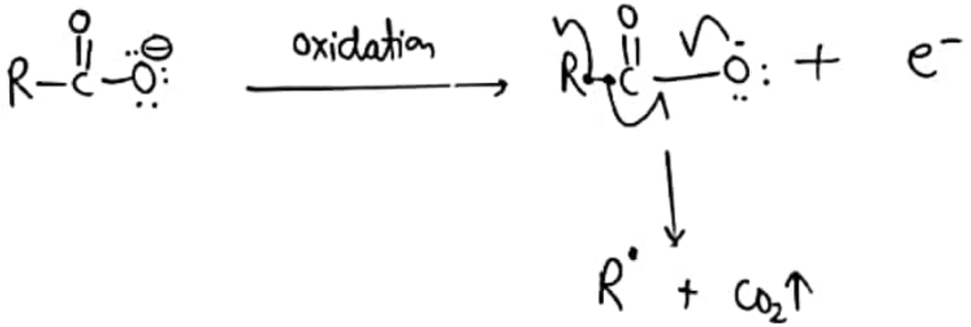
[C] Kolbe's electrolysis of carboxylate salts:

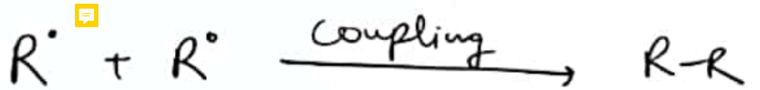


mechanism



on anode:-(oxidation)





⇒ on cathode (Reduction)



ORGANIC REACTION MECHANISM - II (ORM - II)

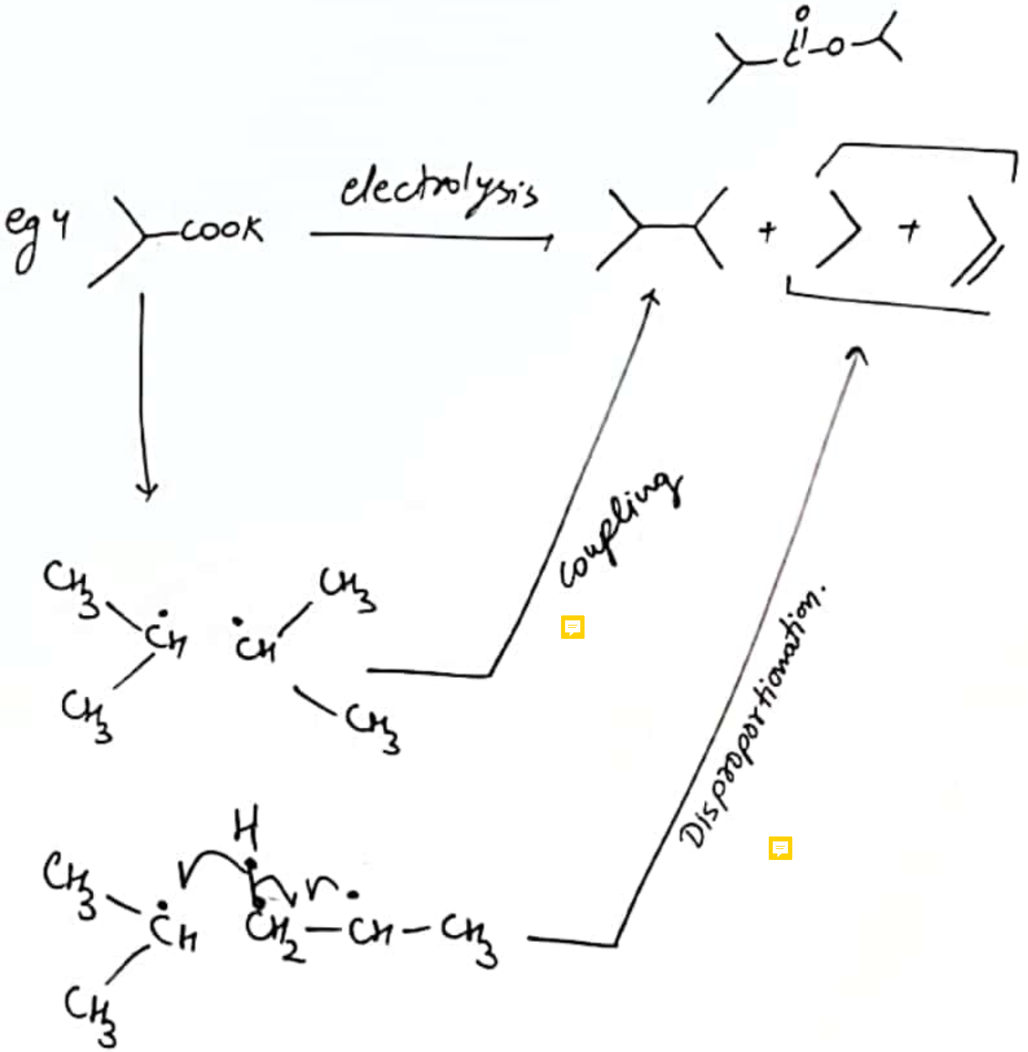
(II) FREE RADICAL SUBSTITUTION REACTION (SFR)

[C] Kolbe's electrolysis of carboxylate salts:

eg 1 $CH_3-COONa \xrightarrow{\text{electrolysis}} CH_3-CH_3$

eg 2 $C_2H_5-COOK \xrightarrow{\text{"}\quad\text{"}} C_2H_5-C_2H_5$

eg 3 $\left[\begin{array}{c} CH_3-COOK \\ + \\ CH_3-CH_2-COOK \end{array}\right]_{mix.} \xrightarrow{\text{"}}$ $CH_3-C_2H_5 + CH_3-CH_3 + C_2H_5-C_2H_5$

eg 4 

ORGANIC REACTION MECHANISM - II (ORM - II)

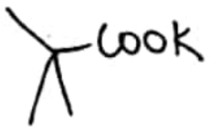
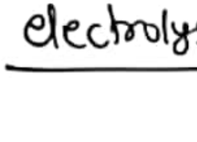
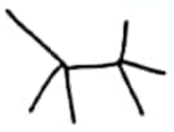
(II) FREE RADICAL SUBSTITUTION REACTION (SFR)

[C] Kolbe's electrolysis of carboxylate salts:

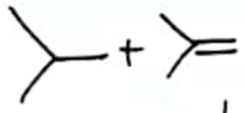
Note :- ⇒ Normally ⇒ Coupling product major.

⇒ When R group of R-COOK is 3° alkyl

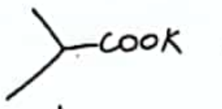
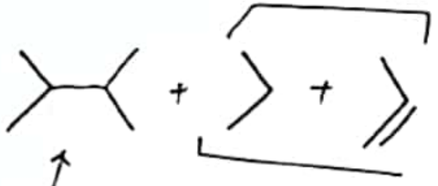
then major product ⇒ Disproportionation product

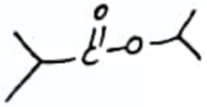
Eg 5   electrolysis 

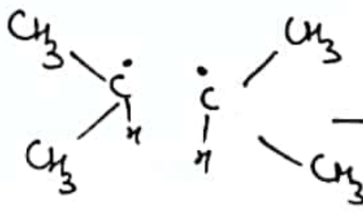
minor



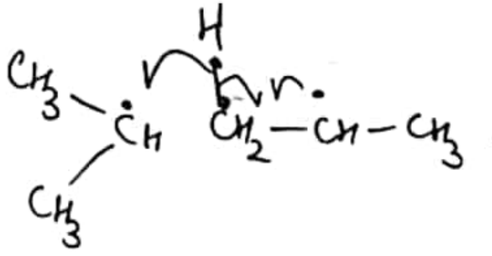
major.

eg 4  electrolysis 





coupling



Disproportionation.

ORGANIC REACTION MECHANISM - II (ORM - II)

(II) FREE RADICAL SUBSTITUTION REACTION (S_{FR})

[C] Kolbe's electrolysis of carboxylate salts:

eg 7



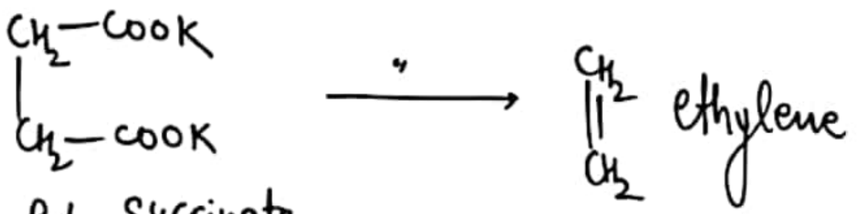
Pot. Glutarate

eg 8



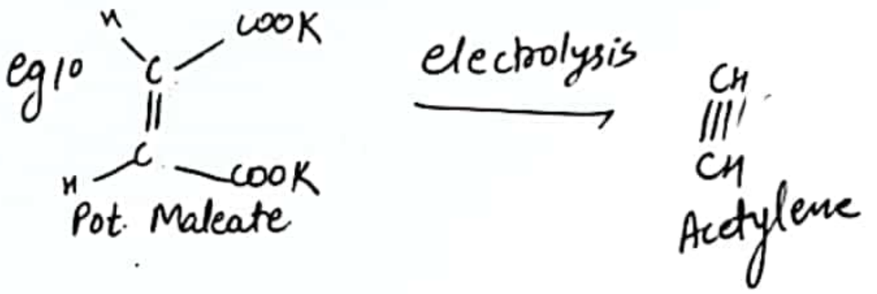
Pot. Pimalate

eg 9



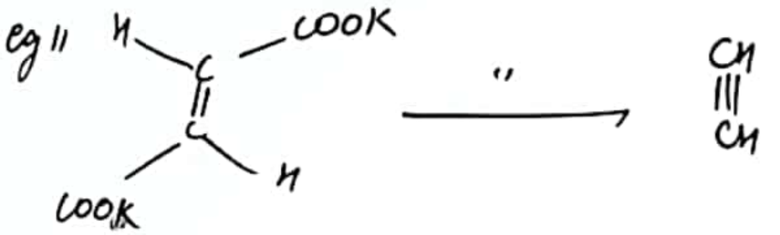
Pot. Succinate

eg 10



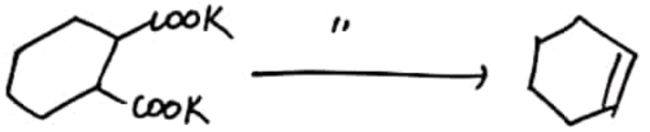
Pot. Maleate

eg 11



Pot. fumarate

eg 12

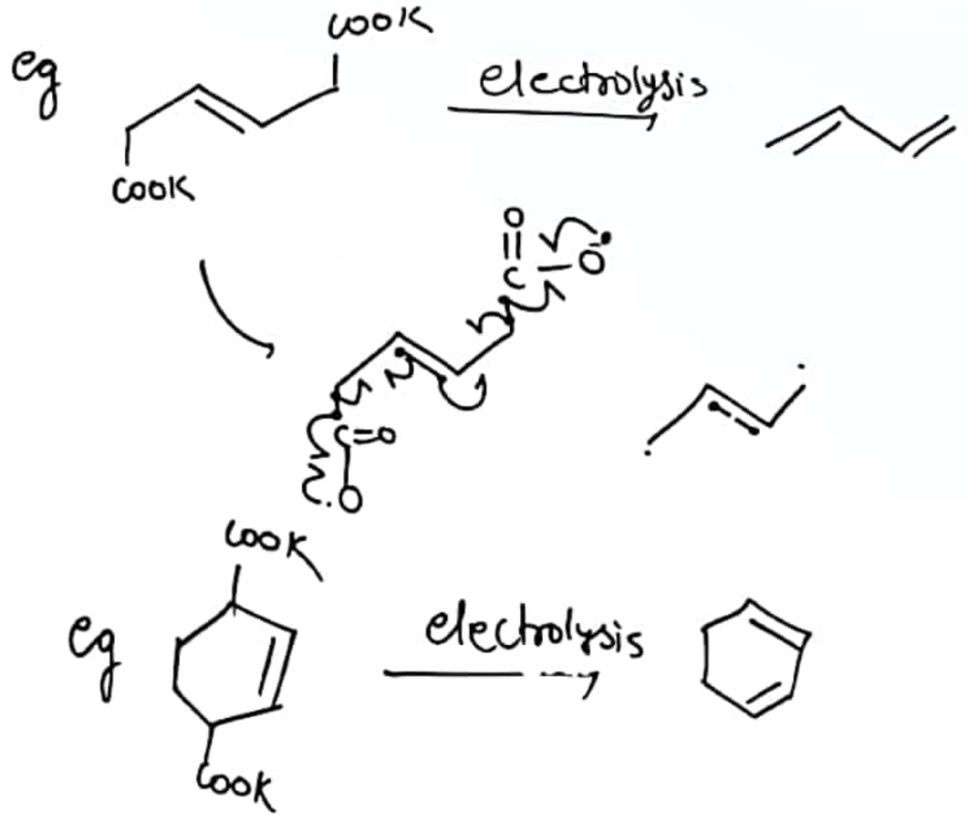


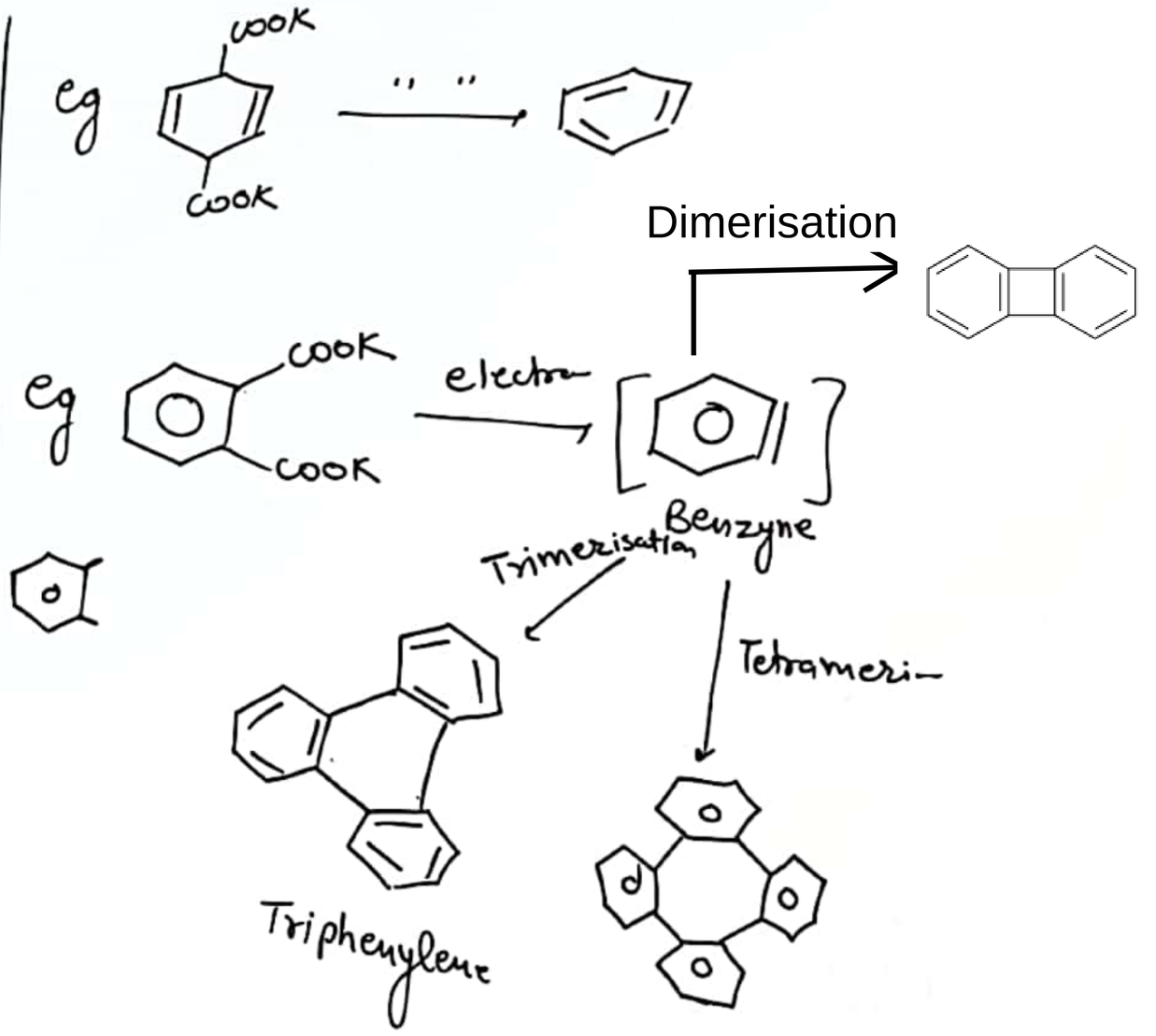
cis or trans

ORGANIC REACTION MECHANISM - II (ORM - II)

(II) FREE RADICAL SUBSTITUTION REACTION (SFR)

[C] Kolbe's electrolysis of carboxylate salts:





ORGANIC REACTION MECHANISM - II (ORM - II)

(II) FREE RADICAL SUBSTITUTION REACTION (SFR)

(D) Hunsdeker's Reaction:-

$$R-\overset{O}{\overset{\|}{C}}-OAg + Br_2 \xrightarrow{CCl_4 \text{ solvent}} R-Br + AgBr\downarrow + CO_2\uparrow$$

Ali. / Aro. mech.

$$R-\overset{O}{\overset{\|}{C}}-OAg + Br-Br \longrightarrow R-\overset{O}{\overset{\|}{C}}-O-Br + AgBr$$

$$R-\overset{O}{\overset{\|}{C}}-O-Br \longrightarrow R-\overset{O}{\overset{\|}{C}}-\dot{O} + Br^{\cdot}$$

$$R-\overset{O}{\overset{\|}{C}}-\dot{O} \longrightarrow R^{\cdot} + CO_2$$

$$R^{\cdot} + Br-Br \longrightarrow R-Br + Br^{\cdot}$$

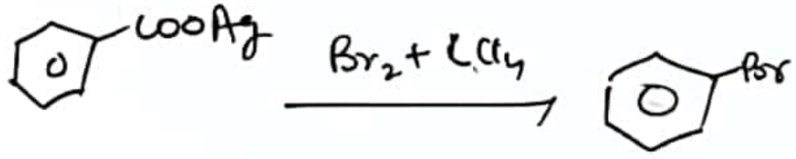
$$R-\overset{O}{\overset{\|}{C}}-O-Br + Br^{\cdot} \longrightarrow R-\overset{O}{\overset{\|}{C}}-O^{\cdot} + Br_2$$

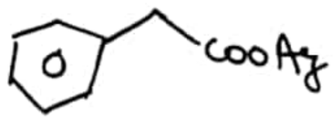
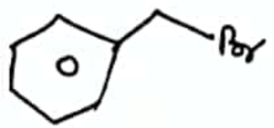
ORGANIC REACTION MECHANISM - II (ORM - II)

(II) FREE RADICAL SUBSTITUTION REACTION (S_{FR})

(D) Hunsdeker's Reaction:-

eg $CH_3-COOH + AgOH \longrightarrow CH_3-\overset{\overset{O}{\|}}{C}-OAg \xrightarrow{Br_2+CCl_4} CH_3-Br + AgBr + CO_2$

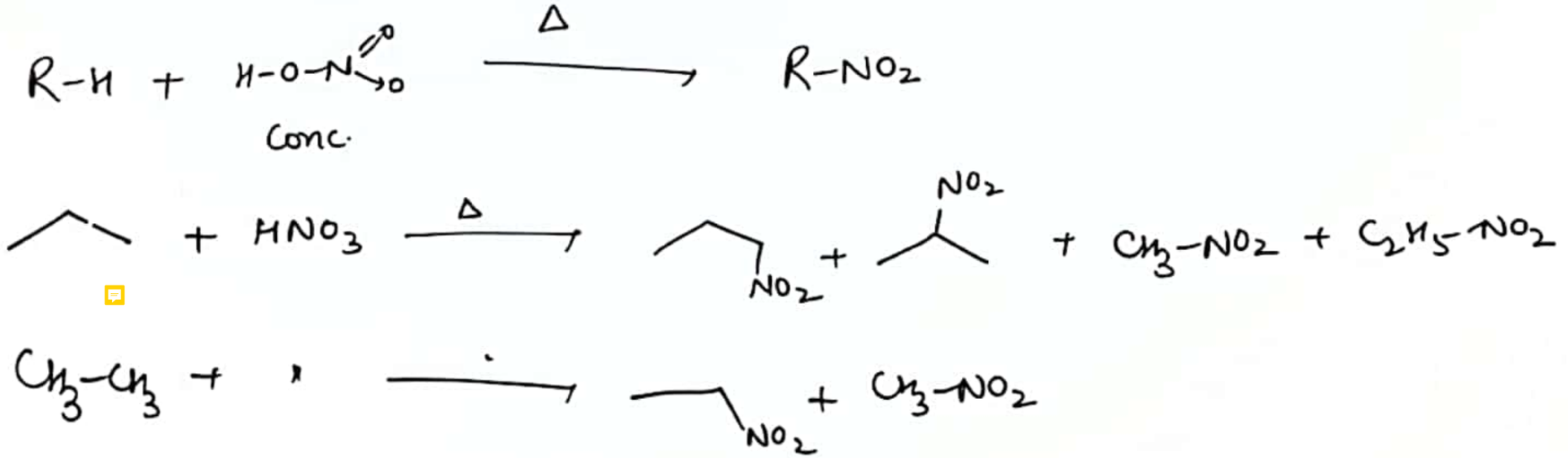
eg  $\xrightarrow{Br_2+CCl_4}$

eg  $\xrightarrow{Br_2}$ 

ORGANIC REACTION MECHANISM - II (ORM - II)

(II) FREE RADICAL SUBSTITUTION REACTION (S_{FR})

(E) Nitration of alkane:-



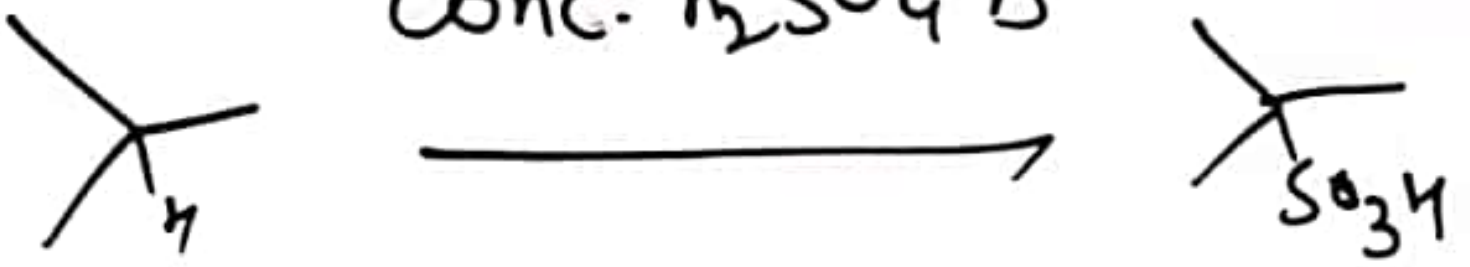
F) Sulphonation of alkane

$$R\text{-}H + \text{conc.}\ H_2SO_4 \longrightarrow R\text{-}SO_3H$$

n-alkane

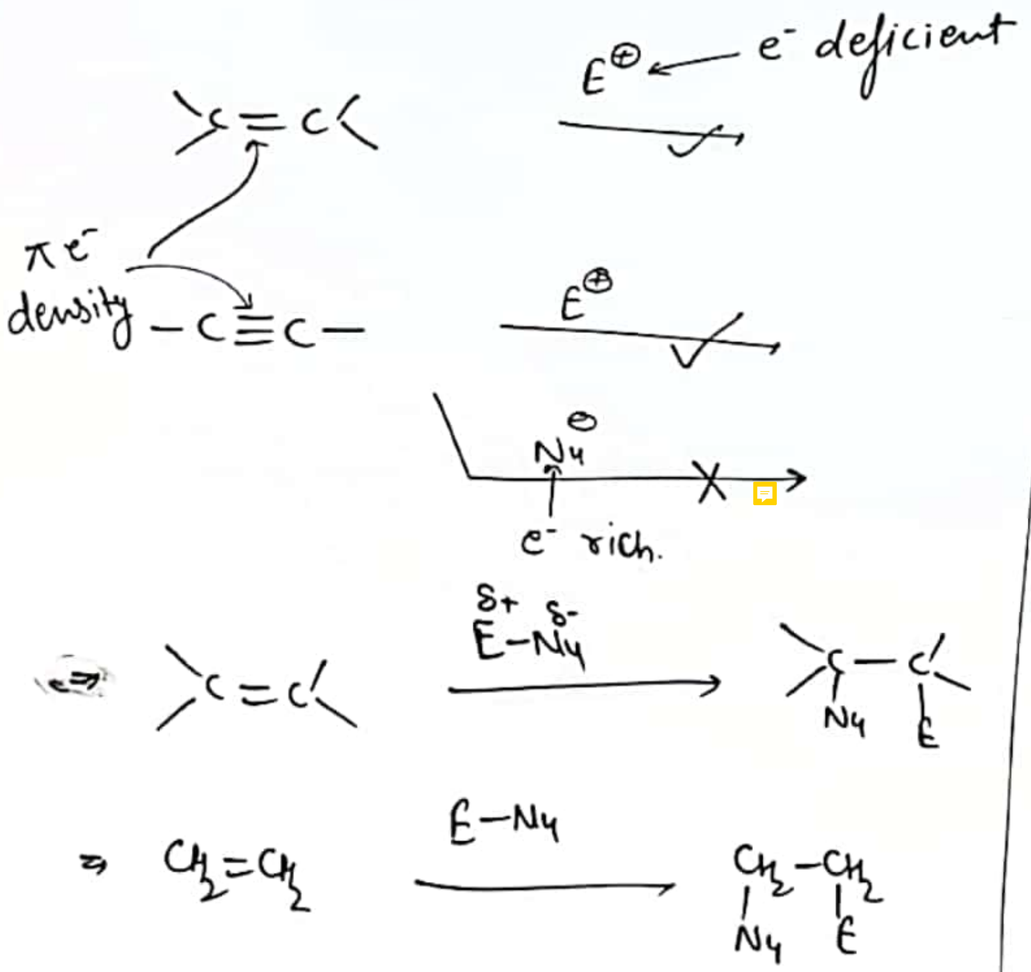
(min. 6 carbon containing)

* Reactivity ⇒ 3° C–H > 2° C–H > 1° C–H

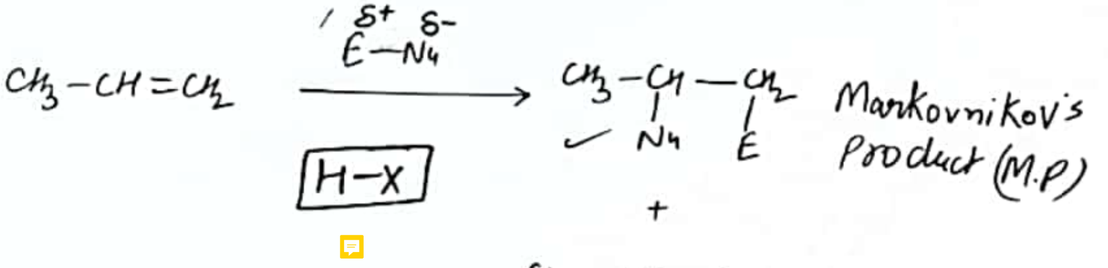
⇒ 

ORGANIC REACTION MECHANISM -II (ORM-II)

(III) Electrophilic Addition Reactions (A_E)



⇒ Markovnikov's addition:-



Markovnikov's Rule:-

On unsymmetrical alkenes and alkyne addition of +ve part takes place on that carbon of carbon-carbo multiple bond which have greater no. of hydrogens an addition of -ve part ....carbon which have lesse no. of H atoms.

ORGANIC REACTION MECHANISM -II (ORM-II)

(III) Electrophilic Addition Reactions (A_E)

π complex



Mechanism

Mech. I

$R-CH=CH_2$ ($>C=C<$)

$\xrightarrow[1^{st}\text{ step}]{E^{\oplus}\ (RDS)}$ $R-\overset{\oplus}{C}H-CH_2-E$ (2° more stable) + $R-CH(E)-\overset{\oplus}{C}H_2$ (1°)

Classical carbocation

$\xrightarrow[2^{nd}\text{ step (fast)}]{N_4^{\ominus}}$ $R-\underset{\underset{E}{|}}{\overset{N_4}{\overset{|}{C}}}H-CH_2-E$ (M.P) (Generally major) + $R-\underset{\underset{E}{|}}{CH}-CH_2-N_4$ (AMP)

Mech. II

$\xrightarrow[RDS]{\ddot{E}^{\oplus}}$ lone pair containing electrophile ($:\ddot{Cl}:^{\oplus}$, $Br^{\oplus}$, $:NO^{\oplus}$, $:CH_2$ etc.)

$\left[R-\overset{\oplus}{C}H-CH_2-E\right]$ (attack from opp. side w.r.t $E^{\oplus}$; $\oplus$ shifts from weaker to E)

$\xrightarrow[\text{(conceted step)}]{N_4^{\ominus}\ \text{(fast)}}$ $R-\underset{\underset{E}{|}}{\overset{N_4}{\overset{|}{C}}}H-CH_2$ (M.P.) (generally major) + $R-\underset{\underset{E}{|}}{CH}-CH_2-N_4$ (AMP)

$\Downarrow$

[Cyclic cation or Non classical carbocation or π complex]

ORGANIC REACTION MECHANISM -II (ORM-II)

(III) Electrophilic Addition Reactions (A_E)

* Factors affecting of reactivity of A_E Rxn :-

(i) Reactivity $\propto$ Stability of intermediate

(ii) " " $\propto$ e^- density on carbon-carbon multi--bond.

(iii) Reactivity ↑es in presence of e^- donating group (+M, +H, +I)

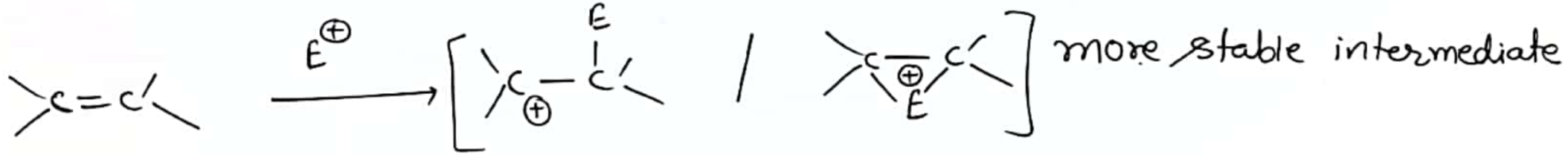
(iv) " " ↓es " " " e^- withdrowing group (-M, -I)

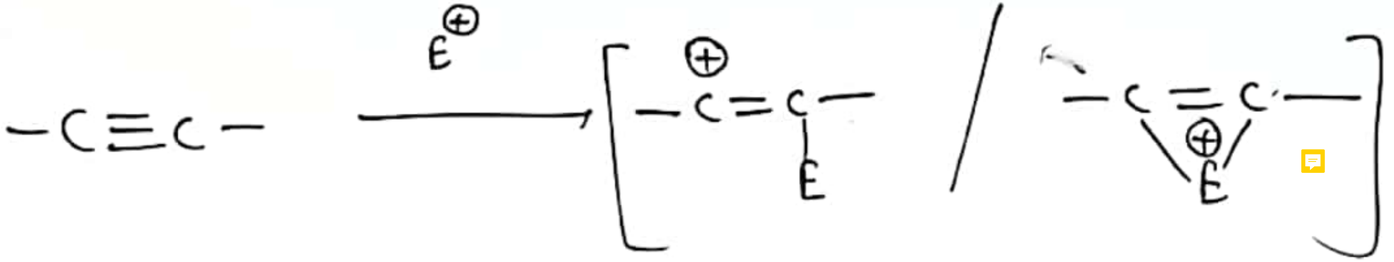
ORGANIC REACTION MECHANISM -II (ORM-II)

(III) Electrophilic Addition Reactions (A_E)

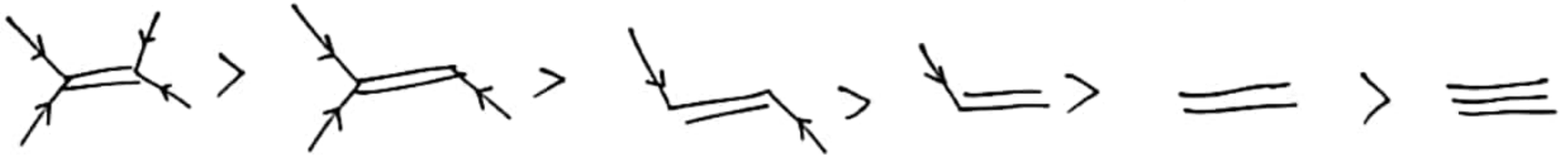
*Factors affecting of reactivity of A_E Rxn :-

eg 1 **Alkene > Alkyne**





eg 2

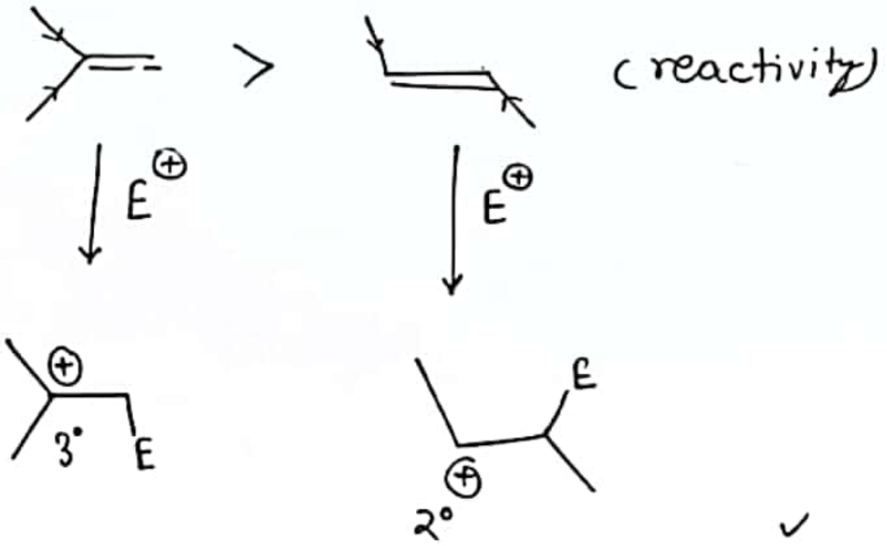


ORGANIC REACTION MECHANISM -II (ORM-II)

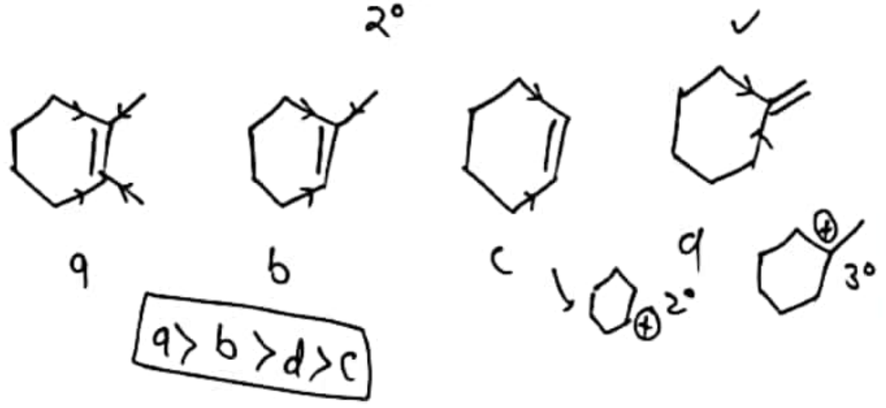
(III) Electrophilic Addition Reactions (A_E)

* Factors affecting of reactivity of A_E Rxn :-

eg 3

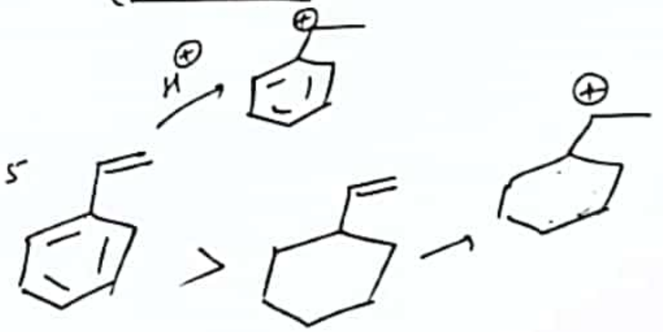


eg 4

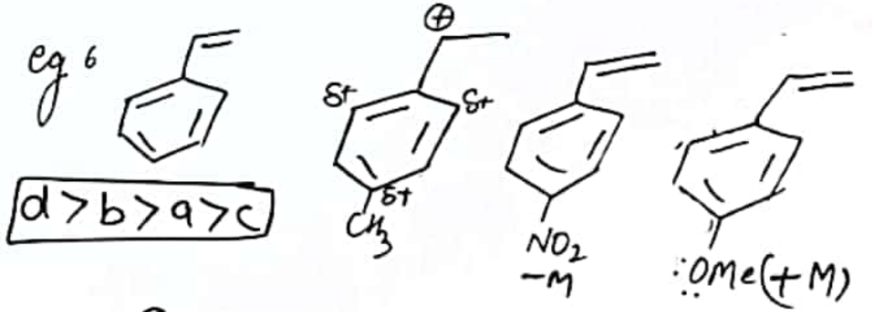


a > b > d > c

eg 5

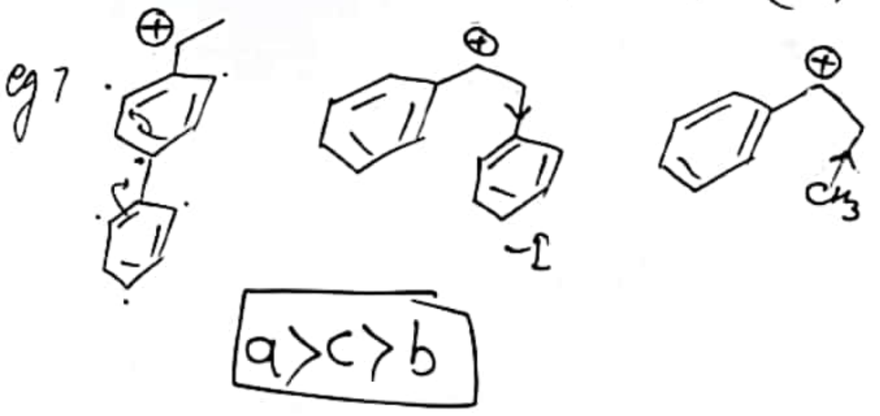


eg 6



d > b > a > c

eg 7



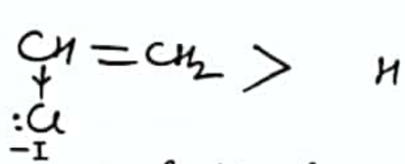
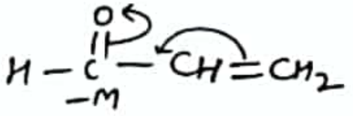
a > c > b

ORGANIC REACTION MECHANISM -II (ORM-II)

(III) Electrophilic Addition Reactions (A_E)

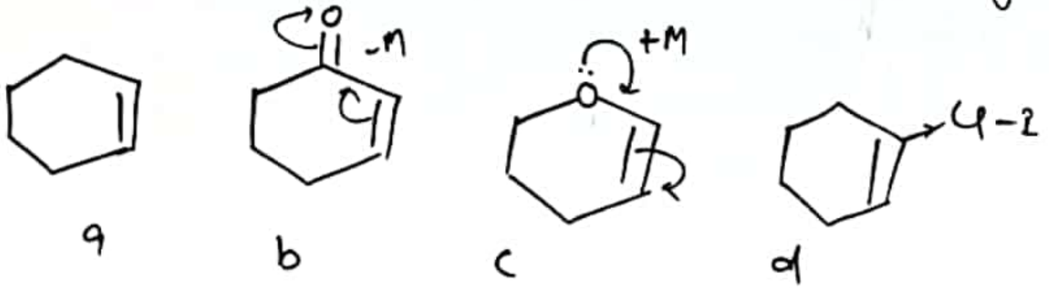
*Factors affecting of reactivity of A_E Rx^n :-

Eg 8

$CH_2=CH_2$ >  > 

Vinyl chloride — Acraldehyde

Eg 9



c > a > d > b

ORGANIC REACTION MECHANISM -II (ORM-II)

(III) Electrophilic Addition Reactions (A_E)

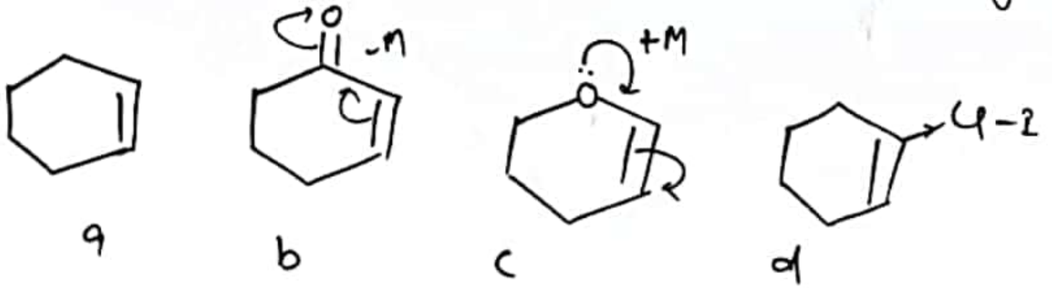
*Factors affecting of reactivity of A_E Rxn :-

eg 8

$CH_2=CH_2$ > $CH=CH_2$ (−I) > $H-C(=O)-CH=CH_2$ (−M)

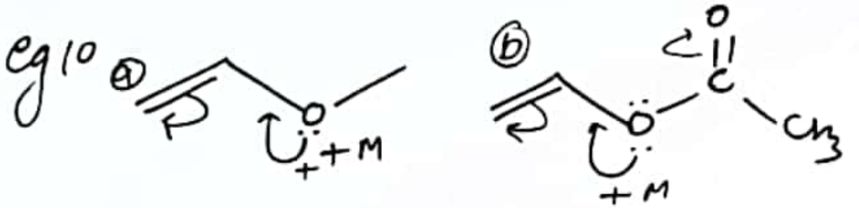
Vinyl chloride — Acraldehyde

eg 9

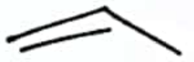


c > a > d > b

eg 10

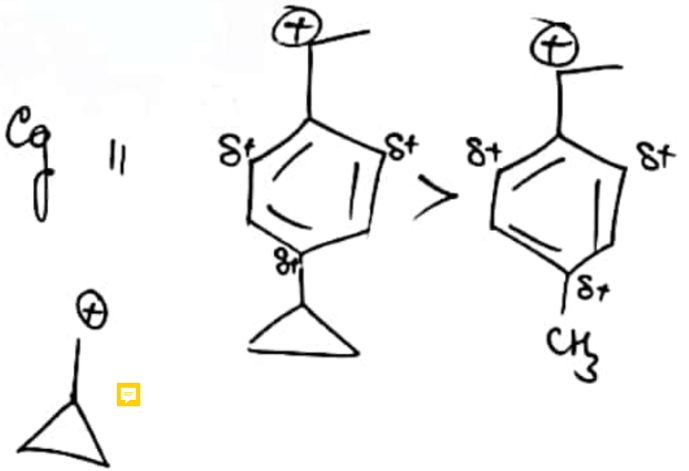


c

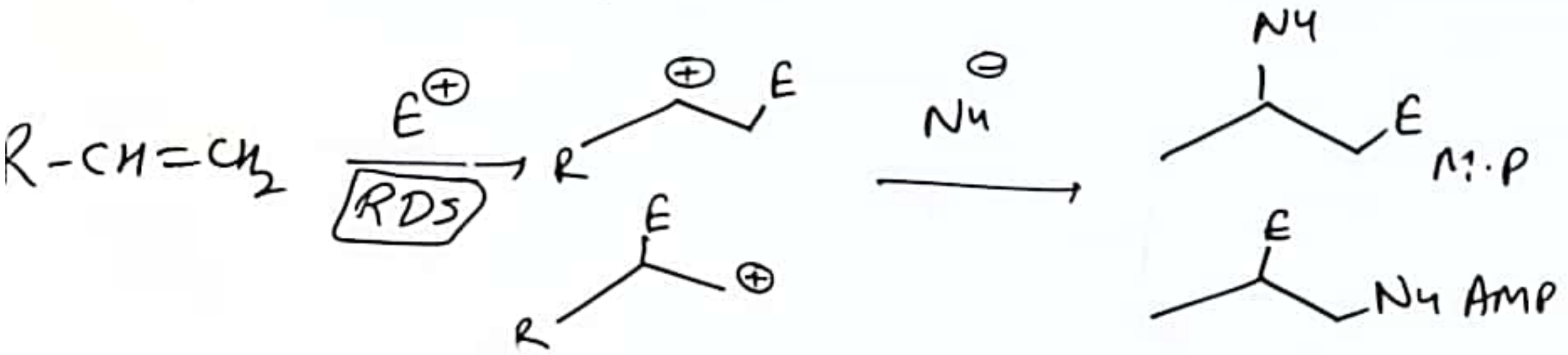


a > b > c

eg 11



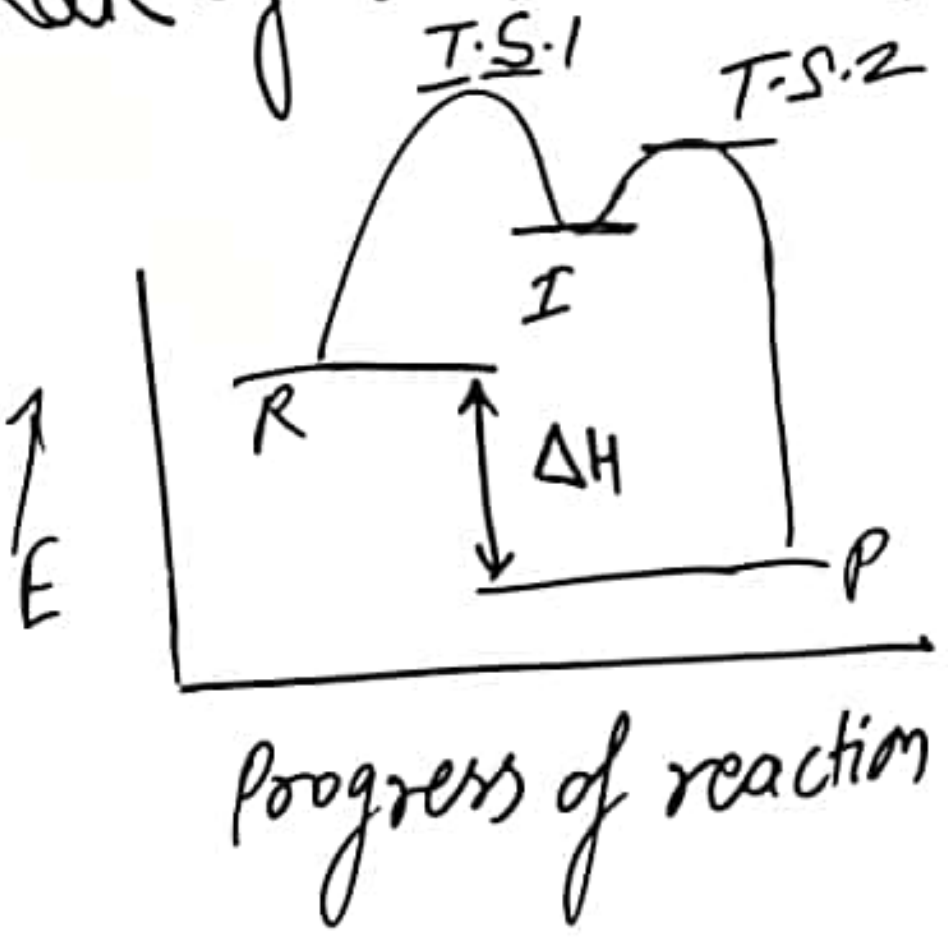
mechanism I



① RDS = I^St Step

② 2nd order Rx^n

③



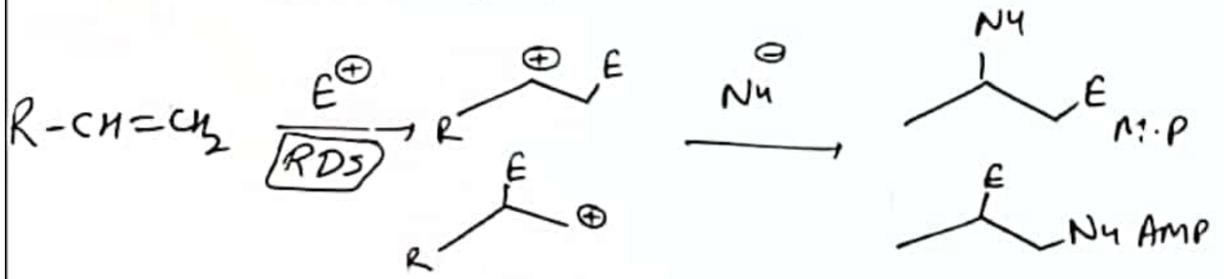
Rate of reaction ∝ [Alkene] [E⊕]

ΔH = enthalpy = -ve

ORGANIC REACTION MECHANISM-II

ORM-II

mechanism I



④ Intermediate = Classical carbonium ion

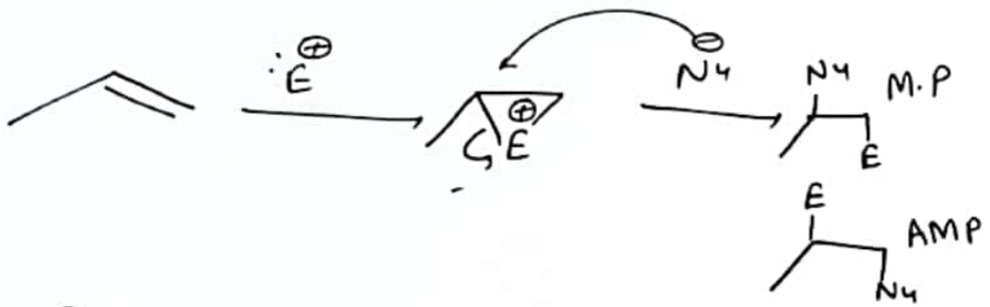
⑤ Rearrangement possible in intermediate

⇒ major product form from more stable intermediate (generally)

⑥ generally M.P = major

⑦ Regioselective reaction

Mechanism · II



④ Intermediate = Non classical Carbocation

⑤ No Rearrangement

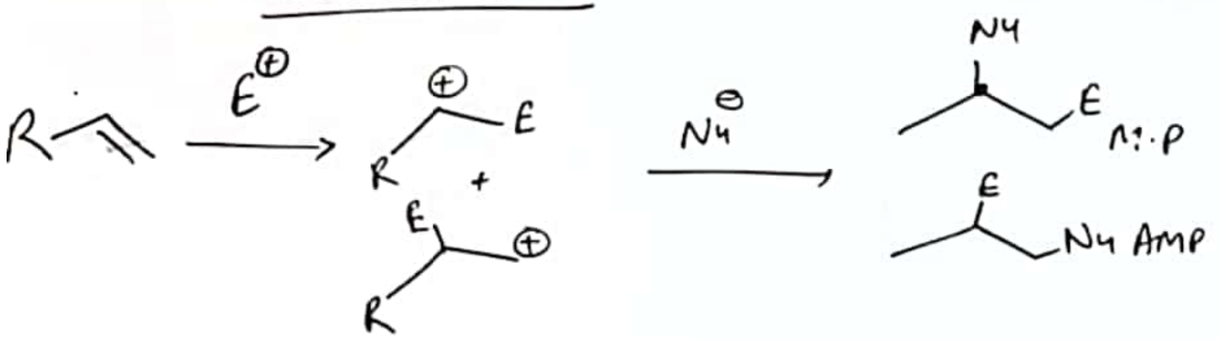
⑥ Same

⑦ Regioselective Rxn

ORGANIC REACTION MECHANISM-II

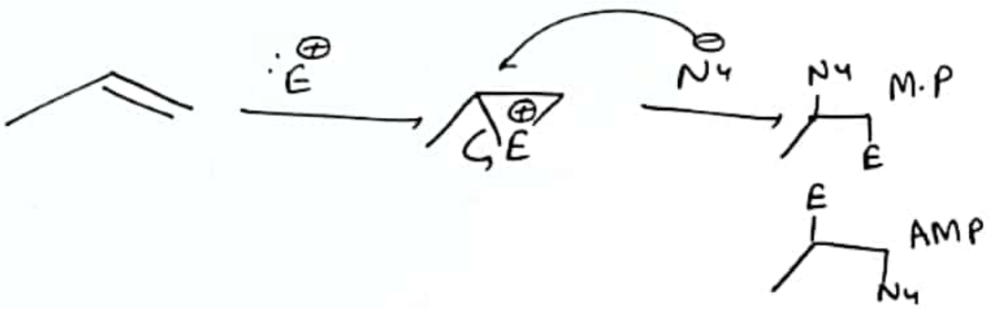
ORM-II

Mechanism - I



⑧ Stereo chemistry of this reaction is not important. (due to formation of planar cation)

Mechanism - II



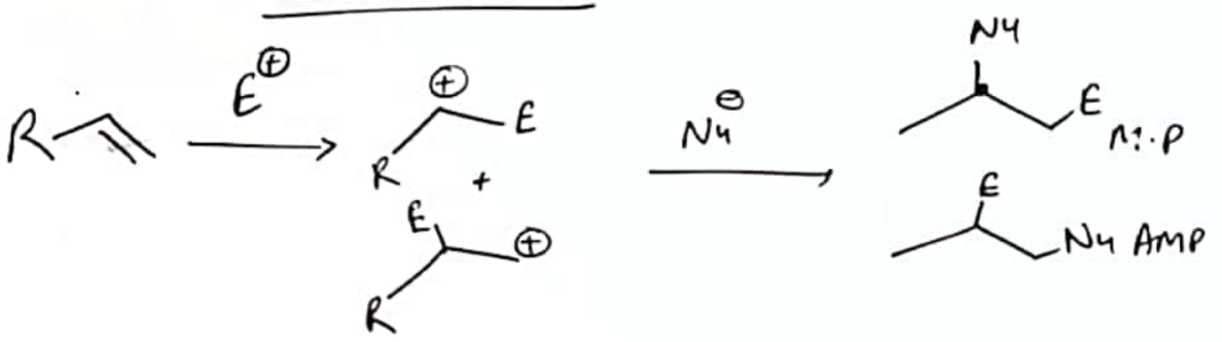
⑧ Stereoselective as well as stereospecific reaction

⇒ Anti addition reaction.
(addition of 2 group from opp. plane)

ORGANIC REACTION MECHANISM-II

ORM-II

Mechanism - I



⑨ Examples.

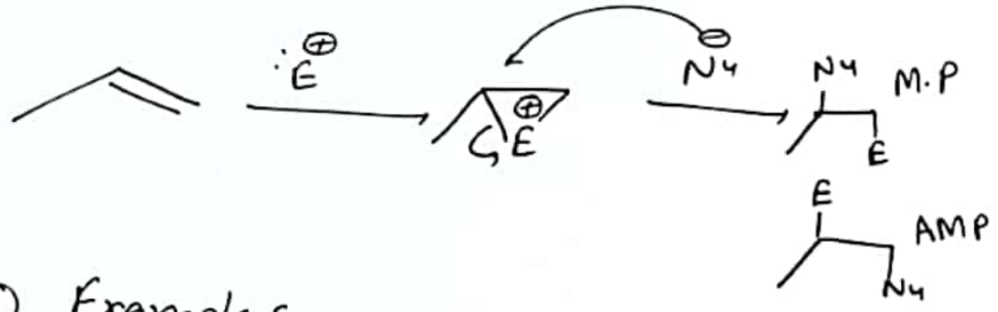
(i) Addition of HX (Hydrohalogenation)

(ii) " " Conc. H_2SO_4

(iii) " " " H_2O (Hydration)

(iv) " " " R-OH etc.

Mechanism - II



⑨ Examples

(i) Addition of X_2

(ii) " " " XX'

(iii) " " " HOX

(iv) " " " $X_2 + H_2O$

(v) " " " NOCl/NOBr

(vi) " " " carbene

(vii) Oxy. mercuration

(viii) Alkoxy mercuration etc

ORGANIC REACTION MECHANISM -II ORM-II

(III) Electrophilic Addition Reaction (A_E)

① Addition of HX (Hydrohalogenation)

→ According to Mech. I :-
- intermediate = Classical carbocation
- Rearrangement = Possible
- Major product = generally M.P
- Non Stereospecific reaction.

Eg! $CH_2=CH_2 \xrightarrow{H-X^{\delta-}} CH_2(X)-CH_3$

$CH_2=CH_2 \xrightarrow{NaX^{\ominus}}$ No Rxn

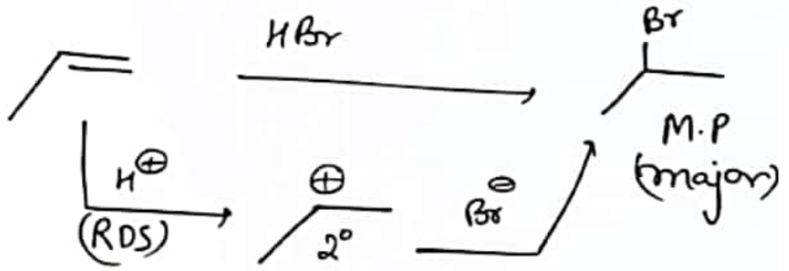
Reactivity of HX → HI > HBr > HCl >> HF (No Rxn)

ORGANIC REACTION MECHANISM -II ORM-II

(III) Electrophilic Addition Reaction (A_E)

① Addition of HX (Hydrohalogenation)

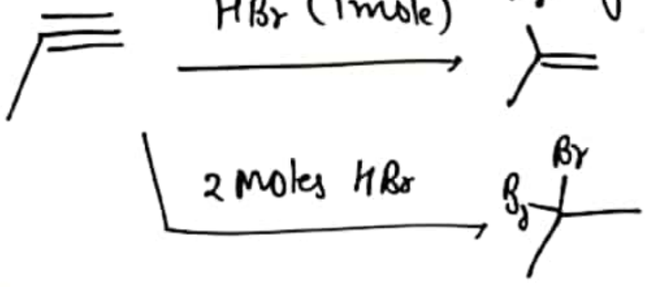
eg 2



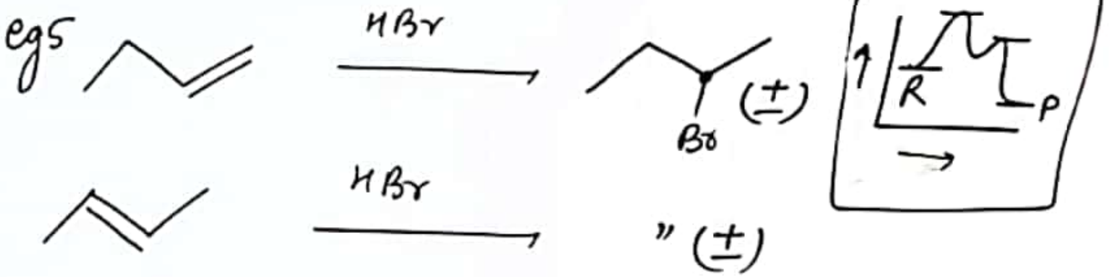
eg 3 $CH \equiv CH$ —(1 mole HBr)→ $CH_2 = CH-Br$

$CH_2 = CH-Br$ —(2 mole HBr)→ $CH_3-CH(Br)_2$ gem. dihalide

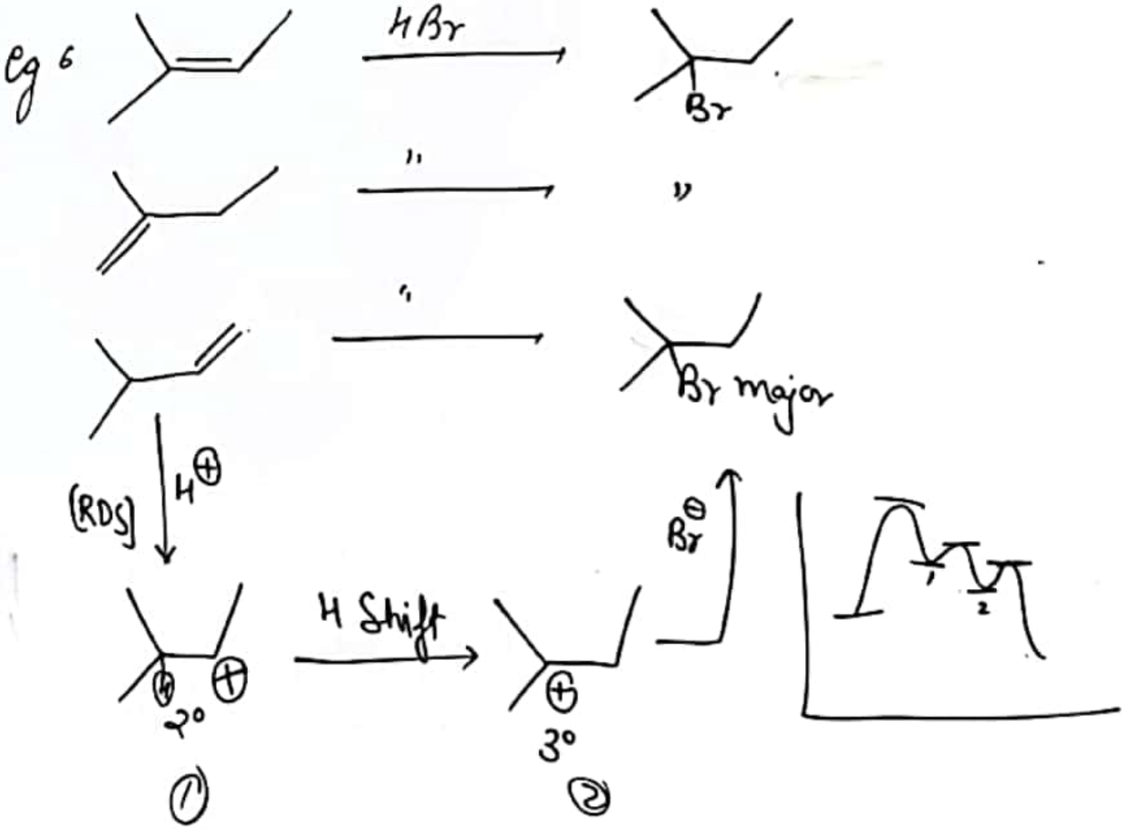
eg 4



eg 5



eg 6

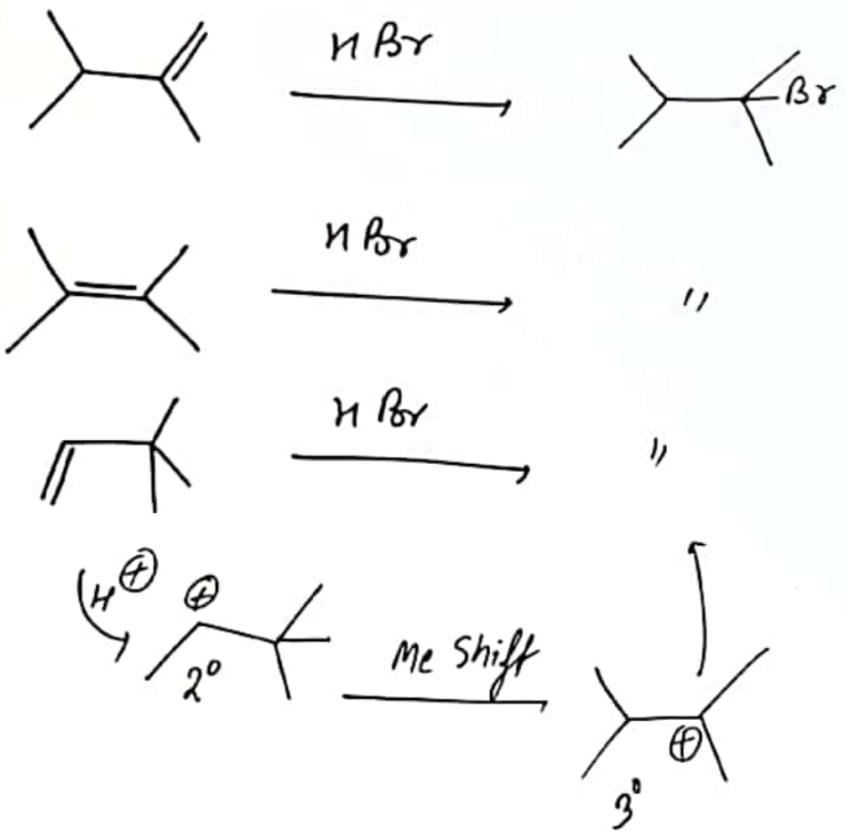


ORGANIC REACTION MECHANISM -II

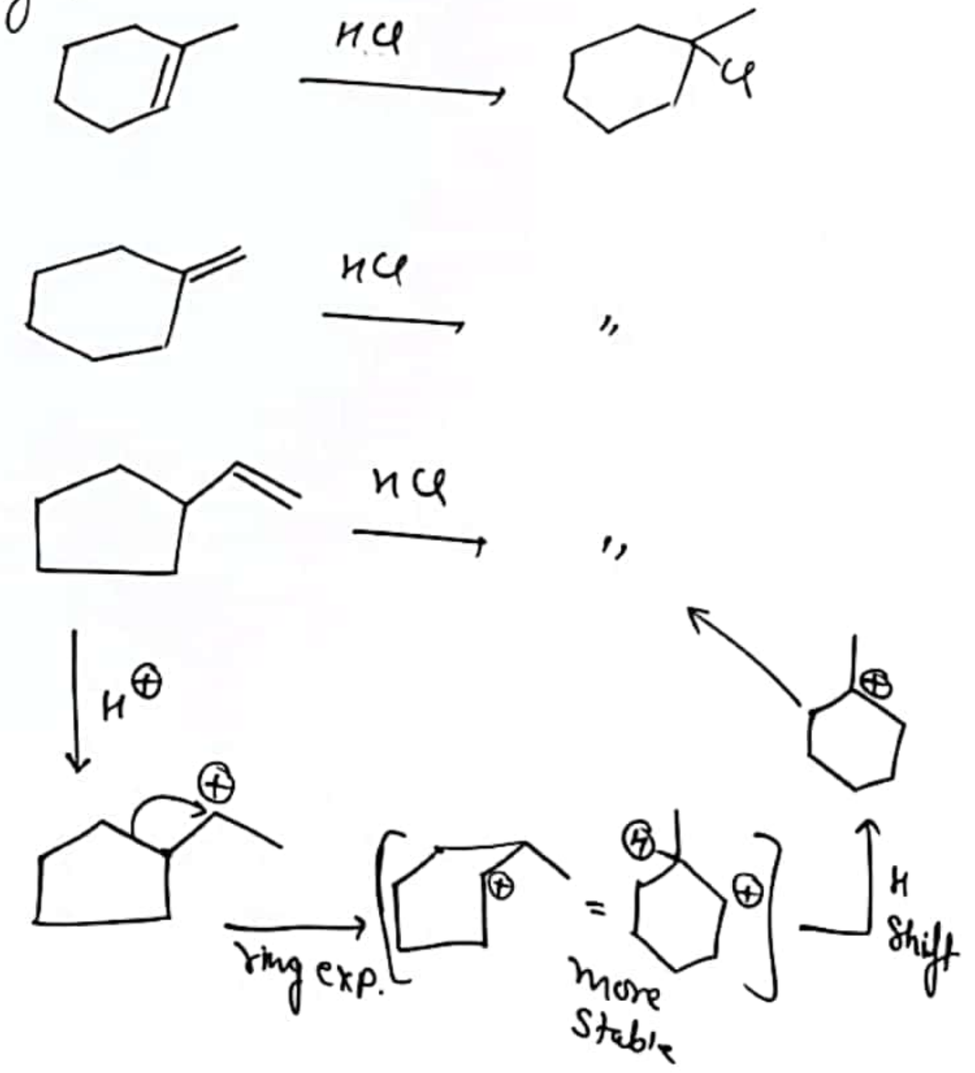
(III) Electrophilic Addition Reaction (A_E)

① Addition of HX (Hydrohalogenation)

eg 7



eg 8

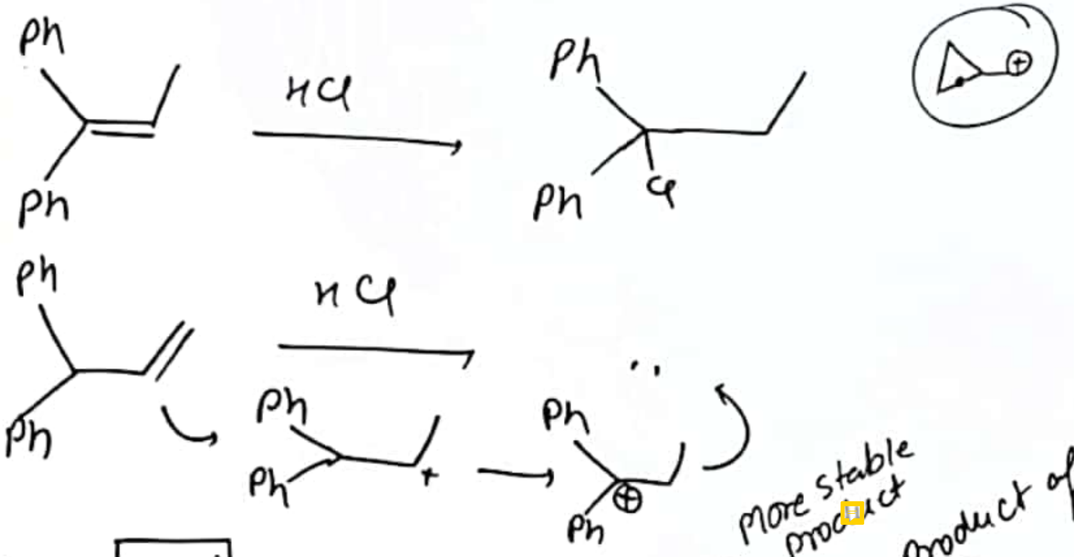


ORGANIC REACTION MECHANISM -II

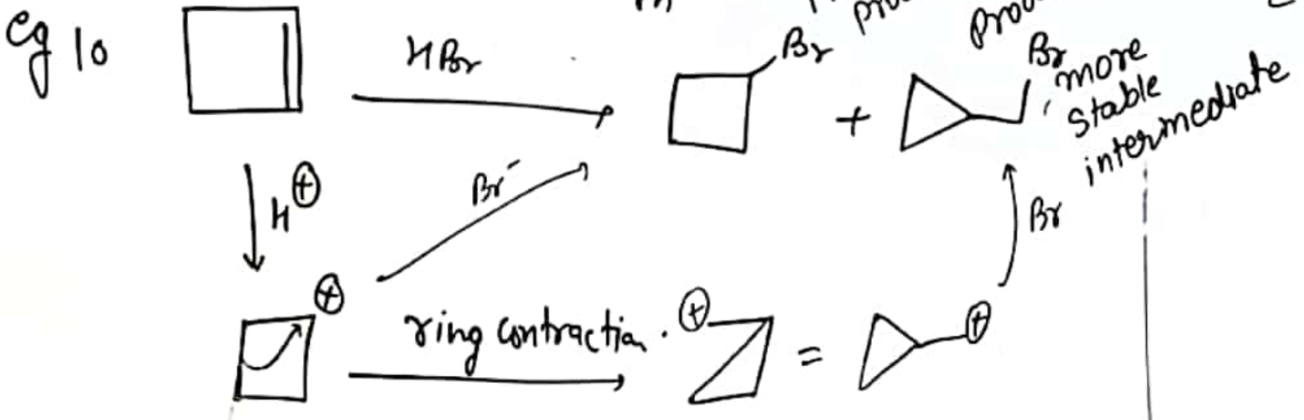
(III) Electrophilic Addition Reaction (A_E)

① Addition of HX (Hydrohalogenation)

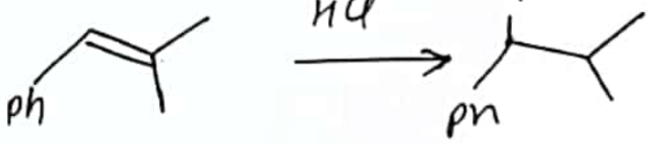
eg 9



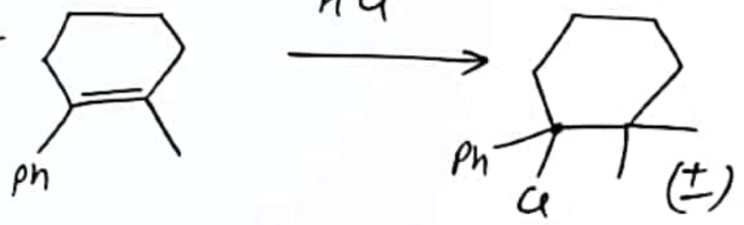
eg 10



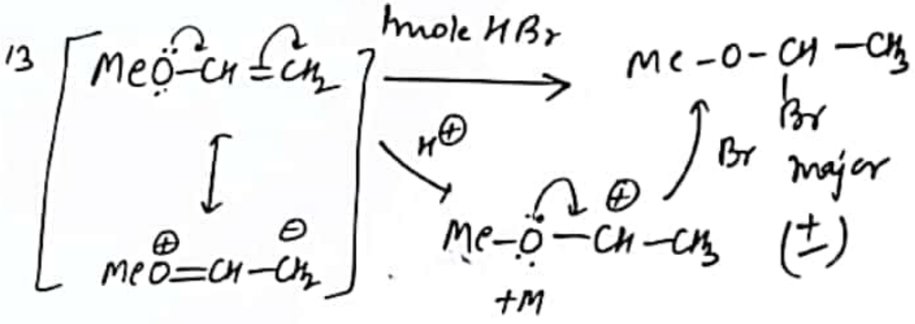
eg 11



eg 12



eg 13



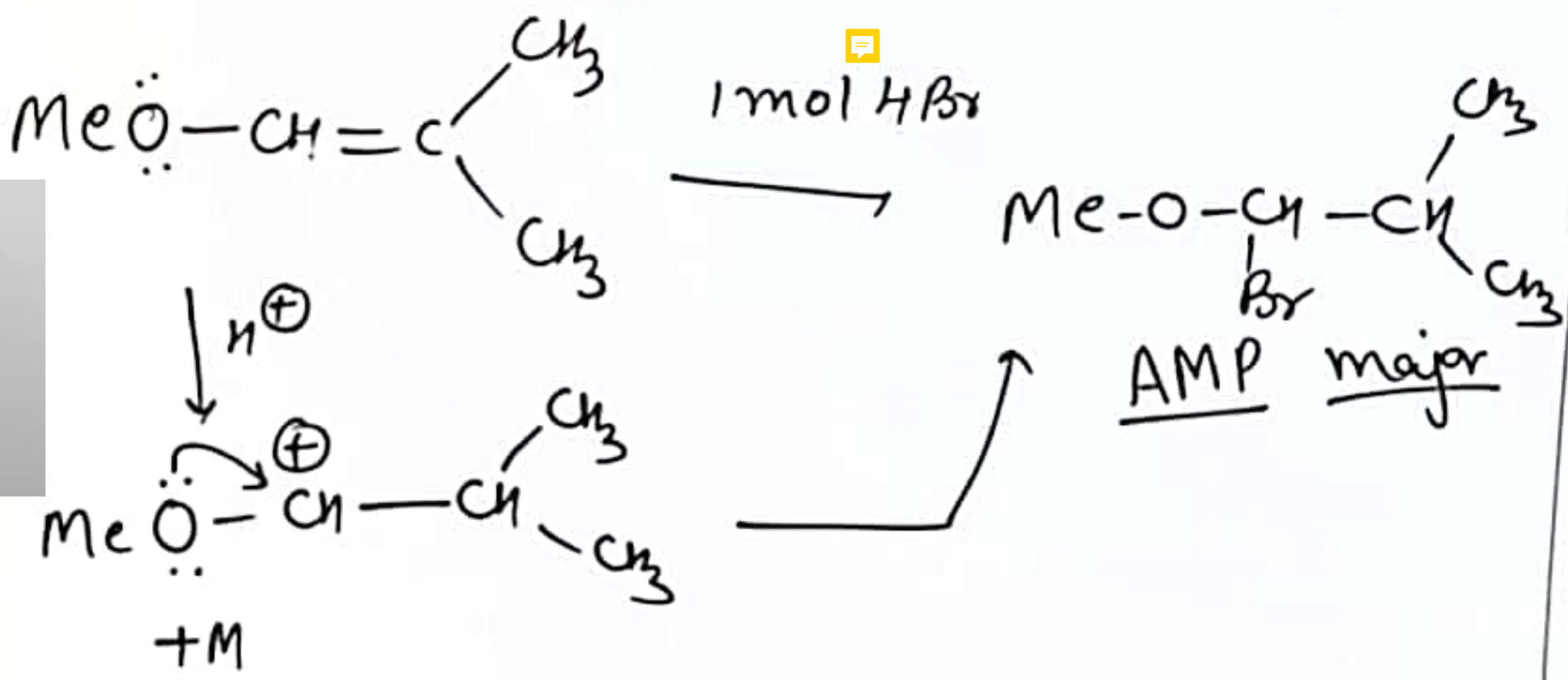


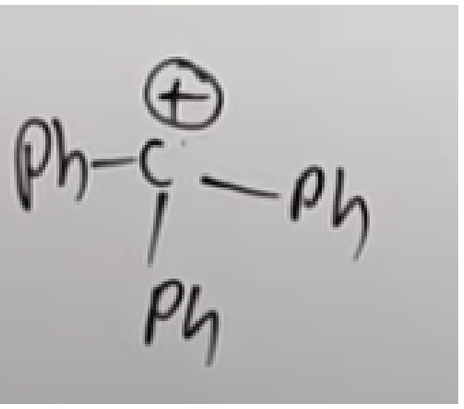
ORGANIC REACTION MECHANISM - II

(III) Electrophilic Addition Reaction (A_E)

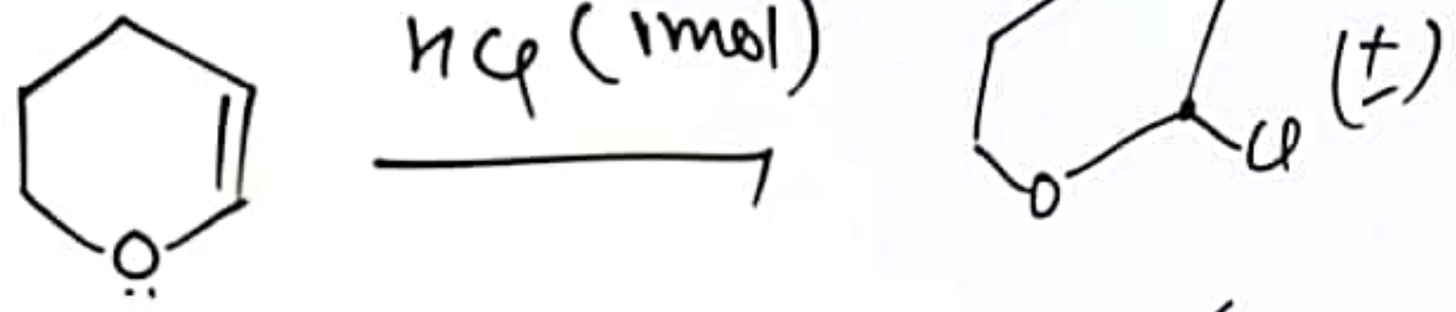
① Addition of HX (Hydrohalogenation)

eg 14

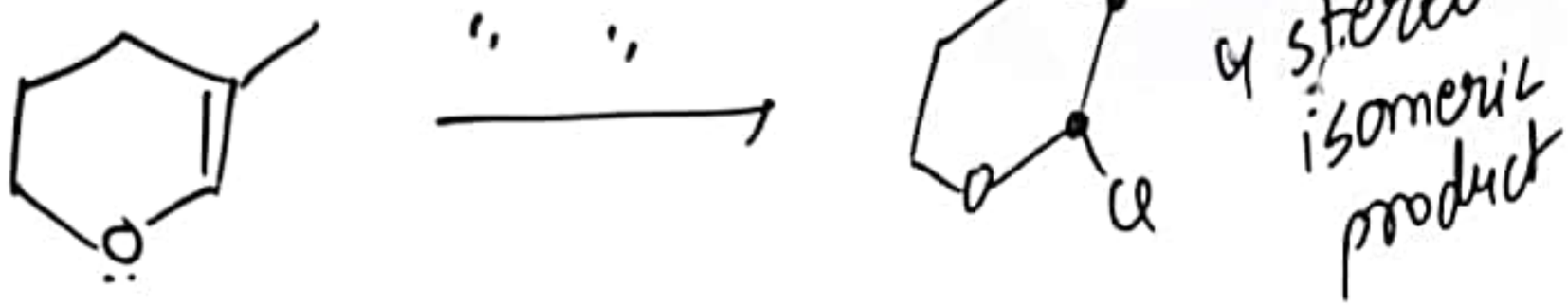




eg 15



eg 16

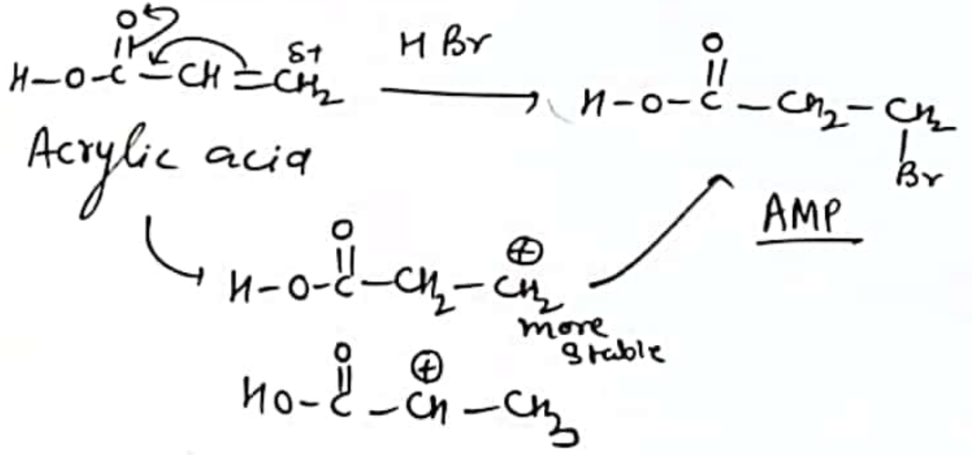


ORGANIC REACTION MECHANISM -II

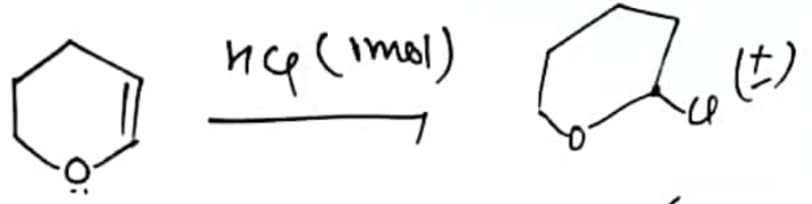
(III) Electrophilic Addition Reaction (A_E)

① Addition of HX (Hydrohalogenation)

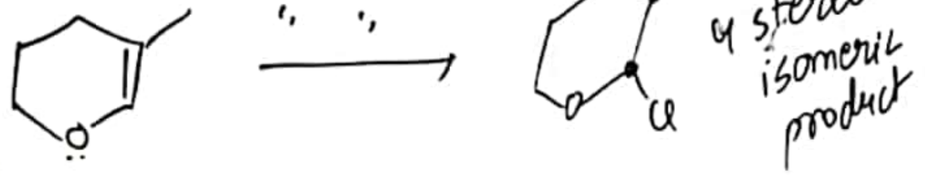
eg 19



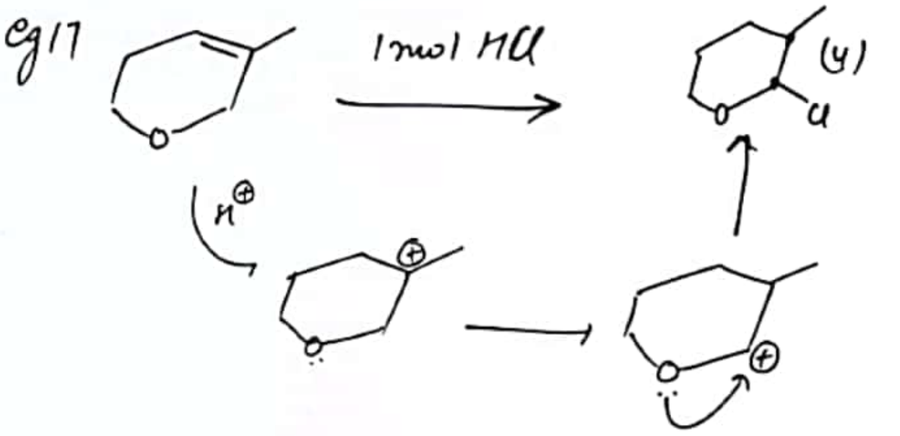
eg 15



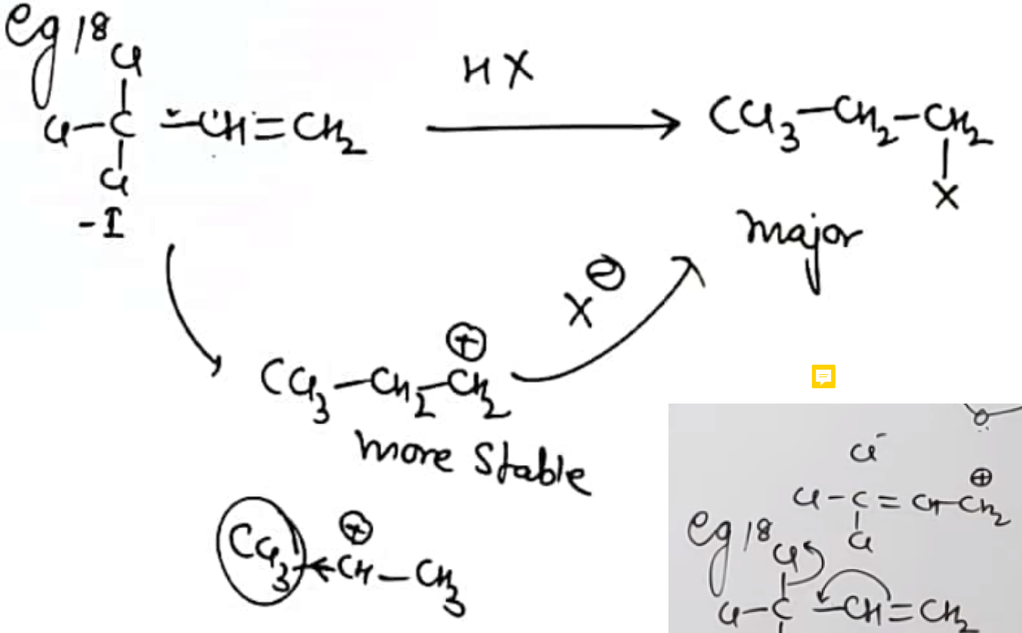
eg 16

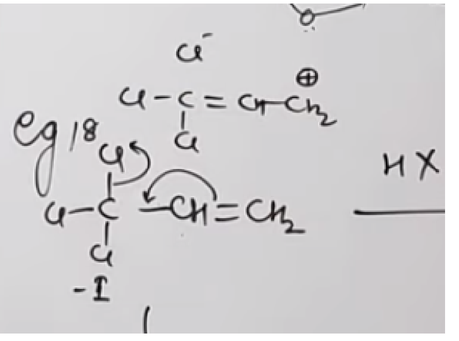


eg 17



eg 18





ORGANIC REACTION MECHANISM-II

(III) Electrophilic Addition Reactions (A_E)

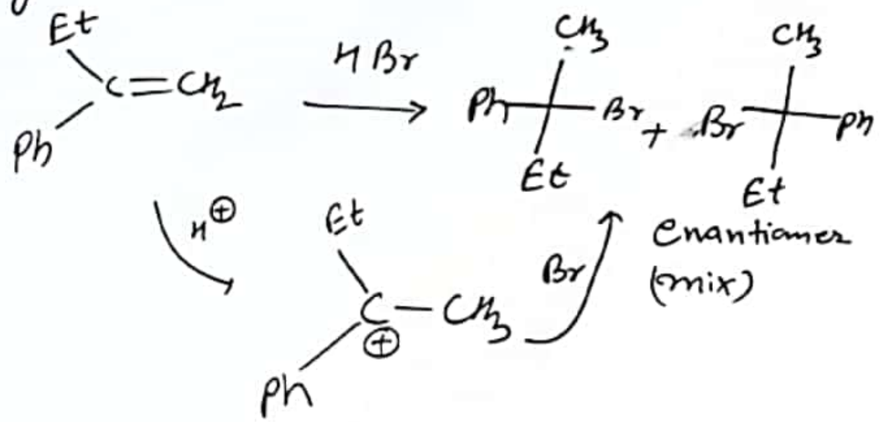
① Addition of HX (Hydrohalogenation) :-

eg 20 $(Cl/Br)-CH=\overset{\delta-}{CH_2} \xrightarrow{HX} (Cl/Br)-\underset{X}{CH}-CH_3$ (M.P.)

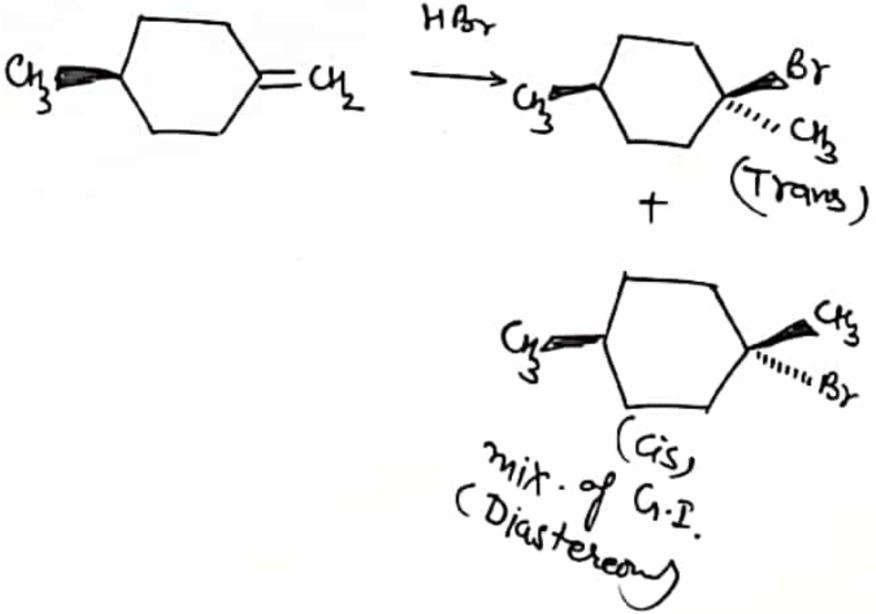
(Vinyl halide)

eg 21 $H-C\equiv C-H \xrightarrow{HBr} Br-CH=\overset{\delta-}{CH_2} \xrightarrow{HBr} Br-\underset{Br}{CH}-CH_3$

eg 22



eg 23

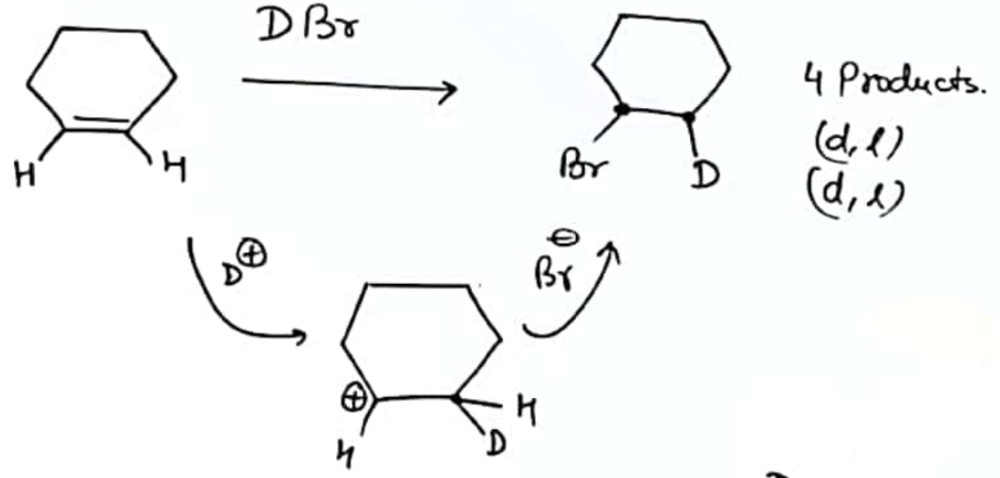


ORGANIC REACTION MECHANISM-II

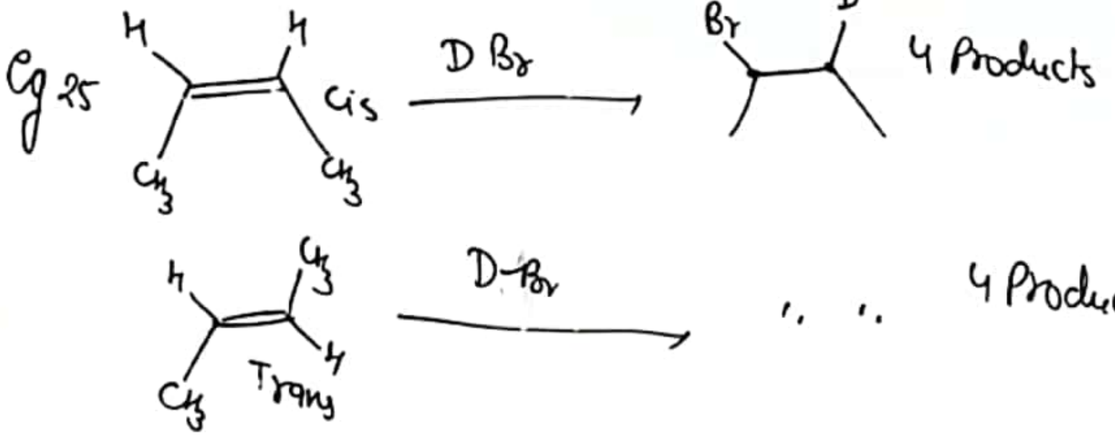
(III) Electrophilic Addition Reactions (A_E)

① Addition of HX (Hydrohalogenation) :-

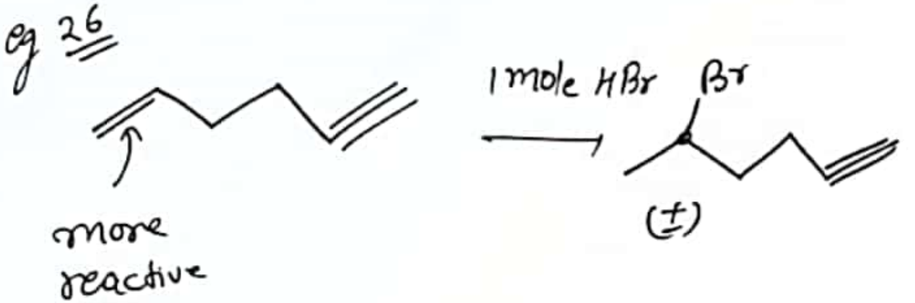
eg 24



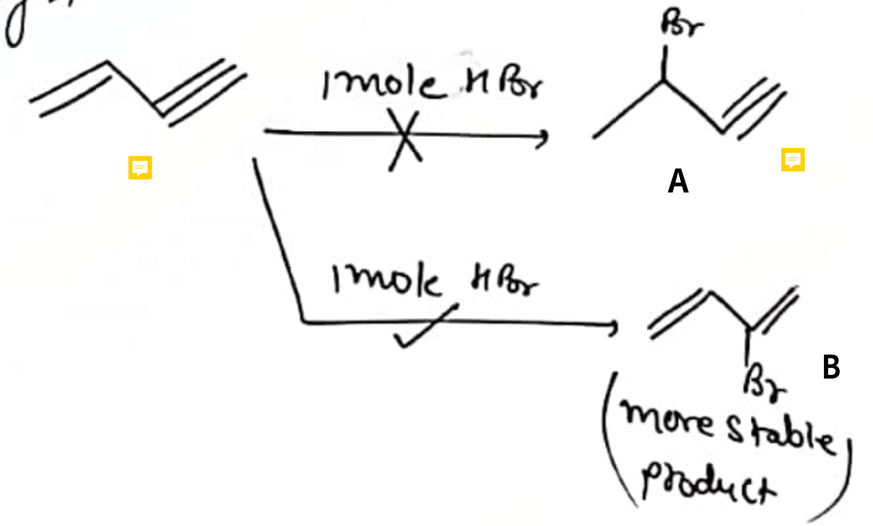
eg 25



eg 26



★ eg 27

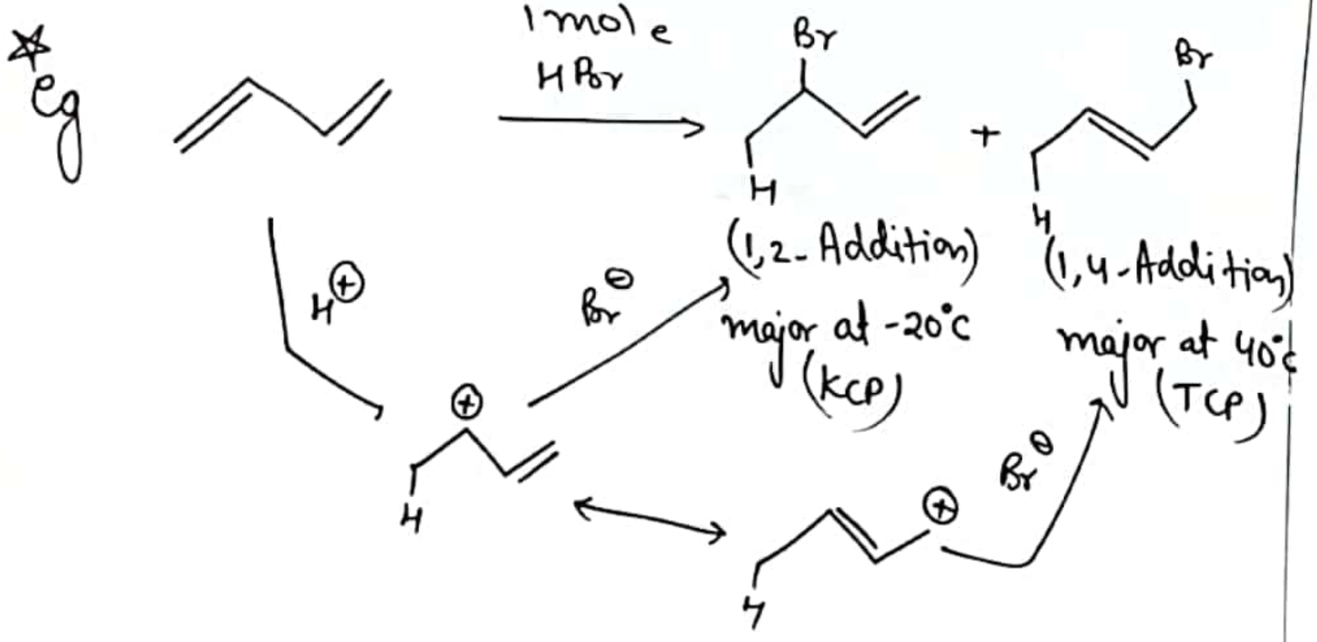


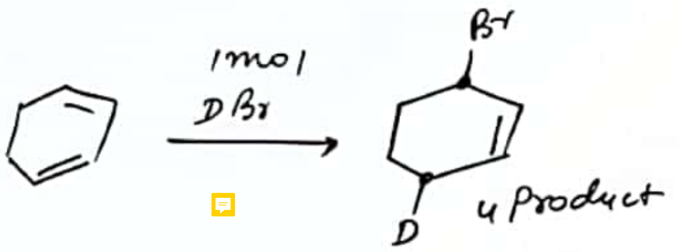
ORGANIC REACTION MECHANISM-II

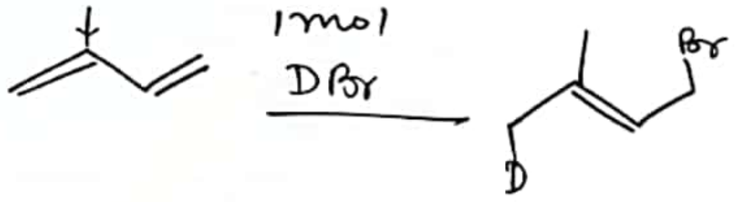
(III) Electrophilic Addition Reactions (A_E)

① Addition of HX (Hydrohalogenation) :-

eg 



eg 

eg 

ORGANIC REACTION MECHANISM-II

(III) Electrophilic Addition Reactions (A_E)

(2) Addition of conc. H_2SO_4 ;—

$H_2O \rightleftharpoons H^{\oplus} + {}^{\ominus}OH$

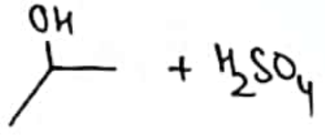
→ According to mech. I

eg 1



ester

$H_2O + H^{\oplus}$



(3) Addition of water (Hydration)

→ Acc. to mech. I

eg 1. $CH_2=CH_2$ (H_2O: ← Nucleophile) → No Reaction.

$H_2O + {}^{\ominus}OH$ → No Rxn.

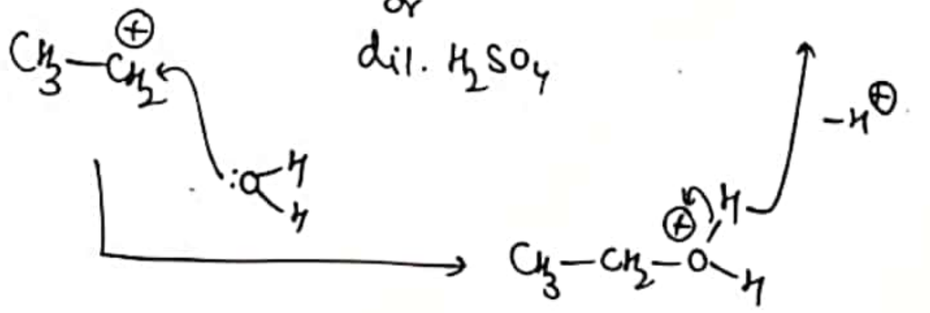
$H^{\oplus}$ (RDS)

$H_2O + H^{\oplus}$ ⇌ CH_3-CH_2-OH Alcohol

or dil acid

or

dil. H_2SO_4

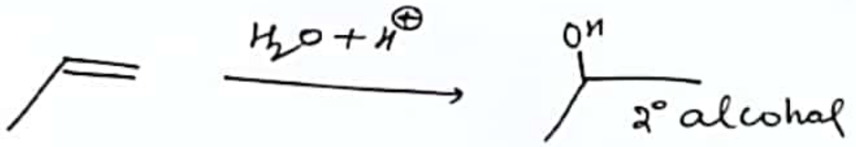


ORGANIC REACTION MECHANISM-II

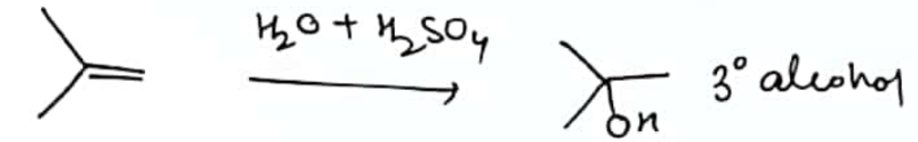
(III) Electrophilic Addition Reactions (A_E)

(2) Addition of water (Hydration)

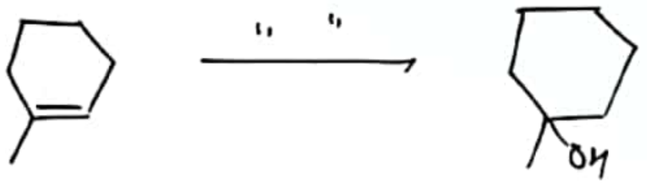
eg 2



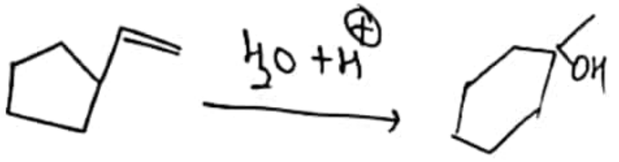
eg 3



eg 4



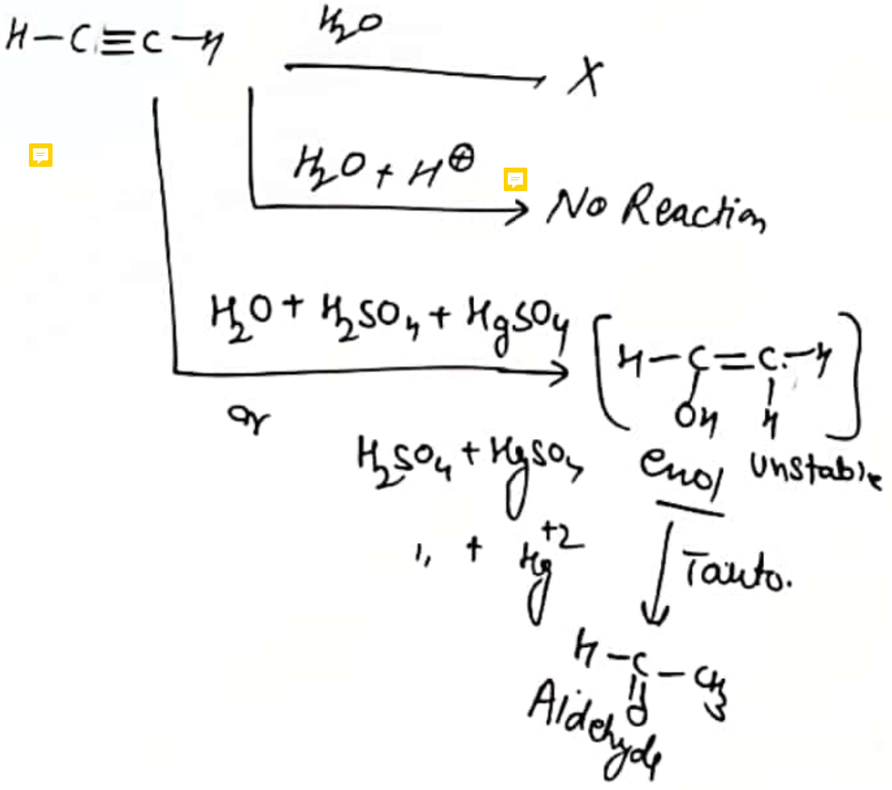
eg 5



eg 6



eg 7

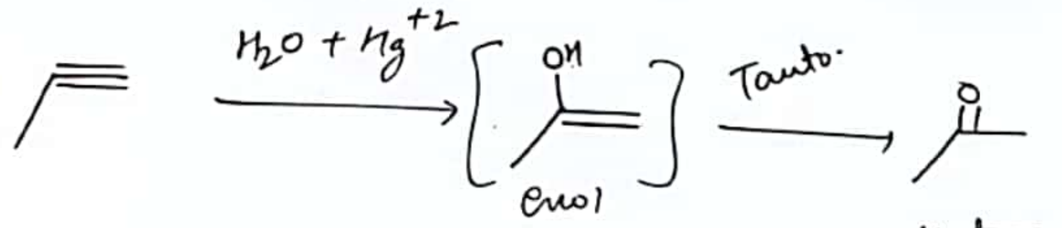


ORGANIC REACTION MECHANISM-II

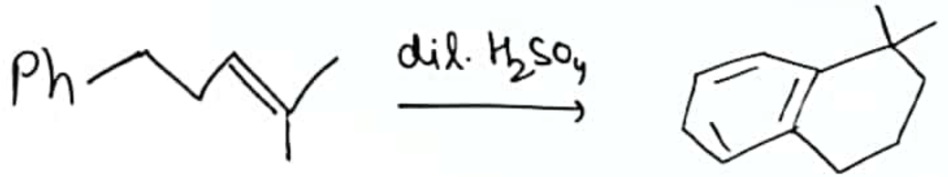
(III) Electrophilic Addition Reactions (A_E)

(3) Addition of water (Hydration)

eg 8

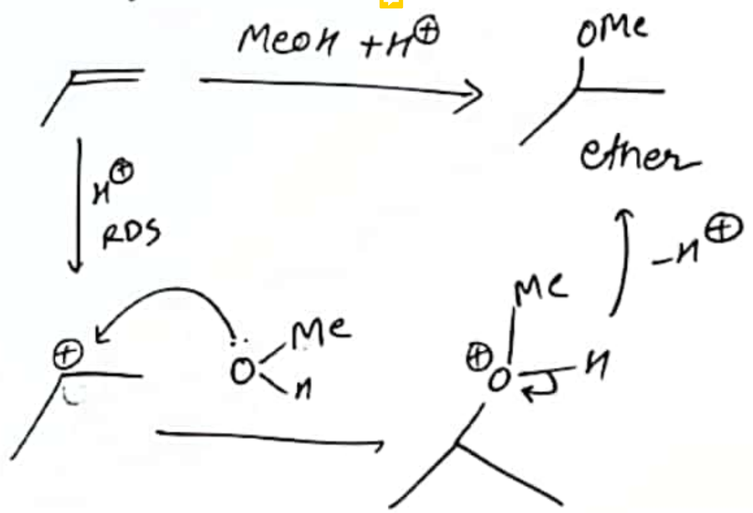


eg 9



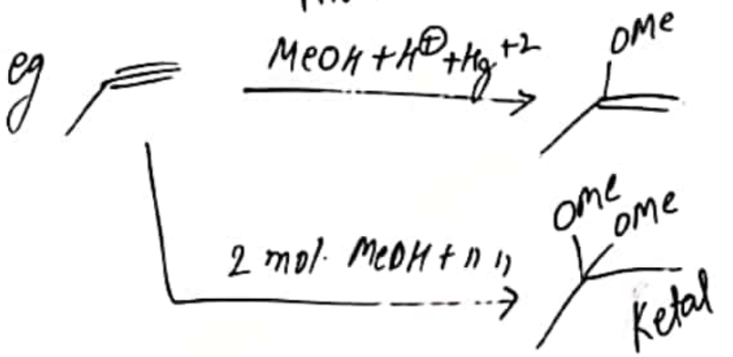
(4) Addition of alcohols

Acc. of mech. I



1 mol

eg



ORGANIC REACTION MECHANISM - II (ORM - II)

(III) Electrophilic Addition Reactions (A_E)

(5) Addition of X_2, X-X' and HOX on carbon - carbon m.b.

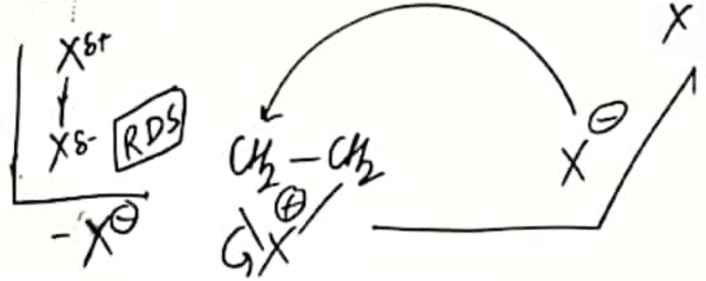
→ According to mech. II

- intermediate = Non classical carbocation
- No Rearrangement
- major product = generally M.P.
- Anti addition
- Stereoselective & Stereospecific reaction.

$:\overset{\oplus}{X}:$

Solvent

Eg: $CH_2=CH_2 \xrightarrow{X_2 + (CCl_4/CH_2Cl_2)} \underset{X \quad X}{CH_2-CH_2}$ Vicinal dihalide



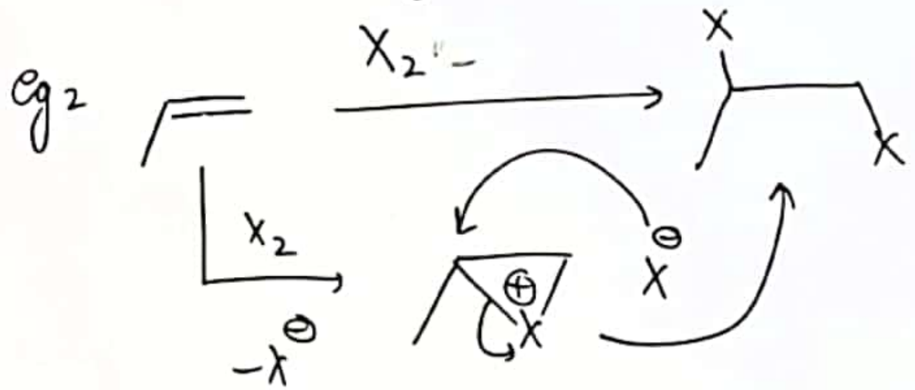
ORGANIC REACTION MECHANISM - II (ORM-II)

(III) Electrophilic Addition Reactions (A_E)

(5) Addition of X_2, X-X' and HOX on carbon - carbon multiple bond

⇒ Reactivity of X_2 :- $F_2 > Cl_2 > Br_2 > I_2$

(Cl_2 > Br_2: useful; F_2: highly exothermic; I_2: reversible reaction ⟷)



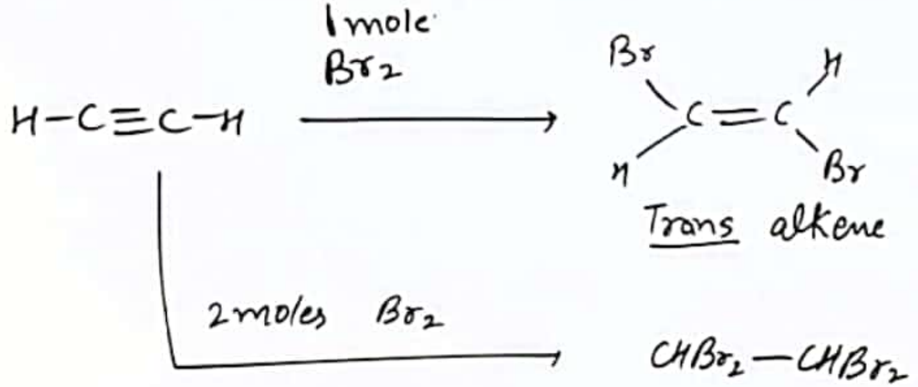
Note Vic. diiodo compounds are generally unstable and form alkene by deiodination reaction.

ORGANIC REACTION MECHANISM - II (ORM - II)

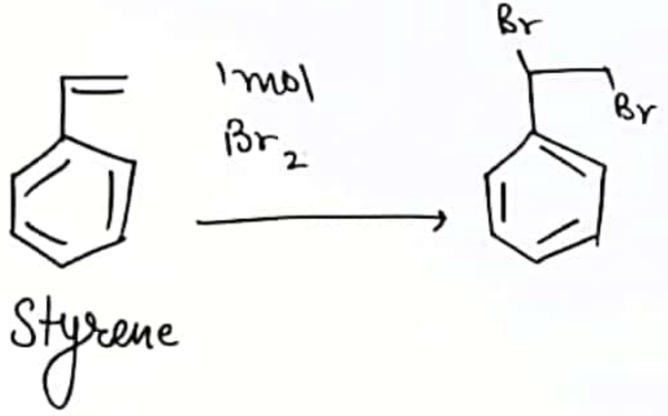
(III) Electrophilic Addition Reactions (A_E)

(5) Addition of X_2, X-X' and HOX on carbon - carbon multiple bond

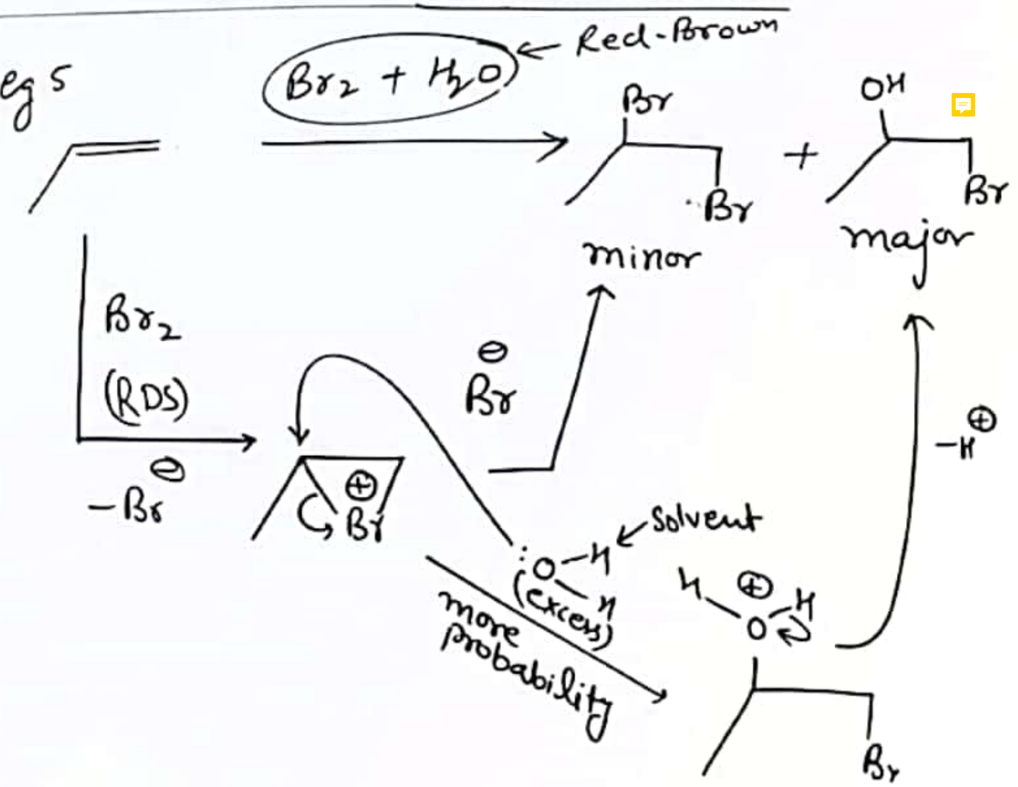
eg 3



eg 4



eg 5

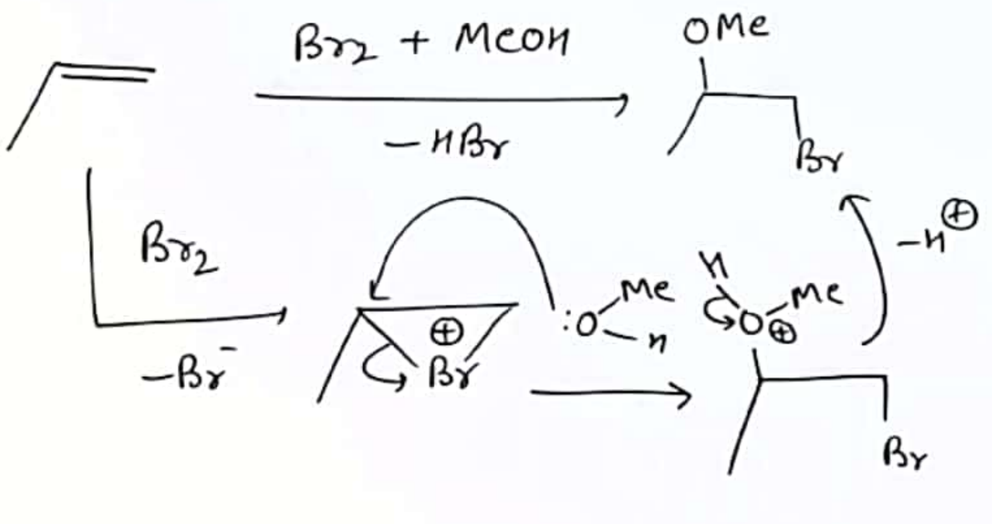


ORGANIC REACTION MECHANISM - II (ORM - II)

(III) Electrophilic Addition Reactions (A_E)

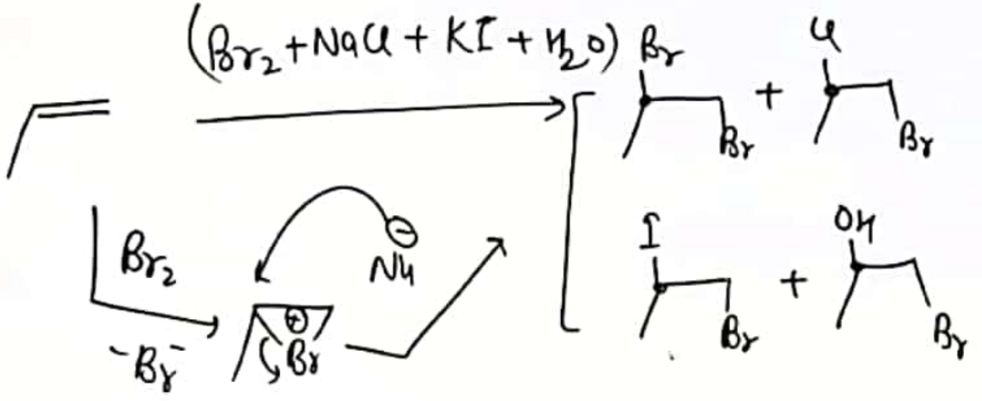
(5) Addition of X_2, X-X' and HOX on carbon - carbon multiple bond

eg 6

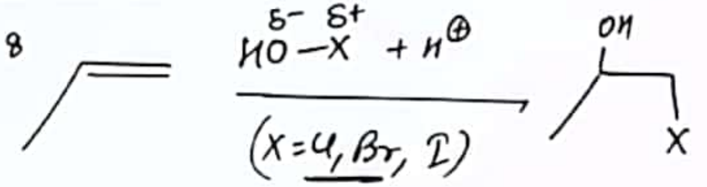


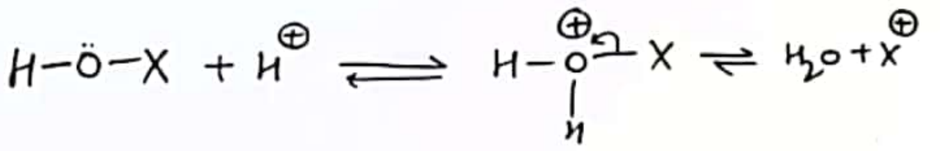
Br_2+H_2O or HO-Br

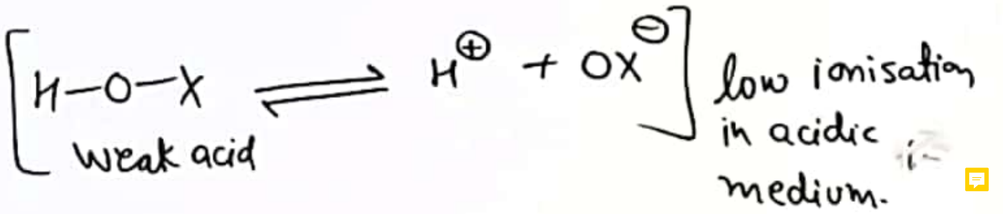
eg 7



eg 8





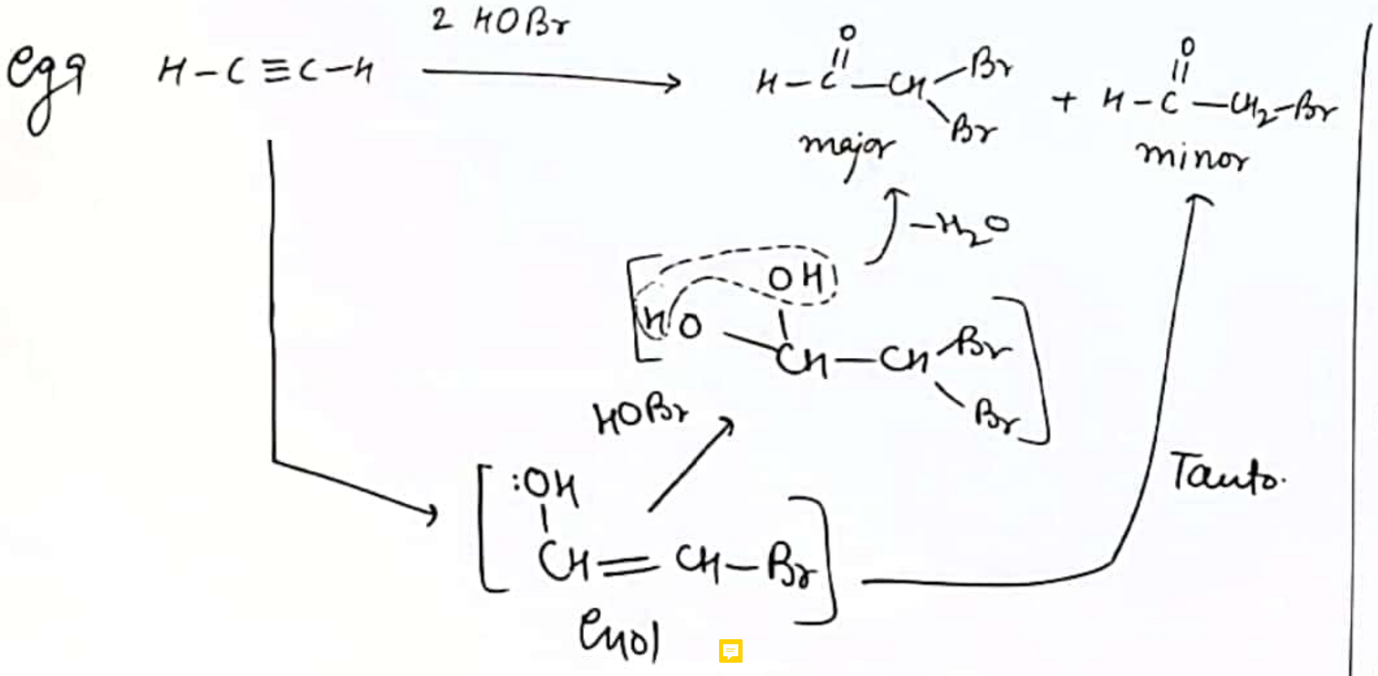


ORGANIC REACTION MECHANISM - II (ORM - II)

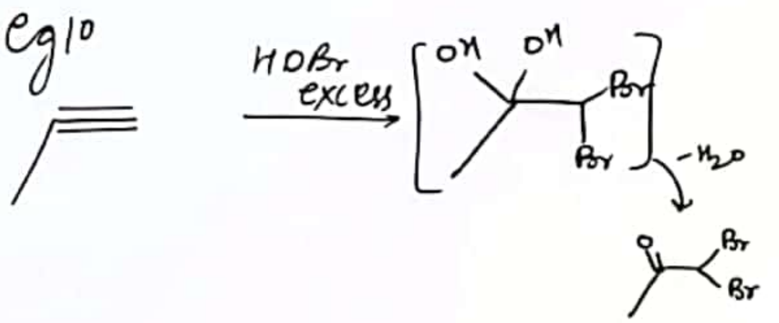
(III) Electrophilic Addition Reactions (A_E)

(5) Addition of X_2, X-X' and HOX on carbon - carbon multiple bond

eg 9



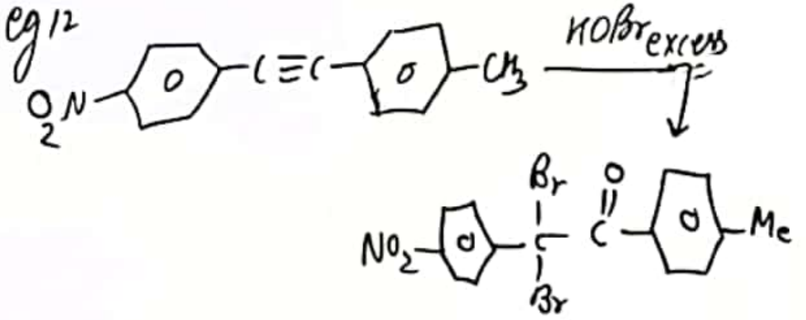
eg 10



eg 11



eg 12

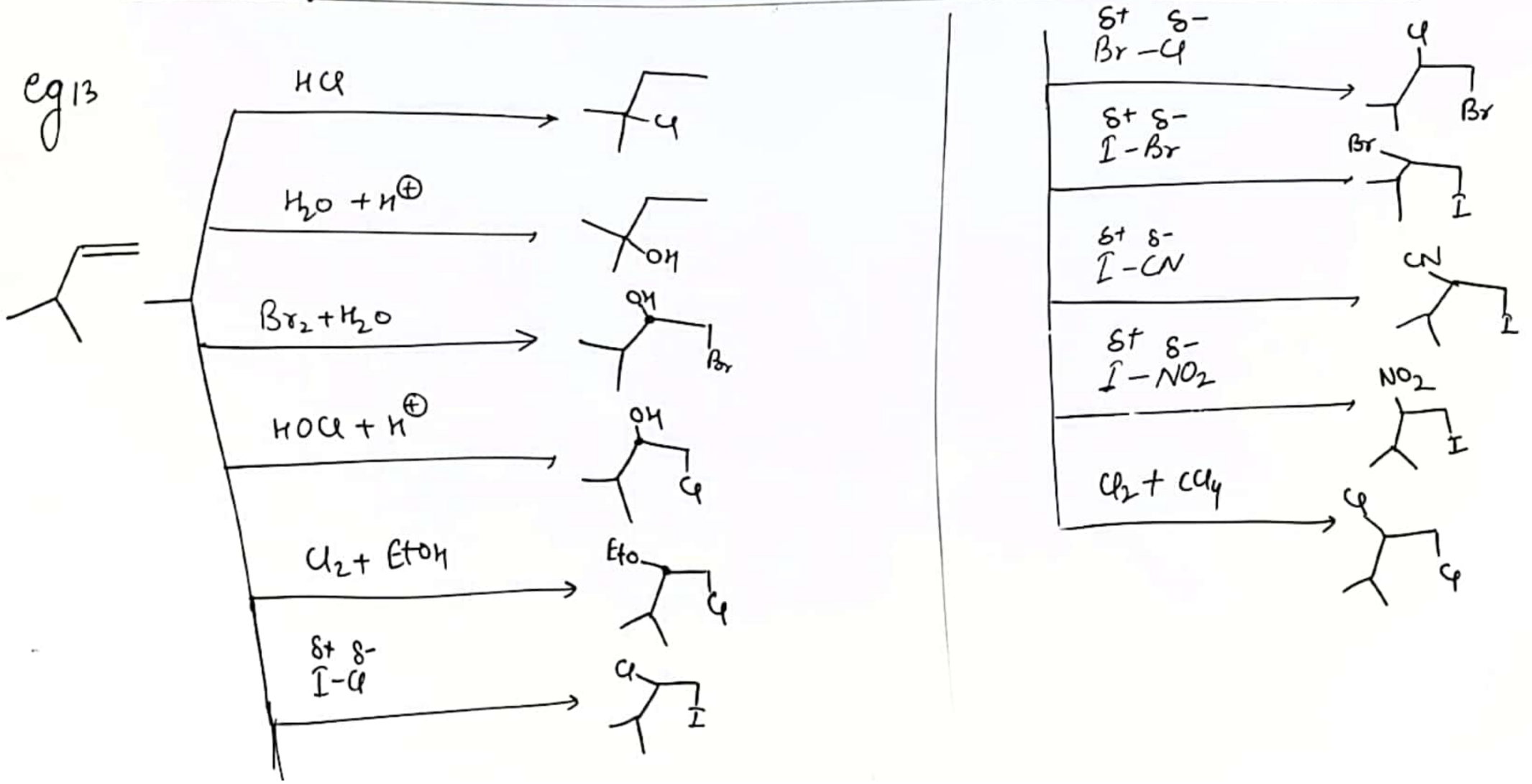


ORGANIC REACTION MECHANISM - II (ORM - II)

(III) Electrophilic Addition Reactions (A_E)

(5) Addition of X_2, X-X' and HOX on carbon - carbon multiple bond

Eg 13

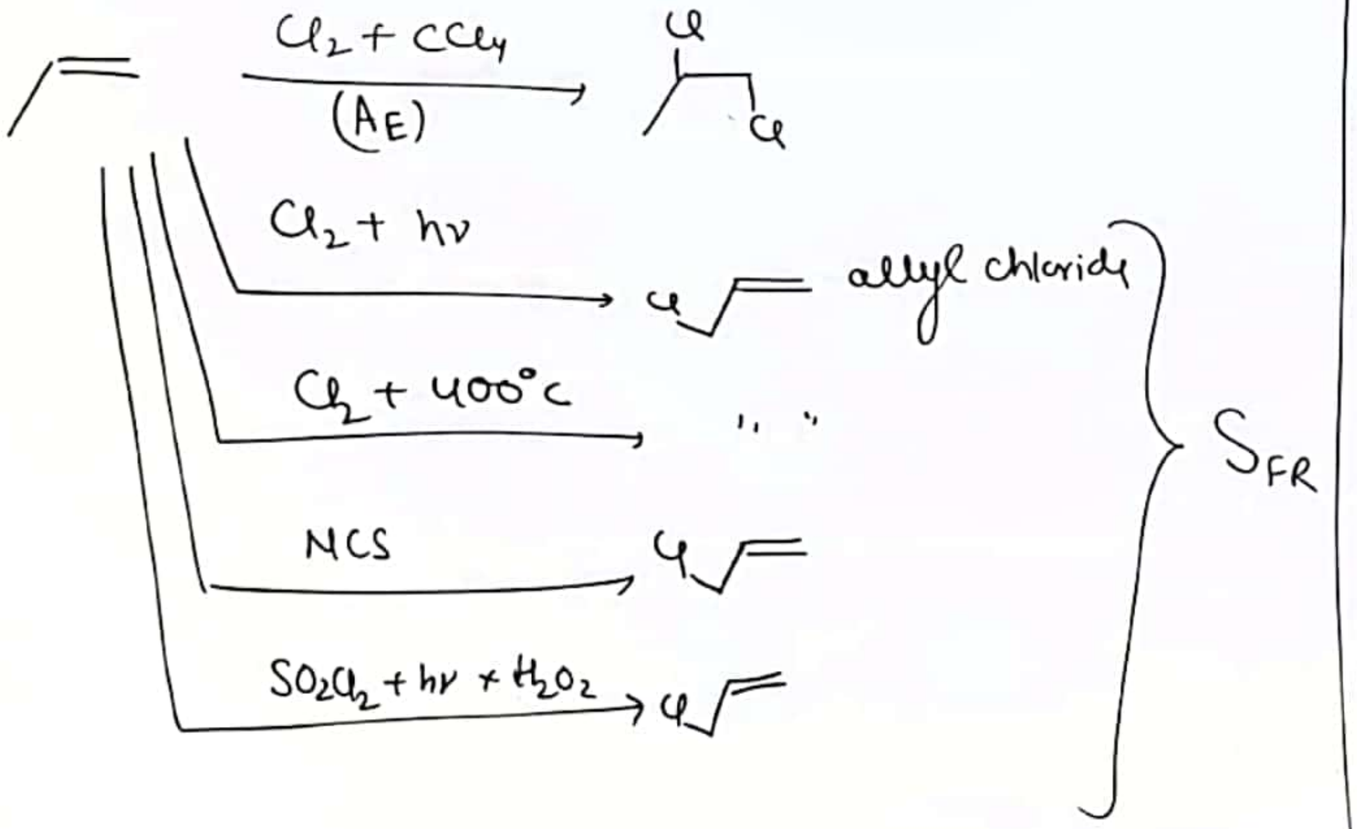


ORGANIC REACTION MECHANISM - II (ORM - II)

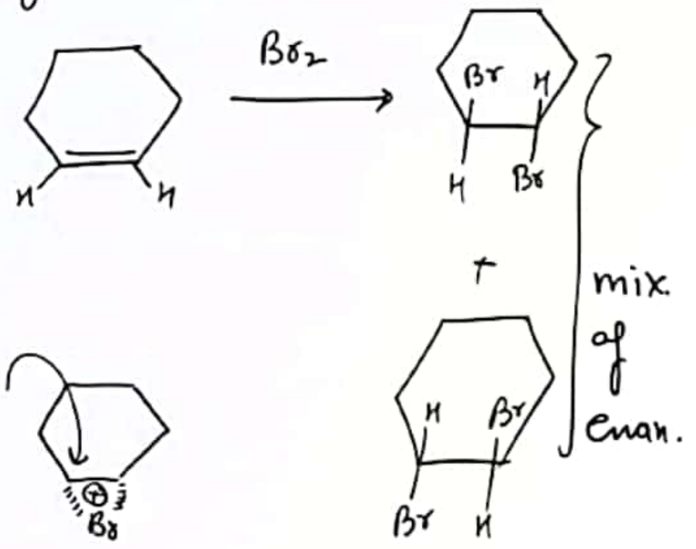
(III) Electrophilic Addition Reactions (A_E)

(5) Addition of X_2, X-X' and HOX on carbon - carbon multiple bond

eg 14



eg 15



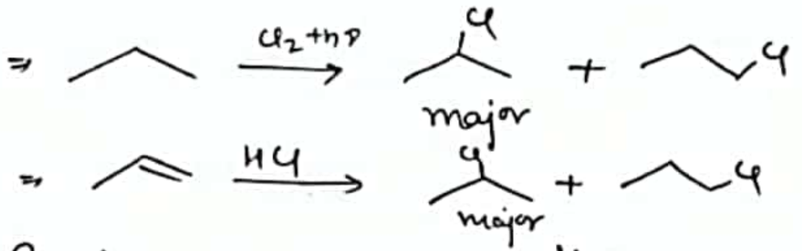
ORGANIC REACTION MECHANISM -II (ORM-II)

(III) Electrophilic Addition Reactions (A_E)

(5) Addition of X_2, X-X' and HOX :-

Some important definitions:-

* **Regioselective reaction:-** Reactions in which more than one structural isomeric products form in unequal amount.



* **Stereoselective reactions:-** Reactions in which more than one stereoisomeric products (some time only one product) form in unequal amount (out of many other possible stereoisomeric products.

Reactant → Cis > Trans
→ Tran > Cis
→ Threo > Erythro

ORGANIC REACTION MECHANISM -II (ORM-II)

(III) Electrophilic Addition Reactions (A_E)

(5) Addition of X_2, X-X' and HOX :-

Some important definitions:-

* **Stereospecific reactions:-** Stereoselective reaction in which one stereo-isomeric reactant form one specific stereo isomeric product in major amount.

Note - All stereospecific reactions are stereoselective but its revers is not alway true.



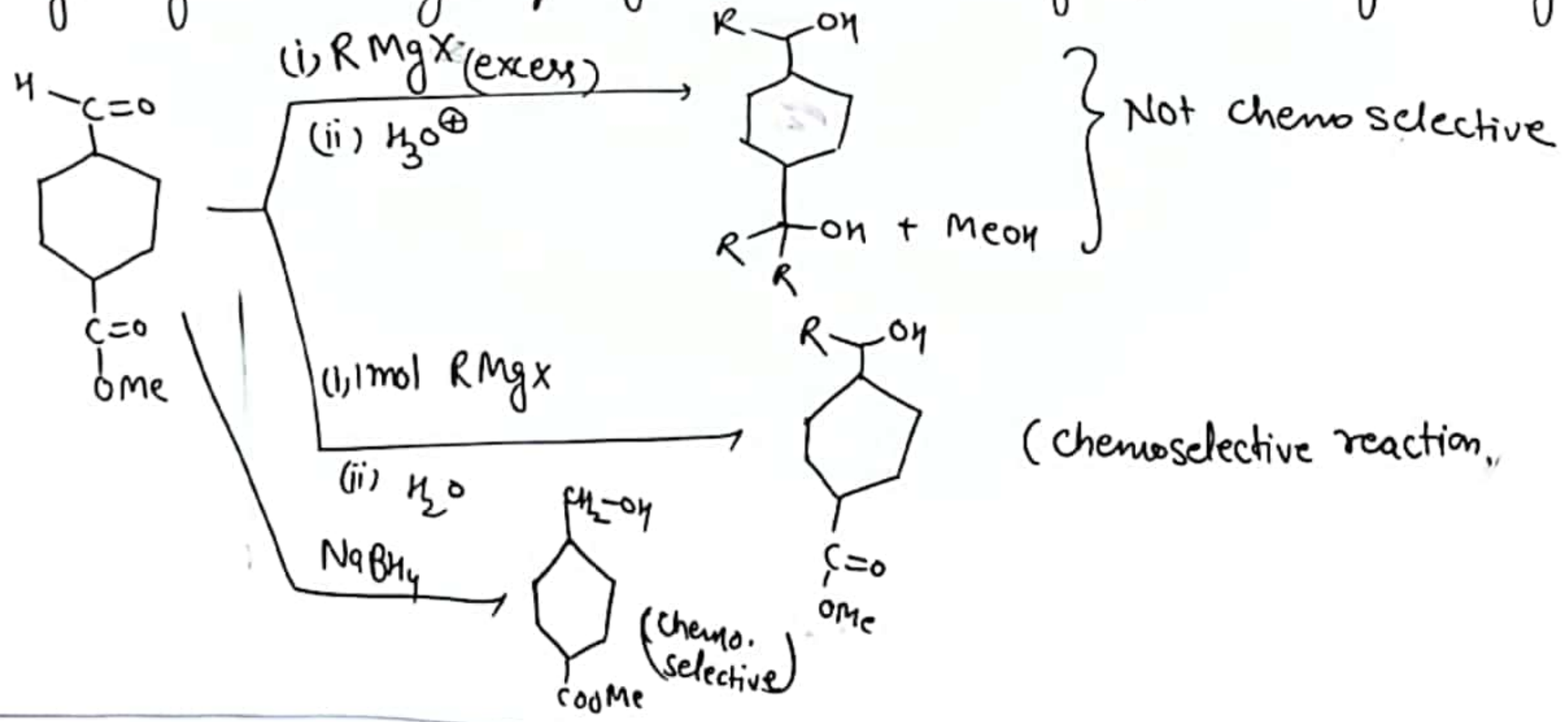
ORGANIC REACTION MECHANISM -II (ORM-II)

(III) Electrophilic Addition Reactions (A_E)

(5) Addition of X_2, X-X' and HOX :-

Some important definitions:-

* Chemoselective reaction:- Reactions in which a specific reagent reacts with specific functional groups of reactant out of presence of many other F.g.)

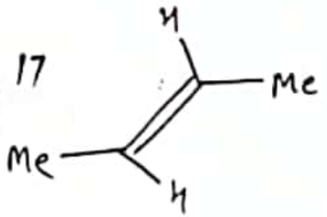


ORGANIC REACTION MECHANISM -II (ORM-II)

(III) Electrophilic Addition Reactions (A_E)

(5) Addition of X_2, X-X' and HOX :-

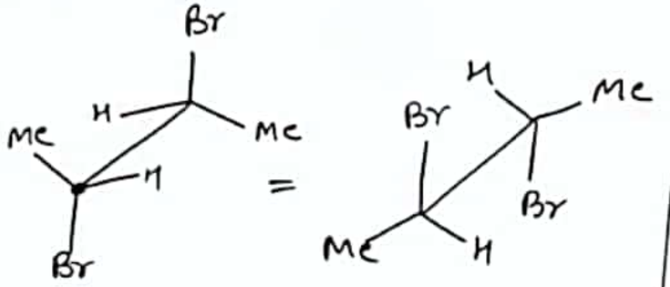
Eg 17



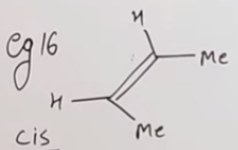
Trans

$Br_2 + CCl_4$

Anti addition



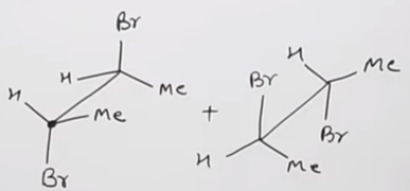
Eg 16



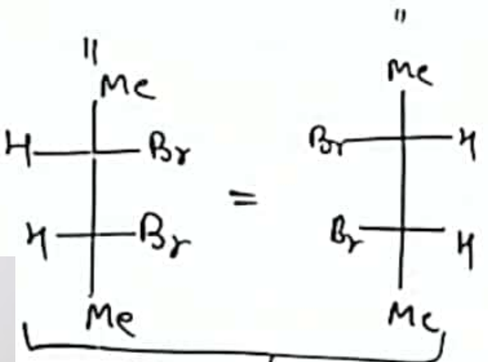
cis

$Br_2 + CCl_4$

Anti addition



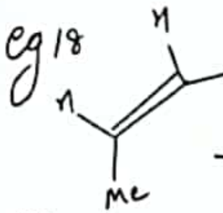
Three Racemic mix.



Meso (Erythro)

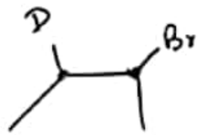
≠ ≠ ≠ ≠

Eg 18



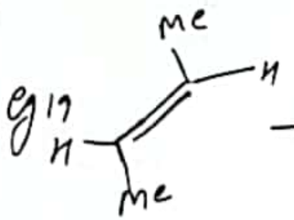
cis

D-Br



Product

Eg 19



Trans

DBr

,, ,,

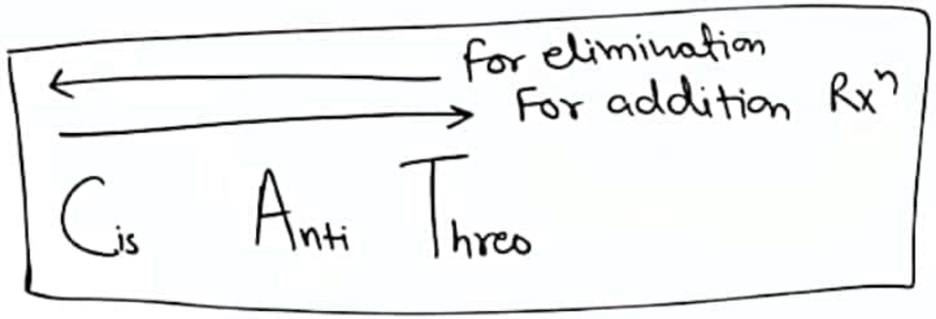
ORGANIC REACTION MECHANISM -II (ORM-II)

(III) Electrophilic Addition Reactions (A_E)

(5) Addition of X_2, X-X' and HOX :-

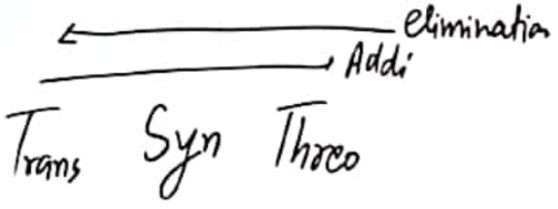
Short trick for stereospecific reactions

[→ For Fischer projection
→ " Two chiral carbon]





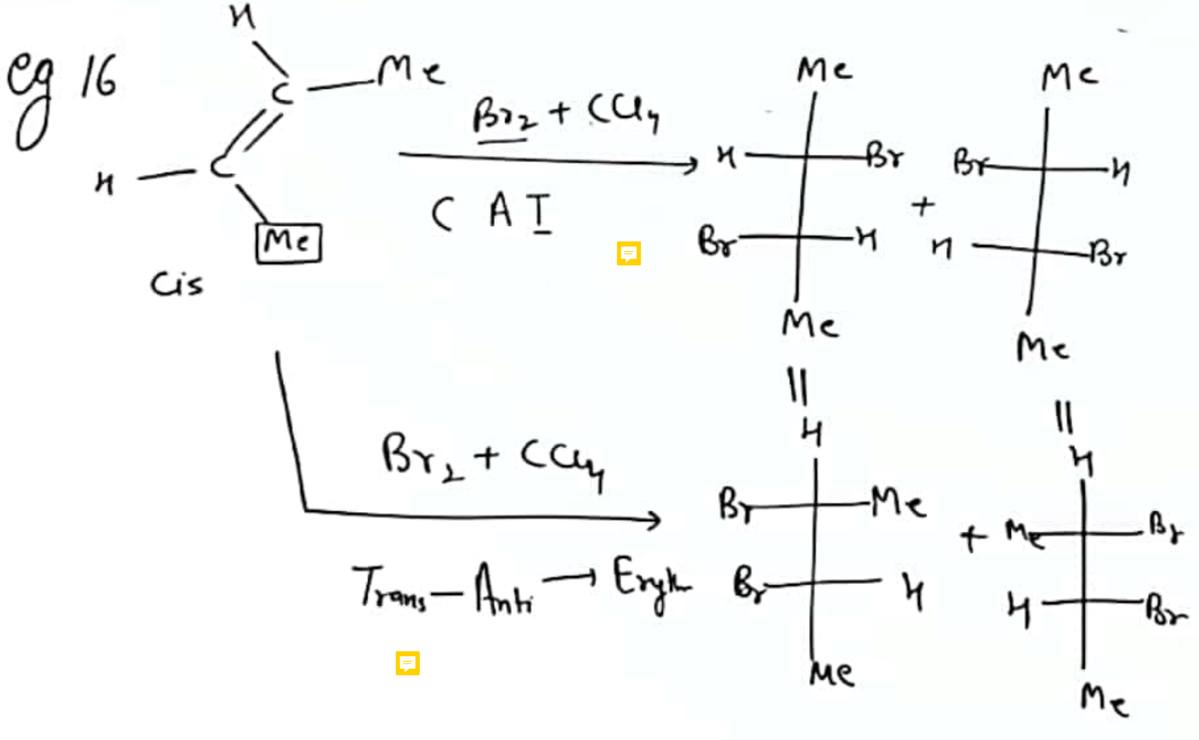




ORGANIC REACTION MECHANISM -II (ORM-II)

(III) Electrophilic Addition Reactions (A_E)

(5) Addition of X_2, X-X' and HOX :-

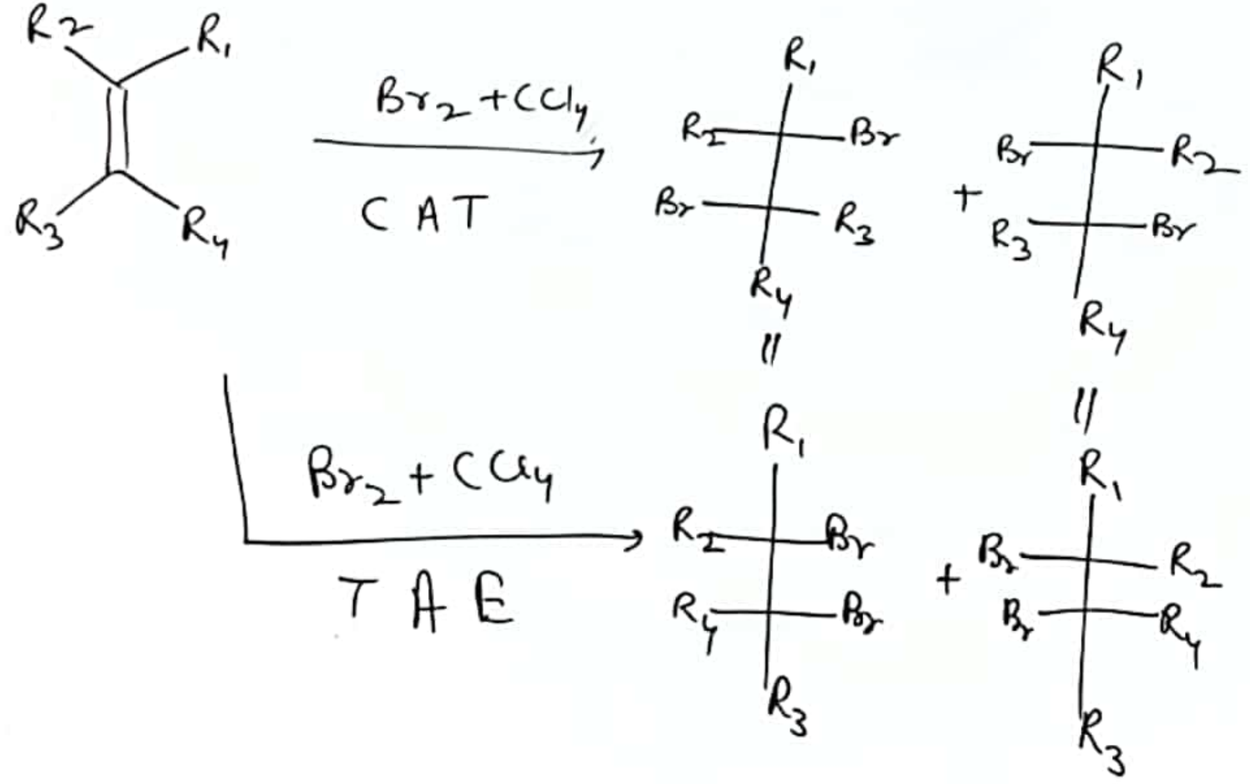


ORGANIC REACTION MECHANISM -II (ORM-II)

(III) Electrophilic Addition Reactions (A_E)

(5) Addition of X_2, X-X' and HOX :-

eg :-

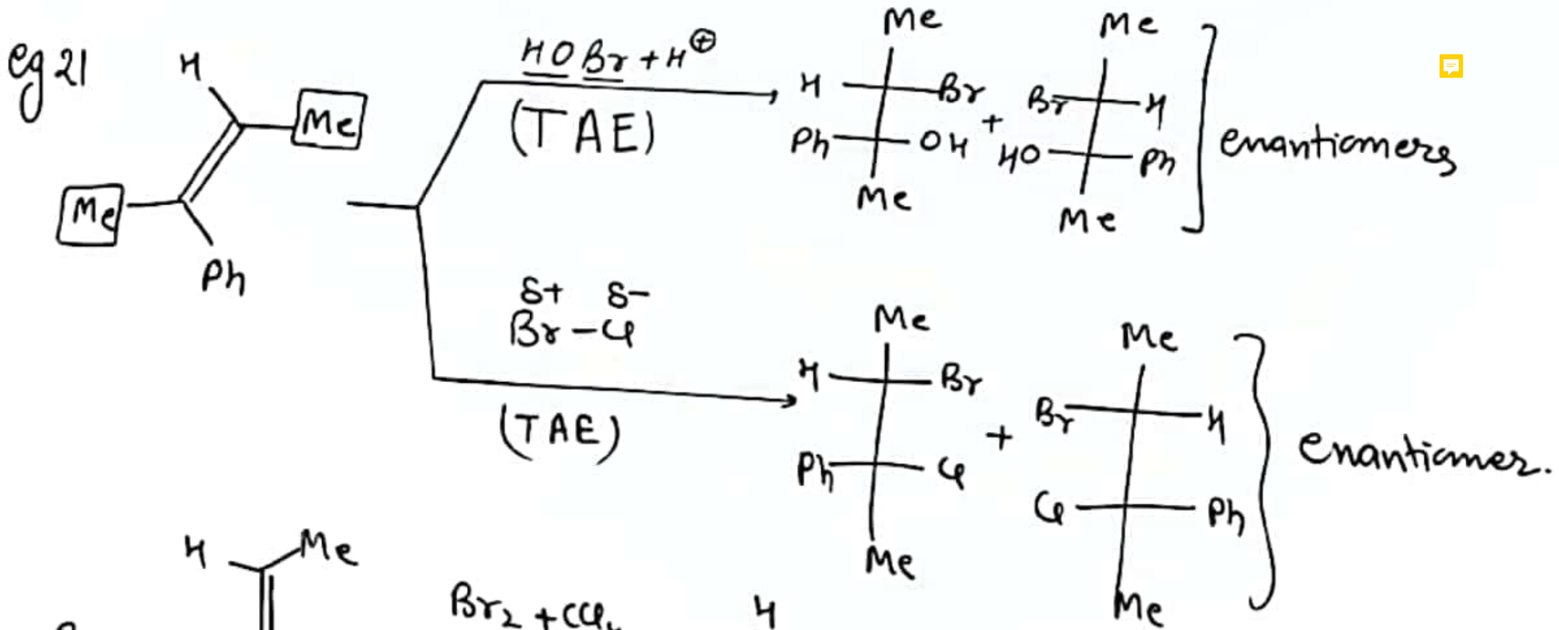


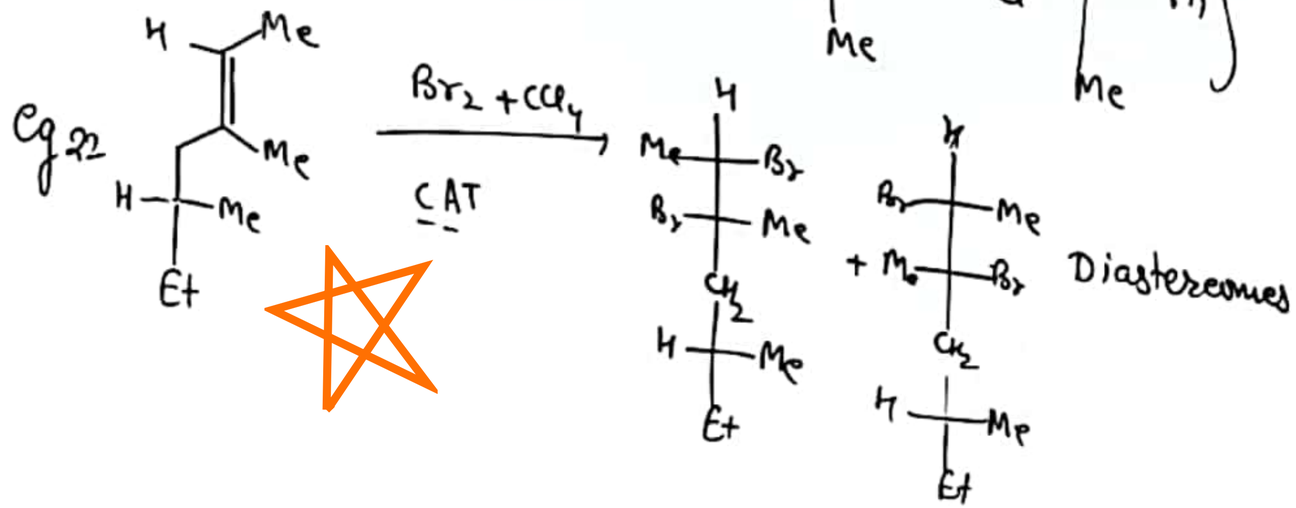
ORGANIC REACTION MECHANISM – II

ORM-II

(III) Electrophilic Addition Reactions :- (A_E)

(5) Addition of X_2, X–X' and HOX :–





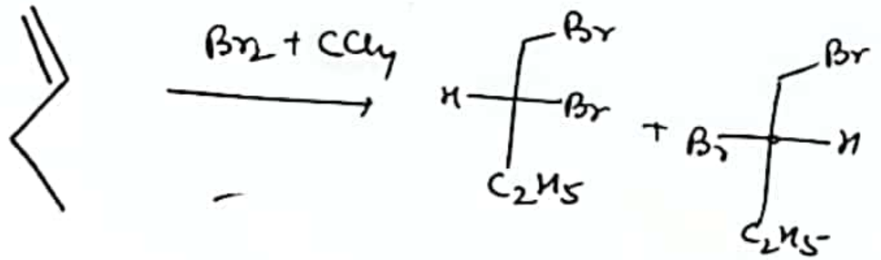
ORGANIC REACTION MECHANISM – II

ORM-II

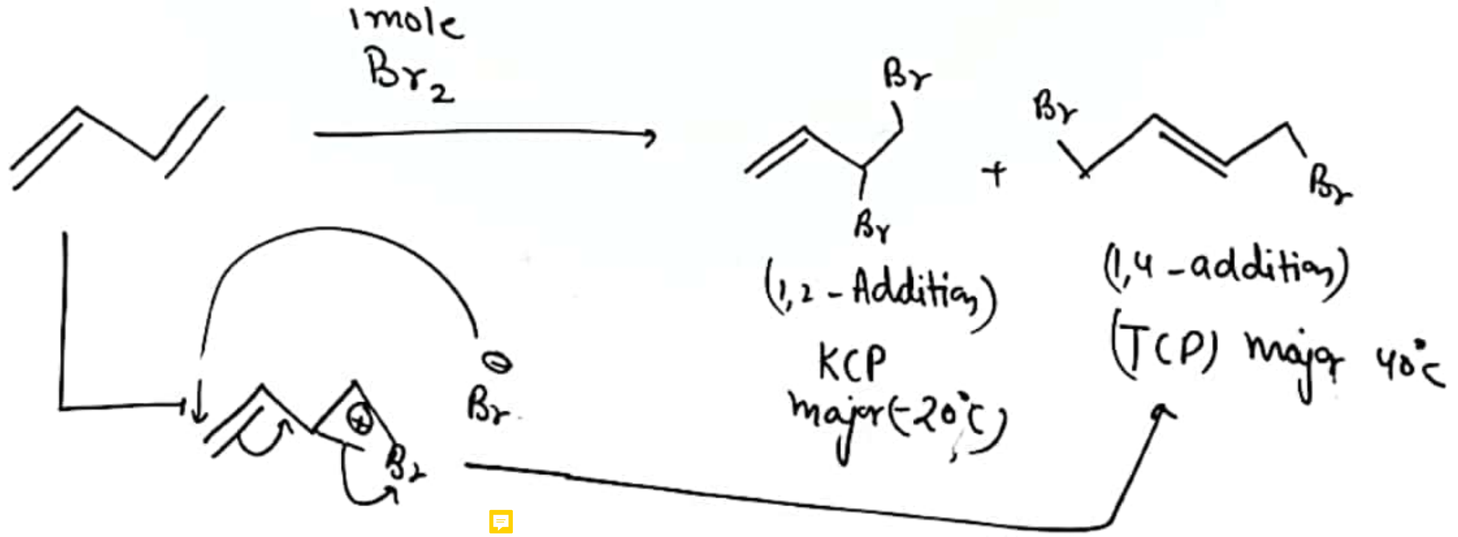
(III) Electrophilic Addition Reactions :- (A_E)

(5) Addition of X_2, X–X' and HOX :–

Eg 23



Eg 24

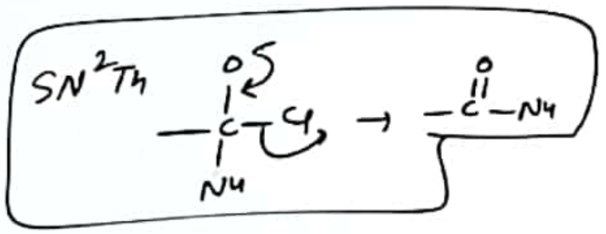


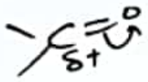
ORGANIC REACTION MECHANISM – II

ORM-II

(III) Electrophilic Addition Reactions :- (A_E)

(6) Addition of NOCl, NOBr



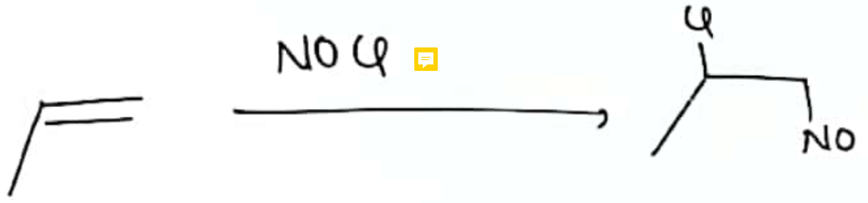


→ According to mechanism – II

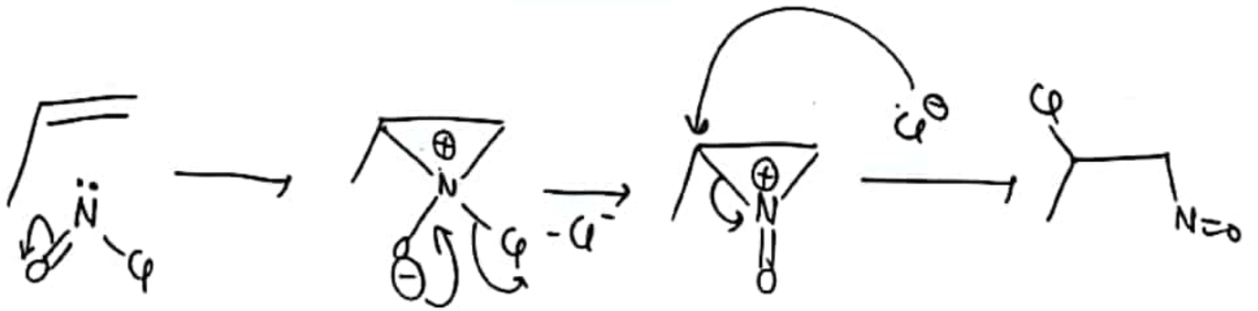
- Nonclassical carbocation intermediate
- No Rearr.
- Anti addition
- Stereo specific

Hint $\overset{\oplus}{NO}$ $X^{\ominus}$

Eg!



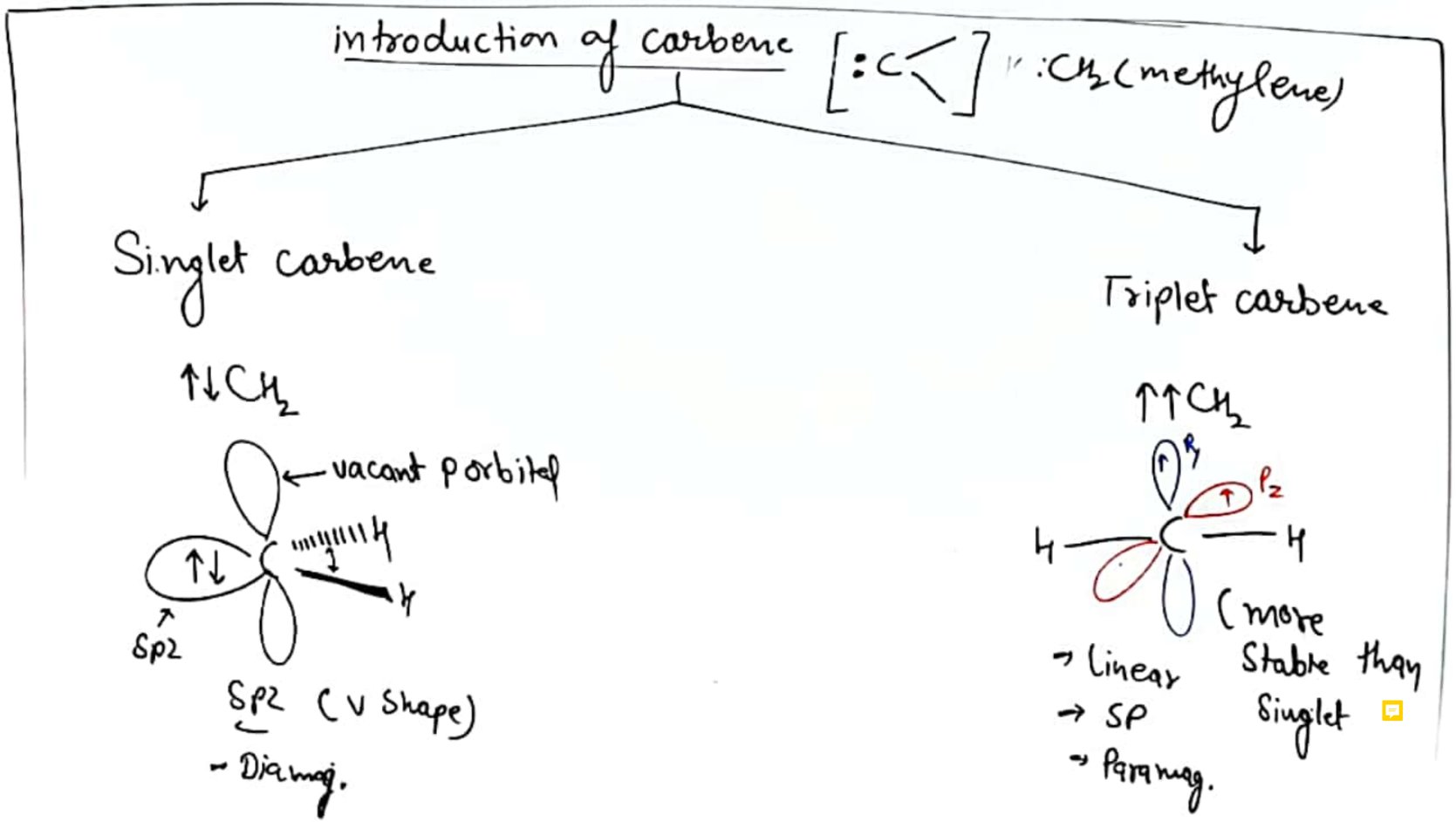
mech.



ORGANIC REACTION MECHANISM – II

(III) Electrophilic Addition Reactions :- (A_E)

(7) Addition of carbene :-



ORGANIC REACTION MECHANISM – II

ORM-II

(III) Electrophilic Addition Reactions :- (A_E)

(7) Addition of carbene:-

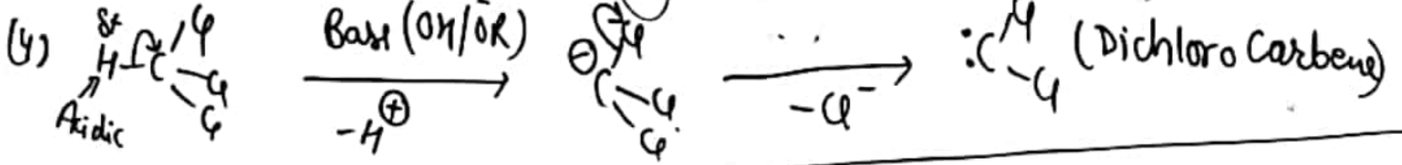
Formation of carbene:-

① CH_2N_2  $\xrightarrow{\Delta,\ h\nu}$ $:CH_2 + N_2\uparrow$

(Diazomethane)

② $CH_2=C=O$ (Ketene) $\xrightarrow{\Delta,\ h\nu}$ $:CH_2 + CO\uparrow$

③ $CH_2I_2 + Zn \longrightarrow$  $\xrightarrow{\Delta}$ $:CH_2 + ZnI_2$

④ 

(Dichloro Carbene)

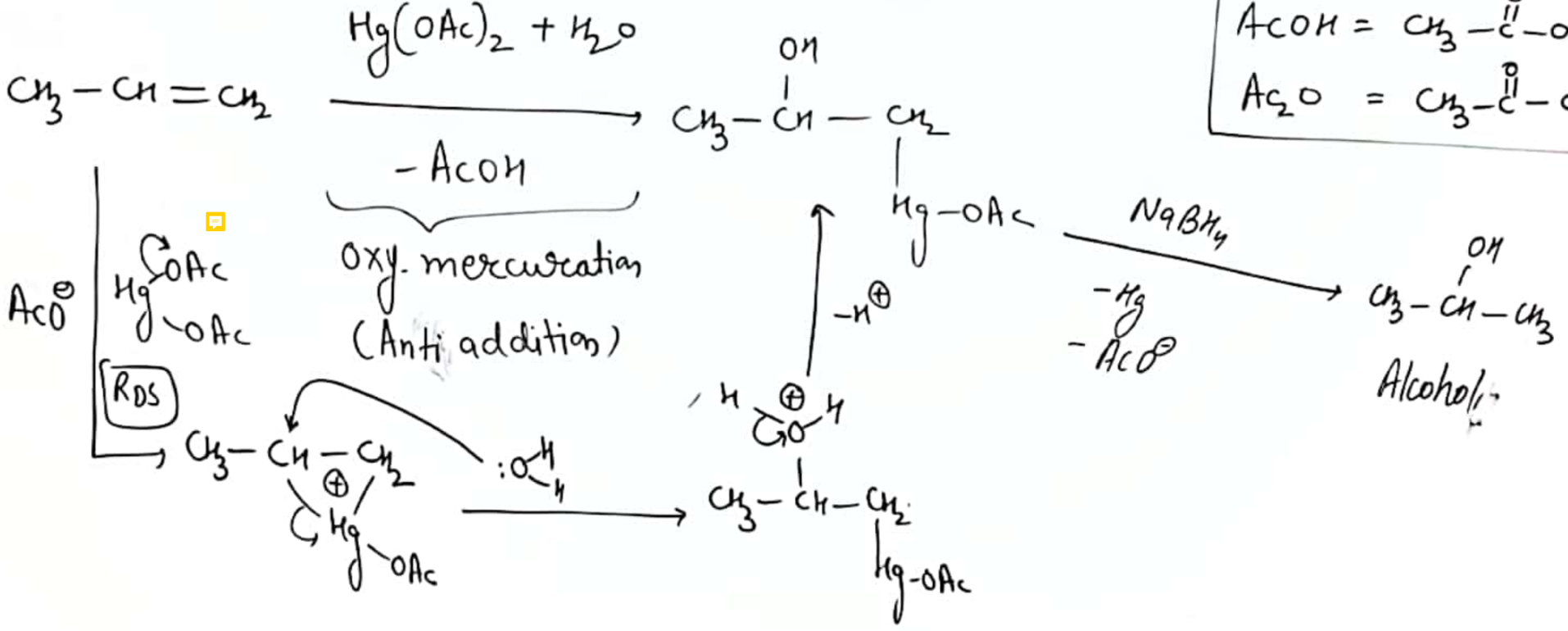
ORGANIC REACTION MECHANISM – II

ORM-II

(III) Electrophilic Addition Reactions :- (A_E)

(8) Oxy-mercuration - demercuration :-

→ Acc. to mechanism - II



Ac–	$CH_3-\overset{O}{\overset{\|}{C}}-$
$AcO^{\ominus}$	$CH_3-\overset{O}{\overset{\|}{C}}-O^{-}$
AcOH	$CH_3-\overset{O}{\overset{\|}{C}}-OH$
Ac_2O	$CH_3-\overset{O}{\overset{\|}{C}}-O-\overset{O}{\overset{\|}{C}}-CH_3$

ORGANIC REACTION MECHANISM – II

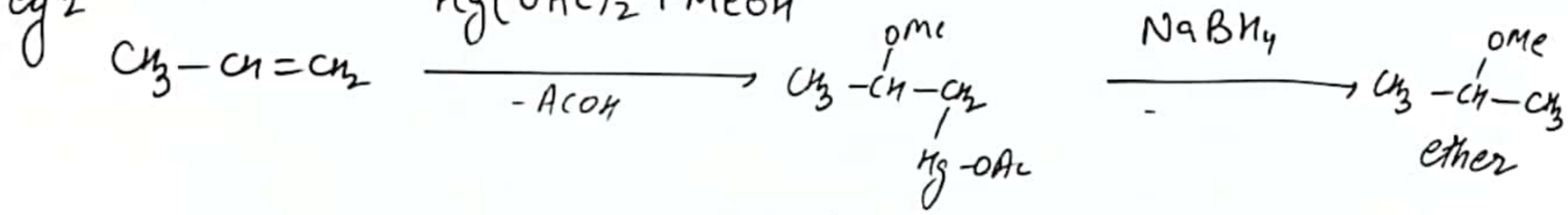
ORM-II

(III) Electrophilic Addition Reactions :- (A_E)

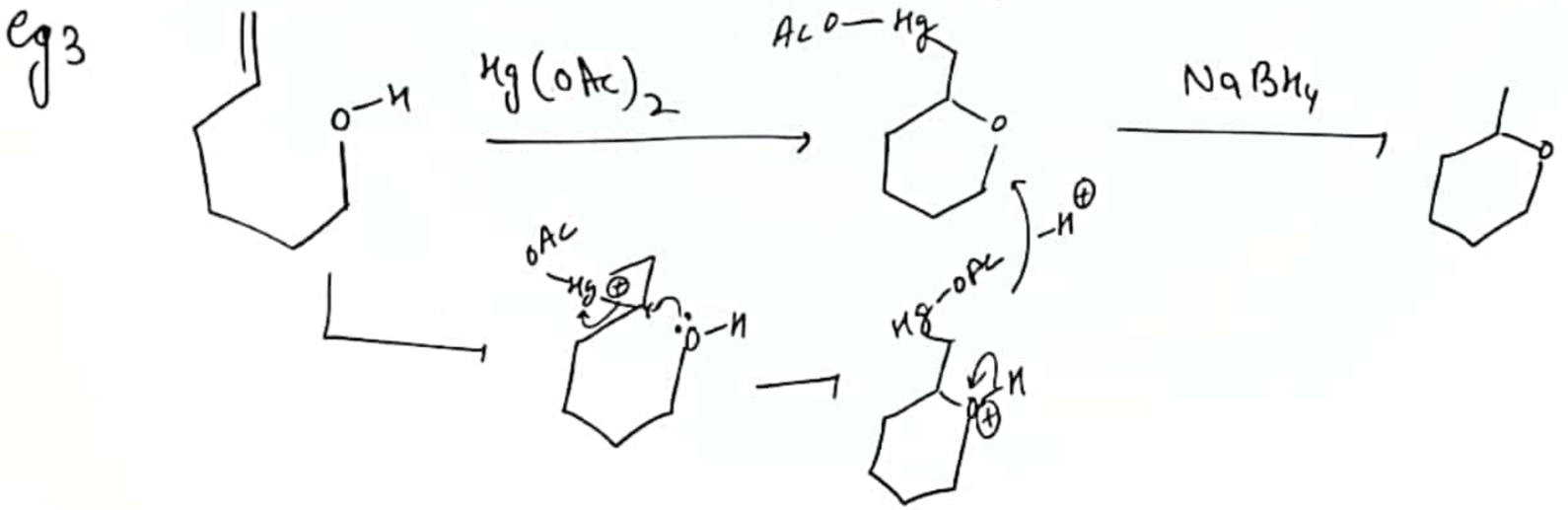
(8) Oxy-mercuration - demercuration :-

→ Acc. to mechanism - II

eg 2



eg 3



ORGANIC REACTION MECHANISM – II

ORM-II

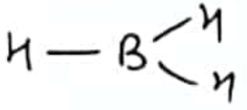
(III) Electrophilic Addition Reactions :- (A_E)

(9) Hydroboration :-

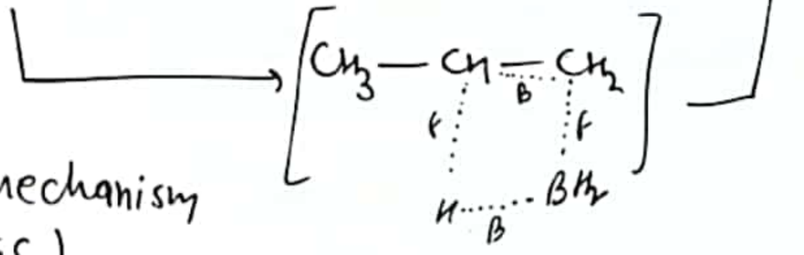


$CH_3-CH=CH_2 \xrightarrow[\text{Syn addition}]{\frac{1}{2}B_2H_6 - THF} CH_3-\underset{H}{CH}-CH_2-\underset{H}{\overset{H}{B}}-H$

$\xrightarrow{2\ CH_3-CH=CH_2} \left[CH_3-\underset{H}{CH}-CH_2\right]_3B$



Electrophile



⇒ Pericyclic mechanism
(Cyclic T.S.)

4 MCTS

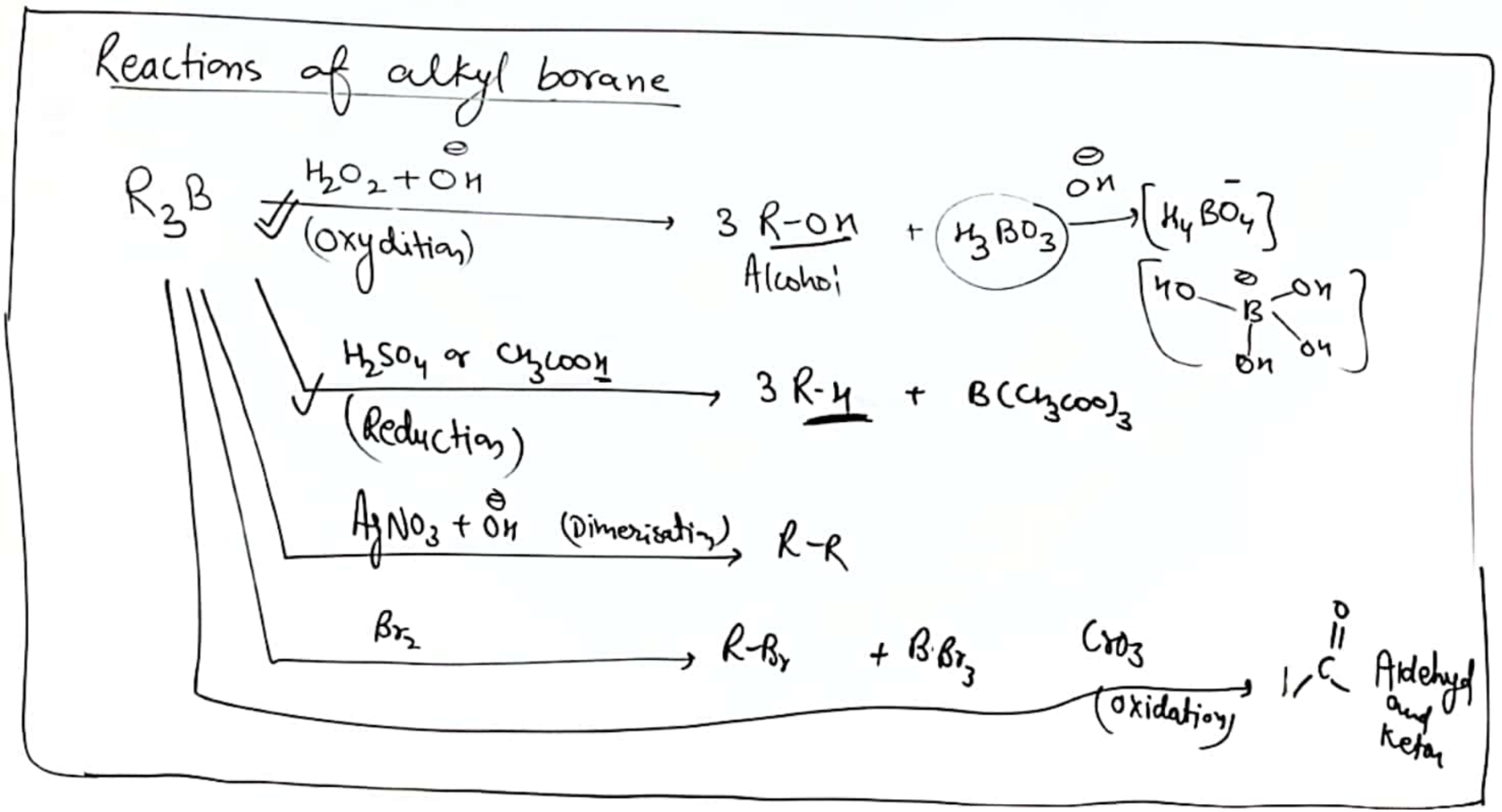
ORGANIC REACTION MECHANISM – II

ORM-II

(III) Electrophilic Addition Reactions :- (A_E)

(9) Hydroboration :-





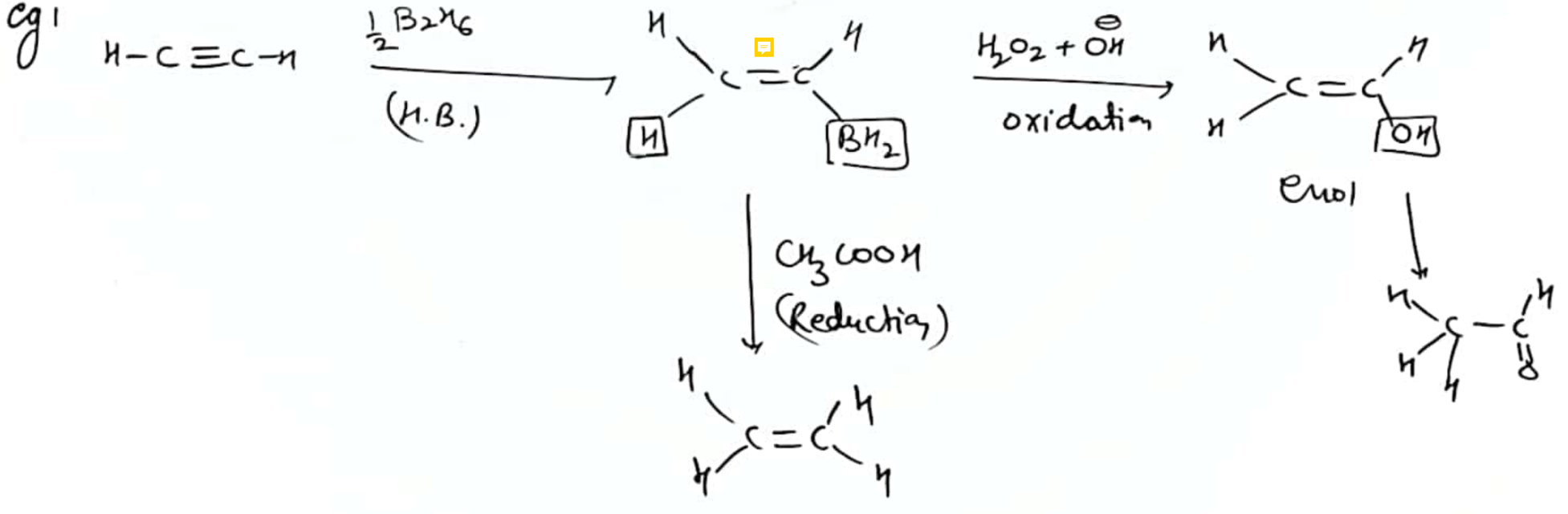
ORGANIC REACTION MECHANISM – II

ORM-II

(III) Electrophilic Addition Reactions :- (A_E)

(9) Hydroboration :-

eg :

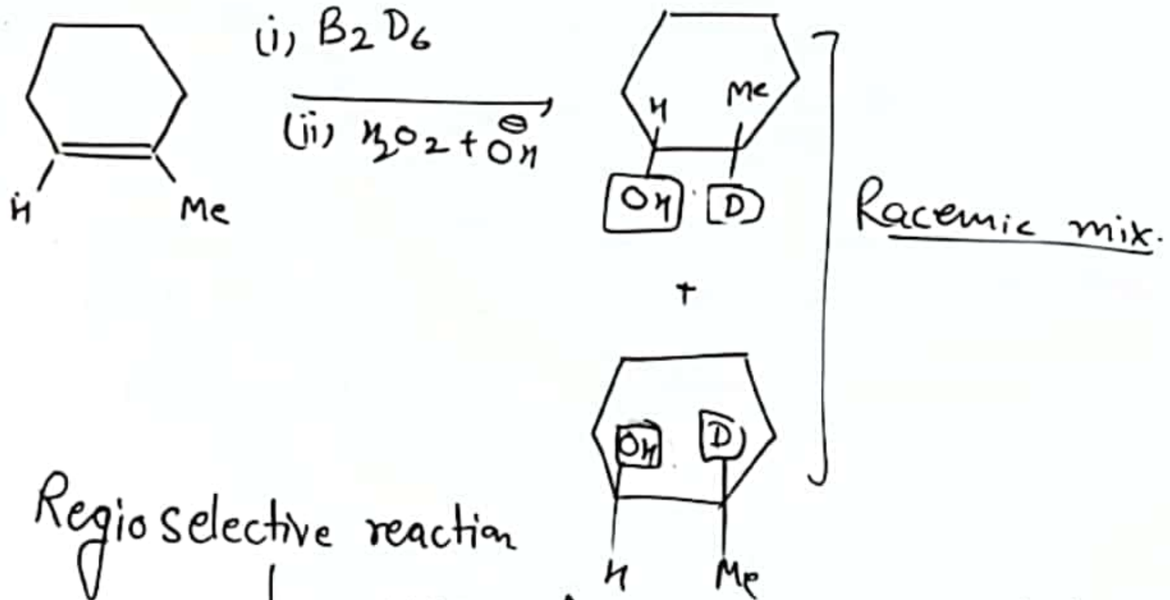


ORGANIC REACTION MECHANISM – II

(III) Electrophilic Addition Reactions :- (A_E)

(9) Hydroboration :-

eg 2



⇒ Regioselective reaction

↳ Addition of B on lesser crowded carbon of C=C / C≡C

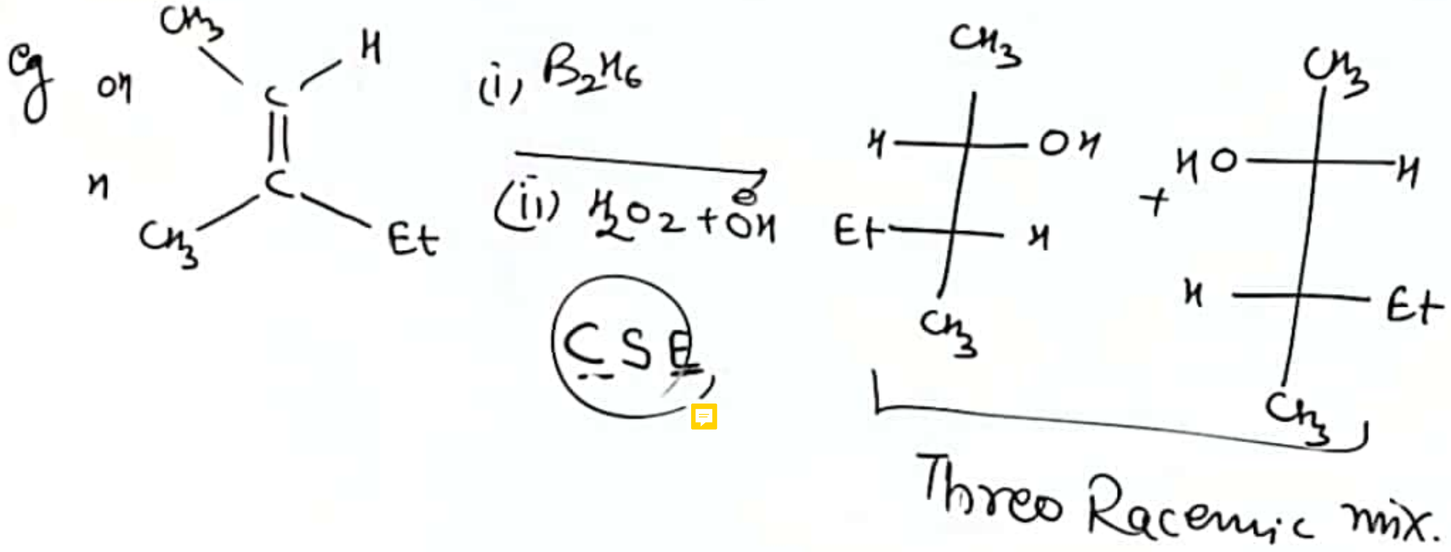
⇒ Stereospecific reaction = Syn addition (CSE & TST)

ORGANIC REACTION MECHANISM – II

ORM-II

(III) Electrophilic Addition Reactions :- (A_E)

(9) Hydroboration :–

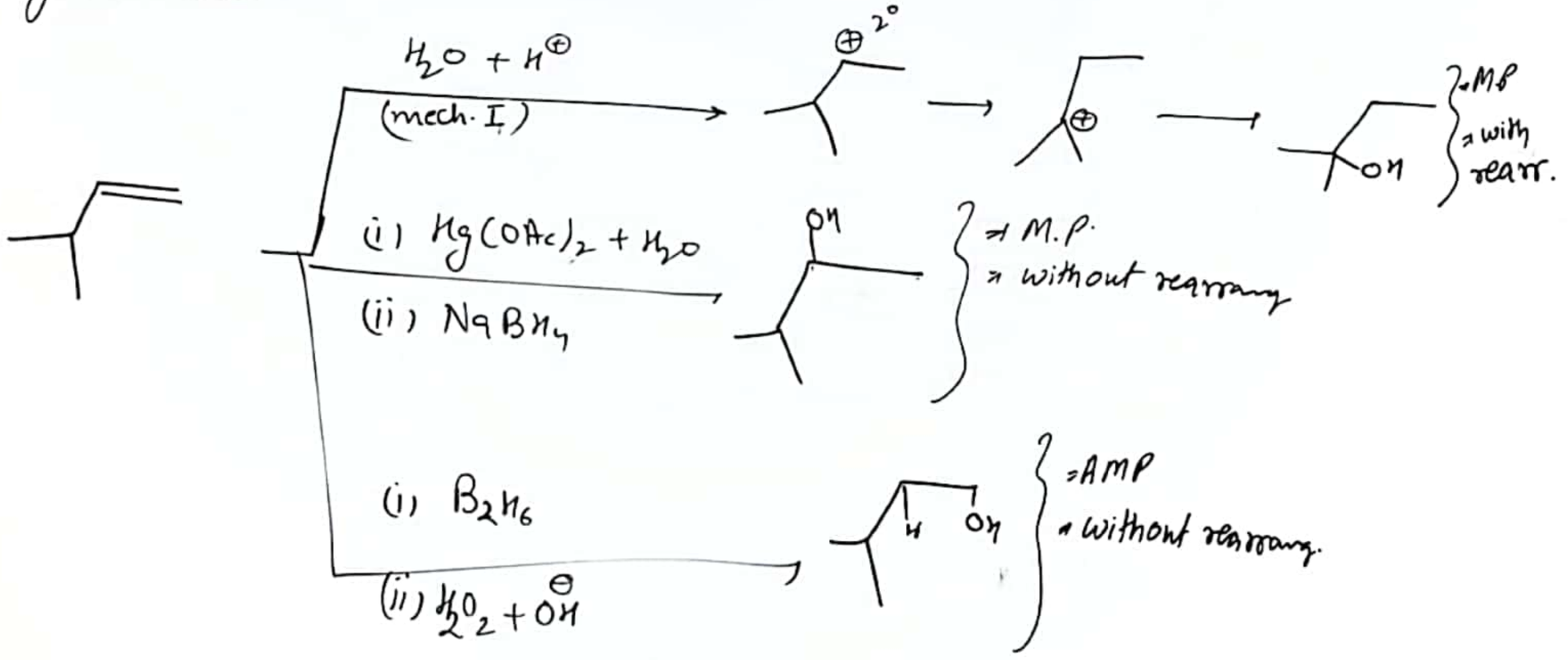


Threo Racemic mix.

ORGANIC REACTION MECHANISM – II

(III) Electrophilic Addition Reactions :- (A_E)

(9) Hydroboration :–



ORGANIC REACTION MECHANISM - II (ORM-II)

(IV) Free Radical Addition Reactions :- (A_{FR})

On unsymmetrical alkenes or alkynes, anti Markovnikov's addition takes place of HBr in presence of peroxide or $h\nu$. (Peroxide effect / Kharash effect)

eg 1 $CH_2=CH_2$

- $\xrightarrow{HBr}$ CH_3-CH_2-Br (A_E)
- $\xrightarrow{HBr + H_2O_2}$ CH_3-CH_2-Br (A_{FR})

eg 2 $CH_3-CH=CH_2$

- $\xrightarrow{HCl}$ $CH_3-\underset{Cl}{CH}-CH_3$
- $\xrightarrow{HBr}$ $CH_3-\underset{Br}{CH}-CH_3$
- $\xrightarrow{HI}$ $CH_3-\underset{I}{CH}-CH_3$
- $\xrightarrow{HCl + R_2O_2}$ $CH_3-\underset{Cl}{CH}-CH_3$
- $\xrightarrow{HI \quad R_2O_2}$ $CH_3-\underset{I}{CH}-CH_3$
- $\xrightarrow{HBr + R_2O_2}$ (or $HBr + h\nu$) AMP major

MP = Majo
$\Rightarrow A_E$
$\Rightarrow C^{\oplus}$
$\Rightarrow$ Rearr. =

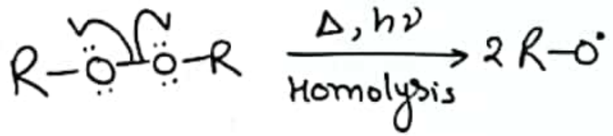
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ORGANIC REACTION MECHANISM - II (ORM-II)

(IV) Free Radical Addition Reactions :- (A_{FR})

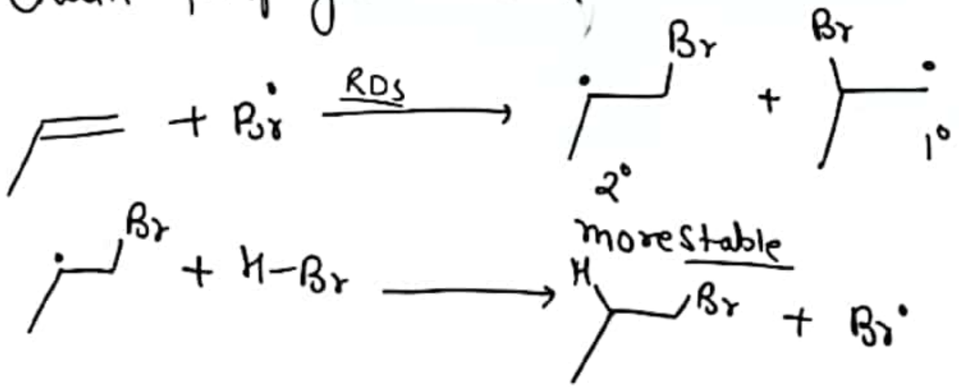
Mechanism

(a) Chain initiation step



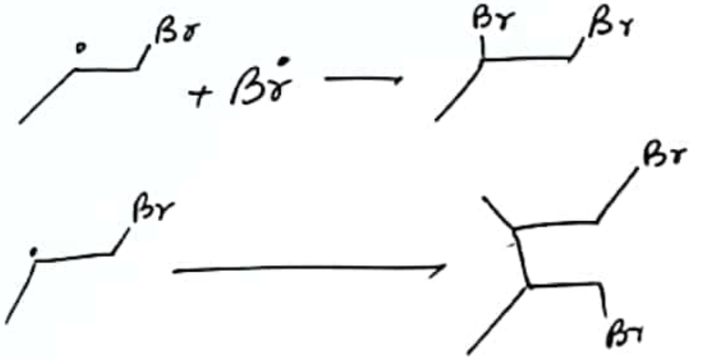
$$R-O^{\bullet} + H-Br \longrightarrow R-O-H + Br^{\bullet}$$

(b) Chain propogation step



(c) Chain termination step

$$Br^{\bullet} + Br^{\bullet} \longrightarrow Br_2$$



Note = Out of four HX only HBr gives free radical addition because only for H-Br the two Chain propagation steps are exothermic (ΔH = -ve), (Thermodynamically more favourable.)

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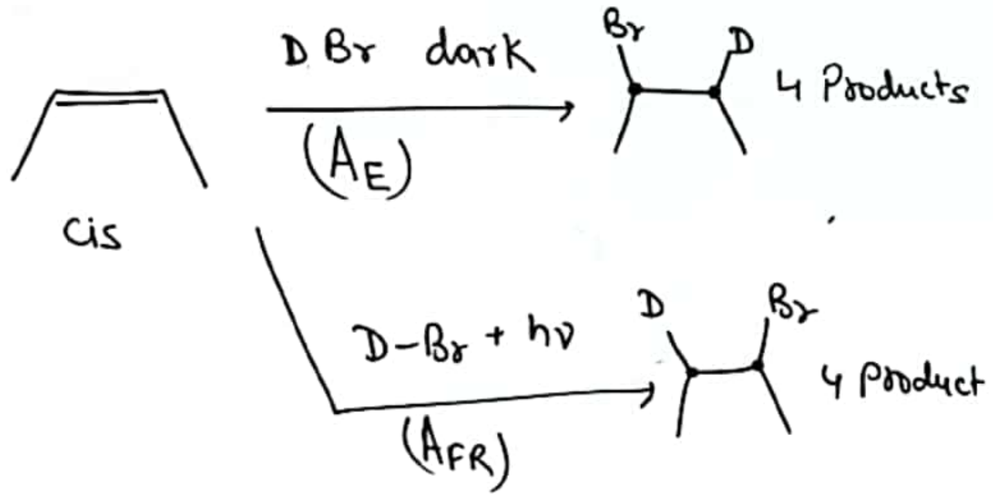
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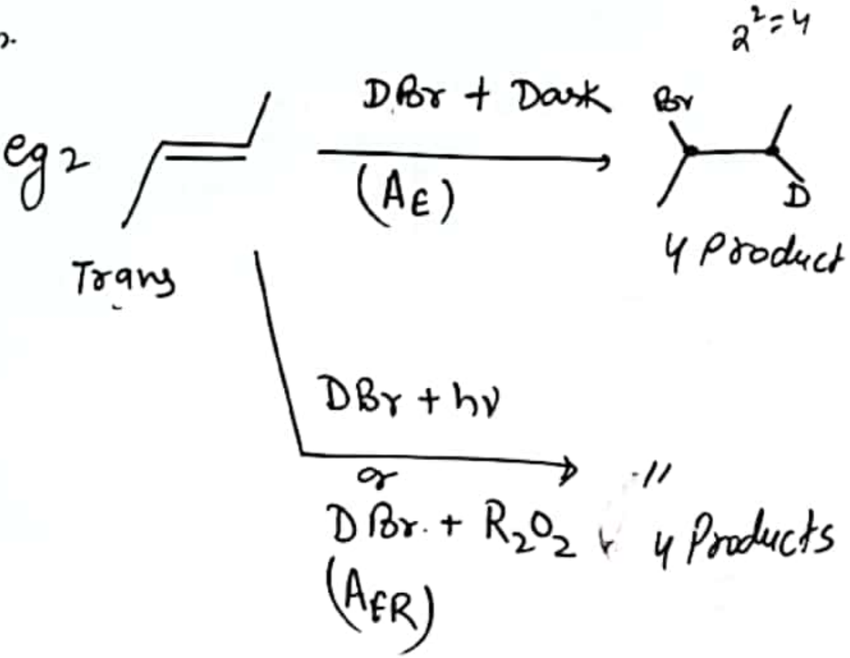
⇒ This reaction (A_{FR}) is Non stereospecific reaction.

⇒ No Rearrangement in free radical addition.

Eg 1



Eg 2

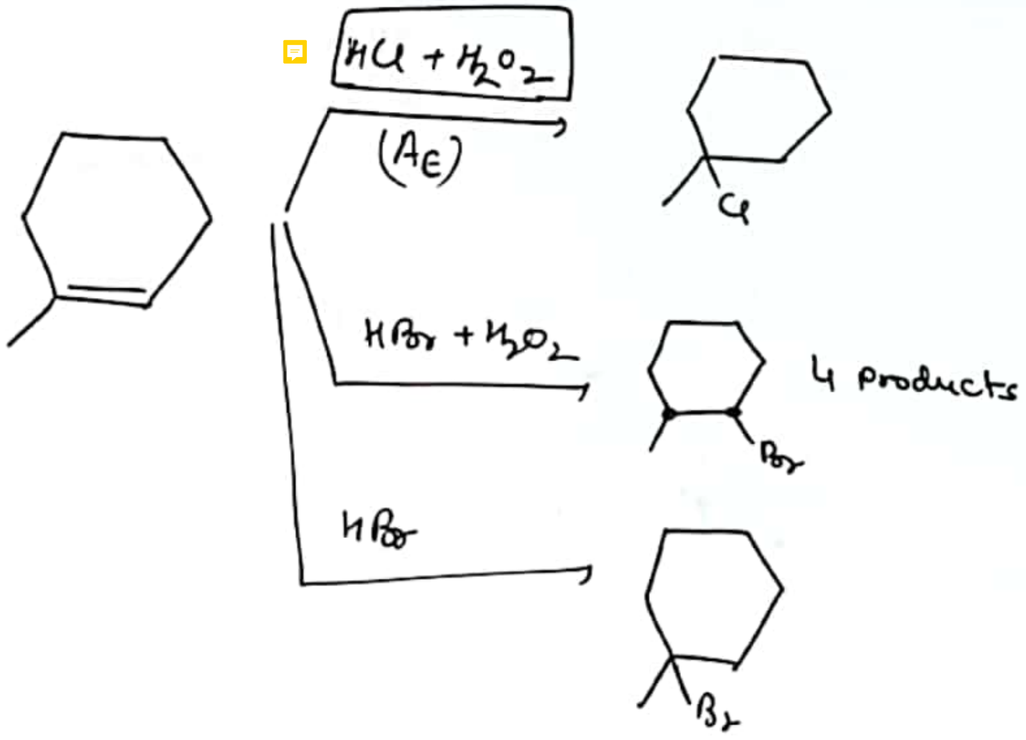


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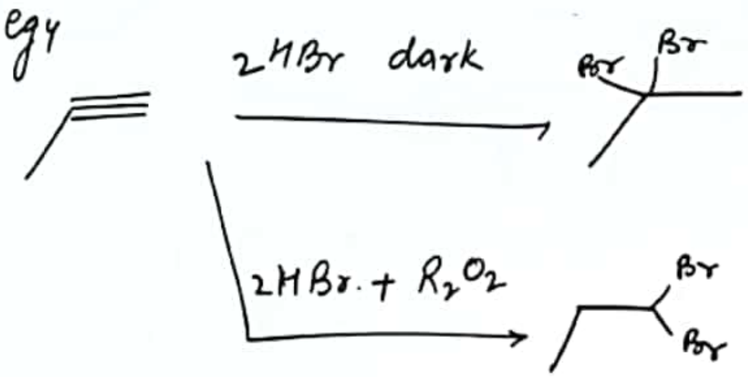
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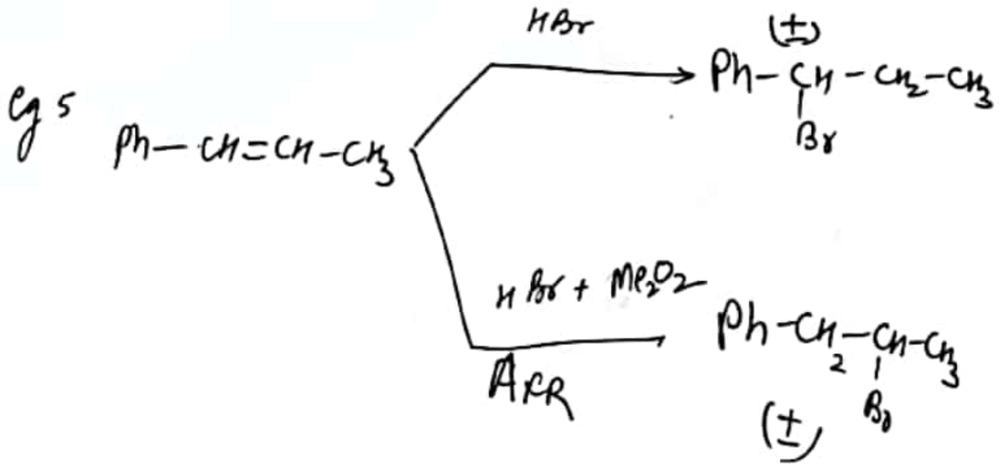
eg 3



eg 4



eg 5

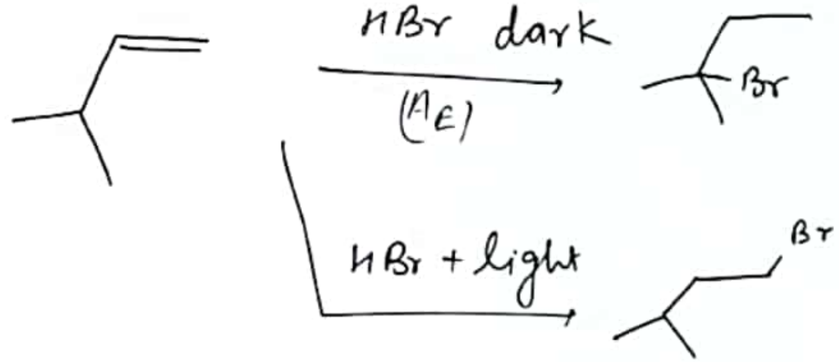


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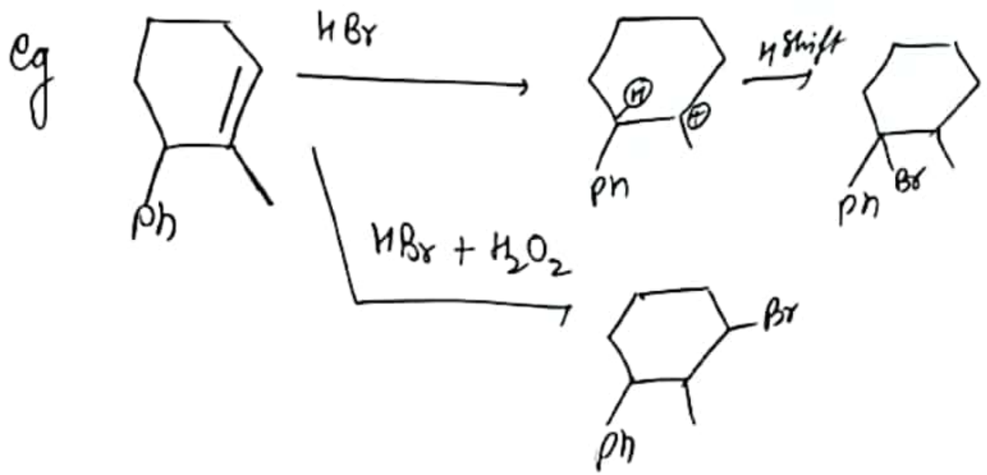
ORGANIC REACTION MECHANISM - II (ORM-II)

(IV) Free Radical Addition Reactions :- (A_{FR})

eg



eg

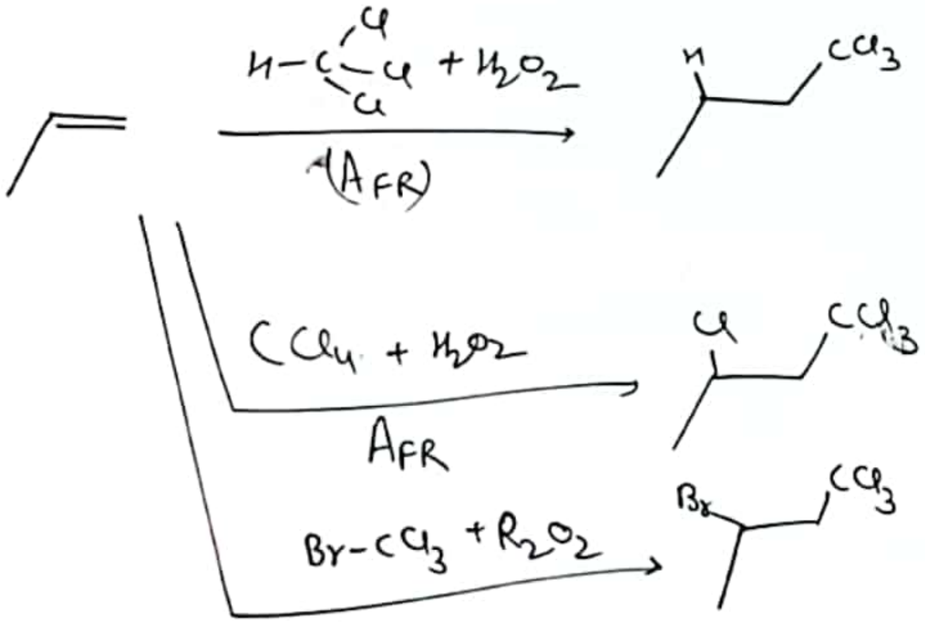


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(IV) Free Radical Addition Reactions :- (A_{FR})

⇒ Other examples of A_{FR}



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ORGANIC REACTION MECHANISM - II (ORM-II)

